Practically-A-Book Review: Rootclaim $100,00
Lab Leak Debate

I watched 15 hours of COVID origins arguments so you don't have to - but you should!

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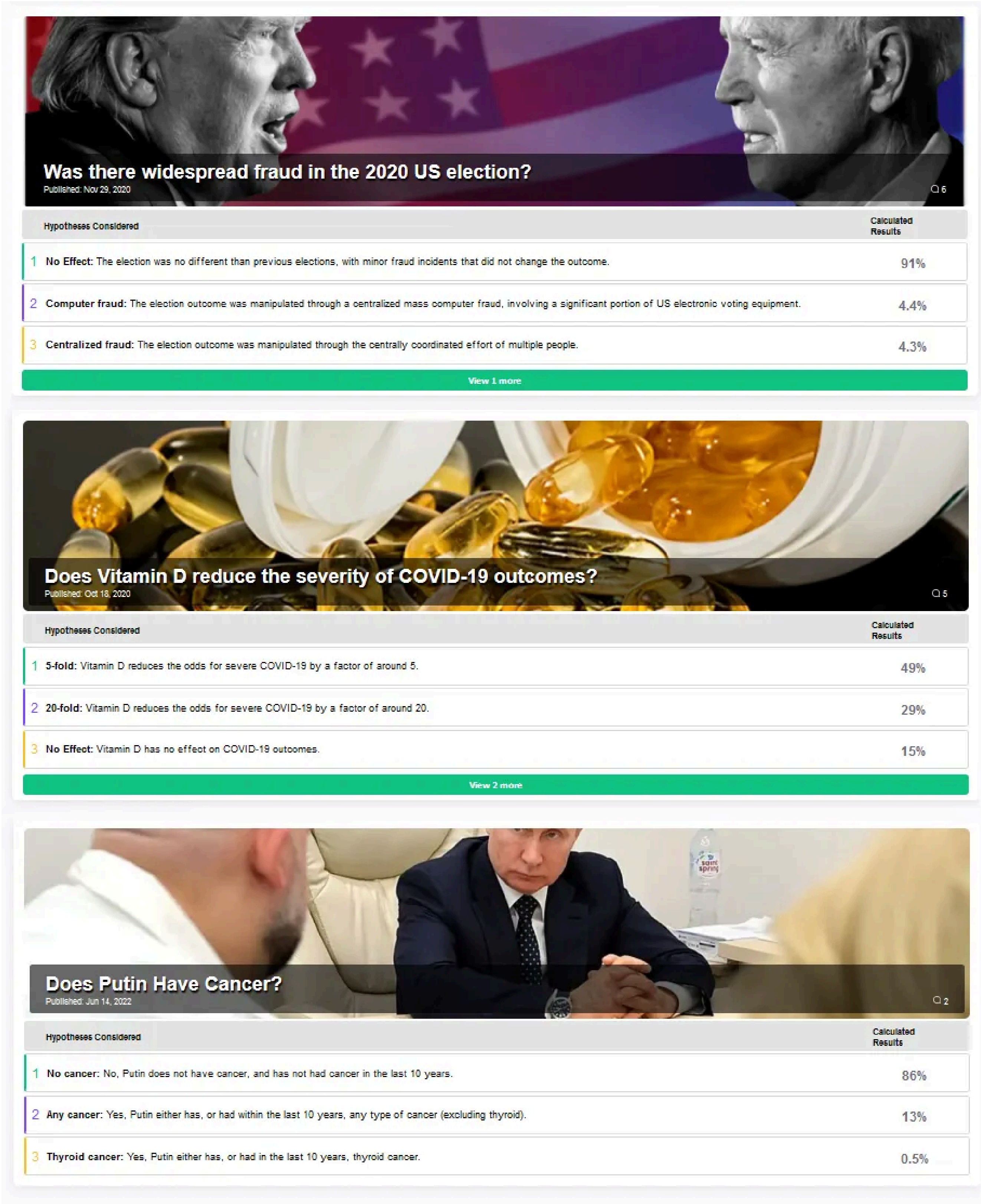
I. Saar's COV-2

Saar Wilf is an ex-Israeli entrepreneur. Since 2016, he's been developing a new forr
reasoning, meant to transcend normal human bias.

His method - called Rootclaim - uses Bayesian reasoning, a branch of math that explains the right way to weigh evidence. This isn't exactly new. Everyone supports
Bayesian reasoning. The statisticians support it, I support it, Nate Silver wrote a wh
book supporting it.

But the joke goes that you do Bayesian reasoning by doing normal reasoning while muttering "Bayes, Bayes, Bayes" under your breath. Nobody - not the statisticians,
Nate Silver, certainly not me - tries to do full Bayesian reasoning on fuzzy real-worl
problems. They'd be too hard to model. You'd make some philosophical mistake converting the situation into numbers, then end up much worse off than if you'd trie
normal human intuition.

Rootclaim spent years working on this problem, until they were satisfied their meth
could avoid these kinds of pitfalls. Then they started posting analyses of different open problems to their site, rootclaim.com. Here are three:



For example, does Putin have cancer? We start with the prior for Russian men ages 60-69 having cancer (14.32%, according to health data). We adjust for Putin's heal lifestyle (-30% cancer risk) and lack of family history (-5%). Putin hasn't vanished from the world stage for long periods of time, which seems about 4x more likely to true if he didn't have cancer than if he did. About half of cancer patients lose their h and Putin hasn't, so we'll divide by two. On the other hand, Putin's face has gotten more swollen recently, which happens about six times more often to cancer patient than to others, so we'll multiply by six. And so on and so forth, until we end up with final calculation: 86% chance Putin doesn't have cancer, too bad.

This is an unusual way to do things, but Saar claimed some early victories. For example, in a celebrity Israeli murder case, Saar used Rootclaim to determine that t main suspect was likely innocent, and a local mental patient had committed the crir later, new DNA evidence seemed to back him up.

One other important fact about Saar: he is very rich. In 2008, he sold his fraud detection startup to PayPal for $169 million. Since then he's founded more compan made more good investments, and won hundreds of thousands of dollars in professional poker.

So, in the grand tradition of very rich people who think they have invented new form of reasoning, Saar issued a monetary challenge. If you disagree with any of his Rootclaim analyses - you think Putin does have cancer, or whatever - he and the Rootclaim team will bet you $100,000 that they're right. If the answer will come out eventually (eg wait to see when Putin dies), you can wait and see. Otherwise, he'll accept all comers in video debates in front of a mutually-agreeable panel of judges

Since then, Saar and his $100,000 offer have been a fixture of Internet debates everywhere. When I argued that Vitamin D didn't help fight COVID, people urged m bet against Saar, and we had a good discussion before finally failing to agree on ter When anti-vaccine multimillionaire Steve Kirsch made a similar offer, Saar took him on it, although they've been bogged down in judge selection for the past year.

Rootclaim also found in favor of the lab leak hypothesis of COVID. When Saar talke about this on an old ACX comment thread, fellow commenter tgof137 (Peter Miller) agreed to take him up on his $100K bet.

At the time, I had no idea who Peter was. I kind of still don't. He's not Internet famo He describes himself as a "physics student, programmer, and mountaineer" who "obsessively researches random topics". After a family member got into lab leak a f years ago, he started investigating. Although he started somewhere between neutr and positive towards the hypothesis, he ended up "90%+" convinced it was false. I also ended up annoyed: contrarian bloggers were raking in Substack cash by promoting lab leak, but there seemed to be no incentive to defend zoonosis.

Unlike Saar, Peter was not especially rich. $100K represented a big fraction of his n worth. But (he wrote me in an email):

> It was a moderately large financial risk for me ... I [expected] a smart and unbias person would vote for zoonosis with, say, 80% odds after seeing all the evidenc both judges voting for lab origin is uncorrelated, that's 20% squared, and it was pretty low odds of a catastrophic financial risk for me.
>
> I wasn't highly worried about losing the debate because I was wrong about the science. I put in enough effort to know I'm probably correct there. My biggest fe was that I'd choke at the debate for some reason, that I'd be too anxious and particularly that I'd be unable to sleep the night beforehand. I have zero prior debate experience to rely upon.

If this seems like a weirdly blase attitude towards risk, Peter told blogger Philipp Markolin that he "is a mountain climber where sometimes there is a 5% chance to c and the stakes are just not that high for a debate."

Unlike the eternally bogged-down Saar-Kirsch debate, here things moved quickly. two contestants put out a call for judges on the ACX subreddit, and agreed on:

- Will van Treuren, a pharmaceutical entrepreneur with a PhD from Stanford and background in bacteriology and immunology.
- Eric Stansifer, an applied mathematician with a PhD from MIT and experience i mathematical virology.

...both of whom received $5,000 as payment for their ~100 [1] hours of work, paid by the two contestants along with their $100,000 table stakes [2].

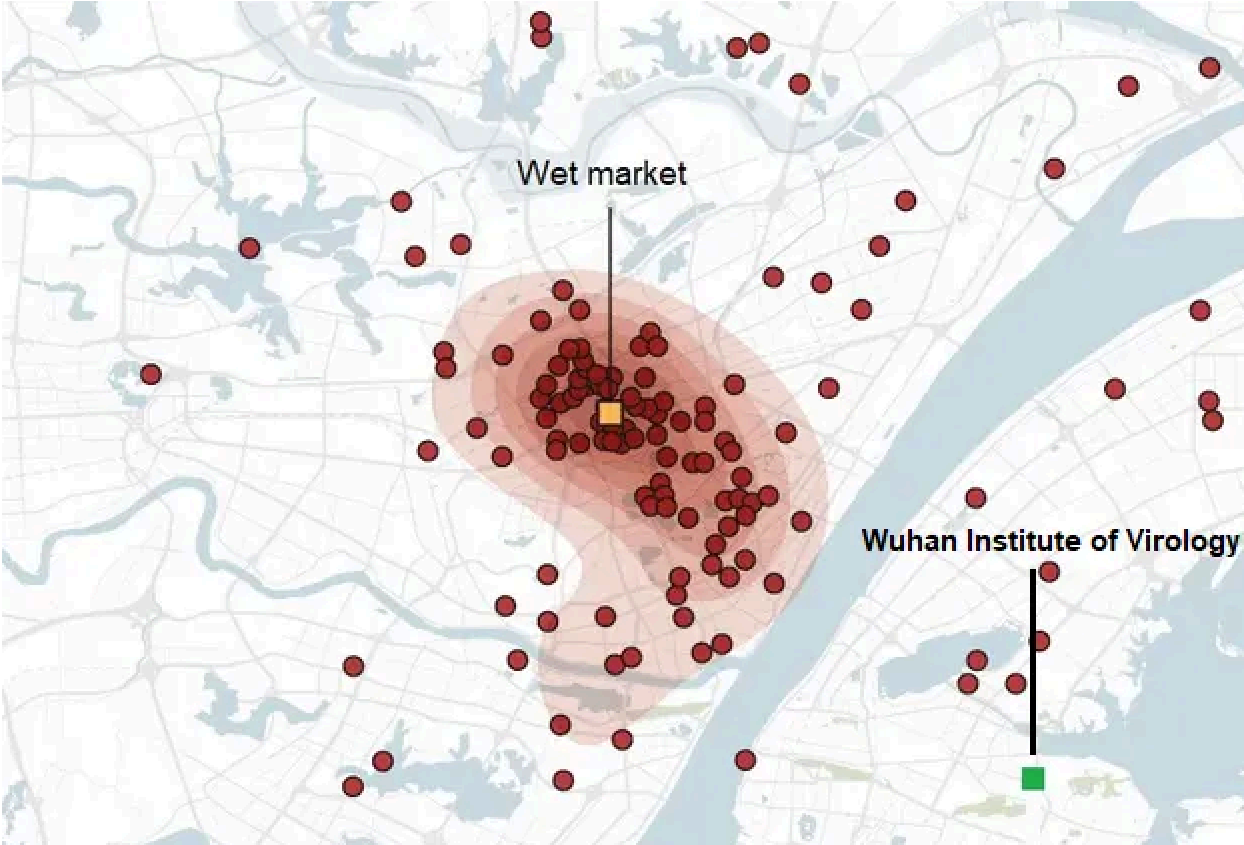
The format would be three sessions, each consisting of hour-and-a-half argument both sides, then three hours for the debaters to answer questions from the judges each other.

II. The Debate

Below, I've included the videos from each session, plus my (long) summary if you prefer text. In the second session (on viral genetics) biotech entrepreneur and lab l expert Yuri Deigin stood in for Saar; Peter continued to represent himself.

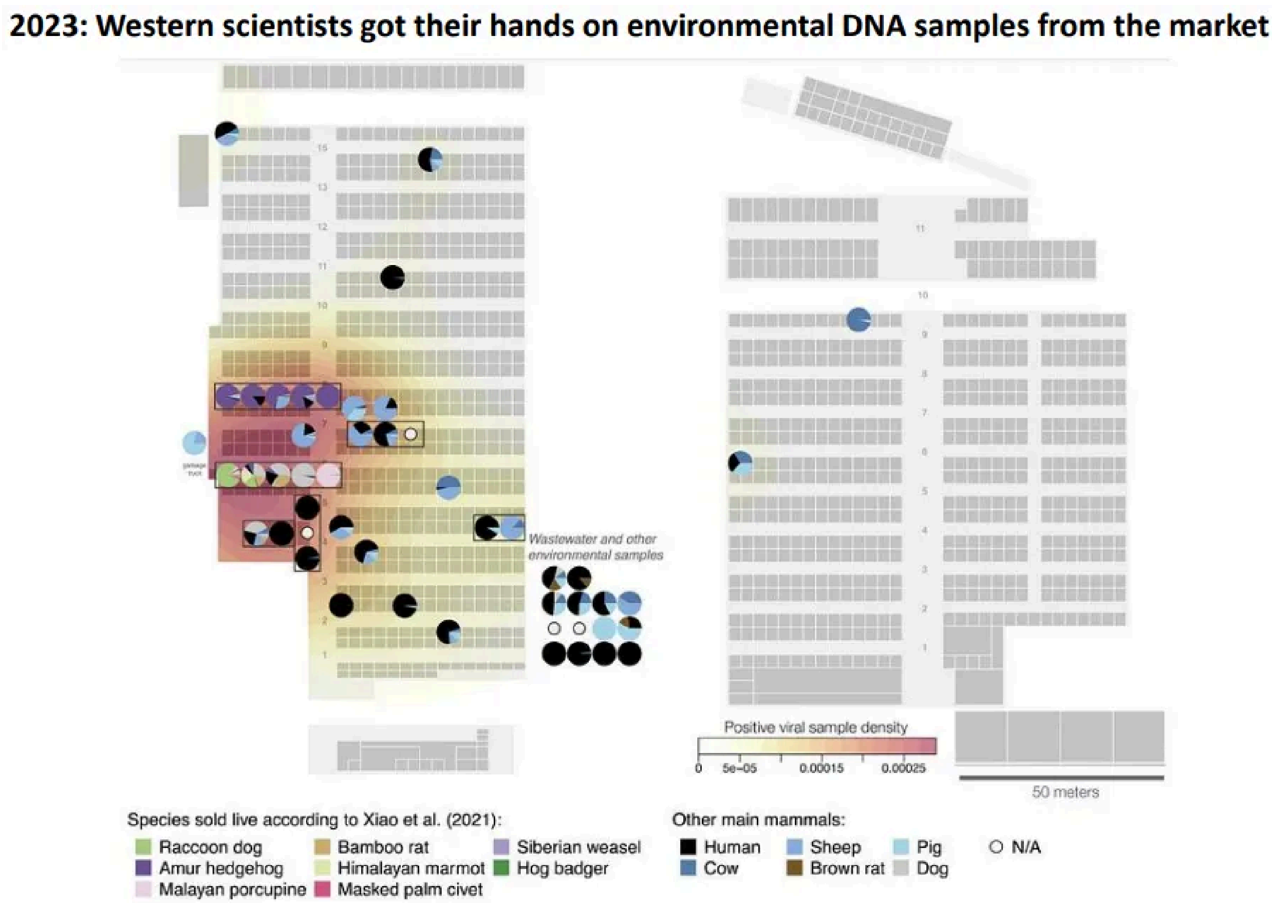
Session 1: Epidemiology

Peter: The first officially confirmed COVID case was a vendor at the Wuhan wet market. So were the next four, and half of the next 40. A heat map of early cases is obviously centered on the wet market, not on the lab. The wet market and the lab a about 6 miles away as the crow flies, or a 15 mile / half hour drive.



Location of COVID cases in December 2020. Source: NYT, slightly edited.

A map of cases at the wet market itself shows a clear pattern in favor of the very southwest corner:



The southwest corner is where most of the wildlife was being sold. Rumor said that included a stall with raccoon-dogs, an animal which is generally teeming with weirc coronaviruses, and is a plausible intermediate host between humans and bats:

Awwww, come on, you can’t stay mad at this little guy.

China said this rumor was false and refused to release any information. Scientists w finally able to confirm the existence of the raccoon-dog shop in the funniest possib way: a virologist had visited Wuhan in 2014, saw the awful conditions in the shop, a took a picture as an example of the kind of place that a future pandemic might star

Source: NPR. To be fair, we have only the scientist's word that this is why he had the picture. But he definitely did have it.

People say it would be a surprising coincidence if a zoonotic coronavirus pandemic just so happened to start in a city with a big coronavirus research lab, and this is tru But it would be an even more surprising coincidence if a lab-leak coronavirus pandemic just so happened to first get detected at a raccoon-dog stall in a wet market!

Saar: It's not clear that the first case was at the wet market; a certain Mr. Chen, wit no connection to the market, seems to have fallen sick on December 8. An SCMP article suggested there were 92 previously-undetected cases suspicious for COVIE far back as November. And even if half of the first forty universally-agreed-upon ca had market connections that means another half didn't.

There was a bias towards detecting cases at the market: because authorities thoug the market was the origin, and because everyone was thinking about zoonosis afte SARS1, they only screened/diagnosed people with a market connection. One of the few non-market-connected COVID cases detected during this period was only detected because he was the relative of a hospital worker; the worker noticed the signs and insisted they go to the hospital despite the lack of a wet market connecti

Although the map of positive samples and cases at the market was centered near t raccoon-dog stall, that could be because that area was sampled more; it's also clos to the mahjong room, where visitors and vendors at the market would go and unwir in a tight, poorly ventilated area.

The next session will focus more on the WIV, but the short version is that they were doing lots of gain of function research. So one story compatible with the evidence i that a worker at WIV got infected with their modified coronavirus and passed it to h contacts. COVID started spreading quietly a few weeks to months before the first market-related case was detected. This accounts for the 92 earlier cases, Mr. Cher case, and the half of officially-detected cases with no wet market association. Ther infected person went to the market, causing a super-spreader event. Some of the infected market patrons went to the hospital, where doctors traced it back to the market and told other doctors to be on the lookout for wet market patrons coming i with weird viral pneumonias. They found some, declared victory, and the few anomalies - like the hospital worker's relative - were forgotten, or assumed to have wet market connections that nobody could find. China quashed all evidence of the research (as was done in previous lab leak cases, eg the USSR) so all we have is th apparent wet market links that Peter found so convincing.

Peter: The supposed pre-wet-market cases are confirmed fakes.

Yes, the WHO did an investigation of whether there might have been COVID cases circulating before the wet market, and identified 92 unusual pneumonias that merit further review. But their final investigation, which included testing samples from the people after good tests became available, found that none of these people really ha COVID.

As for Mr. Chen, he said in an interview that he was hospitalized for dental issues o December 8, caught COVID in the hospital on December 16, and then was erroneou reported as "hospitalized for COVID on December 8". The December 16 date is afte the first wet market cases.

Further, it seems epidemiologically impossible for COVID to have been circulating much before the first cases were officially detected December 11. The COVID pandemic doubles every 3.5 days. So if the first infection was much earlier - let's sa November 11 - we would expect 256x as much COVID as we actually saw. Even if th first couple of cases were missed because nobody was looking for them, the numb of hospitalizations, deaths, etc, in January or whenever were all consistent with the number of people you'd expect if the pandemic started in early December - and no consistent with 256x that many people.

So probably we should just accept that the first reported case - a wet market vend December 11 - was very early in the pandemic. She wasn't literally the first case - th would most likely have been someone who worked at the raccoon-dog shop, whos case might (like 95% of COVID cases) have been mild enough not to come to medi attention. But she was certainly very early.

Although authorities eventually decided COVID spread through a wet market and started deliberately looking for wet market connections, this only happened on December 30. So the earliest cases - including the 40 very earliest cases where ha came from the wet market - weren't biased (at least not through that particular rou So the claim that "the first case, and half of the first 40 cases, had wet market connections" stands as real and convincing evidence.

Although the exact center of the map of positive COVID samples in the wet market was the mahjong room, the samples taken from the mahjong room were not, themselves, positive (cf: although a low-resolution population density map of New York might show Central Park in the exact center of the population density gradient Central Park does not itself have population).

There was no real “super-spreader event” at the wet market. There was a slow burn one case the first day, a few more the next day, a few more the day after that. It’s ha to see how a single visit from an infected lab worker could do that.

So the only way it could possibly be a lab leak is if the lab leaked sometime in late November, infected exactly one lab worker, that worker went straight to the wet market, infected a vendor, then went home, quarantined, recovered, and all other cases were downstream of that first infected wet market vendor. This is unparsimonious.

Saar: The only source saying that Mr. Chen got sick early was an anonymous interview. And even if he was later than the first wet market cases, nobody was able find any wet market connections. This means that whoever infected him was earlier than the index case and not linked to the wet market.

Peter argued that COVID couldn’t have been more than a few weeks old when the f wet market cases were detected. But this was based on its known doubling rate. If pre-discovery COVID had a slower doubling time than known COVID, it could have been around longer. And post-lockdown serology suggested numbers that were lar than claimed at the time. So contra Peter’s claims, the infection could have been go on longer, which wouldn’t require the first lab worker to go straight to the market. It could have been weeks.

Dr. Jesse Bloom’s investigation of the wet market samples, considered the final and most conclusive, failed to find a clear connection between COVID and raccoon-dog or any other animals. Although the concentration of positive samples seemed highe near the raccoon dog stall, if you do a formal statistical analysis of which animals’ D was found near COVID samples most often, raccoon dogs are near the bottom. The top is wide-mouth bass, which can’t get COVID. This is obviously contamination, probably from infected humans touching wide-mouth bass tanks or something.

Although the Chinese data included a negative sample from a mahjong table, it included a mention of poultry being sold nearby, which might mean this wasn't the mahjong room itself, but some other mahjong table at a poultry shop elsewhere in t market, and (dry) mahjong tables might not hold the virus well anyway.

Peter: Raccoon-dogs were sold in various cages at various stalls, separated by air gaps big enough to present a challenge for COVID transmission, and there's no rea to think that one raccoon-dog would automatically pass it to all the others. The statistical analysis just proves there were many raccoon-dogs who didn't have CO\ But you only need one. The raccoon dog shop and the drain leading out of the raccoon dog shop had some of the highest positive sample rates, which is more interesting than a statistical analysis which everyone agrees must be wrong (since i favors bass).

It's unclear why the negative mahjong sample says something about poultry, but based on the stated location, it's definitely the one in the mahjong room.

Session 1.5: Lineages

This was technically part of Session 2, but formed enough of a discrete topic that I found it confusing to intermix it with all the other viral genetics points. I'm spinning out into a separate summary, but the videos are all in the next session.

Yuri: The coronavirus eventually mutated into many different strains. But the first b split, seen in some of the earliest samples, is between two different sub-strains cal Lineage A and Lineage B, which differ by two mutations. In these two mutations, Lineage A is the same as BANAL-52, a bat virus which is the closest-known relative COVID, but Lineage B is different. Since COVID probably evolved from something li BANAL-52, Lineage A must have come first, spread for a while, and then gotten tw new mutations, turning it into Lineage B.

All of the cases at the wet market, including the first detected case, were Lineage E Lineage A wasn't discovered until about a week later, and none of the Lineage A patients had been to the wet market.

Lineage A (left) was used by the Minoan Cretans, but has never been deciphered. Lineage B (right) was used by the Mycaeneans for lists of palace goods.

This matches Saar's story above. The lab leaked to somewhere else in Wuhan, not wet market. The virus spread undetected in the population for a while. During this time, it mutated to Lineage B. Then one of the people with Lineage B went to the w market and started a superspreader event. The authorities sampled the patients, found Lineage B, then started looking elsewhere. Later they detected some of the earlier Lineage A cases.

The market is unlikely to be the origin of the pandemic, because the original Lineag strain wasn't found there.

Peter: Although Lineage A is evolutionarily older, Lineage B started spreading in humans first.

We know this because Lineage B is more common. Throughout the early pandemic until the D614G variant drove all other strains extinct, a consistent 2/3 of the cases were B, compared to 1/3 A. Both strains spread at the same rate, so the best explanation is that B started earlier than A. Since COVID doubles every 3-4 days, probably Lineage B started 3-4 days earlier than Lineage A, which explains why it's always been twice as many cases.

But also, Lineage B also has more internal genetic diversity than Lineage A. In gene older viruses have more genetic diversity (the "molecular clock"). This is further evidence that B started spreading first. Pekar 2022 and Pipes 2021 do analyses wit known parameters for spread rate and diversity, and find 90%+ odds that Lineage I was the first one in humans.

Why did the older strain start spreading later? Probably the virus crossed from bats into raccoon-dogs on some raccoon-dog farm out in the country. It spread in the raccoon-dogs for a while, racking up mutations, including the (less mutated) Linea A strain and the (slightly more mutated) Lineage B strain. Then several raccoon-do were taken to Wuhan for sale, including one with Lineage A and another with Linea B. The one with Lineage B passed its virus to humans earlier. Then 3-4 days later, t Lineage A one passed *its* virus to humans.

Lineage A was first found in a Wuhan neighborhood right next to the wet market (closer to the wet market than 97% of Wuhan's population). Again, it would be a bizarre coincidence if a lab leak pandemic was first detected at a wet market. But it would be an even *more* bizarre coincidence if a lab leak pandemic separated into tv strains, and *both* were first detected at a wet market!

Although no known wet market cases were Lineage A, a positive Lineage A environmental sample was found at the wet market, and everyone agrees most cas went undetected. So maybe the Lineage B raccoon-dog spread its virus to a vendo and that sub-strain mostly stayed in the market. But the Lineage A raccoon-dog spread its virus to a customer, who went back to his house nearby, and that strain spread in the neighborhoods next to the market.

This is the only story that explains the evolutionary precedence of A, the greater spread and older molecular clock of B, and the fact that both strains were first foun very close to the wet market.

Yuri/Saar: Lineage B could be more common and diverse because it got the advantage of a super-spreader event in the wet market.

There are a few scattered cases of intermediates between A and B, and a few othe scattered cases of lineages that seem even more ancestral (ie closer to the bat viru than either. This doesn't make sense in a double spillover hypothesis. But it does m sense if the lineages separated in human transmission somewhere between the lab and the first super-spreader event at the wet market.

Peter: Again, the wet market wasn't a super-spreader event. COVID spread in the w market at exactly its normal spread rate, doubling about once every 3.5 days. Stop calling the wet market a super-spreader event.

The scattered cases of "intermediates" are sequencing errors. They were all found the same computer software, which "autofills" unsequenced bases in a genome to most plausible guess. Because Lineage B was already in the software, depending o which part of a Lineage A virus you sequenced, you might get one half or the other autofilled as Lineage B, which looked like an "intermediate". We know this because the supposed "intermediates" were partial cases sequenced by this particular software. We can confirm this by noting that there are *too many* intermediates! Tha where Lineage A is (T/C) and Lineage B is (C/T), the software found *both* (T/T) "intermediates" and (C/C) "intermediates". But obviously there can only be one real intermediate form, and we have to dismiss one or the other. But in fact we can dism both, because they were both caused by the same software bug.

The scattered "progenitor" cases - those closer to the ancestral bat virus than eith or B - are reversions, ie cases where a new mutation in the virus happened to hit ar already-mutated base and shift it back towards the ancestral virus. We know this because all of these “progenitors” were scattered cases found months after the pandemic started, often in entirely different countries from Wuhan. If these were re progenitor viruses, they would have either fizzled out or exploded into a substantial portion of all cases, not be found one time in one guy in Malaysia. Given the numbe mutations the virus developed over the course of the pandemic, it’s inevitable that some of them would be mutations that bring it closer to the original bat virus, and in fact we find the number of “progenitors” found very nicely matches the number of progenitor-appearing viruses we would expect by chance. And in many cases, we *know* the “progenitors” are newer than the original lineages, because they also hav some of the later mutations that Lineage A or B picked up along the way, alongside their apparent ancestral-bat-virus-like mutations.

Session 2: Viral Genetics

Yuri: Two years before COVID, scientists at the Wuhan Institute of Virology, togethe with colleagues at the University of North Carolina, sent in a grant proposal for the DEFUSE program. This program, intended to locate and better understand potentia future pandemic viruses, involved going into bat caves and collecting new coronaviruses. Once they had them, they would do gain-of-function: specifically, th would add a furin cleavage site to make them more infectious and see what happer

(quick interlude: COVID's spike protein has two sections: one binds to human cells through the ACE2 receptor, the other helps fuse with the cell after binding. In order avoid the immune system, it hides both of these into one spike. But when it reaches cell, it needs to separate them again. It takes advantage of a human respiratory enzyme, furin, to do the separation - this also ensures that it only infects its primary target, human respiratory cells. The part of COVID that lets it get separated by furir called the "furin cleavage site". COVID's bat-virus ancestors were gastrointestinal viruses; the addition of a furin cleavage site was what made them respiratory viruse

We've found two close relatives of COVID: bat viruses called RATG-13 and BANAL- In particular, COVID looks more or less like BANAL-52 plus a furin cleavage site.

There are 1500 sarbecoviruses, members of the family of viruses that includes SAF and SARS2/COVID. None of them except COVID have furin cleavage sites. BANAL-! COVID's closest ancestor, doesn't even have anything resembling one that could mutate into a functional furin cleavage site like COVID's.

Instead, COVID - which mostly just resembles BANAL-52 with a few scattered sing point mutations - has twelve completely new nucleotides in a row - a fully formed fu cleavage site that came out of nowhere. There is nowhere else in the genome that COVID differs from BANAL-52 in such a profound way. It's just BANAL-52 plus a litt bit of random mutation plus a fully-formed furin cleavage site that came out of nowhere.

Further, the furin cleavage site is weird. It uses the protein arginine twice. But insteа of the nucleotides coding for arginine in the usual viral way, both times it uses the codons CGG - the way that higher animals code for arginine. This works fine - it's jı not how viruses do it.

So the obvious conclusion is that WIV, which said in 2018 that it was going to find viruses and add furin cleavage sites to them, found a close relative of BANAL-52 ar added a furin cleavage site. Since they were humans, and most familiar with the human way of encoding arginine, they added it as CGG both times.

COVID seemed surprisingly optimized for infecting humans. Of fifty animals it was tested in, including the usual coronavirus intermediate hosts (pangolins, raccoon-dogs, etc), it was best at infecting human cells. Further, a virus that enters a new species will usually show a burst of mutations as it "figures out" the best way to ad to that species' unique biology. But COVID has had a pretty constant mutation rate humans, from the beginning of the pandemic to the end. That suggests it was alrea adapted to humans. This could be because the lab screened for viruses with existir adaptations, because they passed it through humanized mice in the lab, or becaus adapted in the hundreds of undetected cases that happened between the lab and detection in the wet market.

Usually, research with potentially dangerous coronaviruses is done in BSL-3 or 4, ie high to very-high security. But WIV was irresponsibly doing it in BSL-2, ie medium security. The researchers weren't even required to wear masks. In general, about 1/500 labs will leak any given pathogen they're working on (?!). But because WIV w researching such an infectious virus in such an irresponsible way, the odds of a leal were much higher.

The most likely explanation for all these facts is that WIV went ahead and did the gа of-function research they said they were going to do (the particular DEFUSE grant proposal we know about got rejected, but it proves that Wuhan wanted to do this, a they could easily have gotten funding somewhere else, or done it out of their regula budget). They found a close relative of BANAL-52 and added a furin cleavage site a simple twelve-nucleotide insertion, using the human method of encoding arginine t their genetic engineers were familiar with. Then it leaked, spread for a while in the general Wuhan population, and eventually made it to the wet market where it got detected.

Peter: As mentioned earlier, the DEFUSE grant was rejected. Further, the grant saic that the Wuhan Institute of Virology was responsible for finding the viruses, and the University of North Carolina would do all the gain-of-function research. This was a reasonable division of labor, since UNC was actually good at gain-of-function research, and WIV mostly wasn't. They had done a few very simple gain-of-functio projects before, but weren't really set up for this particular proposal and were happ leave it for their American colleagues.

Even if WIV did try to create COVID, they couldn't have. As Yuri said, COVID looks li BANAL-52 plus a furin cleavage site. But WIV didn't have BANAL-52. It wasn't discovered until after the COVID pandemic started, when scientists scoured the are for potential COVID relatives. WIV had a more distant COVID relative, RATG-13. But you can't create COVID from RATG-13; they're too different. You would need BANA 52, or some as-yet-undiscovered extremely close relative. WIV had neither.

Are we sure they had neither? Yes. Remember, WIV's whole job was looking for nev coronaviruses. They published lists of which ones they had found pretty regularly. They published their last list in mid-2019, just a few months before the pandemic. Although lab leak proponents claimed these lists showed weird discrepancies, this was just their inability to keep names consistent, and all the lists showed basically t same viruses (plus a few extra on the later ones, as they kept discovering more). Th lists didn't include BANAL-52 or any other suitable COVID relatives - only RATG-13, which isn't close enough to work.

Could they have been keeping their discovery of BANAL-52 secret? No. Pre-pandemic, there was nothing interesting about it; our understanding of virology wa good enough to point this out as a potential pandemic candidate. WIV did its gain-of-function research openly and proudly (before the pandemic, gain-of-function wasn as unpopular as it is now) so it's not like they wanted to keep it secret because they might gain-of-function it later. Their lists very clearly showed they had no virus they could create COVID from, and they had no reason to hide it if they did.

COVID's furin cleavage site is admittedly unusual. But it's unusual in a way that look natural rather than man-made. Labs don't usually add furin cleavage sites through nucleotide insertions (they usually mutate what's already there). On the other hand viruses get weird insertions of 12+ nucleotides in nature. For example, HKU1 is ano emergent Chinese coronavirus that caused a small outbreak of pneumonia in 2004 had a 15 nucleotide insertion right next to *its* furin cleavage site. Later strains of CC got further 12 - 15 nucleotide insertions. Plenty of flus have 12 to 15 nucleotide insertions compared to other earlier flu strains.

Sometimes insertions happen because of a mistake in viral replication. Other times virus gets confused between its own RNA and its host's, and splices a bit of the ho RNA into the virus. This would neatly explain why the insertion used the unusual coding CGG for arginine, which is common in animals but rare in viruses. On the ot hand, it's not *that* rare in viruses - COVID uses CGG for arginine about 3% of the ti And human engineers don't necessarily use it any more than that - Peter was able find one example of humans adding arginine to a virus, and 0 out of the 5 arginines added were CGG.

COVID's furin cleavage site is a mess. When humans are inserting furin cleavage si into viruses for gain-of-function, the standard practice is RRKR, a very nice and sim furin cleavage site which works well. COVID uses PRRAR, a bizarre furin cleavage s which no human has ever used before, and which virologists expected to work poo They later found that an adjacent part of COVID's genome twisted the protein in an unusual way that allowed PRRAR to be a viable furin cleavage site, but this discove took a lot of computer power, and was only made after COVID became important. T

Wuhan virologists supposedly doing gain-of-function research on COVID shouldn't have known this would work. Why didn't they just use the standard RRKR site, whic would have worked better? Everyone thinks it works better! Even the virus eventual decided it worked better - sometime during the course of the pandemic, it mutated away from its weird PRRAR furin cleavage site towards a more normal form.

Further, COVID's furin cleavage site was inserted via what seems to be a frameshift mutation - it wasn't a clean insertion of the amino acids that formed the site, it was insertion of a sequence which changed the context of the surrounding nucleotides into the amino acids that formed the site. This is a pointless too-clever-by-half "flourish" that there would be no reason for a human engineer to do. But it's exactly the kind of weird thing that happens in the random chance of evolution.

COVID is hard to culture. If you culture it in most standard media or animals, it will quickly develop characteristic mutations. But the original Wuhan strains didn't have these mutations. The only ways to culture it without mutations are in human airway cells, or (apparently) in live raccoon-dogs. Getting human airway cells requires a donor (ie someone who donates their body to science), and Wuhan had never done this before (it was one of the technologies only used at the superior North Carolina site). As for raccoon-dogs, it sure does seems suspicious that the virus is already suited to them.

The claim that COVID is uniquely adapted to humans is false. The paper that claime that defined how well COVID was adapted to different animals by those animals' difference (on the relevant cell receptors) from humans. So in its methodology, humans came out #1 by default. If you don't do that, COVID is better-adapted to ma other animals.

It's not necessarily true that viruses see a burst of mutations when they enter a nev host. COVID spread to deer and mink, and in neither case was there a burst of mutations. COVID has a pretty simple job of infecting respiratory cells and is alread very good at it, regardless of species.

In Yuri's model, Wuhan Institute of Virology picked up a discarded grant and decide to do the gain-of-function half allotted to a different university, despite their relative inexperience. They skipped over all the SARS-like viruses they were supposed to w on, and all the standard gain-of-function model backbones, in favor of BANAL-52, a virus which would not be discovered for another two years, but which they someho had samples of, which they had for some reason decided to keep secret despite its total lack of interestingness. Then they would have had to eschew all usual gain-of-function practices in favor of inserting a weird furin cleavage site that shouldn't hav worked according to the theory they had at the time, via a frameshift mutation. The they would have had to culture it, a technique beyond their limited capabilities. *The* would have had to leak, and magically show up again in front of the raccoon-dog st at a wet market.

Yuri: WIV wouldn't have needed to keep BANAL-52 "secret" in some kind of siniste way. Plenty of researchers have backlogs of work they haven't published yet. Proba they a found BANAL relative in one of their normal sampling trips, did some preliminary studies on it, and planned to publish it later once they cleaned up their data. Everyone works like this.

The part of DEFUSE saying that they would only work on viruses that were 95% sin to SARS is unclear and might mean something else. It looks more like they say they start with those viruses, but also do some work on novel viruses. BANAL-52 could have been one of the novel viruses.

The furin cleavage site is weird, but the researchers might have done that on purpo to make the virus easier to keep track of, or to test different furin cleavage sites. Depending on the exact BANAL-52 relative they used, it might not even be a frameshift; there's a particular way to spell serine that would make the insertion mo natural.

The claims that COVID can't be cultured in normal media are based on speculative original research by Peter and might not hold up.

Peter: WIV did most of its virus-gathering in a trip to a Yunnan cave between 2010 and 2015. All those viruses have long since been processed and added to the database. There's no sign that they made more trips to Yunnan caves, and no reaso for them to keep that secret. So the idea that they might just have some new viruse they didn't publish doesn't hold up.

But suppose they did make more trips. Given the amount of time between the DEFU proposal and COVID, if they kept to their normal virus-collection rate, they would ha gotten about thirty new viruses. What's the chance that one of those was BANAL-5 There are thousands of bat viruses, and BANAL-52 is so rare that it wasn't found ur well after the pandemic started and people were looking for it very hard. So the chance that one of their 30 would be BANAL-52 is low. Also, they said in DEFUSE tl they planned to go back to the same Yunnan cave. But BANAL-52 was found far aw from that cave, so unless it ranged over a wide area, they probably couldn't have found it even if they got very lucky.

Session 3: Closing Arguments

This third debate was supposed to be about "inference", ie how much Bayesian evidence was provided by each of the facts given so far, and how to fit them into th Rootclaim probabilistic model. I'm going to relegate my summary of the more probabilistic half to the next section of this post, and just include the closing arguments here.

Saar: Peter's case hinges on the idea that it's very improbable that a lab leak pandemic would first show up at a wet market. But this isn't necessarily improbable The Huanan Seafood Market had several factors that made it a likely location for a superspreader event. It was busy, with over 10,000 visitors a day. Many of the peop there (eg the 1,000 vendors) came back daily, letting them reinfect each other. It ha poor ventilation, especially in the high-positivity area near the raccoon-dog stall. It cold wet surfaces on which the virus could survive for long periods. It was indoors, which prevented UV light from killing the virus. Given a small amount of sporadic COVID going around Wuhan, it's not surprising for the first place it started spreadin en masse to be a wet market.

In fact, we have several examples of this. When China was COVID Zero, there woul occasionally be small outbreaks that the authorities would have to contain. Most of these were at wet markets. For example, the big COVID outbreak in Beijing started Xinfadi Market, their local seafood market. This couldn't be an animal spillover, because there were no raccoon-dogs or other weird wildlife there. So it must be th wet markets are natural places for superspreader events. There are several other examples, which make up about half of the total outbreaks in Zero COVID era China plus others in Singapore and Thailand. Since COVID clusters concentrate in wet markets even when there is no animal spillover, we should accept this as a property the virus, and not attribute any significance to the fact that this happened in Wuhan too.

Peter: About 1/10,000 citizens of Wuhan was a wet market vendor. So there's a 1/10,000 chance that the first known COVID case should be a wet market vendor by chance alone. Weibo lists the most popular places for people to check in to their network on their phones, and the wet market was the 1600th most popular place in Wuhan, meaning that if you weight locations by busy-ness, there's a less than 1/160 chance that the first cases would be in the wet market.

Yes, the wet market is indoors, has mediocre ventilation, has repeat visitors, etc. So thousands of other places in Wuhan, like schools, hospitals, workplaces, places of worship. The wet market isn't special in any way. And again, it wasn't a superspread event! COVID spread at the same rate in the wet market as it does everywhere else doubling once per 3.5 days. It doesn't matter what kinds of arguments you can con up with for why the wet market *should have* been the perfect superspreader event location, we can look at it and see that it wasn't. It's an environment that spreads COVID at exactly the normal rate.

Zero COVID era Chinese outbreaks were concentrated in wet markets because they received infected animal products. We know why there was an outbreak in the Xinfa Market in Beijing: it was because the seafood stall got frozen fish from some non-Zero-COVID country, the fish had COVID particles on it, and the vendor got infecte and spread it to everyone else. Something like this is true for the other Chinese wet market based outbreaks we know about it. So this makes the opposite point you thi it does: wet markets start outbreaks because there are infected goods being sold there. Then the virus spreads through the wet market at a completely normal rate.

Saar: The Weibo list of 1600 places bigger than the wet market is likely inaccurate, because it's based on check-in data and people don't check in to seafood markets Most of those 1600 places aren't amenable to superspread. The 70 markets supposedly bigger than Huanan are irrelevant, because they're supermarkets, oper markets, etc. Huanan is the largest seafood market in central China, and a more like place for the first cluster of cases to be noticed.

Markets weren't a common spillover location in SARS1, so the zoonosis hypothesis hasn't "called" this event in a way that should give them a high Bayes factor.

And there's still plenty of evidence for isolated (though not super-spreading) pre-market cases. A British expatriate in Wuhan, Connor Reed, says he got sick in November, three weeks before the first wet market case. Later the hospital tested h samples and said it was COVID. Another paper reports 90 cases before the first we market one.

Peter: Connor Reed was lying. The case wasn't reported in any peer-reviewed pap It was reported in the tabloid *The Daily Mail*, months after it supposedly happened. also told the *Mail* that his cat died of coronavirus too, which is rare-to-impossible. Also, to get a positive hospital test, he would have had to go to the hospital, but he was 25 years old and almost no 25-year-olds go to the hospital for coronavirus. His only evidence that it was COVID was that two months later, the hospital supposedly "notified" him that it was. The hospital never informed anyone else of this extremely surprising fact which would be the biggest scientific story of the year if true. So probably he was lying. Incidentally, he died of a drug overdose shortly after giving t *Mail* that story; while not all drug addicts are liars, given all the other implausibilities his story, this certainly doesn't make him seem more credible. And in any case, he claimed he got his case at a market "like in the media"

The other 90 cases are also fake. A lab leak guy found a paper that mentioned 90 more cases than other papers, and made up a conspiracy theory where the author was trying to secretly communicate that there had been 90 secret cases before an the confirmed cases, even though there was nothing about this in the text of the paper. But actually that paper just counted cases differently than other papers, and they were referring to normal cases after the pandemic officially started.

Again, I'll come back to the discussion about inference later, but for now, here's a ta of both sides' reasoning. This exact presentation comparing both analyses is mine but you can see Saar's version here, and Peter's starting at 45:33 of this video.

	Peter	Saar
Priors		
Years per natural pandemic	20	20
Odds of Wuhan	0.03	0.015
Natural pandemic in Wuhan	0.0015	0.00075
Lab leak at WIV	0.02	0.15
Lab leak starts a pandemic	0.33	0.4
Prior ratio for lab leak (derived from above)	4.4	80
Location in wet market		
First known cases in wet market	0.00025	0.5
Highest case density near wildlife stalls	0.03	
No intermediate host		4
No wild animal vendor sick		
Two lineages		
All market cases B		
First A near market	0.02	
Two basal polytomies	0.25	
Genetic clock	0.1	
Viral genome		
New furin cleavage site through 12 base insertion	5	20
Frameshift PRRAR	0.0005	0.25
Binds to human ACE2		2
CGGCGG	2	5
Reasons WIV couldn't/wouldn't create COVID		
DEFUSE not happening	0.4	0.4
Needed backbone virus	0.001	0.5
Needed to find it interesting	0.1	
Needed reverse genetics for it	0.01	
Live virus instead of pseudovirus	0.5	
FCS in S1	0.5	
Didn't publish anything	0.1	
Have to succeed at creating/culturing COVID	0.1	
Other factors		
No outbreaks in other cities		
Same month as SARS	0.25	
Coverup would succeed	0.1	0.333
Final ratio	**2.0625E-21**	**532.8**
Inverse ratio	**4.848485E+20**	**0.0018768769**

Slightly made up; the two sides didn't express their probabilities in the same way and I had to make editorial decisions to match them. Note that these aren't entirely comparable because Peter is being laxer about out-of-model probability than Saar. Although Saar's final odds here are 533-to-1, this just the central estimate. Rootclaim's real final probability is 94% lab leak. You can see their analysis here.

And The Winner Is . . .

...

...

...

...

...

Peter and the zoonosis hypothesis.

This was a decisive victory. There were two judges, who each gave separate verdic (or were allowed to declare a draw). Both judges decided in favor of Peter.

You can see the judges' own summary of their reasoning here (Will, Eric)

Manifold agreed with the judges. There was a prediction market on who would win. started out 70-30 in favor of lab leak. As the videos came out, zoonosis started doi better and better. I don't want to take the exact final numbers too seriously, since I think some of the later price increases involved hints from the participants' behavic But it's clear which way viewers thought the wind was blowing [4].

Chris Billington

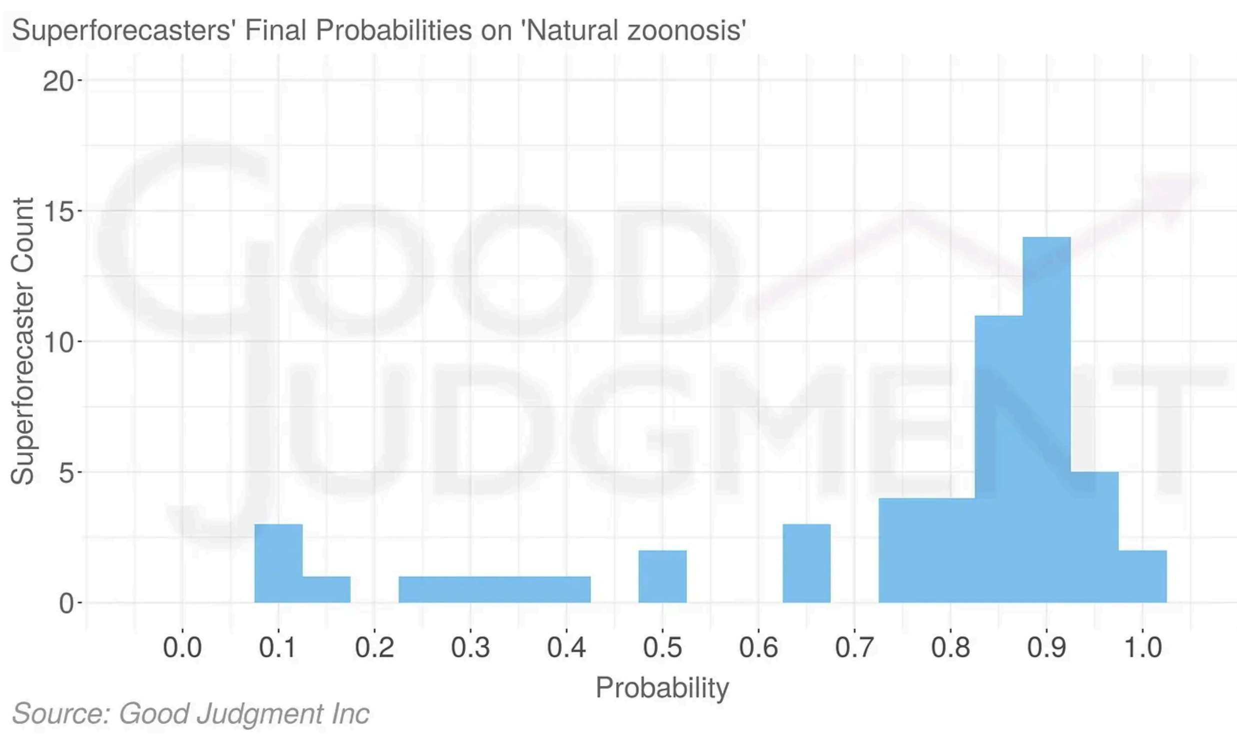
Will Peter Miller win the "Rootclaim Challenge" debate with Saar Wilf on the origins of COVID-19?

Resolved

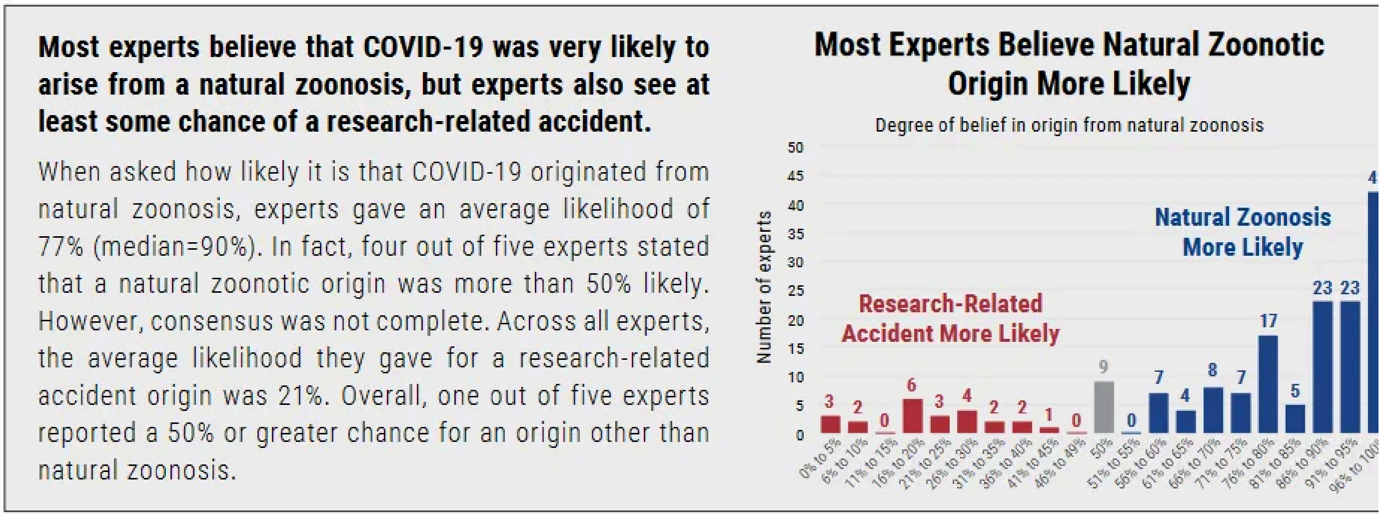
YES

30 201 Ṁ1.7k Ṁ180k resolved Feb 18

Around the same time, the Good Judgment Project - Philip Tetlock's group studyin superforecasters - put out a report on the lab leak hypothesis. After studying it in depth, his forecasters ended up 75-25 in favor of zoonosis. The Rootclaim debate one of ten sources they said they found especially interesting.



And also around the same time, and unrelated to any of this, the Global Catastroph Risks Institute surveyed experts ("168 virologists, infectious disease epidemiologist and other scientists from 47 countries") and found the same thing (though see here for some potential problems with the survey):



For what it's worth, I was close to 50-50 before the debate, and now I'm 90-10 in fa of zoonosis.

III. The Math And The Aftermath

The third debate session was about "inference", how to put evidence together. I put this part off until after disclosing the winner, because I wanted to talk about some c these issues at more length.

The Math: Judges

Both judges included a probabilistic analysis in their written decision. Here's the sa table as above, expanded to add the judges:

	Peter	Saar	Judge Eric	Judge Will
Priors				
Years per natural pandemic	20	20	160	66
Odds of Wuhan	0.03	0.015	0.025	0.01
Natural pandemic in Wuhan	0.0015	0.00075	0.00015625	0.00015152
Lab leak at WIV	0.02	0.15	0.02	0.05
Lab leak starts a pandemic	0.33	0.4	1	0.5
Prior ratio for lab leak (derived from above)	4.4	80	128	165
Location in wet market				
First known cases in wet market	0.00025	0.5	0.0005	0.001
Highest case density near wildlife stalls	0.03			
No intermediate host		4		
No wild animal vendor sick				
Two lineages				
All market cases B				
First A near market	0.02			
Two basal polytomies	0.25			
Genetic clock	0.1			
Viral genome				
New furin cleavage site through 12 base insertion	5	20	20	25
Frameshift PRRAR	0.0005	0.25		0.017
Binds to human ACE2		2	2	
CGGCGG	2	5		10
Reasons WIV couldn't/wouldn't create COVID				
DEFUSE not happening	0.4	0.4	0.06	0.5
Needed backbone virus	0.001	0.5	0.5	0.01
Needed to find it interesting	0.1			
Needed reverse genetics for it	0.01		0.2	0.1
Live virus instead of pseudovirus	0.5			
FCS in S1	0.5			
Didn't publish anything	0.1			
Have to succeed at creating/culturing COVID	0.1		0.5	
Other factors				
No outbreaks in other cities				10
Same month as SARS	0.25			
Coverup would succeed	0.1	0.333	0.1	
Final ratio	**2.0625E-21**	**532.8**	**0.000768**	**0.00350625**
Inverse ratio	**4.848485E+20**	**0.00187688**	**1302.083333**	**285.204991**

I shoehorned the judges' factors into the categories I already had; some of them were actually subtly different from Peter's, Saar's, and each other's. The "priors" category is especially a mess here.

We'll go over these later, but I get the impression that they both thought of probabilistic analyses as an afterthought. For example, Judge Eric wrote 30,000 wo about which considerations moved him, and only then includes the analysis, saying

> I am not convinced that this Bayesian calculation is even an appropriate way to estimate the relative posterior probability of Z and LL; it just seemed fair that aft criticizing Rootclaim's calculations at length I should make an attempt at it myse

Judge Will's decision ran to 10,000 words. He said he independently tried both reasoning it out intuitively, and running the Bayesian analysis, and was relieved whe these two methods returned the same result. He said:

> I am skeptical that the Bayesian decision making/evaluation methods are any mo "objective" than [intuitive reasoning]. I think they maximize legibility, not objectiv and tend to hide the intuitive/heuristic portion in the data inclusion step and valu where it's harder to see . . . I am not skilled in the Bayesian method, and I am sur made significant mistakes. More time and practice would improve and refine my estimates. At the fundamental rules of the universe level, Bayesian analysis mus the best way to evaluate evidence. However, I am unsure that it's a good strateg for a human given our cognitive limitations, and doubly unsure it's truly being us (in the dispassionate sense) where the outcome is social desirability/fame/Twitte likes.

I'm focusing on this because Saar's opinion is that the debate went wrong (for his side) because he didn't realize the judges were going to use Bayesian math, they d the math wrong (because Saar hadn't done enough work explaining how to do it rig and so they got the wrong answer. I want to discuss the math errors he thinks the judges made, but this discussion would be incomplete without mentioning that the judges themselves say the numbers were only a supplement for their intuitive reasoning.

That having been said, let's look deeper into some of Saar's concerns.

The Math: Extreme Odds

Saar complained that Peter's odds were too extreme. For example, Peter said there was only a 1/10,000 chance that a lab leak pandemic would first show up at a wet market.

Peter's argument went something like: obviously a zoonotic pandemic would start a site selling weird animals. But a lab leak pandemic - if it didn't start at the lab - coul show up anywhere. 1/10,000 Wuhan citizens work at the wet market. So if a lab leak was going to show up somewhere random, the wet market was a 1/10,000 chance.

Saar had specific arguments against this, but he also had a more general argument you should rarely see odds like 1/10,000 outside of well-understood domains. In his blog post, he gave this example:

> A prosecutor shows the court a statistical analysis of which DNA markers match the defendant and their prevalence, arriving at a 1E-9 probability they would all match a random person, implying a Bayes factor near 1E9 for guilty.
>
> But if we try to estimate p(DNA|~guilty) by truly assuming innocence, it is immediately evident how ridiculous it is to claim only 1 out of a billion innocent suspects will have a DNA match to the crime scene. There are obviously far bett explanations like a lab mistake, framing, an object of the suspect being brought someone to the scene, etc.

So the real p(wet market|lab leak) isn't the 1/10,000 chance a pandemic arising in a random place hits the wet market, but the (higher?) probability that there's someth wrong with Peter's argument.

Then Saar tried to show specific things that might be wrong with Peter's argument. didn't find his specific examples convincing. But maybe the question shouldn't be whether I *agreed with* him. It should be whether I'm *so confident* he's wrong that I would give it 10,000-to-1 odds.

This makes total sense, it's absolutely true, and I want to be *really, really* careful wit If you take this kind of reasoning too far, you can convince yourself that the sun wo rise tomorrow morning. All you have to do is propose 100 different reasons the sun might not happen. For example:

1. The sun might go nova.
2. An asteroid might hit the Earth, stopping its rotation.
3. An unexpected eclipse might blot out the sun.
4. God exists and wants to stop the sunrise for some reason.
5. This is a simulation, and the simulators will prevent the sunrise as a prank.
6. Aliens will destroy the sun.

...and so on until you reach 100. On the one hand, there are 100 of these reasons. B on the other, they're each fantastically unlikely - let's say 99.9999999999% chanc each one doesn't happen - so it doesn't matter.

But suppose you get too humble. Sure, you might *think* you have a great model of h eclipses work and you know they never happen off schedule - but can you really be 99.9999999999% sure you understood your astronomy professor correctly? Can y be 99.9999999999% sure you're not insane, and that your "reasoning" isn't just random seizings of neurons that aren't connecting to reality at any point? Seems lik you can't. So maybe you should lower your disbelief in each hypothesis to somethi more reasonable, like 99%. But now the chance that the sun rises tomorrow is 0.99^100, aka 36%. Seems bad.

Also, suppose I agree I shouldn't use numbers like 1-in-10,000. What should my rea numbers be? Saar says it's the strength of the strongest hypothesis for why I might wrong. But I can't think of any good reasons I might be wrong here. Should I feel ok after adjusting it down to 1-in-1,000? 1-in-100? 1-in-10?

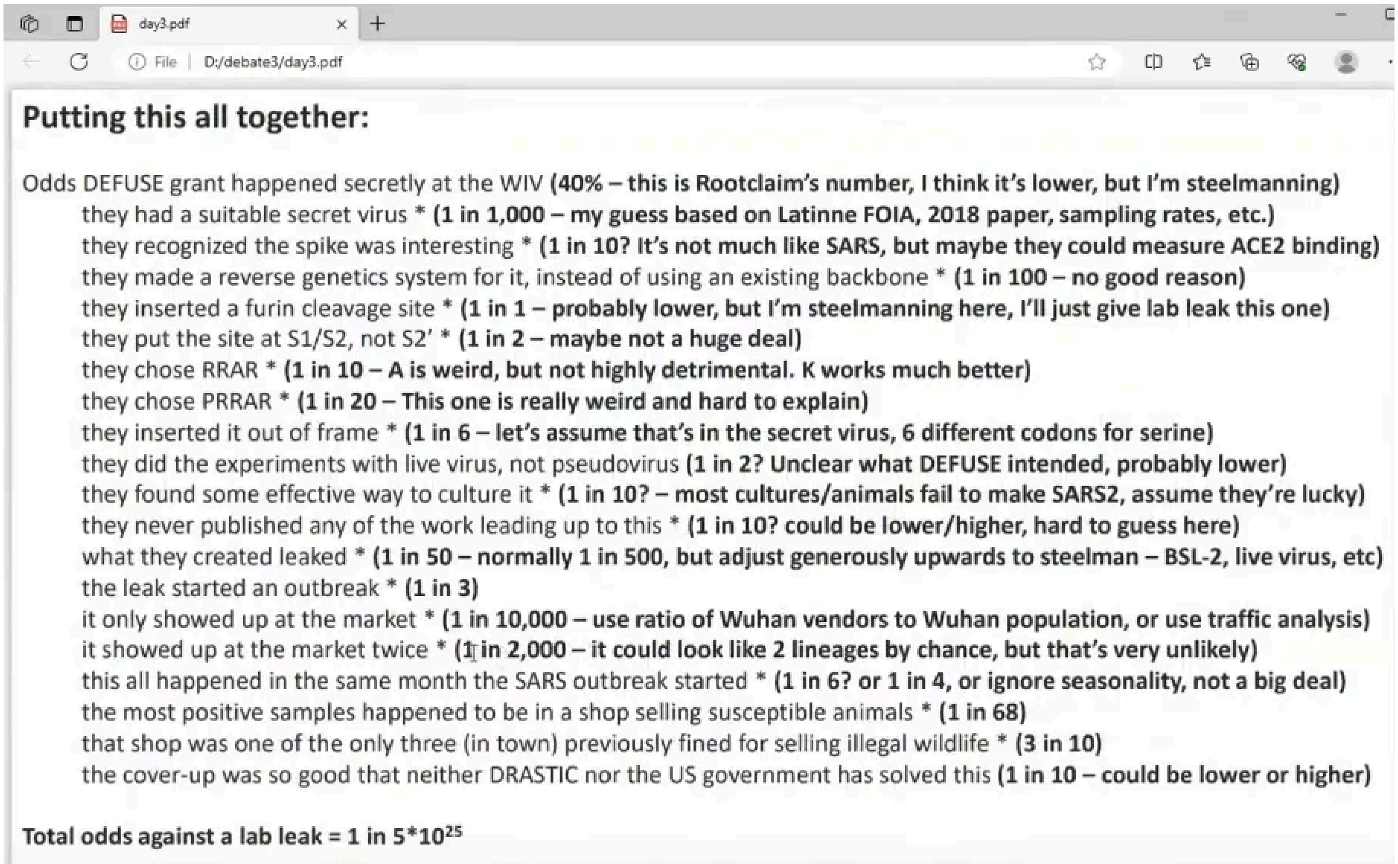
Saar said he “could have” argued that there was only a one-in-a-million chance COVID’s furin cleavage site evolved naturally. Instead, he gave a Bayes factor of 1-i 20, because he wanted to leave room for out-of-model error. But (as Saar understa it) the judges’ probabilistic analysis took Peter’s unadjusted 1-in-10,000 wet marke claim seriously *and* Saar’s fully-adjusted 1-in-20 furin cleavage site claim literally, compared them, found Peter’s evidence stronger, and gave him the victory. You ca see why he’s upset.

(again, the judges deny basing their verdict solely on the probabilistic models)

At some point you have to take your best guess about how confusing a given field i and how wrong you could be - how often do labs overstate their DNA results? How much do you know about how viruses get furin cleavage sites? How sure are you about the demographics of Wuhan? How tightly does your “there have been at leas million days without aliens destroying the sun, therefore aliens will not destroy the s tomorrow” argument hang together? - and choose a number.

The Math: Extreme Odds, Part II

During the debate, Peter showed this slide:



Putting this all together:

Odds DEFUSE grant happened secretly at the WIV **(40% – this is Rootclaim's number, I think it's lower, but I'm steelmanning)**

they had a suitable secret virus * **(1 in 1,000 – my guess based on Latinne FOIA, 2018 paper, sampling rates, etc.)**
they recognized the spike was interesting * **(1 in 10? It's not much like SARS, but maybe they could measure ACE2 binding)**
they made a reverse genetics system for it, instead of using an existing backbone * **(1 in 100 – no good reason)**
they inserted a furin cleavage site * **(1 in 1 – probably lower, but I'm steelmanning here, I'll just give lab leak this one)**
they put the site at S1/S2, not S2' * **(1 in 2 – maybe not a huge deal)**
they chose RRAR * **(1 in 10 – A is weird, but not highly detrimental. K works much better)**
they chose PRRAR * **(1 in 20 – This one is really weird and hard to explain)**
they inserted it out of frame * **(1 in 6 – let's assume that's in the secret virus, 6 different codons for serine)**
they did the experiments with live virus, not pseudovirus **(1 in 2? Unclear what DEFUSE intended, probably lower)**
they found some effective way to culture it * **(1 in 10? – most cultures/animals fail to make SARS2, assume they're lucky)**
they never published any of the work leading up to this * **(1 in 10? could be lower/higher, hard to guess here)**
what they created leaked * **(1 in 50 – normally 1 in 500, but adjust generously upwards to steelman – BSL-2, live virus, etc)**
the leak started an outbreak * **(1 in 3)**
it only showed up at the market * **(1 in 10,000 – use ratio of Wuhan vendors to Wuhan population, or use traffic analysis)**
it showed up at the market twice * **(1 in 2,000 – it could look like 2 lineages by chance, but that's very unlikely)**
this all happened in the same month the SARS outbreak started * **(1 in 6? or 1 in 4, or ignore seasonality, not a big deal)**
the most positive samples happened to be in a shop selling susceptible animals * **(1 in 68)**
that shop was one of the only three (in town) previously fined for selling illegal wildlife * **(3 in 10)**
the cover-up was so good that neither DRASTIC nor the US government has solved this **(1 in 10 – could be lower or higher)**

Total odds against a lab leak = 1 in $5*10^{25}$

Okay, this one is just awful.

It takes the risky gambit above - giving extreme odds to something - then doubles down on it by multiplying across twenty different stages to get a stupendously low probability of 1/5*10^25. If we believe this, it's more likely that we win the lottery th times in a row than that we learn lab leak was true after all.

Eliezer Yudkowsky calls this the Multiple Stage Fallacy. Even aside from the failure mode in the sunrise example above (where people are too reluctant to give strong probabilities), it fails because people don't think enough about the correlations between stages. For example, maybe there's only 1/10 odds that the Wuhan scienti would choose the suboptimal RRAR furin cleavage site. And maybe there's only 1/2 odds that they would add a proline in front to make it PRRAR. But are these really tv separate forms of weirdness, such that we can multiply them together and get 1/20 Or are scientists who do one weird thing with a furin cleavage site more likely to do another? Mightn't they be pursuing some general strategy of testing weird furin cleavage sites?

(For example, Yuri proposed that, because the scientists wanted to understand hov pandemic coronaviruses originate in nature, they might deliberately pick more natu looking features over more designed-looking ones, which would neatly explain mar features seemingly inconsistent with lab leak. Is this a conspiracy theory? Rootclair able to successfully route around this question. If the probability of a feature happening in nature is X, then the probability of it happening in this variant of lab le scenario is X * [chance that the scientists wanted to imitate nature). This gives it a (deserved) complexity penalty without ruling out this (non-zero and potentially important) possibility.)

In any case, Peter didn't care as much about probabilistic analysis as Saar, he didn' make his case hinge on this slide, and he might have been kind of using it to troll Rootclaim (which definitely worked). He might not have been making any of the mistakes above. But anyone who took this slide seriously would end up dramaticall miscalibrated.

The Math: Big Pictures

Another of Saar's concerns with the verdict was that Peter was an extraordinary debater, to the point where it could have overwhelmed the signal from the evidence

It's hard to watch the videos and not come away impressed. Peter seems to have a photographic memory for every detail of every study he's ever read. He has some k of 3D model in his brain of Wuhan, the wet market, and how all of its ventilation duc and drains interacted with each other. Whenever someone challenged one of his points, he had a ten-slide PowerPoint presentation already made up to address tha particular challenge, and would go over it with complete fluency, like he was recitin memorized speech. I sometimes get accused of overdoing things, but I can't imagi how many mutations it would take to make me even a fraction as competent as Pet was.

Saar’s closing argument included the admission:

> Peter, I think everyone can agree, has much more knowledge on [COVID] origins than we do. He's invested much more time. He may be a much more talented researcher. He's much more into the details. He probably knows the best in the world on origins at this point.

Once you’ve described your opponent that way in your closing argument, what’s le of your case?

Saar thought a lot was left. Throughout the debate, he tried to make a point about h getting the inference right was more important than winning sub-sub-sub-debates about individual lines of evidence. Although Peter won most specific points of contention, Saar thought that if the judges could just keep their mind on the big picture, they would realize a lab leak was more likely.

I’m potentially sympathetic to arguments like Saar’s. Imagine a debate about UFOs Imaginary-Saar says “UFOs can’t be real, because it doesn’t make sense for aliens come to Earth, circle around a few fields in Kansas, then leave without providing an other evidence of their existence.” Imaginary-Peter says “John Smith of Topeka saw UFO at 4:52 PM on 6/12/2010, and everyone agrees he’s an honorable person who wouldn’t lie, so what’s your explanation of that?” Saar says “I don’t know, maybe he was drunk or something?” Peter says “Ha, I’ve hacked his cell phone records and geolocated him to coordinates XYZ, which is a mosque. My analysis finds that he’s there on 99.5% of Islamic holy days, which proves he’s a very religious Muslim. And religious Muslims don’t drink! Your argument is invalid!” On the one hand, imaginary Peter is very impressive and sure did shoot down Saar’s point. On the other, imaginary-Saar never really claimed to have a great explanation for this particular U sighting, and his argument doesn’t depend on it. Instead of debating whether Smith could or couldn't have been drunk, we need to zoom out and realize that the aliens explanation makes no sense.

The problem was, Saar couldn't effectively communicate what his big picture was. Neither deployed some kind of amazingly elegant prior. They both used the same k of evidence. The only difference was that Peter's evidence hung together, and Saar evidence fell apart on cross-examination.

I think - not because Saar really explained it, but just reading between the lines - Sa thought the un-ignorable big picture evidence was the origin in a city with a coronavirus gain-of-function lab, and the twelve-nucleotide insertion in the furin cleavage site.

To some degree, Peter just ate the loss on those questions. No matter how you slic it really is a weird coincidence that the epidemic started so close to Asia's biggest coronavirus laboratory. Peter tried to deflect this - he pointed out there were other BSL-3 and BSL-4 laboratories in Beijing, Shanghai, Shenzhen, etc. But this was a ra question where he unambiguously came out looking worse - the other cities' labs h much less coronavirus-specific research. Wuhan really was unique (aside from the other big coronavirus lab in North Carolina).

Peter did better when he tried to control the damage: there are a couple hundred million people in the South Asian areas where people eat weird animals exposed to virus-infected bats, Wuhan has a population of about 12 million, so maybe 1.5% of a potential zoonotic pandemics should start in Wuhan. Peter tried to argue that Wuha was a local trade center, so maybe we should up that to 5 - 10%. 5 - 10% coincidences aren't that rare. Even 1.5% coincidences happen sometimes.

Likewise, the furin cleavage site really does stand on a genetic map. I didn't feel like either side did much math to quantify how weird it was. Naively, I might think of this "30,000 bases in COVID, only one insertion, it's in what's obviously the most interesting place - sounds like 30,000-to-one odds against". Against that, a virus w a boring insertion would never have become a pandemic, so maybe you need to multiply this by however much viral evolution is going on in weird caves in Laos, and then you would get the odds that at least *one* virus would have an insertion interest enough to go global. Neither participant calculated this in a way that satisfied me (though see here for related discussion). Instead, Peter tried to undermine the furin argument by showing that, as surprising as the site was under a natural origin, it wo be an even more surprising choice for human engineers. Saar argued it wasn't - bu because of his policy of giving adjusted-for-model-error odds, he only gave this a factor of 30 in his analysis. Since Peter gave it a *higher* factor of 50 in *his* analysis, i looked from the outside like Saar had already conceded this point, and the judges were mostly happy to go with Saar's artificially-low estimate.

The Math: Double Coincidences

Saar brought up an interesting point halfway through the debate: you should rarely high Bayes factors on both sides of an argument.

That is, suppose you accept that there's only a 1-in-10,000 chance that the pander starts at a wet market under lab leak. And suppose you accept there's only a 1-in-10,000 chance that COVID's furin cleavage site could evolve naturally.

If lab leak is true, then you might find 1-in-10,000 evidence for lab leak. But it's a fre coincidence that there was 1-in-10,000 evidence for zoonosis [5].

Likewise, if zoonosis is true, you might find 1-in-10,000 evidence for this true thing. But it's a freak coincidence that there was 1-in-10,000 evidence for lab leak.

Either way, you're accepting that a 1-in-10,000 freak coincidence happened. Isn't it more likely you've bungled your analysis?

I was following along at home, and I definitely bungled this point; I had some high Bayes factors on both sides. I adjusted some of them downward based on Saar's g point, but how far should we take it?

Here I remember The Pyramid And The Garden: you can get very strong coincidenc
if you have many degrees of freedom, ie buy a lot of lottery tickets. So for example,
suppose there are fifty things about a virus. You should expect at least one of those
have a one-in-fifty coincidence by pure chance.

What about more than that? You might be able to get away with this by saying there
are an infinite number of possible conspiracy theories, and some from that infinite s
are brought into existence when a strong enough coincidence makes them plausibl
For example, it's really weird that John Adams and Thomas Jefferson both died on
50th anniversary of the Declaration of Independence. If I wanted, I could form a
conspiracy theory about a group of weird assassins obsessed with killing Founding
Fathers on important dates, and then Jefferson and Adams' deaths would be 1/10,0
evidence for that theory. But this is the Texas Sharpshooter Fallacy, which Saar
warned against several times.

I don't know if "the virus started in Wuhan, which is where they're doing this resear
gets a Texas Sharpshooter penalty, or how high that penalty should be. But the furi
cleavage site doesn't - people were talking about lab leak before anyone noticed it.

The Aftermath: Peter

Peter seemed satisfied with the result, in an understated sort of way:

> It seemed like an interesting experiment in monetizing the debunking of a
> conspiracy theory. I think there's usually a big asymmetry where it's easy to get
> spreading bullshit (like, the top anti-vaxxers during the pandemic all made a milli
> dollars a year on substack), but it's almost impossible to make money on debunl
> it. The Rootclaim challenge seemed like one rare case where the opposite was tr
>
> Beyond that, I don't know what it's good for. It does seem like there could be a
> positive social impact from more people understanding that the lab leak hypothe
> is (almost certainly) false.

The Aftermath: Saar

Saar says the debate didn't change his mind. In fact, by the end of the debate, Rootclaim released an updated analysis that placed an even higher probability on la leak than when they started.

In his blog post, he discussed the issues above, and said the judges had erred in no considering them. He respects the judges, he appreciates their efforts, he just thin they got it wrong. Although he respected their decision, he wanted the judges to correct what he saw as mistakes in their published statements, which delayed the public verdict and which which Viewers Like You did not appreciate:

Shump

If Rootclaim/Saar Wilf loses the 100k challenge on COVID origins, will they be a sore loser?

Resolved
YES

3 35 Ṁ670 Ṁ5.4k resolved Feb 29

I ran an early draft of this post by him. There was some miscommunication about th exact publication date, so he hasn't had time to write up a full response, but he has some quick thoughts (and I'll link the full response when he writes it). He says:

> We will provide a full response to this post soon, but the main problem with it is fairly simple:

There is general agreement that the main evidence for zoonosis is HSM (Huanar Seafood Market) forming an early cluster of cases. The contention is whether it i amazing 10,000x evidence, or is it negligible. All other evidence points to a lab le and if HSM is shown to be weak, lab leak is a clear winner.

We provided an analysis of why it is negligible that is as close to mathematical proof as such things can be. Read it here.

Scott and I exchanged a few emails on this issue and Scott preferred to discuss more intuitive analyses of HSM, using rules of thumb that likely served him well i the past.

While I believe I managed to mostly explain where these failed, and Scott understands HSM is far weaker evidence than he initially thought [6], he still has a very strong intuitive feeling (based on years of dealing with probabilities) that thi is some exceptional coincidence, and that prevents him from properly updating l posterior.

At the end of the day, this cannot be settled without going through our semi-forr derivation, understanding it, and either identifying the problem with it or accepti it (and thereby accepting lab-leak to be more likely).

Here is a quick summary of the mistakes made by those claiming HSM is strong evidence:

1. The first mistake is conflating Bayes factors with conditional probabilities. 1/10000 is the supposed conditional probability p(HSM|Lab Leak), That shou be divided by the conditional probability of HSM under Zoonosis. Markets w not identified as a high-risk location prior to this outbreak (This will be elaborated in the full response), and in SARS1 the spillovers were mostly at restaurants and other food handlers that deal more closely with wildlife. While it's cool to point to the raccoon dog photo, that was a result of a

> retrospective search (we don't know what other photos they took which in retrospect would be brought up as premonition). Unbiased data shows mark are not a likely spillover location for zoonosis. We originally estimated p(HSM|Zoonosis)<0.1. Following more research we did to answer Scott's questions, this is more likely <0.03.
>
> 2. Even if it's unclear how strong HSM is as an amplifier, it is clearly not some random location, with its high traffic (that alone should be at least 10x), man permanent residents, and the recurrence of early clusters in other seafood markets. It's highly overconfident to claim it is less than a 1% location - It means Wuhan is somehow unique in that its largest seafood market is differ from others and won't form an early cluster. So the conditional probabilities under both hypotheses are fairly similar
> 3. The obvious lack of evidence for a wildlife spillover in HSM further reduces t factor, making HSM negligible evidence.

Saar remains as committed to Rootclaim as ever. He’s even still committed to $100,000 bets on Rootclaim findings, settled via debate. He’d even be willing to re-debate Peter on lab leak! (Peter declined)

He does want to make some changes to the debate format - specifically, switching from video to text, and checking in more often with the judges to get feedback on t thought processes.

The switch from video to text seems reasonable. Saar was clearly flummoxed by Peter’s memory and agility, and wants a format where ability to remember/think on your feet is less important, and where you can do lots of research before having to think up a response.

The part with the feedback seems to be Saar wanting even more of an opportunity identify disagreements with the judges early, and get a chance to tell them beforeh about issues like the ones above.

I'm sympathetic to both these changes, but I don't think they would have changed outcome of this debate.

The Aftermath: Rootclaim

Rootclaim is an admirable idea. Somebody called it "heroic Bayesian analysis", and like the moniker. Regular human reasoning doesn't seem to be doing a great job puncturing false beliefs these days, and lots of people have converged on somethir something Bayes as a solution. But the something something remains elusive. Whil everyone else tries "pop Bayesianism" and "Bayes-inspired toolboxes", Rootclaim asks: what if you just directly apply Bayes to the world's hardest problems? There's something pure about that, in a way nobody else is trying.

Unfortunately, the reason nobody else is trying this is because it doesn't work. The too much evidence, and it's too hard to figure out how to quantify it.

Peter, Saar, and the two judges all did their own Bayesian analysis. I followed along home [7] and tried the same. Daniel Filan, who also watched the debate, did one too. Here's a comparison of all of our results:

	Peter	Saar	Judge Eric	Judge Will	Scott	Daniel
Priors						
Years per natural pandemic	20	20	160	66	20	
Odds of Wuhan	0.03	0.015	0.025	0.01	0.015	0
Natural pandemic in Wuhan	0.0015	0.00075	0.00015625	0.00015152	0.00075	0.
Lab leak at WIV	0.02	0.15	0.02	0.05	0.03	0
Lab leak starts a pandemic	0.33	0.4	1	0.5	0.4	
Prior ratio for lab leak (derived from above)	4.4	80	128	165	16	
Location in wet market						
First known cases in wet market	0.00025	0.5	0.0005	0.001	0.002	0.
Highest case density near wildlife stalls	0.03				0.5	0.
No intermediate host		4			4	
No wild animal vendor sick					1.5	
Two lineages						
All market cases B					3	
First A near market	0.02				0.1	
Two basal polytomies	0.25					
Genetic clock	0.1					
Viral genome						
New furin cleavage site through 12 base insertion	5	20	20	25	50	
Frameshift PRRAR	0.0005	0.25		0.017	0.25	0
Binds to human ACE2		2	2			
CGGCGG	2	5		10	2	
Reasons WIV couldn't/wouldn't create COVID						
DEFUSE not happening	0.4	0.4	0.06	0.5	0.66	0
Needed backbone virus	0.001	0.5	0.5	0.01	0.5	0
Needed to find it interesting	0.1				0.5	
Needed reverse genetics for it	0.01		0.2	0.1		
Live virus instead of pseudovirus	0.5					
FCS in S1	0.5					
Didn't publish anything	0.1					
Have to succeed at creating/culturing COVID	0.1		0.5			
Other factors						
No outbreaks in other cities				10		1
Same month as SARS	0.25					
Coverup would succeed	0.1	0.333	0.1		0.5	
Final ratio	**2.0625E-21**	**532.8**	**0.000768**	**0.00350625**	**0.0594**	**0.03**
Inverse ratio	**4.848485E+20**	**0.00187688**	**1302.083333**	**285.204991**	**16.835017**	**25.227**

Again, most people didn't use these exact categories, I'm putting them in this format to make them easy to compare, and any errors are mine....

The six estimates span twenty-three orders of magnitude. Even if we remove Peter (who's kind of trolling), the remaining estimates span a range of ~7 OOMs. And eve we remove Saar (limiting the analysis to neutral non-participants), we're still left wit factor-of-50 difference.

50x sounds good compared to 23 OOMs. But it only sounds good because everyor except Saar leaned heavily towards zoonosis. If raters were closer to even, it would become problematic: even a factor of 50x is enough to change 80-20 lab leak to 8 20 natural.

Saar's perspective is that true theories should have many orders of magnitude mor evidence than false theories (most of the Rootclaim analyses end up with normal-sounding percentages like "94% sure", but that's after they correct for potential mc error). If that's true, a 50x fudge factor shouldn't be fatal.

But 23 orders of magnitude is fatal any way you slice it. The best one can say is tha maybe this is no worse than normal reasoning. Among normal people who don't us Rootclaim, many are sure lab leak is true, and many others are sure it's false. If we interpret "sure" as 99%, then even normal people without Rootclaim are a factor of 10,000x away from each other. If we interpret "sure" as including more nines than tl maybe normal people are 23 OOMs away from each other, who knows? In this mod Rootclaim is no worse than anything else; it's just legible enough that we notice the discrepancies. We've gotten inured to people failing to agree on difficult issues. Ma Rootclaim gets credit for showing us exactly where we fail, and putting numbers on the failure.

Still, this is faint praise for a method that hoped to be able to resolve these kinds of disagreements. In the end, I think Saar has two options:

1. Abandon the Rootclaim methodology, and go back to normal boring impure reasoning like the rest of us, where you vaguely gesture at Bayesian math but certainly don't try anything as extreme as actually using it.
2. Claim that he, Saar, through his years of experience testing Rootclaim, has som kind of special *metis* at using it, and everyone else is screwing up.

Saar gestured at (2) in the debate, repeatedly emphasizing that Rootclaim was diffi and subtle. But he mostly talked about things like the Texas Sharpshooter Fallacy, which all participants already knew about and were trying to avoid. Maybe he shoul go further.

This wouldn't necessarily be special pleading. When psychoanalysts claim their therapies work, they don't mean that someone who just read a two page "What Is Psychoanalysis?" pamphlet can do good therapy. They mean that someone who sp ten years training under someone who spent ten years training and so on in a lineag back to Freud can do good therapy. When scientists say the scientific method work they don't mean that any crackpot who reads an Intro To Science textbook can figu out the mysteries of the universe. They mean someone who's trained under other scientists and absorbed their way of thinking can do it.

If Saar wants to convince people, I think he should abandon his debates - which wouldn't help even if he won, and certainly don't help when he loses - and train five people who aren't him in how to do Rootclaim, up to standards where he admits they're as good at it as he is. Then he should prove that those five people can reliab get the same answers to difficult questions, even when they're not allowed to comp notes beforehand. *That* would be compelling evidence! [8]

The Aftermath: Pseudoscience

Suppose we accept the judges' decision that COVID arose via zoonosis. Does that mean lab leak was a "conspiracy theory" and we should be embarrassed to have ev believed it?

The term "conspiracy theory" is awkward here because there were definitely at leas two conspiracies - one by China to hide the evidence, one by western virologists to convince everyone that lab leak was stupid and they shouldn't think about it. Saar cited some leaked internal conversations among expert virologists. Back in the earl stage of the pandemic, they said to each other that it seemed like COVID could hav come from a lab leak - their specific odds were 50-50 - but that they should try to obfuscate this to prevent people from turning against them and their labs. So the b we can say here is that maybe the conspiracies got lucky on their 50-50 bet, and tl thing they were trying to cover up wasn't even true.

Still, it's awkward to use "conspiracy theory" as an insult when the conspiracies we real. Maybe a better question is whether lab leak is "pseudoscience".

The argument against: lots of smart people and experts believed it was a lab leak. There were all those virologists giving 50-50 odds in their internal conversations. E Peter says *he* started out leaning lab leak, back in 2021 when everyone was talking about it.

The argument in favor: since 2021, experts (and Peter) have shifted pretty far in fav of zoonosis. They've been convinced by new work - the identification of early case: the wet market surveys, the genetic analysis.

What category of noun does the adjective"pseudoscientific" describe? It doesn't necessarily describe *theories*: Newtonian mechanics wasn't pseudoscience when Newton discovered it, but if someone argued for it today (against relativity), that *w* be pseudoscientific. It doesn't even describe *arguments*: "we don't have enough d to confirm global warming" was a strong argument against global warming before there were good data, and a pseudoscientific one now. Might we place the locus of pseudoscientificness in people, communities, and norms of discussion?

Peter's position is that, although the lab leak theory is inherently plausible and didn start as pseudoscience, it gradually accreted a community around it with bad epistemic norms. Once lab leak became A Thing - after people became obsessed v getting one over on the experts - they developed dozens of further arguments whic ranged from flawed to completely false. Peter spent most of the debate debunking these - Mr. Chen's supposed 12/8 COVID case, Connor Reed's supposed 11/25 CO' case, the rumors of WIV researchers falling sick, the 90 early cases supposedly “hidden” in a random paper, etc, etc, etc. Peter compares this to QAnon, where an early “seed” idea created an entire community of people riffing off of it to create m and more bad facts and arguments until they had constructed an entire alternative epistemic edifice.

If we don’t accept the judges’ verdict, and think lab leak is true, are we worried the zoonosis side has some misbehavior of its own? Yuri and Saar didn’t talk about tha much. High-status people misbehave in different ways from low-status people; I th the zoonosis side has plenty of things to feel bad about (eg the conspiracies), but pseudoscience probably isn’t the right descriptor.

The Aftermath: Ebb And Flow

During the debate, Peter accused the lab leak side of being constantly left flat-foot by new evidence. Sure, it had seemed plausible back in 2020, but they’d had to scramble to explain a steady stream of pro-zoonosis papers.

Afterwards, Saar and Yuri got some new evidence of their own. A Chinese team appeared to have found a T/T intermediate strain of COVID in Shanghai, possibly imported from very early in Wuhan. If true, it would provide new evidence against a double spillover, instead supporting Lineage A mutating into B in humans.

(You can see Peter’s response here - basically that we’re not sure it’s a true intermediate and not a reversion - if it were true, how come the two strains on eithe side of it got millions of cases, and it just got one guy in Shanghai? But if it were tru would still be compatible with zoonosis - Lineage A would have spread from an ani and quickly mutated into B, the first A case would have been someone who left the wet market for a nearby area, and the first B case would have been someone who stayed in the wet market.)

Also, a new Freedom of Information Act request got early drafts of the DEFUSE gra proposal with new details, of which the most explosive was a comment by the

American half of the team, reassuring the Chinese half that even though the propos focused on American work to please funders, they would let the Chinese side do sc "assays". Lab leakers say this disproves the argument that, because DEFUSE said tl work would be done in the US, the Wuhan Institute of Virology couldn't/wouldn't dc advanced gain-of-function research.

(I asked Peter his response - he said the original draft of DEFUSE also said that the Chinese side would do "live virus binding assays", and this isn't the kind of gain-of-function research necessary to make COVID.)

In an email, Saar and Yuri suggested it was an "interesting coincidence" that all the new evidence that came out after the debate favored their side. I've decided agains updating on these considerations - either Peter's version or Saar/Yuri's. My impress is that anyone who starts out believing something at time t will also believe all the n evidence after time t favors that thing.

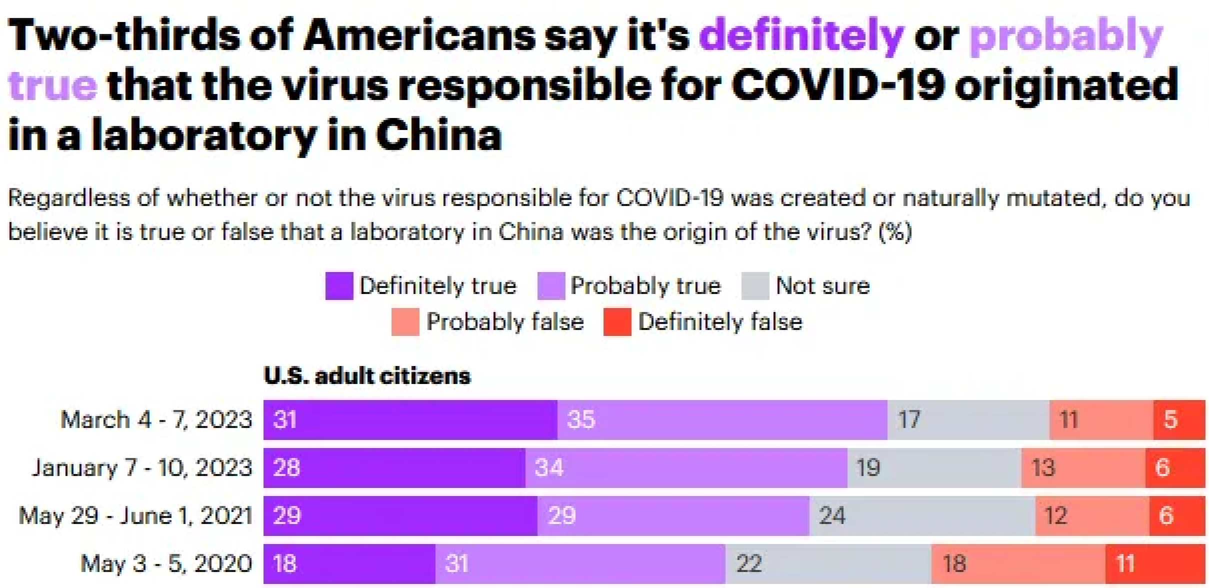
There's also a pattern I want to discourage, where one side will come up with some new trivial finding, or re-dredge up and re-package something that everyone alreac everyone else had already considered, then release it as THE SMOKING GUN! Ther they release another SMOKING GUN!, and another, and after five or six SMOKING GUNS, they say their opponents are stubborn and refuse to yield to evidence, since they've obstinately ignored every single SMOKING GUN! without changing their probability even a little bit.

Overall I don't think it's useful to update on the exact contours of the ebb and flow new evidence. Just treat new evidence the same as old evidence, updating your model the same amount as everything else.

The Aftermath: Debate

Some skeptic blogs picked up this story last month, and one of the points they mac was that even if this one turned out well for their side, in general they're against this kind of thing.

Part of their argument was that debating “conspiracy theories” just helps spread ar legitimize them. I’ve made fun of this position before, and I’ll make fun of it again nc According to polling, about 66% of Americans believe lab leak, compared to 16% w believe natural origin and 17% who aren’t sure. That means that people with an opinion on the issue are more than 4:1 in favor of lab leak. At some point you have t start debating! What are you waiting for? If you hold off so long that finally every sir person in the world except you believes lab leak, would you still be sitting there, pristine in your imperturbability, saying from your lofty height “I refuse to engage, because that would be providing the rest of you oxygen”?



The other part of the argument was that saying “I will debate all comers for an $X b is annoying, and we shouldn’t encourage that kind of thing. Certainly this technique has been used by bad actors - for example, the Holocaust denial group Institute Fo Historical Review offered a $50,000 prize to anyone who could prove the Holocaus happened (it was eventually won by an Auschwitz survivor whose “proof” was that saw his family led to the gas chambers; IHR failed to accept this; the survivor sued won). Likewise, anti-vaccine multimillionaire Steve Kirsch has offered to bet $500,C on the results of a debate about vaccines not working [9] (Saar took him up on it and they're continuing to hammer out the specifics).

I assume the concern is that (if the court system hadn't stepped in), the Institute fo Historical Review could have kept denying any evidence they were given, then kept taunting people with "We've offered $50,000 for proof that the Holocaust happene nobody has ever won our money, so the proof must not exist". Or Kirsch could keep saying "Nobody will bet me $500,000 on vaccines, guess they're scared and think they don't have evidence" (when in fact it's just that most people don't have the tin courage, and risk tolerance to do this, especially when there's no guarantee the rigl person will win the debate). In order to deny these people this weapon (the argume goes) we need to make it common knowledge that this strategy isn't legitimate. An taking people up on their offer, having a great debate that leaves everybody more enlightened and serves as a model for rational discourse, then having the right side win in the end - seems like the opposite of delegitimizing this strategy.

I guess I classify this with all the other examples in Less Utilitarian Than Thou. Cool Machiavellian plot you have there, but maybe the fact that you're losing 16%-66% should make you question whether you're really as smart as you think you are, and whether your plan to suppress all discussion for the greater good is really the mastermind-level strategy you hoped it would be.

It's good to assert the true fact that these kinds of challenges are often dumb/rigged/useless, and that "nobody has yet responded to my challenge" isn't a valid argument that someone's necessarily right. But I stop short of trying to set so kind of social norm that nobody may respond to anyone else's challenges, even if tl think that person is being honest and has organized the challenge well (as Saar wa and did). That almost seems like *itself* legitimizing the whole thing, in the sense of accepting that if someone loses a challenge then it means something important. I would rather place the illegitimacy where it belongs (a challenge really doesn't prov anything, separate from the arguments made in it) and let people do what they war

I want to see more debates like this. I learned more watching the 15 hours of Rootcl debate than I think I would have researching on my own for 15 hours. But a lot of things had to come together to make this work. Most of all, this debate worked out because the judges were two very smart scientists with relevant expertise. To get s good judges, lots of things had to fall into place.

First, the debate itself had to be expensive enough that neither side begrudged pay the extra $5,000 per judge to hire the best people.

And second, the debate had to be about a topic where lots of intelligent people haven't yet made up their minds. If the debate was about flat earth, I would despair finding good judges. Either the judges would already be convinced the Earth was round (which the flat Earth side would understandably refuse to accept). Or they would be 50-50, which would mean they were extremely weird people whose reasoning couldn't be trusted. Flat Earth is an extreme example, but even a debate about COVID vaccines would be pushing it here.

(since writing this, I learned Peter had made this same argument and analogy in a b post on Kirsch; sorry for the unintentional plagiarism)

I think I would genuinely update on the conclusion of any other Rootclaim debate w the same caliber of participants as this one, but not necessarily on whatever Steve Kirsch or the Institute of Historical Review comes up with, nor the next person to hi the strategy of "I'll pay you $100,000 if you prove me wrong!" [10]

The Aftermath: Conclusion

This was one of my favorite topics to write about this year, for a few reasons.

First, on the object level, I learned a lot about the origins of COVID, which is a great story. I feel like I know much more now about this disease that came out of nowher and ruined all of our lives for a few years. It's a weird rabbit hole, which I'm not yet entirely out of. I have a weird urge to visit Wuhan as a tourist, see the Wuhan Institu of Virology, stroll through the Huanan Central Seafood Market (unfortunately closec maybe eat a raccoon-dog.

Second, some of the lessons of this debate are actionable. I've written before abou how we should learn the lessons of lab leak even if it turns out to be false this time; that hasn't changed. But this was a good reminder to also learn the lessons of zoonosis, for the same reason. We need more attention on closing wet markets anc tracking weird Chinese wildlife. The DEFUSE proposal wanted to immunize bats - is this still a worthwhile idea? The virologists got a bad rap for wanting to gain-of-function exactly the pathogen that caused the century's worst pandemic, but in a w that speaks well of them - they clearly knew what to be worried about. Has anyone mumbled an apology and asked them if they have any other useful predictions?

Third, John Nerst has written about erisology, the study of disagreements. This was surely one of history's greatest erisological studies. Two very smart people spent fifteen hours hashing out every argument and counterargument in good faith, then quantified all of their beliefs in a way that lets us figure out exactly where they diffe and by how much. This isn't entirely a victory - as a newly minted member of team zoonosis, I still can't trace exactly why Saar is so sure I'm wrong. But if the COVID origin story fascinates me as this peek deep into a pestiferous underworld of siniste laboratories and reeking wet markets, something about this debate felt like analogc peek into the creepy subconscious swamps where disagreements begin.

Fourth, for the first time it made me see the coronavirus as one of God's biggest an funniest jokes. Think about it. Either a zoonotic virus crossed over to humans fifteer miles from the biggest coronavirus laboratory in the Eastern Hemisphere. Or a lab l virus first rose to public attention right near a raccoon-dog stall in a wet market. Eit way is one of the century's biggest coincidences, designed by some cosmic joker w wanted to keep the debate acrimonious for years to come.

But fifth, if the coronavirus' story is a comedy, all of this - Rootclaim, the debate, th $100K - is a tragedy. Saar got $100 million, decided to devote a big part of his life t improving human reasoning, and came up with a really elegant system. He was so confident in his system, and in the power of open discussion, that he risked his mo and reputation on an accept-all-comers debate offer . Then some rando who nobo had ever heard of accepted the challenge, turned out to be some kind of weird deb savant, and won, turning what should have been Rootclaim's moment of triumph in bitter defeat. Totally new kind of human suffering, worthy of Shakespeare.

I look forward to the movie, especially seeing who plays the dashing young blogger who helped the participants meet.

Other Resources

- Daniel Filan's running Twitter commentary of the debate
- Daniel's Bayesian analysis Google Doc
- Peter's blog, with some articles on his COVID investigation
- Rootclaim site
- Philipp Markolin's article (preachy, annoying, but has good interview with Peter
- Another debate paralleling this one on Less Wrong, starting with Roko for lab le and with viking_math and EZ97 on the zoonosis side.
- Both sides emailed me excitedly about this claim by Marc Johnson to have fou COVID regressing back to its original gut virus form. Johnson claims the regression pattern suggests COVID spent significant time in some non-bat ani before progressing to humans, although this doesn't distinguish between an intermediate host (eg raccoon-dog) or an animal-based culture medium at a la (eg human airway cells). Yuri thinks you might be able to run COVID ancestors through different animals and media to see which one produces COVID-characteristic mutations, which might help resolve this debate.
- Interview with Saar about Rootclaim (not the lab leak debate) in Israeli news

- Michael Weissman has an excellent Bayesian analysis favoring lab leak, which I unfortunately didn't include in my table because it would have taken too long t translate his language into mine.
- Judge Eric has a site about the debate and a blog including a response to Mich Weissman.
- If you want to try doing your own analysis in the same style as the ones above, put a calculator up here. You'll have to copy it to your own Google Drive before adding your own numbers. If you do try it, please link your results here (after setting share settings to "anyone with the link") so I can try to aggregate everybody's work later. The calculator has not been exhaustively tested and mi be wrong/buggy, please let me know if you notice problems.

---

1 15 hours of watching/moderating the debate, and the rest was a combination of researc (to double-check participants' claims) and writing their verdicts.

2 Potential conflict of interest notice: I handled escrow for this bet, and was paid $1,000.

3 Making these tables was tough, both because everyone's analysis had a slightly differer structure, and because you have to be careful not to accidentally double-count factors. Peter and I got in an argument over whether having a prior for a given pandemic being l leak, a factor for the pandemic starting in Wuhan in particular, and a factor for how likel the virus was to escape from WIV was implicitly double-counting things. I eventually se for something like his presentation, but I've probably managed to mess it up somehow. Sorry in advance if this analysis turns out not to make sense.

4 A separate market on the lab leak hypothesis itself shifted less, from about 70% to 60% This could either be because bettors thought Peter was a great debater but wasn't actu right, or because most people in this (very large) market didn't even watch the debate. I general I'm not optimistic about markets with no plausible way of ever being resolved.

5 In a discussion on this point, Saar says he's suspicious even of 1-in-10,000 level eviden on *one* side, because even if you're right, you might be ignoring outside-of-model error.

accept that this is a valid point, but counter with this article on how Strong Evidence Is Common. For example, suppose I win the lottery, I'm told I win the lottery, the lottery company gives me a big check, I cash the check, and I become rich. Given that there we 1-in-100-million odds against me winning the lottery, the lottery company giving me the check and so on must be at least 100-million-to-1-level evidence - otherwise I should refuse to believe I really won the lottery, even as I enjoy my newfound wealth! This is a k of stupid and trivial point, but I still think it's useful to remember. The reason 1-in-10,00(level odds are rare in arguments isn't because they're rare in real life, it's because if ther an argument it means there's a smart person on the other side who's not interpreting th odds the same way you do, so you might be wrong.

6 I wouldn't really describe our interaction this way. I started with a 10,000x Bayes factor c the market, but it was extremely lightly considered and not really adjusted for out-of-mc error. Based on our discussions, I divided by four based on Saar's good point that the market represented less than 100% of the possible zoonotic spread opportunity in Wuh (I cashed this out as it representing 25% of opportunity, though with high error bars). Tl I divided by an extra factor of five representing some sort of blind outside view adjustm based on how strongly Saar holds his position (this was kind of also a decision to explici include potential outside-the-model error because that would make discussing it with S easier).

7 The end probability on my analysis (95% zoonosis) is pretty close to my genuinely held probability (~90% zoonosis). But this is a coincidence - my original calculation had som errors, it returned 99.999% zoonosis, and I planned to ignore it and continued to believe ~90% zoonosis. I don't claim to be good enough at explicit probabilistic analysis that yo should take my results seriously.

8 Or he should enter forecasting contests. I hope to talk to him more about specifics of th option.

9 Some people have since claimed that Kirsch's debate offer isn't honest - see https://twitter.com/SkepticJonGuy/status/1665071649555906561 and https://twitter.com/SwaledaleMutton/status/1721170306788630713.

10 It would also be interesting to have a respected central institution that runs these kinds debates, just as longbets.org is a respected central institution that runs bets.

907 Comments

Write a comment...

Comment deleted Mar 29, 2024

Comment deleted

EMSKE Phytochem Apr 3, 2024

>In fact your prior...

The 1977 fku pandemic is feeling very neglected here.

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4542197/

REPLY

S

Comment removed Mar 28, 2024

Comment removed

REPLY (1)

S

polscistoic Mar 31, 2024 *Edited*

The (applied) point of the discussion is the hope that the CCP will react to this (for them) embarassing and apparently never-ending discussion by cleaning up both the wet markets the research laboratories studying stuff like this.

...and from that angle, the ideal outcome of the discussion is 50-50 for the wet market vers the laboratory leak - hypothesis.

More generally, there are two motivations for caring about the probabilities for a lab leak ve zoonosis:

(1) You portray yourself as an observer of the world, and simply are intellectually curious ak what goes on in it. You care about the probabilities only because of the off-chance that dur your lifetime CCP will fall from power or something equally dramatic will happen in China, w makes the authorities open the archives so it is possible to check if what you thought was probably the cause, was actually the cause.

(2) You portray yourself as a participant in the world, and want to influence policy-makers a your fellow man to limit threatening risks. The probabilities you and/or others assign to pos causes serve as calls to action to do something to change/reduce the probability of similar pandemics in the future.

These motivations are very different. They are likely to influence your own stated probabilit as well as how you interpret why others offer the probabilities they do.

REPLY

S

Comment removed Mar 28, 2024

Comment removed

REPLY (1)

S

Peter Gerdes Mar 28, 2024

I don't think that's really a problem. Saar has an idea he wants people to take a look at so h paying to get eyeballs. It's just like Pepsi advertising on TV with more social benefit.

I actually underestimated the signalling value of it since I'm so used to rich people (my pillo guy) announcing this kind of shit but then adding fine print that makes it impossible to win.

REPLY (1)

Comment removed Mar 28, 2024

Comment removed

REPLY (1)

Peter Gerdes Mar 28, 2024

Certainly, but that's not bad for the listener. Indeed, I'm well served if both sides a
strongly incentivized to win (and Saar is motivated here by wanting his idea to loo
good).

REPLY

Comment removed Mar 28, 2024

Comment removed

REPLY (1)

Sun Kitten Mar 29, 2024

I think it's different in biology (assuming you're speaking from a non-biological engineering point of view?). It's been a while since I did any myself, but making mutations in any organis not something you mess around with. You insert your chosen mutation in the simplest poss well established way because even then most of the time it won't work. Messing around for sake of it is best kept to other things, like giving genes and alleles funny names.

REPLY

Comment removed Mar 28, 2024

Comment removed

REPLY (3)

Comment deleted Mar 28, 2024

Comment deleted

Leo Abstract Mar 28, 2024

I have a memory of Trump barring travel from china fairly early, while also saying it's probably fake, while also calling it the China Virus, and while also saying that it'll blow soon. This is consistent with my model of Trump saying lots of things for various reaso none of which are primarily anything like "provide parsimoniously accurate information

I'm open to seeing a well-cited timeline of his actions and statements, but I don't think of us care enough to create or find such a thing.

REPLY (1)

S

JoshuaE Mar 28, 2024

The Chinese travel ban occurred in late January of 2020 after an increasing numb of cases were discovered in the US. In mid march San Franscisco and the Bay are imposed the first shelter in place order (although behaviors were changing in the area for the prior two weeks and on March 20th NY declared a stay at home order (effective march 22) https://en.wikipedia.org/wiki/Timeline_of_the_COVID-19_pandemic_in_the_United_States_(2020). One thing is that the behavior in ear 2020 involved a series of relatively (within the lifetime of most people) unprecede steps and while it is easy in hindsight to argue about what should have been done neither China nor the US really understood the consequences and in particular we constantly make decisions that were days to weeks behind where they should hav been.

REPLY (1)

S

Leo Abstract Mar 28, 2024

Yeah I personally have no blame for any decision-makers. That itself seems a unpopular opinion, but it's just incentives playing out. What's the age of the electorate? How scared of dying are first-world elderly? Boom, tells you everything you need to know.

REPLY

S

Leo Abstract Mar 28, 2024

I agree that it's unimportant per se. I'd argue that the other specifics throughout were also unimportant: masks, vaccine efficacy, vaccine safety, lockdown efficacy, ivermectin efficac vitamin d efficacy, etc. None of the mechanisms seemed to matter. What mattered more wa "are people lying to us?" and the answer to that was "yes, unambiguously" (on both sides c each and every one of these debates).

REPLY

S

Tophattingson Mar 28, 2024

The consensus prior to 2020 was that shutting borders would only serve to delay respirato disease outbreaks by ~2-4 weeks because, under exponential growth, blocking 99% or eve 99.99% of travel still means its not long until covid hops the border. For China to have acte

maliciously in not closing the border would require that 2021 values and beliefs travelled ba
in time to 2019.

REPLY (1) S

Comment removed Mar 28, 2024

Comment removed

REPLY (3) S

Tophattingson Mar 28, 2024

Locking down internal travel is just something communist regimes do. In China it i part of the Hukou system. In these sorts of regimes, internal travel restrictions are historically the norm, and permitting internal travel is the aberration. Therefore Ch reinforcing internal travel restrictions is far less interesting than it sounds.

REPLY S

Kenny Easwaran Mar 28, 2024

They wanted to delay spread by 2-4 weeks.

REPLY S

JoshuaE Mar 28, 2024

They should have locked down the whole country at that time but that would have been too extreme in 2019 so they did an extreme but less extreme thing of locking down the city but of course it had already spread to all of China by that point so th lockdown was only moderately effective. Again everyone was learning and reactin early 2019/2020 and typically they were making decisions that were weeks late du not thinking about exponential curves.

REPLY S

Comment removed Mar 28, 2024 *Edited*

Comment removed

REPLY (3) S

gwern Mar 28, 2024 *Edited*

Yes, I don't understand this (paraphrased) claim by Peter: (EDIT: this seems to be Scott ma a mistake in paraphrasing Peter)

> He also told the Mail that his cat got the coronavirus too, which is impossible.

'Impossible', thus implying the man was lying? I was under the impression that, quite aside cats having tons of coronaviruses in general (FCoV being a particularly serious threat to yo cats, which also seems to be a remarkable case study of the harms of the FDA), that it was just not 'impossible' for domestic pet cats to get the coronavirus too, it was routine for ther get COVID-19, and even other cat species in *zoos* have tested positive and this was true early in the COVID-19 pandemic and quite well publicized and well known (eg April 2020 https://www.nationalgeographic.com/animals/article/tiger-coronavirus-covid19-positive-tes bronx-zoo). This was a topic of interest to me at the time because I like cats and have a ca and was wondering what the implications of me being inevitably infected might be for my c and so I remember this quite well despite my general attempt to remain ignorant of as many COVID-19 matters as possible... And double-checking now to see if all of these reports we somehow false positives or faked, I continue to see everyone like the CDC stating that it is totally possible and routine for cats in close contact with infected humans (you know, like a *pet* cat) to be infected with COVID-19: https://www.cdc.gov/healthypets/covid-19/pets.ht

Given that Peter has supposedly spent years autistically researching every last detail and tl detail in particular in order to discredit that British dude, I'm experiencing sudden Gell-Man Amnesia here about the rest of his claims, as well as the supposed experts evaluating Pete claims if they didn't flag that (I have not checked).

REPLY (6) S

Comment removed Mar 28, 2024

Comment removed

REPLY (1) S

gwern Mar 28, 2024

This doesn't justify Peter's claim. Both of my links confirm that cats do exhibit physical symptoms and disease traits like 'dry coughs' and 'loss of appetite', amo many other symptoms like 'vomiting' or 'diarrhea', and in the case of the zoo cats, those symptoms were why they were tested at all for COVID-19 and the blood samples could then test positive for COVID-19 in addition to them being sick. Thu is clearly possible, and not 'impossible' for a cat to have COVID-19 beyond a circu test result, and it is established fact according to the CDC, Bronx Zoo and every c source I have checked that cats really can 'have COVID' under any 'conventional understanding of disease' and do not simply merely carry it while testing positive (which a lot of *other* animals do - just not domestic cats and a bunch of their fel relatives).

REPLY S

Simon stats Mar 28, 2024

indeed, cats are very susceptible to covid

REPLY (1)

Nyctereutes Dec 2, 2025

Far more susceptible than racoon dogs, in fact

REPLY

Andrew Currall Mar 28, 2024 *Edited*

Generally speaking I think that:

* Peter makes a lot of very solid points; more than Saar does.

* Peter frequently manages to completely dismiss some of Saar's claims; Saar hardly e does this to Peter.

* Peter is just generally a better debator.

But...

* Saar is arguing honestly and presenting his actual views.

* Peter is dishonestly making the strongest case he think he can for his side, which includes deliberately saying incorrect and misleading things.

I trust Saar much more, despite probably broadly thinking that the lab leak is <50%.

REPLY (3)

Thomas Kehrenberg Mar 28, 2024

Yeah, I found this comment from Peter on the Manifold Market a bit concerning:

> So what's my incentive to come up with the most reasonable values possible, if opponent just says any data I present is fake/meaningless/actually supports his th via some twisted argument?

> By the end of the debate, it was more like, "fuck it, if he's just going to make up numbers, I'm going to stick with largest numbers for my case".

https://manifold.markets/chrisjbillington/will-bsp9000-win-the-rootclaim-chal#2C7VearXa03px5PI8p6A

REPLY (1)

Steady Drumbeat Mar 29, 2024

The context for that is Saar asserting that the market data carries 0 weight a: evidence for Zoonosis. Peter is saying that he hewed to an argument which w *clearly* unreasonable instead of contesting numbers which Peter thought w reasonably subject to debate.

REPLY S

Whatever Happened to Anonymous Mar 28, 2024

>Saar is arguing honestly and presenting his actual views.

>Peter is dishonestly making the strongest case he think he can for his side, whic includes deliberately saying incorrect and misleading things.

This makes sense when you remember that one of the debaters is just some guy a the other one is very wealthy. If 100k is a very meaningful part of your net worth, y can't afford to not play to win.

REPLY S

Charles Pye Mar 28, 2024

"making the strongest case he can for his side"- isn't that how a debate is suppos to work? It's certainly how the legal system works. It's a competitive process, you not supposed to help out the other side. The *judges* are the ones who are suppo to be neutral and get at the truth of the matter.

REPLY (1) S

whale Mar 28, 2024

Yes, and this is the result:

https://www.youtube.com/watch?v=__reD_phfbg

REPLY S

Écorché Mar 28, 2024

During the debate, Peter also cited a paper showing that the first Beijing superspreade event (another wet market, what are the odds?) had originated in frozen fish.

REPLY (1) S

Écorché Mar 28, 2024

Oh, it's in the transcript above. Peter says:

"We know why there was an outbreak in the Xinfadi Market in Beijing: it was becau the seafood stall got frozen fish from some non-Zero-COVID country, the fish hac

COVID particles on it, and the vendor got infected and spread it to everyone else. Something like this is true for the other Chinese wet market based outbreaks we k about it."

REPLY

S

tgof137 Mar 28, 2024

I think it's an error with Scott's paraphrasing -- Connor said that has cat died, not that got sick. Here's his quote:

"Pictured: Connor and the cat that hangs around his apartment in Wuhan, China. Suddenly, I'm feeling better, physically at least. The flu has lifted. But the poor kitten h died. I don't know whether it had what I've got, or whether cats can even get human fl feel miserable"

From a dailymail article:

https://www.dailymail.co.uk/news/article-8075633/First-British-victim-25-describes-coronavirus.html

I am well aware that cat's can catch the coronavirus (though my cat never got visibly il of the 4 times covid's been in my household). But the prior on an pet cat dying from it should be low.

Another commenter above suggested that "maybe the cat just tested positive". This is November or early December 2019, before covid was known to exist and before covid tests were invented. The only way that Connor himself would even know if he had it is went to a hospital and they saved a sample and then the hospital tested it later. Conno claims that the hospital called him weeks/months later and told him he tested positive, there's no other record of that. Either way, they most definitely did not save a swab fro his cat and PCR test that.

The prior on a 25 year old going to the hospital from covid should also be pretty low. T also can't be a sampling bias thing, given the predicted number of covid patients at th time, there aren't going to be many and even less are going to speak English -- even ir case growth model that Rootclaim tried to present, I calculated there would be at most that time)

The last interesting thing is that Connor says that he shopped at the market!

Specifically, he wrote:

"My local paper back in Llandudno, North Wales, has been in touch with me. Maybe I caught the coronavirus at the fish market.

It’s a great place to get food on a budget, a part of the real Wuhan that ordinary Chine: people use every day, and I regularly do my shopping there.

Since the outbreak became international news, I’ve seen hysterical reports (especially the U.S. media) that exotic meats such as bat and even koala are on sale at the fish market. I’ve never seen that."

So, that's not 100% explicit as to referring to Huanan market, but Huanan market was one in the news, and there was a specific rumor about Koalas being sold at Huanan market, so it seems pretty clear that's the market he's referring to.

So, either:

A. He's lying about the whole story and just fabricating it from stuff he's seen in the ne

B. He's a super early patient who got infected at Huanan market, and he's great evider for the zoonosis theory.

C. His story is true except that he actually shopped at a different market other than Huanan, and that makes him the super secret early case that proves lab leak.

I interpret this as most likely A. Rootclaim sees C as most likely.

In the third debate, I even tried to lay out a quick bayesian analysis for how to decide between these options. I think that's sort of the problem with Rootclaim's method -- tl decided from a high level view that they think that lab leak is true, then they went look for evidence to confirm that view, with no good filter on what evidence should be trust i.e. they decided that the cases in the WHO report are 100% fraudulent but Connor Re case is 100% likely to be covid. It all just seems like motivated reasoning, to me.

REPLY (1)

gwern Mar 28, 2024

> I think it's an error with Scott's paraphrasing -- Connor said that has cat died, n that it got sick. Here's his quote:

This isn't about Connor, this is about Peter. Did Peter claim what Scott paraphrase him as claiming, and apparently no one in the debate called him on it? (I'd just che myself, but I've now searched the YT auto-transcript of like 6 or 7 of these YT vid and haven't been able to find a quote matching it; and I'm definitely not watching whole thing, so hopefully someone who *does* know where in this rats-nest of lin Peter would have said it, can go rewatch it and tell us what he said.)

REPLY (2)

tgof137 Mar 28, 2024 Edited

I'm Peter. I claimed that it was an untrustworthy account because the cat die
not because the cat got sick (as well as all the other issues I listed above).

IIRC Saar later made some counter claims that Connor shopped at a different
market and that the cat died of some feline coronavirus. None of which is supported by the original articles, just a purely motivated reading, I think.

REPLY (2) S

Seta Sojiro Mar 28, 2024

Off topic, but I find myself less interested in the epistemology behind COVID's origins (which is still quite fascinating) and more interested in w
you're planning to do with your research and persuasion superpower. Th
are lot of people who capable of being intellectually rigorous but not persuasive verbally. And also vice versa. To have both is quite rare. Most
traditional careers after a physics degree seem like a waste.

REPLY (1) S

tgof137 Mar 29, 2024

I certainly intend to research other things in the future, and I hope t
done with this topic soon.

I have never thought of myself as persuasive at all, have never deba
before or given many presentations or speeches. It just seemed like
couldn't be that hard -- Rootclaim's case is so thoroughly wrong --
went for it. But mostly it was stressful and not very fun, so I'll proba
stick to writing, coding, researching, and things like that.

REPLY S

gwern Mar 29, 2024 *Edited*

> I claimed that it was an untrustworthy account because the cat died, n
because the cat got sick (as well as all the other issues I listed above)...N
of which is supported by the original articles, just a purely motivated reading, I think.

We are going off on a tangent here from my original question about the paraphrase, but now that I check... how is that not supported, and how i
your description supported by the original articles, exactly? The source l
seems to be the Daily Mail account https://www.dailymail.co.uk/news/art
8075633/First-British-victim-25-describes-coronavirus.html which inclu

photos of the cat, which is clearly a kitten (an important detail to omit), a
all of the references to the cat are:

>> Day 9: Even the kitten hanging around my apartment seems to be fee under the weather. It isn’t its usual lively self, and when I put down food doesn’t want to eat. I don’t blame it – I’ve lost my appetite too...Day 11: Suddenly, I’m feeling better, physically at least. The flu has lifted. But the poor kitten has died. I don’t know whether it had what I’ve got, or wheth cats can even get human flu. I feel miserable.

I do not understand how you go from his claim "I don’t know whether it h what I’ve got, or whether cats can even get human flu. " to your slide https://www.youtube.com/watch?v=Y1vaooTKHCM#t=1h10m30s saying "Claims his cat died from covid" (which you also repeat aloud to empha it). You then repeated it again in the slides & aloud (without any addition source so presumably still the Daily Mail source) at https://www.youtube.com/watch?v=CTK1hWOzwns&t=1510s as "He says cat died from covid...this is what I read in the article". Are we looking at t same 'article'? Because I don't see how you could have read that in the I Mail article given the only references to the cat in it which I quote above.

Connor explicitly doesn't claim the kitten got what he got (much less go COVID-19), so a claim that he didn't make cannot make him 'untrustwor

Nor is the cat dying anything to make him 'untrustworthy': it is entirely ordinary for kittens to die of respiratory illnesses like various coronavirus to which they are especially vulnerable. (Lots of cat diseases to go arou unfortunately.) This is a major source of kitten mortality in litters, pounds notorious as to have their own names like 'kennel flu'), and cat colonies, cats in general, and there is nothing all that odd about it happening or a kitten dying of it; and it dying is that much more likely if not treated (whi apparently wasn't inasmuch as Connor was busy being sick and being a expat probably couldn't call on anyone for a local vet). So it should be n surprise that kittens die of COVID-19: https://bvajournals.onlinelibrary.wiley.com/doi/full/10.1002/vetr.247 Anyw young kittens die like he describes of FCoV *all the time* (you may have heard about the Cyprus outbreak last year), so there's nothing odd abou the death itself.

REPLY (1) S

tgof137 Mar 29, 2024

See some of the articles Scott dug up, for more inconsistencies:

https://www.astralcodexten.com/p/practically-a-book-review-rootclaim/comment/52739539

Both of those articles say Connor was hospitalized for two weeks, v previous ones saying he was sick for 2 weeks and then went to a hospital briefly.

The first article also goes on to quote him saying, "I honestly don't t the virus is as serious as all that. The people who have died must ha had underlying health problems", which is a weird description for hospitalized for 2 weeks.

I suppose you can't entirely determine whether Connor's story is inconsistent or this is bad reporting.

Same with "the feline coronavirus". How would he know what the ca had? He says he was housebound, wasn't going to the vet. I'm not s the usual timeline for FIPV. In the Dailymail, he's asking if the cat go "flu" he had. Then in the Wales online article, he's saying he's not s if he caught it from the cat. But if he's certain it was "the feline coronavirus", that doesn't even infect people, right?

The path to this all being important requires both that "Connor is trustworthy" and also "Connor is lying about shopping at Huanan" (else I misinterpreted which market he was talking about) and "We'v settled on a specific growth rate for the market outbreak" for which evidence would contradict and it's somewhat dependent on "Conn got covid on Nov 25th, not during his second wave of illness in early Dec".

I think I'm very likely to win either "Connor is untrustworthy", with "Connor shopped at Huanan" as a backup.

With a certain motivated reasoning, though, you can conclude that Connor is great evidence to contradict zoonosis at the market, and therefore "SMOKING GUN FOR LAB LEAK!"

I also think I have a strong enough model, between case growth rat genetic diversity, Wuhan serology, etc, to simply put low prior odds dubious covid infections in November as being likely to be true.

Likewise, if Connor really got sick in November, where are the interviews with 100+ Chinese people that also did, that you would statistically expect? There are interviews available with some of the earliest people sick at the market. Did China somehow retroactively censor every other similar account from before?

It's the same thing with e.g. positive covid tests in Italy in Septembe did a laborious deep dive through all those papers just in case Saar brought them up, but you can also determine they're unlikely to be if you have a good enough understanding of the other evidence.

In a larger sense, I think there's a certain personality type that gets drawn to the patterns and another that gets fixated on outliers. Thi dynamic even created a new theory, at the debate:

https://drive.google.com/file/d/1u_Ne53YLJuSPm75jjCRkeQJHGvpF r/view?usp=sharing

I wrote some code to model and graph mutations that were reversic There were 2 dots out of place in my graph. I sort of noticed when making it, but it didn't really register in my brain as that important.

Saar noticed the dots later and concluded they were "SMOKING GL FOR LAB LEAK!" and they coincided with SMOKING GUN DELETED EARLY CASES and SMOKING GUN ANTARCTICA SOIL SAMPLES (w both also turn out to be misleading and unimportant claims under further investigation).

So I dug into the code and the data to figure out why my diagram di match Pekar's, and found it was a mundane explanation for the 2 dc and corrected the diagram.

Once you've been through this process dozens of times with the lak leak crew, you kind of develop an understanding that they're just throwing out one bad claim after another, and those rarely ever holc

I think this is actually a great heuristic, that's better than bayesian reasoning, for spotting conspiracy theories.

One of the first places I might have developed this approach was in 2020, when I was trying to figure out if the election was stolen, i.e. f curiosity and to know who to bet on in the political betting markets.

The 2020 election fraud people made a lot of claims that the electic was stolen. People who doubted it asked "where is the evidence?"

So someone made a site called "here is the evidence"

https://hereistheevidence.com/

It looks like it currently has 1,687 claims as to why the 2020 electior was stolen.

My process, back in 2020, was to go through maybe 20-30 electior fraud claims and evaluate them. Every single one I evaluated was ei clearly false or just ambiguous and misleading. So I concluded that not have to read a list of 1,000 and I bet that Biden would still win (I forget the rules of the market, perhaps I was still taking a risk on a successful coup).

It's kind of similar with lab leak -- I've evaluated hundreds of claims and found the vast majority either blatantly or subtly wrong. Had an them been convincing, I would have dropped out prior to the debate and moderated my stance on covid origins, maybe even flipped it. S credulously relied on many of them in the debate, many more transparently false than Connor Reed's covid case.

From Saar's probabilistic perspective, it's more like: the details for Connor don't even matter. Even if Connor didn't have covid, someor else did, and if he dug hard enough, he's sure he could find that pe And because he thinks that, you can't apply high odds to the marke outbreak. In fact, he thinks you can't apply any odds at all to it, he thinks all existing case data is bayes factor 1 between market and la

I think Saar has concluded that lab leak is almost certainly true base on the location in Wuhan and the furin cleavage site. Therefore, lots non-market early cases must exist somewhere, and he doesn't eve have to look.

If someone made "here is the lab leak evidence dot com", with 1,70 different examples of early cases, I would go through 100 and notic that 95 are clearly wrong and 5 are ambiguous and whoever curate the site is probably just not very careful.

Saar would look at the list and say, "wow, that's 1,700 different clain even one of them is correct, then it's surely a lab leak! What are the odds that 1,700 claims are all wrong?"

I could be wrong about all this, of course. If hospital records for Cor Reed ever turn up, I will apologize, change my mind, and evaluate whether that changes the origins picture based on those details.

REPLY (1) S

gwern Mar 29, 2024 *Edited*

> See some of the articles Scott dug up, for more inconsistenc

Well, that's a long comment and I'm sure probably of interest t other people, but most of it is irrelevant to my questions here. I didn't ask for 'inconsistencies'. I asked a simple question: wher did he say (as you have repeatedly claimed now, in many place while dragging in dozens of extraneous claims and arguments) "COVID-19 killed my cat"? In both sources, he does *not* say t In this new source, he says in total with reference to his cat: https://www.walesonline.co.uk/news/wales-news/welshman-thought-first-brit-catch-17684127 (emphasis added)

>> "My kitten caught the *feline coronavirus* and developed pneumonia and died, but *I don't think I caught it from her*. I th that was just coincidence."

That is, he is explicitly saying his kitten caught and died of FCo and *not* COVID-19, and he is denying any suggestions that hi mystery sickness might've been some non-COVID-19 caught f her (which is probably right, I've never come across anything a FCoV infecting humans).

> Same with "the feline coronavirus". How would he know what cat had?

He knew it had 'feline coronavirus' (FCoV) for the reasons I kne that the kitten probably died of perfectly ordinary FCoV/FIP bef ever saw this additional article: because that kills a huge numb healthy kittens with exactly the described symptoms and often quite rapidly once they become visibly sick with respiratory symptoms. (I would not be surprised if the reason he was firme and more precise in his second claim is because he had time t look up more information on FCoV or someone read his Daily M comments and told him that it was definitely FCoV.)

On a side note, Scott, the whole drug approval situation for FIP interesting and might be worth a post: https://www.theatlantic.com/science/archive/2020/05/remdesiv cats/611341/ https://en.wikipedia.org/wiki/GS-441524 https://en.wikipedia.org/wiki/Feline_infectious_peritonitis

> He says he was housebound, wasn't going to the vet. I'm not sure the usual timeline for FIPV.

The usual timeline is days to weeks for a large fraction of cats f the wet form. In other words, exactly what he reported.

> In the Dailymail, he's asking if the cat got the "flu" he had.

Which is a reasonable question to ask in a diary entry while delirious and sick and ignorant of veterinary medicine. (Becaus he had COVID-19, she absolutely could have gotten it, as we n know, but she probably didn't because she died so it was far m likely to be FCoV.)

> Then in the Wales online article, he's saying he's not sure if h caught it from the cat. But if he's certain it was "the feline coronavirus", that doesn't even infect people, right?

Er, no, that's not what he said? He says the exact opposite. He says he *didn't* catch it from her: "I don't think I caught it from her."

So let me ask again: you have repeatedly said that Connor said COVID-19 killed his cat. The two sources you have offered by Connor do not say that and in fact say the opposite of that. Wh does Connor say COVID-19 killed his cat?

REPLY (1) S

tgof137 Mar 29, 2024 *Edited*

I wrote my impression of Connor's case after reading and presenting the Daily Mail article (and perhaps 1 or 2 other sources, I forget). It sounds like we are in disagreement as whether my presentation of his case was accurate.

I had not read the Wales online article until today. I suppos that makes me update more towards thinking his cat did d from some other virus, but I'm not sure why I would also update towards the combination "he had covid and also di not visit Huanan market" or "this anecdote is sufficiently trustworthy to decide the covid origins debate".

I provided a longer reply to explain why I think this questio likely is not relevant to the covid origins debate. It's unfortunate that you are disinterested, but perhaps it will benefit someone else.

REPLY (1) S

gwern Mar 30, 2024 *Edited*

> It sounds like we are in disagreement as to whether presentation of his case was accurate.

Er, yes, we are. That is indeed the problem here, your presentation does not seem to be "accurate". You ha repeatedly directly contradicted what Connor said in claims about Connor, and continued to maintain it in y comments here despite presumably re-reading the sources you used, and after I excerpted what Connor in fact say.

> after reading and presenting the Daily Mail article (a perhaps 1 or 2 other sources, I forget)

The Daily Mail article where the cat quote already provided directly contradicts your summary?

How does his Daily Mail "But the poor kitten has died don't know whether it had what I've got, or whether c can even get human flu." turn into your "Claims his ca died from covid"?

> I had not read the Wales online article until today.

? You simply skipped over one of the only sources ab what Connor said until now, despite all that time talkir about Connor in the debate and including him and the in your slides etc...? (This is not some huge investmer time - the sum total of his cat-related statements is ~ sentences by my count, and the rest of the material is that much either, and looks like it's shorter than the comments you've left in just this one thread...)

> I provided a longer reply to explain why I think this question likely is not relevant to the covid origins deb

I think it's relevant to how one interprets *this* debat

REPLY (3) S

Abhishek Mar 30, 2024

Gwern, you are 100% right. Peter has a history o
using carefully crafted, intentional misdirections.

Big part of his argument against DEFUSE was he
claimed that the GoF work would be done in UN(
https://twitter.com/abhishek_s_1/status/1774112ɛ
34471600

However, a draft of the proposal (not the final
submitted proposal) had a comment from Peter
Daszak that he would do a lot of the assay work i
China but was mentioning UNC to make DARPA
comfortable giving them the grant.

When Scott asked Peter about this, he tried to
misdirect Scott and did so successfully.

"(I asked Peter his response - he said the origina
draft of DEFUSE also said that the Chinese side
would do “live virus binding assays”, and this isn’
kind of gain-of-function research necessary to m
COVID.)"

This is a misdirection because the GoF work com
under one of the assay works but you would have
read the DEFUSE proposal to figure this out.

REPLY S

everythingism Mar 31, 2024

"I think it's relevant to how one interprets *this*
debate."

Clearly both sides made some errors and missed
certain details but it's also important to make sur
we are judging this equally.

Based on Connor Reed's own statements it seem
incredibly likely he was trying to say he shopped
the Huanan Seafood Market and could have been
infected there.

Here's what Reed said: "Maybe I caught the coronavirus at the fish market. It's a great place t get food on a budget, a part of the real Wuhan th ordinary Chinese people use every day, and I regularly do my shopping there. Since the outbre became international news, I've seen hysterical reports (especially in the U.S. media) that exotic meats such as bat and even koala are on sale at t fish market. I've never seen that."

https://www.dailymail.co.uk/news/article-8075633/First-British-victim-25-describes-coronavirus.html

Unless people want to try to argue there was ANOTHER fish market in Wuhan that had been covered in US media and where bats and koalas falsely alleged to have been sold, it seems overwhelmingly likely he's describing the Huanan Seafood Market.

https://www.nzherald.co.nz/world/food-market-a centre-of-deadly-coronavirus-outbreak-admits-selling-live-koalas-snakes-rats-and-wolves/WMMILWGWLTKZVLSK5VK6W5H7G4/

This information has been out there for years and DRASTIC and other lab leak proponents have ign it, even as they promoted Reed's story as eviden there were earlier cases unlinked to the market outbreak. Saar did not mention it in the debate a also tried to use Reed's story to make that argum

Is that "intentional misdirection" as the other commenter is saying about Miller? How is it poss they missed this given that, as you said, there ar limited number of articles about Reed in the first place, and this literally appears in a Daily Mail sto and quotes him directly? I'd guess just motivatec reasoning rather than misdirection but it obvious greatly reduces the significance of Connor Reed evidence against the market origin scenario.

REPLY

S

gwern Apr 1, 2024 *Edited*

Anyway, it seems like Peter is done with this comment thread, and is unable to explain or justi his description of Connor, and I am not missing a sources or justifications he had for it.

So I'll sum up my part of the thread: I left a comn questioning an apparent error by the winner (Pet about cats & COVID-19, which turned out to just mistaken summary by Scott, now corrected in OI however, since I had put in some work to look at Peter's slides and statements about Connor & ca and how this discredited Connor by making impossible claims, I thought I'd do an epistemic spot-check of what Peter actually did say about Connor & cats. (I like cats, so I didn't mind the prospect of reading more about cats & COVID.) I started with his first slide point which Scott has r paraphrased and so could be regarded as a quas random choice on my part: "Claims his cat died f covid". Quotes are some of the easiest things the are to use and to factcheck, and so good for epistemic spotchecks of miscitations. ('Miscitati is common https://gwern.net/leprechaun#miscita , unsurprisingly usually favors the thesis of the misciter, highly skewed in specific papers https://twitter.com/ianhussey/status/1641745134 596547 https://twitter.com/ianhussey/status/1641745136 805314 with papers ranging as high as 50% fals descriptions of all cited sources, and in my experience miscitation is a red flag for bad resea So miscitation research tends to justify the old sa _falsus in uno, falsus in omnibus_.)

This turns out to take all of 15s to fact-check: ho
to Google 'connor covid cat daily mail', open hit #
"First British victim, 25, describes coronavirus"
https://www.dailymail.co.uk/news/article-8075633/First-British-victim-25-describes-coronavirus.html , C-f ' cat ', and the first hit is ar
image caption quoting Connor, "Pictured: Conno
and the cat that hangs around his apartment in
Wuhan, China. Suddenly, I'm feeling better,
physically at least. The flu has lifted. But the poo
kitten has died. I don't know whether it had what
got, or whether cats can even get human flu. I fe
miserable". This obviously does not support, or
outright contradicts, Peter's claim and is an
immediate red flag. Perhaps the rest of the articl
No. The 'day 11' quote is the same, and further
reading just shows a sentence or two about how
kitten died of what sounds like a classic FCoV
infection
(https://en.wikipedia.org/wiki/Feline_coronavirus)
has the usual terminal outcome.

Perhaps Peter has some *other* source where
Connor changes his mind later on and is now
claiming that cats definitely can get 'human flu' a
his kitten definitely did die of it? No, in the Wales
article (not findable via 'connor covid cat' but hit
for 'connor reed covid cat'), he says specifically
it was "feline coronavirus" (FCoV). OK, it's not
looking good... So let's ask Peter. This is a dead
simple question that he should have in his notes,
least, if not his slides.

And... you can see the results in this thread.
Thousands of words about everything *but* how
interprets "I don't know whether it had what I've
or "My kitten caught the feline coronavirus and
developed pneumonia and died" and turns that in
his assertion that Connor "Claims his cat died fro
covid" & this helps discredit Connor. Obviously P

just screwed it up and it was convenient for his c
so he never double-checked. Either he was in too
much of a rush to actually read the Daily Mail art
or he did actually read that Wales article but he
didn't know what FCoV is or bothered to look up
phrase and so just *assumed* that 'feline
coronavirus' must have meant 'COVID-19 but in a
cat'. (It's actually quite funny how many words ar
how many other topics get dragged in to not resp
to such a simple question - where it would've be
so much easier to just say "hm yeah I guess I
misread it in my rush since it was, after all, such a
minor point in a big complicated debate. It happe
My bad! I'll correct that if it ever comes up again.

So: 1/1 epistemic spotchecks failed.

I think this is good to know, not because Peter
should be embarrassed. He shouldn't be, IMO.
Peter's job here was not to tell the truth about
Connor, so it's not a problem that he didn't. His j
was to make whatever mouth-noises convinced t
judges to give him $100,000. He has succeeded
splendidly at this. Congratulations! (To put it ano
way, Peter has now earned approximately $99,0(
more from 'arguing online' than I ever have.) If
anything, his victory is *more* impressive if he's
making basic miscitations and no one notices un
long afterwards.

The people who should be embarrassed here are
Peter, but *his opponents* and the judges. The p
of an adversarial format is that your opponents a
supposed to keep you honest. Checking Peter's
citations is about the very least critical examinati
one could do. It takes literally ~15s to check his c
claim, and requires no scientific knowledge or sk
beyond elementary school-level English literacy.
yet, Saar and his team could not do that (and nei

did the judges, despite being paid a substantial amount of money to judge).

Did they lack adequate incentive? Apparently, whatever that is, '$100,000' is still not enough incentive. If $100,000 doesn't pay for a few seco checking the top-level slide claims like "*did* Co claim his cat died from COVID?", I don't know ho much money it would take to incentivize checkin Peter's claims to a level that, say, a newspaper w not be embarrassed by. And this is the simplest, easiest possible type of claim to check; claims al all the *other* stuff like software bugs or mahjon parlor locations or state-level censorship operations...

So what I've learned from this is that the Rootcla adversarial debate format is not fit for its purpos and has blatantly failed to incentivize adequate criticism of positions.

Can this be fixed? Honestly, I am pretty negative about adversarial debate formats in general, regardless of Bayesian gloss. Even excluding the pathological instances like 'debate clubs', 'debat does not have a good historical track record. You know what sorts of organizations are really fond debates? Ones like the medieval Catholic Church contemporary Creationists. We don't do science debate formats. (I am also pretty down on approaches to AI safety involving 'debate', althou at least there you have the benefit of being able *force* AI agents to spend arbitrary amounts of t examining every part of a debate tree.)

If Rootclaim wants to continue experimenting wit this mechanism, instead of considering alternati with better chances of success like (either adversarial or cooperative) prediction markets or (cooperative) Delphi panels, I have two suggesti for them:

1. refocus on questions that will be definitively kr in the short-term. As matters stand, Rootclaim seems to be obsessed with failing to create a tra record. 'Did Putin have cancer?' 'Was COVID a la leak?' Come on. You may never get a definitive answer to these. And if you look at what they consider a 'track record' https://www.rootclaim.com/rootclaim-track-reco it's stuff like "was XYZ murdered like we concluc after all? well, later on it turned out someone has surveillance tape of a guy sitting in a car! Case closed."

2. adopt a more truly adversarial attitude towards intellectual redteaming. For example, add a simp attention check, like requiring the judges to add randomized number of deliberate errors to both sides in a debate; report how many were found b the adversaries at the end with the verdict. (For debate this size, I think around a dozen would be adequate.) I predict that most of the errors will n be found, and this will, correctly, cast considerak doubt on the evidential value of the debate: if the debators cannot find the deliberate errors reliabl why do you think they can find any of the debato self-biased errors adversarially chosen to fool listeners, or the underlying errors in the original sources, or that the final verdict based on all this would be reliable? For an even more rigorous deb format, one could add a requirement about passi the attention check: if the winner doesn't find at least half of the errors added to their adversary's case, they forfeit either their winnings or their winnings & their stake to charity. (Otherwise, one no incentive to check any more carefully than se necessary to beat an incompetent opponent and loaf.)

REPLY (4)

S

Continue thread →

tgof137 Mar 28, 2024

Alright, here's where I first made the claims in week 1:

https://www.youtube.com/watch?v=Y1vaooTKHCM#t=1h10m30s

Here's where Saar brought up the discussion again in week 3:

https://www.youtube.com/watch?v=XWMmj8Zprnl#t=55m30s

At 57 minutes, Saar infers that it was not Huanan market.

It is ambiguous, Connor just says "the fish market". I'm inferring that it's Huai market because he says it's in the news and the thing about koalas. Saar is inferring that it's some other market, but I don't see why.

On the next slides Saar says it was "the feline coronavirus". That appears to b completely made up, if you compare it to the actual quote I provided above.

Saar then claims that there's evidence Connor went to Zhongnan hospital. There's no evidence of that, as far as I know, other than Connor's claim. And not sure if Connor ever even named which hospital. There's no official record the hospital contacting him and telling him he got sick.

And then here's where we discussed it at the 3rd debate:

https://www.youtube.com/watch?v=CTK1hWOzwns&t=25m10s

REPLY (2)

S

gwern Mar 29, 2024 *Edited*

Thanks for the links. I did see those in the transcripts but I wasn't sure if they were the key ones or if the auto-transcripts had errors. I don't know what the 'fish market' or 'Zhonang hospital' have to do with anything I've asked, but moving on, I can't see where anyone in those 3 videos claims "cats can't get COVID-19", and you do seem to acknowledge that they c in the third video when discussing how he could've tailored his story pos hoc in 2020 (which would be a strange claim for you to bring up had you ever said anywhere in the debates something like "cats can't get COVID like Scott's paraphrase).

So I think Scott Alexander needs to correct his paraphrase in OP: you di
say "cats can't get COVID-19", you seem to instead be saying that "cats
never (ie. rarely) die of COVID-19" (which may sound similar but of cour
are quite different).

I do, however, still have a question about what you did say about cats; se
sibling comment https://www.astralcodexten.com/p/practically-a-book-review-rootclaim/comment/52734427

REPLY (1)

Scott Alexander Mar 29, 2024

Author

My mistake; I've changed the wording in the post.

REPLY

Scott Alexander Mar 29, 2024

Author

During our discussions, Saar sent me this article, in which Connor does claim it was "the feline coronavirus":
https://www.walesonline.co.uk/news/wales-news/welshman-thought-firs
brit-catch-17684127

This doesn't make sense to me - how did Connor know what disease his
had? He said he was housebound that whole period, so he couldn't have
gone to the vet. Also, it seems to contradict his previous story about thinking it was COVID-19.

This article (https://www.foxnews.com/health/man-coronavirus-china-whiskey-honey) says he was hospitalized for two weeks at Zhongnan, w
doesn't match any of his other stories.

I think this is more evidence he's making some of this up.

REPLY (1)

everythingism Mar 29, 2024

If nothing else this quote should throw Connor's credibility into que
slightly: "I think to be fair the Chinese government have been quite transparent with what's going on."

REPLY

everythingism Mar 29, 2024

You're correct that cats are susceptible to SARS-CoV-2 -- there's even a serological s
of cats in Wuhan at the time of the outbreak that found they were infected.
https://pubmed.ncbi.nlm.nih.gov/32867625/

However IMO there are a number of other reasons to question the significance of Conn Reed's story. For one thing, there's no actual verification that he was sick with COVID during that timeframe in the first place, which is why he was not included in the officia cases in spite of all these media stories. Other people have claimed to be unconfirmed early cases such as athletes attending the Wuhan Military Games in October, but these generally believed to be cases of misidentification.

In his story Connor Reed says he might have been infected at "the fish market" that wa written about in the Western media, where bats and koalas were allegedly sold. This o describes one market in Wuhan, the Huanan Seafood Market, where news sites incorre reported the sale of koalas (it ended up being a mistranslation of another species).

https://www.nzherald.co.nz/world/food-market-at-centre-of-deadly-coronavirus-outbreak-admits-selling-live-koalas-snakes-rats-and-wolves/WMMILWGWLTKZVLSK5VK6W5H7G4/

https://mothership.sg/2020/01/wild-animal-wuhan-market/

So if we take him at his word, Connor Reed may actually be backing up the market orig although it's strange that he seems to not know the name of the market itself. He apparently lives some distance from the Huanan Market. All the details in his story cou be picked out of media reports.

There was a wave of influenza going through Wuhan at the time so it seems possible t he just had a bad case of flu and confused it with COVID-19 (although he claims to hav been tested). It would be interesting to follow up with Reed's friends and contacts in Wuhan at the time and ask if they know exactly what happened.

REPLY S

Theodric Mar 28, 2024

Doesn’t a lot of Saar’s argument also rest on evidence that would be impacted by Chinese meddling?

The correct response to “Chinese meddling in the data” is to reduce the likelihood that we “know” the correct answer, rather than to just say “therefore Peter wrong”.

Recall that a wet market origin is also very embarrassing for China. The official CCP propaganda is that it came from some foreign source.

REPLY (1)

S

Comment removed Mar 28, 2024

Comment removed

REPLY (1)

S

Theodric Mar 28, 2024 *Edited*

Which of Peter's arguments on the Turin cleavage site rest on "China being truthf
The participants largely seem to agree on what the site *is* but disagree on whetl
or not the site is more likely to be of natural evolution or a man-made insertion.

For DEFUSE, yes to some extent Peter is taking WIV at their word. But he also poi
out several reasons why it's actually likely that they weren't doing the right type o
experiments on the right strain to cause the epidemic. And why if they were doinç
those experiments, it's likely they would have said so (since before the epidemic t
had no reason to cover up).

Meanwhile Saar's theory *depends on* WIV / CCP lying about what they did after
getting the DEFUSE grant. And he doesn't actually have any proof that they did, h
just inferring that they did from his conclusion that the furin cleavage sites were
probably man-made. His argument also relies on either WIV/CCP pre-emptively
covering up their activities that eventually caused COVID (while being open about
their other research and desire to be involved in gain-of-function studies) OR this
research occurring at exactly the right time that it had gotten done but not publisl
before the virus leaked and they went into cover-up mode.

Conditional on this being an unintentional lab leak, yes it's very likely it would occı
Wuhan vs. somewhere else. Conditional on it being zoonotic, there's not really an
reason it would be in Wuhan vs some other Chinese city (but no particular reason
wouldn't be in Wuhan). How much you weight you put on that "coincidence" will
largely be swamped by how likely you consider zoonotic transmission and lab leal
the first place (which is really hard to sus out, since both are rare).

REPLY (1)

S

Comment removed Mar 28, 2024

Comment removed

REPLY (2)

S

Theodric Mar 28, 2024

Again, your entire argument here is that there is a cover-up that is effect
enough to make zoonotic origin plausible enough to convince a lot of sm
people who've looked deeply into it, but not so good that Saar et al coul
prove it to be false. That's a fine line! Meanwhile you expect Peter to pro
negative (CCP didn't lie in precisely that way).

I would also reiterate: a wet market outbreak still looks very bad for Chin
and we can see that the Party line is that *neither* theory is true. Why w
the CCP stage an elaborate cover up that still paints them in a bad light,
cooking up some evidence that it was from a Sri Lankan tourist or whate

I don't have infinite time but I'll address two points where I think you're b
uncharitable to Peter's argument.

2. Peter is not saying the WIV is "perfectly transparent". He is however saying they were open about their research, and until the leak of Covid would have had no reason to cover up specifically the line of research ar exact strain that would later lead to an outbreak, because they would ha had no way to know beforehand that that strain and that research was th one that would leak.

Unless you believe it was a deliberate bioweapon, in which case you'd sc of expect them to be completely opaque rather than "very open and the opaque only after a possible screw up".

The WIV going full CYA mode strikes me as likely whether or not they we at fault. There was uncertainty, the people involved are paranoid, and th know they were obvious suspects.

5. I agree that it's possible a human scientist *could have* made the FCS we see it. However, Saar's initial position was that it was *basically impossible* for the FCS to be natural. When Peter gave several reasons that type of FCS could occur naturally, Saar shifted to "well okay maybe researchers were deliberately trying to mimic or boost natural evolution" That strikes me as dishonest argumentation and kind of conspiratorial - the exact type of epicycle conspiracy theorists use to dismiss contrary evidence.

REPLY (1) S

Comment removed Mar 28, 2024

Comment removed

REPLY (1) S

Theodric Mar 28, 2024

Peter won the debate and convinced Scott, and Peter's eviden
seems to be reliant on a growing body of recent research and expert opinion shifting back toward zoonosis. Anyway, ignore n throwaway line, point was this would need to be a very good bu not perfect cover up, and a cover up that still leaves the CCP looking bad, which is weird.

I have to disagree with the rest of your assessment. The initial argument was that the 12 nucleotide mutation, and inclusion of sequence similar to a human sequence was too unlikely, too "o nowhere", to be natural in origin.

"It looks natural, but that's because it was intentionally designe look natural and/or it was mutated in a lab by manipulating natu processes" is a plausible argument, but it's incompatible with t first argument.

If you concede the second argument, that preserves the possil that the virus was lab-made, but it must reduce the probability it is lab-made, because you're conceding that the virus is no lo obviously distinct from a naturally mutated zoonotic virus.

REPLY S

everythingism Mar 31, 2024 *Edited*

You are missing some important context which is that many of the allegations you mentioned (about a "severe situation" at WIV for instanc seem to be exaggerated or made up.

https://fallows.substack.com/p/on-that-propublica-chinese-lab-leak

The "severe situation" story by Katherine Eban and Propublica seems to based on a bad translation, with China experts questioning it and Propul eventually issuing an update on the article itself. And there are a numbe stories about the lab like this that end up being the result of bad reportir misinterpreted evidence etc.

When you look into who is behind these stories many of them seem to b coming from the Trump State Department under Mike Pompeo, which conducted an investigation into the lab back in 2020. "State Departmen investigators" is the source given for many of the key lab leak stories, su as the claims about sick scientists with COVID symptoms published by

Michael Shellenberger and the Wall Street Journal or the Sunday Times story including the claim that a Chinese scientist might have been execu by being thrown off the roof of the lab.

There's actually an account by another State Department official who wa involved in the investigation about what happened:

https://christopherashleyford.medium.com/the-lab-leak-inquiry-at-the-state-department-96973cff3a65

He says he was open to the lab leak but that certain officials like David Asher and Miles Yu became very fanatical about it. According to him the were already believing unhinged conspiracy theories like the virus being racially targeted bioweapon and wanted to charge China under the Biological Weapons Convention, which he thought was a bad idea. Eventually they ended up accusing him of being part of the conspiracy a trying to cover up their work. The same people, along with Trump, Pomp and Richard Dearlove, the former head of MI6, appeared in Sherri Marks documentary "What Really Happened in Wuhan."

There's some interesting background to all this which is that both Miles and Mike Pompeo have connections to the New Federal State of China, a weird regime change group that says it wants to overthrow the Chinese government. This group was the first to promote the lab leak theory in a major way starting all the way back in mid January 2020.

https://twitter.com/wanderer_jasnah/status/1631438683688648705

https://mediamanipulation.org/case-studies/cloaked-science-yan-report

https://www.nytimes.com/2020/11/20/business/media/steve-bannon-china.html

Mike Pompeo received an award from the group and Miles Yu has spoke glowingly of its founder Guo Wengui, who is now facing trial for defraudi the overseas Chinese community out of $1 billion, in speeches he's give So it seems quite possible this group and others in the anti-CCP Chinese diaspora are influencing the lab leak narrative to some extent.

https://twitter.com/dfriedman33/status/1631346935905886224

https://twitter.com/mrspanstreppon/status/1745855771005317279

https://twitter.com/gujingc/status/1707281401857663145

Only problem is that when the ODNI issued a report on the lab last year, seemed to greatly downplay the significance of many of these allegatior

Turns out there was no specific biosafety incident known about, the peo at the lab who got sick probably just had normal seasonal illness and the was no indication they were the ones working on coronavirus research (discussed towards the end of this article).

https://archive.is/BEwU2

All of this therefore needs to obviously be taken with a lot of skepticism. There are geopolitical motives, propaganda and BS stories on both sides unfortunately, not just from the Chinese government.

REPLY S

Dolores Ipsum Mar 29, 2024

You have arbitrarily excluded the third possibility that your friend is simply mistaken. His conclusion remains extremely unlikely regardless of (your judgment of) his personal honest

REPLY (1) S

Comment removed Mar 30, 2024

Comment removed

REPLY (1) S

Joe Blevin Jul 26, 2025

You have a blog complaining about wokeness, I know what media you and your di friends follow. You're not nearly as smart as you think you are, is my guess.

REPLY S

Deiseach Mar 28, 2024

"Either a zoonotic virus crossed over to humans fifteen miles from the biggest coronavirus laboratory in the Eastern Hemisphere."

As they say, 'you couldn't make it up'. Or, 'truth is stranger than fiction'. It does *seem* very suspicious co-incidence that a new virus that would turn into a world-wide pandemic just happe to pop up out of nowhere on the doorstep (as it were) of a specific institution dedicated to doin this kind of research, but there was no connection. And yet we seem to be forced to accept that is so.

Congratulations and gratitude for sitting through 15 hours of video to produce this post! And I d appreciate the Linear B/Lineage B joke.

REPLY (6) S

Comment deleted Mar 30, 2024

Comment deleted

John Schilling Mar 30, 2024

Cities are designed to make distance and geography irrelevant. They aren't perfect ab it, but the intent is that your interactions are defined by your social and economic netw and the physical infrastructure allows you to hop between those nodes without concer for what boring irrelevant-to-you terrain is between them. You probably barely notice t stuff when you're on the freeway, and you can't see it at all from the subway,

Again, that's only an ideal which is imperfectly approached. But what we need to know isn't "what's the distance from the WIV to the HWSM", it's "how many WIV lab techs h families with a taste for fresh seafood or exotic bushmeat?" If the answer is "lots", ther 15 miles won't stop them and/or their wives from schlepping their virus-laden selves o to the market.

Or maybe there's some class or cultural divide that means nobody working in the WIV' labs is going to shop at the HWSM; I don't know. That information, which would be ver informative here, is hard to come by. The geographic distance, which isn't nearly as informative, can be read off any online nap; a few clicks and calculations, So that's wh most people use; looking for their lost keys under the streetlight.

My prior, starting from ignorance, is that the odds of an average WIV lab tech('s wife) visiting the HWSM, is the same as the odds of an average Wuhan resident visiting that market. Which is something we can calculate.

REPLY S

Randomstringofcharacters Mar 28, 2024

Isn't it a reverse correlation issue though, the lab is situated there because it's an area whe coronaviruses were found in the past.

REPLY (1) S

Saar Wilf Mar 28, 2024

No. Wuhan is both far from bat habitats and far from south China, where wildlife consumption is popular.

REPLY (4) S

Simon stats Mar 28, 2024

I would also add that concerns about a lab leak in Wuhan are not post hoc. There were concerns about lab safety in Wuhan in the scientific literature published fron 2015 onwards. If you were to ask people in the know prior to the pandemic - wher the world would there most likely be a lab accident with a coronavirus, Wuhan wo have been top of the list given that it has the world's largest collection of bat coronaviruses, and did incredibly dangerous experiments on them at BSL-2.

REPLY (1)

Alex Zavoluk Mar 28, 2024

Concerns about the wet market are not post hoc either. An Australian virolog posted pictures and video from visiting the HSM back in 2014, noting it woul an ideal breeding ground for coronavirus pandemic.

REPLY (2)

Simon stats Mar 28, 2024

Actually, Wuhan was used as a negative control for coronavirus antibodi a WIV study prior to the pandemic. i.e. It was thought a v unlikely place f spillover prior to the pandemic.

Shi Zhengli has said she never thought a spillover would occur in Hubei

REPLY

Simon stats Mar 28, 2024

Also, the Australian virologist you refer to (Eddie Holmes) was in Wuhan because he was visiting the Wuhan Institute of Virology! He collaboratec with the WIV on RatG13, one of the closest known relatives to covid, in 2 (and then apparently forgot that he had done this_

REPLY (1)

Alex Zavoluk Mar 28, 2024

...and therefore... what? I don't understand your point here at all. I k he was visiting WIV. We already knew about RatG13. So?

Gonna combine 2 threads to reduce confusion:

> Shi Zhengli has said she never thought a spillover would occur in Hubei

Did she think a spillover would occur in the WIV? It's kind of weird t have this sentence right after your prior one... are we just cherry pic experts now? Some people identified wet markets, including HSM, ahead of time. Some people identified WIV ahead of time. Seems lik kind of a wash to me.

REPLY (1)

Simon stats Mar 28, 2024

I was highlighting an irony here. He wasn't visiting Wuhan beca it was a zoonosis hotspot, he was visiting Wuhan because it wa coronavirus research centre. What he took a photograph of do matter.

Shi Zhengli is the world's leading expert on coronavirus spillov and didn't think it would happen in Hubei. So, I think we should weight her claims more than Holmes'.

REPLY (2)

Simon stats Mar 28, 2024

I am making the point that Holmes never said or implied th "Wuhan is a very likely place for the next coronavirus outbreak". He was actually there because of the WIV

REPLY

Alex Zavoluk Mar 28, 2024

> he was visiting Wuhan because it was a coronavirus research centre.

Yes, we already knew this. Why do you think he went to H He randomly was in the mood for pangolin meat? No, it wa because the WIV researchers he was visiting wanted to sh him the market *because it was a likely place for a panden to start.*

> Shi Zhengli is the world's leading expert on coronavirus spillover, and didn't think it would happen in Hubei. So, I th we should weight her claims more than Holmes'.

Shi also said that it didn't come from the WIV. This is just cherry-picking.

REPLY (1)

Simon stats Mar 29, 2024

My point is that Holmes taking a photo in the Huanan market is not evidence for the proposition that prior t pandemic the Huanan market was thought to be a ver likely location for an outbreak. That would be true if h had gone to that market specifically to point that out. he wasn't at the Huanan market because he thought i was an especially likely location for a zoonosis. He wa there on a day trip from visiting the most dangerous coronavirus research lab in the world.

Wuhan being used as a negative control for sars antibodies by the WIV provides confirmation of this p

REPLY (1) S

Alex Zavoluk Mar 29, 2024

> But he wasn't at the Huanan market because h thought it was an especially likely location for a zoonosis.

This is in fact exactly what happened https://www.timesofisrael.com/report-wuhans-huanan-market-identified-as-pandemic-risk-5-years-prior-to-covid/

> Evolutionary biologist and virologist Eddie Holr said he was taken to the Huanan seafood market Chinese health officials five years prior to the pandemic, to see an example of a place where a virus transmission could occur from animals to humans, The Telegraph reported.

> "The Wuhan Centre for Disease Control took u there, and here's the key bit, because the discus was: 'where could a disease emerge?' Well, here the place – that's why I went," Holmes told the Br newspaper.

> "I've been to a few of these markets, but this w big one – it felt like a disease incubator, exactly t sort of place you would expect a disease to emer he added.

> Wuhan being used as a negative control for sa
antibodies by the WIV provides confirmation of t
point.

You mean like this?

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6
078/

> As a control, we also collected 240 serum sam
from random blood donors in 2015 in Wuhan, Hu
Province more than 1000 km away from Jinning (
1A) and where inhabitants have a much lower
likelihood of contact with bats due to its urban
setting.

This just says that people who live in cities aren't
likely to contact live bats. It has nothing to do wit
the relevant theory, which is that animals were
brought to wet markets in Wuhan and infected
workers there (who are obviously dissimilar from
random Wuhan residents in a relevant way). It
doesn't have anything at all to do with the pande
risk of HSM.

REPLY (1) S

Continue thread →

Randomstringofcharacters Mar 28, 2024

There's extensive bat populations in the cave systems of Enshi in western Hubei
aren't there? And the whole province is within the range of the rufous horseshoe b
There wouldn't be large populations in the area around Wuhan itself since it's hea
urbanized, but hunters would bring them into the city to sell them given that its th
provincial capital and richest city in the nearby area by a far margin.

Also raccoon dogs are common across rural areas of central and eastern China ar
are another possible source.

Obviously I don't know what the internal reasoning of the Chinese CDC was for
deciding where to put their research into coronaviruses. But Wuhan, as 7th larges
city in China, fairly centrally located, and in the region of the relevant animals wou
seem a good choice. Chengdu is the only larger city in central China, but is much

further west and has less tech sector, so presumably would be harder to get gooc
people

REPLY

Jim Haslam Mar 29, 2024

You are on the right side for the wrong reasons...

https://jimhaslam.substack.com/p/6-if-you-believe-the-wiv-engineered

REPLY

Vampyricon Aug 8, 2024

Wuhan is *in* southern China.

REPLY

Ethan Mar 28, 2024

You also couldn't make it up that the virus appeared in a wet market that was identified yea
ago as a plausible place for a pandemic to appear. These kinds of coincidences are the rule
rather than the exception, for real-world events.

REPLY (1)

Ethan Mar 28, 2024

Re-reading your comment, I realize you meant that as a joke, sorry.

REPLY

Argentus Mar 28, 2024

I was once at Disney World (over a thousand miles from my hometown of population 3000)
met a person from my hometown who also happened to be there that day. We had 0 idea tl
other was going.

This is a lot less weird than if I met them at a random gas station in rural Mongolia, but it's s
pretty weird!

REPLY (2)

Ghillie Dhu Mar 29, 2024

While on a high school trip, my classmates & I ran into someone else from our class (n
associated with the extracurricular organization that had arranged the trip) on the seco
level of the Eiffel Tower.

REPLY

James C. Mar 30, 2024 *Edited*

I once was on random free city tour in another country ~5,000 from my home with may
10 people in total and met another person who lived within five miles of me back home
(same suburb, so not like a city of millions either).

REPLY (1)

Andries Aug 22, 2024

I once was in London and met nobody from my home town, or school, or current c
of residence. Everybody was amazed at this coincidence ...,

There are an almost infinite number of dimensions on which one can post hoc det
coincidences: birthday, favourite author, birthday of mother, birthday of brother, ..
thus aren't amazed at all the coincidences that DON'T happen. But when we're
amazed by a coincidence we forget the millions of dimensions along which a strik
coincidence COULD have struck us.

Also, the group of people likely to go to the sort of place YOU go to, is tiny compa
to the whole population of the world, your country, etc. So it is less strange that y
meet them. Yes, I met a fellow artist who lives one kilometer from me in South Afri
at the Marlene Dumas show in Venice two years ago, but like the artist, both of us
from South Africa, both of us as teachers want to keep abreast of what's happeni
the art world ... Etc.

REPLY

10240 Mar 29, 2024

How many labs do coronavirus research?

I've seen the argument that it's suspicious that COVID appeared near one of China's only t
BSL-4 labs, but in this debate the *proponents* of the lab leak hypothesis said that the WI
recklessly did coronavirus research in a BSL-2 lab, not in the BSL-4 lab. (The WIV has a BS
lab as well as BSL-3 and BSL-2 labs.) It's less suspicious that COVID popped up near a
coronavirus lab if dozens of labs do such research than if this is one of the few such labs in
world.

REPLY (1)

Aristophanes Mar 29, 2024

Yeah, the emphasis on the pro-lab leak side puts less emphasis on the BSL-4 designat
and much more weight on "there are/were two labs that were at the forefront of bat
coronavirus research, WIV and one in North Carolina." Plus WIV seems to have been d
GoF research.

REPLY

S

Biorealism77 Apr 8, 2024

Several recent papers challenge the core arguments relied on for Huanan Seafood Market origin in the Rootclaim debate. Also, Miller incorrectly claimed the N501Y mutation would re from passage in hACE2 mice (he mixed them up with BALB/c mice).

1. Samson et al (2024) estimate likely point of emergence between August and October 20 Rendering the December 2019 cases and January 2020 market samples irrelevant to origin

They refer to horizontal gene transfer of the spike gene from BANAL bat CoVs in Laos to pangolin CoVs occurred in mid 2018. "Horizontal gene transfer" can occur in the wild (rarel can be induced in he lab artificially "transgenic". In other words a nice code word for lab engineered virus leak.

Samson S, Lord É, Makarenkov V (2024) Assessing the emergence time of SARS-CoV-2 zoonotic spillover. PLOS ONE 19(4):

https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0301195

2. Dietrich Stoyan, Sung Nok Chiu, Statistics did not prove that the Huanan Seafood Whole Market was the early epicentre of the COVID-19 pandemic, Journal of the Royal Statistical Society Series A: Statistics in Society, 2024;,

https://academic.oup.com/jrsssa/advance-article-abstract/doi/10.1093/jrsssa/qnad139/7557954

3. Lv et. al. (2024) found new intermediate genomes so the multiple spillover theory is unlik (it was anyway given lineage A and B are only two mutations apart as François Balloux, Virg Courtier-Orgogozo and Jesse Bloom have observed). Single point of emergence is more lik with lineage A coming first (as Nick Longrich points out below). The market cases were all lineage B so not the primary cases. Their findings are consistent with Caraballo-Ortiz (202 Bloom (2021).

Jia-Xin Lv, Xiang Liu, Yuan-Yuan Pei, Zhi-Gang Song, Xiao Chen, Shu-Jian Hu, Jia-Lei She, Liu, Yan-Mei Chen, Yong-Zhen Zhang, Evolutionary trajectory of diverse SARS-CoV-2 varia at the beginning of COVID-19 outbreak, Virus Evolution, Volume 10, Issue 1, 2024

t.co/50kFV9zSb6

4. Jesse Bloom (2024) published a new analysis showing that genetic material from some animal CoVs is fairly abundant in samples collected during the wildlife-stall sampling of the Huanan Market on Jan-12-2020. However, SARS-CoV-2 is not one of these CoVs.

Jesse D Bloom, Importance of quantifying the number of viral reads in metagenomic sequencing of environmental samples from the Huanan Seafood Market, Virus Evolution, Volume 10, Issue 1, 2024,

t.co/rorquFs1wm

5. Michael Weissman (2024) shows a model with ascertainment collider stratification bias f early Covid case location data much better than the model that all cases ultimately stemme from the market. George Gao, Chinese CDC head at the time, acknowledged this to the BB last year - they focused too much on and around the market and may have missed cases o the other side of the city).

Michael B Weissman, Proximity ascertainment bias in early COVID case locations, Journal c the Royal Statistical Society Series A: Statistics in Society, 2024;

https://academic.oup.com/jrsssa/advance-article-abstract/doi/10.1093/jrsssa/qnae021/7632556

6. Chen et al (2024) discuss how using the Grunow-Finke assessment tool lab origin looks least as likely.

Chen, X., Kalyar, F., Chughtai, A. A., & MacIntyre, C. R. (2024). Use of a risk assessment to determine the origin of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Ri Analysis, 1–11. https://doi.org/10.1111/risa.14291

7. The anonymous expert who identified coding errors in Pekar et. al. leading to an erratum year has found another significant error. Single spillover looks more likely. t.co/GAPihZu51P

8. The argument that an engineer wouldn't make the furin cleavage site with the features o SARS-CoV-2 overlooks it resembles that of MERS in several structural and functional ways, the sequence looks quite similar. In 2019 WIV researchers were involved in MERS research. Andreas Martin Lisewski discusses similarities with a MERS infectious clone described in 2 here. t.co/fAVUlJu0TK

9. Broad Institute biologist Alina Chan also observes the S1/S2 FCS PRRA insertion in SARS CoV-2 generates a Class IIS restriction enzyme site (BsaXI). This was used by WIV and Ral Baric at UNC previously. The full DEFUSE proposal available since the debate strengthens t argument of Bruttel et al. Specifically, the use of BsmBI, 6 fragments, and leaving the sites https://usrtk.org/covid-19-origins/scientists-proposed-making-viruses-with-unique-feature of-sars-cov-2-in-wuhan/

REPLY S

Simon stats Mar 28, 2024

What you say about raccoon dogs here is mistaken. Raccoon dogs are not a plausible intermedi host for sars-cov-2 on the basis of information that has been known since 2021. There are seve considerations.

1. Xiao et al (2021) - https://www.nature.com/articles/s41598-021-91470-2%E2%80%8B%E2%80%8B%E2%80%8B , which includes a co-author of Worobey et al (2022 leading zoonosis paper states in table 1 that the raccoon dogs were wild caught in Hubei, not farmed as you assert in the piece. This alone rules out raccoon dogs as plausible hosts for two independently sufficient reasons. Firstly, there is unanimity in the literature that the bat ancestra virus to SARS-CoV-2 is in southern Yunnan or South East Asia. Everyone agrees with this, incluc Shi Zhengli. If a species was wild caught in Hubei, then there would be no explanation of how it acquired the ancestral bat virus, given that Hubei is 1000 miles from southern Yunnan.

Secondly, a mystery of sars-cov-2 is how it acquired the furin cleavage site that makes it so transmissible. There are 850 known sars-like coronaviruses, and only one with a furin cleavage According to private messages exchanged by proponents of zoonosis, the furin cleavage site cc not have been acquired in the market because the density of animals was too low (only 3-4 per cage). When avian influenza acquires a furin cleavage site that occurs on farms with thousands chickens densely packed, i.e. not in the wild and not when there are a handful of animals in cage a market. https://usrtk.org/covid-19-origins/visual-timeline-proximal-origin/

2. Wang et al (2022) https://academic.oup.com/ve/article/8/1/veac046/6601809 also confirms tl the raccoon dogs were wild caught in Hubei. What's more, Wang et al (2022) tested 15 wild raccoon dogs of suppliers of Wuhan markets, including the Huanan market, in January 2020 an found them to be negative for SARS-CoV-2. On average, 38 raccoon dogs were sold across the markets in Wuhan from 2017 to 2019. So, the 15 raccoon dogs likely comprised nearly the whole inventory of raccoon dogs that would have been supplied to the Huanan market at the time.

In short, the raccoon dogs supplied to the market were tested and were negative. It is very strar that the raccoon dog narrative has persisted for this long given this public information. Even the strongest proponents of the raccoon dog hypothesis have walked back their bold claims that raccoon dogs are thost.

Xiao et al (2021) has a list of species sold at the Huanan market. I would encourage you to read list and suggest which animals you think are plausible, and I will tell you why they are not actuall plausible.

REPLY (3) S

ardavei Mar 28, 2024

I don't think there's a consensus that the ancestral virus is from Yunnan or SEA. No plausib immediate ancestor has been found anywhere. For what it's worth, high seroprevalence for SARS-related viruses have previously been found in Hubei:

https://www.science.org/doi/10.1126/science.1118391

Second, the line between wild-caught and farmed animals in the wildlife trade is blurry. For instance, there was a wildlife farm in Hubei where SARS was detected in palm civets. The fa contained a mix of wild-caught and farm-breed civets:

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1153763/

I am still not convinced that racoon dogs were the intermediate host, since SARS-CoV-2 evidently has a very broad host range.

REPLY (4) S

Philipp Markolin, PhD Mar 28, 2024

While it is true that Sarbecoviruses have a wide range, direct bat ancestors to SARS-C 2 have a much narrower range considering bat and viral dispersion patterns. Still unde investigations though, and politically sensitive topic.

REPLY (1) S

ardavei Mar 28, 2024

My point was specifically about SARS-COV-2, which can infect mammals from se different orders. And we don't know whether the restricted bat lineages are descendent from an ancestral generalist lineage, or if SARS-CoV-2 is descendent from a more restricted lineage.

REPLY (1) S

Philipp Markolin, PhD Mar 28, 2024

Si et al has a preprint out (zhengli shi is a co-author as well) adressing some by analyzing what makes RBD have broad or narrow ACE2 tropism. Seems br tropism is ancestral

REPLY S

Simon stats Mar 28, 2024

There are SARS-like coronaviruses in bats in Hubei, but they are too distant from sars-cov-2 to be the ancestor virus.

Shi Zhengli of the Wuhan Institute of Virology stated in July 2021:

“We have done bat virus surveillance in Hubei Province for many years, but have not fc that bats in Wuhan or even the wider Hubei Province carry any coronaviruses that are closely related to SARS-CoV-2. I don't think the spillover from bats to humans occurre Wuhan or in Hubei Province”. (Cohen interview)

This is confirmed by wider bat sampling efforts by the Wuhan Institute of Virology and EcoHealth Alliance prior to the pandemic. (Fan et al 2019 - WIV team; Hou et al 2010, Latinne)

The WHO Joint Study report noted:

"Tests on samples of more than 1000 bats from Hubei Province showed that none was positive for viruses related to SARS-CoV-2" (p106)

The samples tested had 74% to 90% sequence identity to SARS-CoV-2,(annex F table compared to 96.1% for RatG13 (from Yunnan), and 96.8% for BANAL-52 (from Laos). (Temmam)

From January 7th to 18th 2020, Wang et al (2022) collected samples from 334 bats around Wuhan. All were negative for SARS-CoV-2 or its progenitors.

Hassanin et al (2024) reported the "the results of several field missions carried out in 2017, 2021 and 2022 across Vietnam during which 1,218 horseshoe bats were samplec (Hassanin) They conclude that "For SARS-CoV-2 however, phylogeographic indicators provided very high support for an origin in the zone covering northern Laos, southern Yunnan and north-western Vietnam".

Pekar et al (2023) argue that the ancestral host for SARS-CoV-2 is 'very unlikely' to be from Hubei. *Pekar et al are the main proponents of the zoonosis hypothesis*

"Considering the large distance separating the closest-inferred bat virus ancestors fro where SARS-CoV-1 and SARS-CoV-2 each emerged (Fig. 3i,j) and the high dispersal velocities necessary to traverse the aforementioned distance (Extended Data Fig. 18), very unlikely that the lineages descending from the closest-inferred bat virus ancestor SARS-CoV-1 and SARS-CoV-2) (which reached, respectively, Guangdong Province anc Hubei Province solely via dispersal of these viruses through their bat reservoirs."

REPLY S

Simon stats Mar 28, 2024

Another point - If the animals came from a farm, why were there approx 1-2 spillovers in Wuhan city sufficient to cause an ongoing outbreak? If the farm had hundreds of raccoon dogs, why were they not numerous spillovers at other locations the farm was supplying? This is much easier to explain on the lab leak hypothesis.

REPLY (1) S

Kenny Easwaran Mar 28, 2024

What is the standard structure of exotic meat suppliers to wet markets? Are they usually big operations that supply many markets on a weekly basis, or are they us small mom-and-pop operations that turn over their entire inventory in a single transaction to a single market? I'm sure I'm presupposing a million false things in t way I've phrased this question, but I think some of this information would be relev to knowing how much and whether your question is a significant one to ask.

REPLY (1)

Simon stats Mar 29, 2024

Hello, farm size varies a lot - for SARS-1 civet farms had around 100 civets pe farm, but some bamboo rat farms have tens of thousands of animals. So, a lo depends on what the intermediate host was and how big the farm was.

The go to source on the suppliers to the Huanan market in particular is this washington post article by michael standaert https://www.washingtonpost.com/world/asia_pacific/china-covid-bats-caves hubei/2021/10/10/082eb8b6-1c32-11ec-bea8-308ea134594f_story.html

which is cited by Worobey 2022. They say that the suppliers of the Huanan market were farms in Hubei.

They reported that reported that authorities closed 290 farms in western Hul that accounted for between 450,000 and 780,000 animals. This suggests tha there were 1,551 - 2,689 animals per farm. The number of spillovers posited I zoonosis proponents is around 8. None of these spillovers have actually beer ascertained, they are the product of a (bad) model, and none of the official confirmed cases in the wet market worked at wild animal stalls.

REPLY (1)

Kenny Easwaran Mar 29, 2024

Is there any reason to expect that the farms that were closed were relati equal in size, rather than some being tens of thousands of animals and s being only a few dozen? Is there any reason to expect that the farms tha were closed had a distribution of sizes that is representative of the set o potentially relevant farms?

REPLY (1)

Simon stats Mar 29, 2024

There is a lack of information on this front. The farmed live animals at Huanan were mainly from Enshi and "A clampdown on Enshi's wil trade at wet markets began on Dec. 23, 2019, according to state me eight days before China publicly acknowledged the new virus. The I

start in Enshi doesn't mean officials found something amiss: It coul
have been preventive, as rumors emerged of market vendors falling
mysteriously ill in Wuhan. But it means evidence regarding Enshi's
wildlife trade was erased before the world was aware of the existen
a novel coronavirus."

It would be very nice to know what the specific farms supplying the
market were, but we don't have much information. But certainly one
to posit that the farms could not have been very large, otherwise th
would have been numerous zoonotic spillovers at other markets, wh
were not observed

REPLY

S

Simon stats Mar 28, 2024

So, do you think the Wang et al (2022) and Xiao et al (2021) papers are fraudulent or mistaken? Have you thought about calling for a retraction. Wang et al literally sought o the suppliers of raccoon dogs to the Wuhan markets and tested the wild caught racco dogs. All negative.

REPLY (1)

S

ardavei Mar 28, 2024

They only tested for viruses, not antibodies. If you tested 100 random humans toc your chance of finding any specific endemic virus (e.g SARS-COV-2) would be qu low.

They also only state that some raccoon dogs were wild-caught, not that all were.

REPLY (1)

S

Simon stats Mar 28, 2024

As noted above, they tested the entire inventory that would have been suppli to Wuhan and said they were all wild caught - they were not farmed. Xiao et a (2021) also says the raccoon dogs for sale at Huanan were wild caught. Xiao (2021) includes a co-author of Worobey et al (2022), the main zoonosis pape

They tested their tissue and blood samples by PCR. Zoonosis proponents po that the approx 20 raccoon dogs for sale at the Huanan market in the month November 2019 caused approx 8 zoonotic spillovers, of which 2 led to an ongoing outbreak (per Pekar et al (2022). Despite testing of the suppliers to market in January, they were all negative.

REPLY (2) S

Simon stats Mar 28, 2024

There are, to recap, three independently sufficient reasons to rule out raccoon dogs (1) wild caught, so no explanation of FCS acquisition; (2) v caught in Hubei, so ~impossible that they were exposed to the ancestral virus; (3) the animals at the market suppliers were tested and tested negative.

REPLY (1) S

ardavei Mar 28, 2024

You seem very focused on a very narrow scenario for zoonosis, that very similar to the consensus position for the SARS-CoV-1 outbreak you accept any scenario for one hypothesis, but any scenario for another, of course you will end up favoring the second hypothesis. I you could prove definitively that it couldn't be raccoon dogs, that's evidence against a zoonotic origin. But it's far from devastating. Th were plenty of other susceptible animals at the market. Additionally, was a dense area with an unusual amount of traffic from people wit close contact to wildlife. I think you really need evidence of enginee in order to believe in a lab leak, and I with a molecular biology background, find the evidence on that profoundly unconvincing.

I would also argue that since it's CCP policy that the outbreak didn' happen in Wuhan, we should be skeptical of statements from Chine researchers working in China.

REPLY (1) S

Simon stats Mar 28, 2024

I agree it's not conclusive proof that zoonosis didn't happen. S and Peter both suggest that raccoon dogs are a plausible intermediate host. So do various scientists. They are incorrect.

Yes, there were plenty of other susceptible animals sold at the market. Which one do you think is plausible? I think if you actu looked into it, zoonosis looks implausible for any of them. Fundamentally, there is no known trade route from Yunnan/SE to the Huanan market, but a well-established route via scientis

REPLY (1)

Simon stats Mar 28, 2024

Given that we have been on the wrong end of a PR campa from scientists for the last 3 years saying that raccoon do are the most likely host, I think it is interesting that they ar wrong.

REPLY (1)

ardavei Mar 28, 2024

There's also been lots of balanced views among seric scientists, and even a few favoring a lab origin (Jesse Bloom comes to mind). And on the other hand, there been a steady campaign against a zoonotic origin fro the CCP and various grifters (Matt Ridley comes to m Just because there is a convergence on some views doesn't mean that there is a concerted campaign to p false or misleading information.

REPLY

Sei Mar 28, 2024

Xiao does not seem to declare that all raccoon dogs sold at the market wild-caught.

>Approximately 30% of individuals from 6 mammal species inspected (labelled W in Table 1) had suffered wounds from gunshots or traps, implying illegal wild harvesting (Table 1).

>Raccoon dog fur farming is legal in China; however, due to a drop in fur prices, raccoon dogs are now frequently sold off in live animal markets, augmented by wild-caught individuals."

If raccoon dogs in any specific population were subject to any particular epidemic that burned itself out due to herd immunity, it's possible that n virus would be sampled two months later.

REPLY (1)

Simon stats Mar 29, 2024

Xiao et al Table 1 marks them as wild caught on the basis of injuries vendor interviews.

As noted above, on average, 38 raccoon dogs per month were sold across the four markets in Wuhan from 2017 to 2019. So maybe like were sold at the Huanan market in November 2019, which is less th one per day. Wang et al 2022 states

"We immediately started a surveillance investigation on the

origin of SARS-CoV-2 during 7–18 January 2020. Consequently,

the lung, liver, and intestinal tissue samples were collected from

mammals, which were captured in the rural area (Changxuanling an Yaoji towns) of Wuhan by three local traders for

vendors at animal markets including Huanan Seafood market

during 7–18 January. "

I.e. these are locally captured wild animals. The 15 raccoon dogs lik comprised nearly the whole inventory of raccoon dogs that would h been supplied to the Huanan market at the time.

REPLY

Ethan Mar 28, 2024

"If a species was wild caught in Hubei, then there would be no explanation of how it acquire the ancestral bat virus, given that Hubei is 1000 miles from southern Yunnan." How often d viruses travel a thousand miles? I was under the impression it's common.

"Even the strongest proponents of the raccoon dog hypothesis have walked back their bolc claims that raccoon dogs are thost." Who are these strongest proponents?

REPLY (1)

Simon stats Mar 28, 2024

It is true that viruses often travel more than 1000 miles, the point is that they do not dc this via bats which do not migrate that far. e.g. it seems that SARS traveled ~1000 mile from Yunnan to Guangdong via the civet trade. But with SARS-CoV-2, for animals wild caught in Hubei, there is no explanation of how they would have been exposed to the ancestral bat virus.

eg Angie Rasmussen is in many ways the face of the zoonosis hypothesis and a co-au on many zoonosis favoring papers. She said to the Atlantic "This is a really strong indication that animals at the market were infected. There's really no other explanation makes any sense."

Then she said this after the Bloom critique "As a co-author of the "raccoon dogs saga' report, it's disappointing to see a journalist outright lie about our work. We never claim to find infected animals."

https://twitter.com/angie_rasmussen/status/1651976430773731336

REPLY (2) S

Biff Wiss Mar 28, 2024

> She said to the Atlantic "This is a really strong indication that animals at the mai were infected. There's really no other explanation that makes any sense."

> Then she said this after the Bloom critique "As a co-author of the "raccoon dog saga" report, it's disappointing to see a journalist outright lie about our work. We never claimed to find infected animals."

I'm not sure what this is supposed to prove. There's no contradiction.

If I go into a bathroom and smell shit, there's a really strong indication that someo took a dump in there. There's really no other explanation that makes any sense. B that doesn't mean I found a turd in the toilet.

REPLY (1) S

Simon stats Mar 29, 2024

I was responding to this: "Even the strongest proponents of the raccoon dog hypothesis have walked back their bold claims that raccoon dogs are thost." are these strongest proponents?"

She said "there is no explanation that makes sense apart from an infected raccoon dog". She then said that this doesn't imply that a raccoon dog was infected.

REPLY (1)

S

Biff Wiss Mar 29, 2024

No. She's saying that all signs point to an infected raccoon dog, and tha they have not found an infected raccoon dog.

A detective finds a body. All signs point to a gunshot wound. He has not found a bullet or a gun. It doesn't mean he's changed his mind, or that it now necessarily more likely that the guy lying on the ground with a big h in his head died of strangulation.

REPLY (1)

S

Simon stats Mar 29, 2024

We might have to agree to disagree here.

REPLY

S

everythingism Mar 29, 2024

Except that animals were not all wild caught in Hubei. This may be part of what tripped up Saar during the debate -- lab leak supporters have spent the past four years repeating things that aren't necessarily true and creating an epistemic bubk for themselves.

In reality there are many ways a virus from southern China could have made its wa a market selling wildlife in central China. For instance the markets in Wuhan seem have sourced raccoon dogs and civets from farms in a mountainous region called Enshi about 630km to the west of Wuhan, nearly halfway to Yunnan province. Ens has some of the largest cave systems in the world and there's no testing of bats c farmed animals from there that's been released so far. This is the region, western Hubei, where animals were found infected with SARS-CoV-1 back in the early 200 https://www.independent.co.uk/asia/china/covid-coronavirus-bats-caves-hubei-b1940443.html

We also know that the Huanan Market also imported at least some animals such a bamboo rats directly from southern Yunnan province, as well as Guangxi. And the are only the sources we know about...it's unclear exactly where the raccoon dogs sold at the time of the outbreak came from.

Meanwhile the evidence at the market does seem to point to raccoon dogs. The s where they were sold appears to be the most likely source of the outbreak within market. Positive samples were found on a cage used for holding them, as well as

tool used for stripping fur off the animals. That doesn't prove they were the speci
that spread the virus to humans but there's definitely not a good basis to rule it ou

REPLY (1)

S

Simon stats Mar 29, 2024

As noted above, there is consensus in the literature that the ancestral bat viru not in Hubei. So, you are going against all of the scientific literature here, including Pekar et al (2023), who are leading proponents of zoonosis.

The bamboo rats you are referring to I believe were not live animals but froze refrigerated animal products. See the Gao teams' report - https://www.nature.com/articles/s41586-023-06043-2

Worobey et al cite the washington post article which says that:

“A person with knowledge of the Wuhan market supply chains, who spoke on condition of anonymity to protect his contacts, told The Post that live animals sold at markets in Wuhan were sourced from Hubei, particularly Enshi and Xianning prefectures, as well as from Hunan and Jiangxi provinces.”

Further confirmation for this is provided by the fact that:

“A clampdown on Enshi's wildlife trade at wet markets began on Dec. 23, 201 according to state media, eight days before China publicly acknowledged the new virus. The head start in Enshi doesn't mean officials found something an It could have been preventive, as rumors emerged of market vendors falling mysteriously ill in Wuhan. But it means evidence regarding Enshi's wildlife tra was erased before the world was aware of the existence of a novel coronaviru

I don't understand why you think the raccoon dogs were not wild caught loca Xiao et al Table 1 says this explicitly, as does Wang et al 2022

REPLY (1)

S

everythingism Mar 29, 2024

Yes but did you read that Washington Post article? It's actually the same this article I posted in the comment above: https://www.independent.co.uk/asia/china/covid-coronavirus-bats-caves hubei-b1940443.html

It seems to be saying that many animals were from farms in Enshi where civets and raccoon dogs were raised. That's a significant distance from Wuhan and yet you're pointing to testing of the 15 raccoon dogs from wi the Wuhan city limits itself.

As far as the ancestral virus not being from bats in Hubei that may be tru
but like I mentioned SARS-CoV-1 also somehow infected animals in west
Hubei. So there are certainly some unexplained details but I don't think a
of this makes a spillover to humans from raccoon dogs or civets impossi
at all. Remember the zoonotic origin doesn't need to be something as si
as it spilling over from bats to one species of animal that then infects humans at the market. It could have spread between multiple species jus
COVID did after infecting humans.

Also the reported shutdown of these markets in western Hubei a week before the Huanan Market was closed is interesting and that alone woulc
seem to call the lab leak explanation into question.

REPLY (1) S

Simon stats Mar 29, 2024 *Edited*

The 15 raccoon dogs were traced from suppliers of raccoon dogs ir
Huanan market. It's not that they randomly sampled raccoon dogs i
Wuhan. They did this over 11 days. Around 38 were sold on average
month. so, the raccoon dogs they tested likely accounted for the er
raccoon dog inventory in Wuhan at the time.

The wapo article is about the farms supplying the market, but Xiao
says that not all of the animals sold at the market were farmed. Worobey et al also says this!

fwiw I think the infected animals for SARS in western Hubei is dispu
though haven't looked into it.

Shutting down of the farms is not evidence against lab leak. Some p
of the government clearly thought it was a zoonosis and acted accordingly in Dec and Jan, which is eg why they disinfected the market in Jan.

REPLY (1) S

everythingism Mar 29, 2024

Regarding the 15 raccoon dogs all it says is that they were "captured in the rural area (Changxuanling and Yaoji towns) of Wuhan by three local traders for vendors at animal markets including Huanan Seafood market..."

I don't know about you but to me that does not say "these were only sources for raccoon dogs" or even "this is where raccoon dogs sold at Huanan Market at the time of the outbreak were sourced from." And that interpretation would seem to be contradicted by Michael Standaert's Washington Post article al talking about how they were sourced from Enshi as well as the Wuhan Forestry Bureau naming a farm in Hebei province.

There are a lot of things that don't line up with the lab leak explanation as well BTW. There's no indication the lab had a clo enough virus, no evidence anyone there got COVID or that they ever did the experiments some people suspect created the viru such as DEFUSE.

The lab still being open and the same team of people there conducting similar research as in the past, sampling sarbecoviruses in animals and testing their infectivity in human seems like a strange outcome if they caused a global pandemi for either of these two scenarios there are contradictions, miss info and unanswered questions, but at least one of them has so actual on-the-ground evidence supporting it.

The lab leak can't just be a "god of the gaps" explanation, ther needs to be some positive evidence that it actually happened a that's where it really comes up short. And this is where I think Rootclaim is completely wrong -- a complex exercise in determining the likelihood should not overrule real-world evide

REPLY (1) S

Simon stats Mar 29, 2024 *Edited*

My point about the 15 is about the numbers, i.e. it's half th inventory. Their investigation went for 11 days, so it was m if not all of the inventory.

The wapo article doesn't say the raccoon dogs were sourc from Enshi. btw even if you think the animal was supplied a farm in Hubei, there's still lots of confusing elements to t zoonosis story. You have to posit that an animal was sourc from Yunnan and taken to Hubei, where it sustained such intense transmission that it acquired the FCS, then like 10 only 10 of these animals were supplied to Wuhan and caus

only one spillover than led to an epidemic. Despite that, th
is a miniscule amount of covid in the relevant animals stall

"There are a lot of things that don't line up with the lab lea explanation as well BTW. There's no indication the lab had close enough virus, no evidence anyone there got COVID that they ever did the experiments some people suspect created the virus such as DEFUSE."

You are conflating 'no evidence' with 'didn't happen'. And think you mean 'direct evidence'. unfortunately we don't h their lab books, we have no idea what they have collected since 2015. Hey, maybe they could clear it all up by puttin their massive coronavirus database online.

There is also no direct evidence for zoonosis. There's no direct evidence that nature had a close enough virus, ther no direct evidence that anyone got infected by an animal. There's no infected animal.

So, you think if they were covering up for a lab leak, they would have shut down the lab and stop them doing coronavirus research?

There is evidence for a lab leak, from a Bayesian point of v The location in Wuhan is evidence for a lab leak. DEFUSE i evidence for a lab leak. The apparent only single introduct into the population is evidence of a lab leak. The lack of ar infected animal is evidence of a lab leak. The lack of spillo at other locations is evidence of lab leak.

If you reject Bayesianism and only want direct evidence, tl neither side has direct evidence, so you should just be agnostic

REPLY (1) S

everythingism Mar 29, 2024

Neither explanation is proven (and BTW I worry that people are ignoring other scenarios by focusing so m on these two) but there is simply far more evidence fo the market origin, and nothing to rule it out.

Closer viruses to SARS2 have been found in nature th
the closest virus the lab is known to have had. They o
discovered a limited number of SARS viruses in all the
years of sampling bat caves. If the outbreak started tl
you'd expect to see infections among the lab workers
family and contacts, not at a wet market selling anima

And yes it does seem difficult to imagine the CCP allowing Shi's team of scientists at the lab to carry on doing the same work if they had caused a pandemic, killing millions of people and eventually leading to ma civil unrest in China itself. Has anyone tried calculatin the Bayesian likelihood of this? The CCP simply does react to problems in such a lenient way, so in order to that this happened it seems like either you'd have to believe they are engaged in an elaborate charade or t CCP themselves never realized that a lab leak happer The much simpler answer would seem to be that it jus didn't happen, entirely consistent with all the evidenc

REPLY (1) S

Simon stats Mar 29, 2024

Closer viruses have indeed been found in other locations, but this is evidence in favour of lab lea because the relevant countries include countries WIV was sampling in. The closest known virus is Laos. The WIV has been sampling in Laos since 2006, and was actively sampling there from 2017 2019. In contrast, on the zoonosis story, the clair must now be that a raccoon dog (?) was infectec Laos, transported to a farm in Hubei, sold to Wuh and only Wuhan.

If they shut down the lab, everyone would know t they had caused the pandemic! That would make them look very bad. The CCP sent in the military general in charge of biodefense in China to run tl WIV after the pandemic started.

As to the lenient point, we don't really know wha
implications for the scientists or their families wil
or might be. There is also speculation that Zhou
Yusen, who worked with the WIV on coronaviruse
2019 was killed in May 2020.

REPLY (1)

Continue thread →

everythingism Mar 29, 2024

1) It's not clear if that report was indicating that ALL animals were wild-caught or just some There was an interesting incident back in 2020 where a Chinese environmental NGO wrote the Wuhan Forestry Bureau (responsible for regulating the wildlife trade in Wuhan) asking about animal sales at a single store at the Huanan Market: https://cbcgdf.wordpress.com/2020/05/31/breaking-news-cbcgdf-policy-and-research-department-received-a-response-about-the-dazhong-livestock-game-stall-at-the-wuhan-south-china-huanan-seafood-market/

The Forestry Bureau responded with a list of sources for the store -- which included raccoo dogs sourced from a farm where mink and foxes were also bred. So in at least some cases, they were sourced from farms.

2) This study doesn't say that all raccoon dogs were wild-caught and testing 15 animals frc one location is a ridiculous number to draw a conclusion from.

Overall the wildlife trade in China is difficult to track with animals being raised on farms but poached from the wild, often shipped long distances, with some such as pangolins traffick illegally. We never got a full accounting of animal sales at the Huanan Market -- in spite of t post above by the Chinese NGO, this information was left out of the eventual WHO report. There was no mention of the sources for raccoon dogs and civets, two of the most-suspec species, and China denied there was any evidence live mammals had been sold at the mark at all anytime in 2019.

REPLY (1)

Simon stats Mar 29, 2024 *Edited*

The article you link to refers to an advertising sheet from 2015. Xiao et al (2021) is a st from 2017-2019 involving observation of the animals and interviews with sellers.

15 is not a ridiculous number to draw a conclusion from. If the monthly sales were 38 p month across four markets in Wuhan, then the January testing by Wang et al (2022) o

supplier is half of the inventory that would be supplied to that specific market in a mor

Worobey et al said of the raccoon dogs photographed in Dec 2019 ""appear to be loca wild-caught common raccoon dogs rather than farmed raccoon dogs and that their plu coats are consistent with those observed in the winter". edit - reference is the worobey preprint - https://zenodo.org/records/6299600 fig 3

REPLY (1)

S

everythingism Mar 29, 2024

No the article I linked involves an environmental NGO asking the Wuhan Forestry Bureau about that advertising sheet, which lists nearly a hundred species for sale, exceeding the total number of species you're allowed to sell in China.

The Wuhan Forestry Bureau responds by giving a list of sources for that store at t market, and it includes raccoon dogs from a specific farm. This is the ministry tha was responsible for regulating the animal trade in Wuhan.

Meanwhile the Xiao study you mentioned simply indicates that some of the anima observed appear to be wild-caught. So this is a good example of why it's importa not to draw such a strong conclusion when there are all these different facts swirl around. Let's at least wait until China gives a source for the raccoon dogs that we sold on the western side of the market -- they've never provided this information amazingly.

REPLY (1)

S

Simon stats Mar 29, 2024

Ah I see, that's interesting I hadn't seen that.

Still, the referenced farm is in the east of China, which is even further from th relevant bats than Hubei. In general, fur farms in China are overwhelmingly in east of China.

REPLY (1)

S

everythingism Mar 29, 2024

Yeah I agree but this particular store, located on the east side of the mar was already less likely to be the origin. The question is why this informat sources for the most suspected species, was left out of the WHO report There are no sources listed for raccoon dogs or civets at all, in spite of t clearly having been sold there and in spite of China already having both sampling data and seemingly also sales records showing they were.

WHO investigators later said CCP officials pushed them to exclude any mention of these animals from the 2021 WHO report and denied there w any evidence of any live mammal sales at the market during 2019. So Ch seems to have censored almost all of the information (beyond the 15 loc tested animals you mentioned) about the most likely intermediary specie

REPLY

The Author Mar 28, 2024 *Edited*

Roko (the one infamous for the basilisk) has argued that what might have appeared as a shift in expert consensus is in fact "manufactured consensus." This is worth considering. Link: https://www.lesswrong.com/posts/bMxhrrkJdEormCcLt/brute-force-manufactured-consensus-i hiding-the-crime-of | I also believe an algorithm somehow sorting, weighting evidence cleverly could solve the problem of "too much evidence." The CIA, in fact, at least has used one such algorithm, invented by Richard Heuer.

REPLY (3)

__browsing Mar 28, 2024

I think the fact that the expert class went out of their way to suppress the lab leak hypothes and that lab leaks of this nature have occurred in the past, is probably damning enough. My impression is that Roko hasn't really introduced any new arguments that Peter didn't cover the debate, and Peter isn't being funded by the CDC, to my knowledge.

REPLY (2)

Ch Hi Mar 28, 2024

Everybody plays "Cover Your Ass" whether they're innocent or guilty. So that's not a v argument. Nobody likes to be accused of having made a mistake, whether they have o not.

REPLY (2)

__browsing Mar 28, 2024

Is this supposed to excuse any possible degree of pre-emptive collusion and frau cover up for institutional incompetence?

REPLY (1)

Tatu Ahponen Mar 28, 2024

No, it's supposed to say that suppressing a hypothesis is not damning evider of the hypothesis being correct. The hypothesis might be correct or incorrec suppressing it is bad anyway.

REPLY (1)

__browsing Mar 28, 2024

I agree. That's why I'm saying it's a 'damning' indictment of our reigning technocrats.

REPLY (1)

Dan L Mar 28, 2024

I regret to inform you that technocracy is unusually *good* on this point. If you can reliably find a strain of human that doesn't reflexive hide their misdeeds, we sure could use them!

REPLY (1)

__browsing Mar 30, 2024

Our governing technocratic class has colluded with a raft of delusional left-wing ideologies when it comes to explaining outcome disparities between sexual, racial and class demographics, not to mention anti-nuclear-power activism anc latest phase of hysteria over police killings or trans murders. Conspiratorial thinking is not actually more common on the rigl it's just that most left-wing conspiracy theories were culturally normalised and written into law decades ago.

REPLY (1)

Dan L Mar 30, 2024

Ah. In fairness, I should have seen that coming from "reigning", that's on me.

REPLY

Dave Mar 28, 2024 *Edited*

>So [playing "Cover Your Ass"] is not a viable argument.

It *is* a viable argument in the sense that it dampens and modulates the strength other arguments.

Arguments about what the WIV can or cannot do, arguments about the number o cases centered around the wet market, and arguments about which DNA sequenc were found and when they were found, all need to have their Bayesian impacts divided by a selection-bias-from-a-Maoist-Totalitarian-Government factor.

REPLY (1)

Ch Hi Mar 28, 2024

OK, but you've got to include EVERYONE, not just the people you don't like. I wasn't *just* the various segments of the government of China, it's also the investigators, people who have a public positions ahead of time (and "public' be a argument with another expert).

Just about everyone who made a public statement on any factor related to COVID during the early days had at least one mistake that they wanted to "de emphasize". There were "public experts" in the US claiming that we could ke COVID out after it had already been circulating for months. And the border closings were a farce. The quarantines were not really enforced, and lots of people were allowed to essentially skip them. (Not that it really mattered sinc COVID was already circulating.)

REPLY (1)

Dave Mar 29, 2024

This would be the perfect placement for the Peter Parker-Harry Osborn meme.

I already distrust the FDA, the CDC, the WHO, and the CCP. You don't ne to sell that distrust to me.

REPLY

John Schilling Mar 29, 2024

The expert class went out of their way to suppress the lab leak hypothesis in part beca Trumpist Republicans were promoting the lab leak hypothesis and the expert class is reflexively opposed to anything associated with Donald Trump. And in part because discussion of the lab leak hypothesis, regardless of its factual basis, would imperil the funding and prestige of themselves and their colleagues.

Neither of these depend on the lab leak hypothesis being true or probable, so the fact of the suppression carries little information regarding the truth of that hypothesis.

More generally, innocent people reflexively engage in coverups often enough that "Coverup, thus Guilty!" is a weak argument at best. The bit where innocent people welcome thorough investigation of every accusation made against them because they confident it will prove their innocence, is sadly disconnected from reality.

REPLY (3)

S

everythingism Mar 29, 2024

Agree on the second part but scientists also dismissed the lab leak hypothesis because they genuinely found it to be less likely than zoonotic origin. The private Slack messages everyone likes to cite where scientists initially discussed the possibility it came from a lab end with them saying that. https://jabberwocking.coi read-the-entire-slack-archive-about-the-origin-of-sars-cov-2-there-is-no-evidei of-improper-behavior/

Also important to realize that the idea of lab origin was pushed by a number of different groups, not just Trump Republicans. The first version of it was actually fr the CCP themselves, who tried to claim SARS2 was a bioweapon the USA attacke them with at the Wuhan Military Games. From there it was picked up by anti-CCP groups in the Chinese diaspora, which had a surprising degree of influence on cei Trump officials, and the storyline changed to focus on the Wuhan Institute of Viro Other governments such as Iran and Russia came up with their own versions blam the USA.

So there's really an ocean of BS on this topic and you can't blame scientists for sa it's less likely when it actually does appear to be less likely. None of this resulted i the idea itself being suppressed. It's probably gotten something like a hundred tin or more coverage in the media as the animal trade origin. Governments and other groups are incentivized to promote some version of this storyline for geopolitical reasons -- in China's case to evade responsibility for the pandemic, in the USA's c to more directly blame China. Meanwhile the real origin gets lost in the confusing back and forth of competing claims and accusations.

REPLY (1)

S

__browsing Mar 30, 2024 *Edited*

"It's probably gotten something like a hundred times or more coverage in the media as the animal trade origin"

Really? You want to dig into the archives of NYT and show me a hundred-to-(ratio of articles propounding the lab leak theory vs. zoonosis?

I also think 'Biorealism' made a fair point at the end of the comments section

"There is at least one argument they used knowing it was misleading. They u
Ron Fouchier's argument (without acknowledgement) that WIV would have u
a well known reverse genetics system despite Andersen saying that wasn't th
case on at least two occasions. He noted they had been created "on a whim"
on 20 Feb if anyone thought these were hard to create from scratch a group
researchers had just created one in a week. They should have acknowledged
this.

The Nature reviewer clearly took too hard a line as well. Their initial manuscri
was far more balanced. Moving from lab origin was not "necessary" to "not
plausible" was not justified. As David Relman emailed Francis Collins there ne
to be an impartial assessment which acknowledged the lack of evidence eith
way. A Dept of Defense Working Paper dated 26 2020 noted their conclusion
were based on "not on scientific analysis but, on unwarranted assumptions".'

REPLY (1) S

everythingism Mar 30, 2024

Just doing a quick search on Google, even NYT, which has been criticize
lab leak supporters for their supposed biased coverage, has about twice
many stories that mention "Wuhan Institute of Virology" vs "Huanan
Market." And overall in the media the number of articles on the animal tr
origin -- i.e. looking into the animal trade in China in relation to the outbr
-- is almost nothing vs. those covering the lab leak controversy.

This is strange because even if you think it leaked from a lab there are
specific people in China, vendors and workers at the market, who could
us much more about what happened, and yet the focus has shifted almo
entirely to American scientists and politicians. You then have to think ab
the fact that many experts believe the lab leak never happened at all!

As far as the Proximal Origin paper I think it's fine to say the conclusion
should have been less strong, maybe they should have said "lab origin is
less likely" rather than "not plausible." But what's funny is you can also f
a quote from Richard Ebright from the same time period (but before
Proximal Origin was published) where he says "Based on the virus genor
and properties there is no indication whatsoever that it was an engineere
virus."

So clearly this was not an unreasonable conclusion to come to, unless yo
are saying Ebright has no idea what he's talking about, and where it falls
apart IMO is this idea that in writing this paper scientists were somehow

suppressing the lab leak theory. Just weeks after the paper was publishe
the president of the USA was repeating stories about how someone at tl
lab had started the pandemic. The State Department launched an
investigation that resulted in the stories about "sick WIV scientists with
COVID symptoms." Steve Bannon ran a PR campaign to promote Li-Men
Yan's bioweapon allegations. Congress issued multiple reports and held
hearings about it. We're now going to see yet another investigation by R
Paul, who doesn't exactly seem neutral. You can't say this idea was reall
suppressed that much...instead there's a narrative people repeat over ar
over about how it was suppressed.

REPLY (1) S

__browsing Mar 30, 2024

"Just doing a quick search on Google, even NYT, which has been criticized by lab leak supporters for their supposed biased coverage has about twice as many stories that mention "Wuhan Institute of Virology" vs "Huanan Market.""

If you use those exact terms, yes, you get 450 results vs. 170 in fav of the Wuhan lab being mentioned. If you search for "wet market" v "lab leak", you get ~40,000 results for the former vs. ~16,000 for th latter, so the balance of coverage flips. If you search for the same te on google outside of the NYT website, you get about 3 times more mentions of the Wuhan lab or 'lab leak' than you do for the wet mar but on the other hand, "chinese animal trade" returns 89 million res as opposed to 274,000 for "chinese biolab". So I'm not sure the zoonotic origin story is being downplayed, precisely.

REPLY (1) S

everythingism Mar 30, 2024

"If you search for "wet market" vs. "lab leak", you get ~40,000 results for the former vs. ~16,000 for the latter, so the balance coverage flips."

We're getting into the weeds here but if you look at what these results actually consist of, you'll notice most of the wet market stories have nothing to do with the pandemic. I get a story abo Anthony Bourdain opening a market in New York for instance. I counting only five stories on the first page of results that actua have to do with the wet market in Wuhan, vs. about half of the leak stories being relevant. That's why I was searching with the exact names instead.

Anyway obviously it's difficult to say what the exact search terı should be to compare, but based on researching this topic the number of in-depth stories about the animal trade in China is v low. Whether that's more down to laziness by journalists or the of information out of China could be debated, but at this point Kristian Andersen's Slack messages have probably gotten mor coverage than many important details about the outbreak.

REPLY S

__browsing Mar 30, 2024 *Edited*

My argument is not "cover-up, therefore lab leak is true". My argument is "cover-ι therefore untrustworthy"- that the effort to suppress the lab leak hypothesis was crime in itself, and the expert class are still 'guilty' in that sense.

REPLY S

Écorché Mar 30, 2024

My understanding was that the Proximal Origins paper was already published by t time Trump made a public statement in early April, but I don't have a ref handy.

REPLY (2) S

everythingism Mar 30, 2024

Yeah that's correct, although it had been promoted earlier by other GOP politicians such as Tom Cotton, and outside the government by former Trump officials like Steve Bannon. Actually I don't think Trump himself was ever too invested in this idea. There's a hilarious interview he did with Sherri Markson where it quickly becomes clear he doesn't have much more knowledge than a random person. It was more a project of Mike Pompeo, the Secretary of State and former CIA director, and a group of people working under him

REPLY (1) S

Écorché Mar 30, 2024 *Edited*

Proximal Origins was in review by the time of Cotton's editorial.

I think the conclusion might not have been fully exaggerated yet though.

REPLY (1)

everythingism Mar 30, 2024

Ultimately the Proximal Origins controversy becomes much less interesting IMO if there was no actual inside knowledge about what happened, and even Emily Kopp of US Right to Know told me she doesn't think that's the case, although she does think scientists engaged in a "coverup" of some kind. Nothing ever prevented anyo from writing a different paper saying lab origin is plausible and man did.

REPLY

John Schilling Mar 30, 2024

Hence my specifying "Trumpist Republicans" rather than "Donald Trump". As everythingism notes, Trump himself was not really the problem, but there we plenty of people in Trump's orbit advancing the Lab Leak theory.

And advancing it well beyond what the evidence would have supported at tha early date. It was absolutely reasonable for the "expert class" to push back against that, and at least understandable that they would overcorrect in the c direction.

REPLY

Kimmo Merikivi Mar 28, 2024

This is one of the situations where I don't feel any more confident trusting the expert consensus than looking at the papers and/or debates and trusting my own reasoning to evaluate the arguments.

A good example I personally like to use of this definitely being the case would be from COV as well: during early 2020, the Finnish Institute and Wealth and Welfare (THL) officially clai IFR of [alpha-variant, in immunonaive population, before best practices for treatment were discovered, as average within a population with a lot of elderly people] COVID to be <<.01% purportedly repesenting the best available scientific evidence, while at the same time some .5% of entire population of Bergamo in Italy had died (and similar IFR could have been calculated from Diamond Princess data, etc). Here, rational evidence easily trumps claims made based on scientific evidence: a child could see if explained to that no matter how ma unidentified cases there might be, there's no way for IFR to be lower than total excess mort within the population. And additionally, I have a clear model where the people working in th institute with their virology doctorate doctorate degrees went wrong: they rely (indeed, mig be contractually obligated to, I haven't checked) on scientific studies, but as none had bee published at such an early stage, they pattern-match to the most closely analogous study, which might be about flu or something. Most people, even experts, just aren't epistemologi nuanced enough to use rational evidence, as Scott has often written about in "The Phrase ' Evidence" Is A Red Flag For Bad Science Communication".

When expert scientists are discussing statements within established science, I would trust claims to not misrepresent the science, and since the scientific method approximates Baye theorem pretty well, I have a high degree of trust for the claim. This isn't such a case! Scier works, but it advances one funeral at a time more or less, and there's no way this debate ha gotten to a point of established science, indeed the polls lean to one direction but also specifically show the question ISN'T settled.

REPLY S

polscistoic Apr 1, 2024 *Edited*

The point made in the link is classic the-social-aspect-of-opinion-formation. The world of domain experts can be analysed as a community (more-or-less closely integrated), comple with opinion leaders, followers, rebels, go-betweens, outcasts and every other (social) role related to the goings-on in any community.

The negative spin on these (social) processes in the linked post is unneccessary, though. P because such social processes are inevitable/unavoidable, partly because they may also be conductive in getting the joint reseach process from A to B. (The importance of all social factors on opinion formation is Janus-faced.)

(...and ACX/self-declared rationalists is also to a certain extent a community, of course.)

REPLY S

Tobias Schneider Mar 28, 2024 *Edited*

I have no horse in this particular race, but I do have a lot of expertise in some of the areas rootclaims "investigates" (especially the stuff related to Syria and chemical weapons) - where tl analysis is so shoddy and laughable it's indistinguishable from Youtube conspiracies - and the biggest surprise to me here is that anybody really bothers with rootclaim in the first place? The more you learn...

REPLY (4) S

The Author Mar 28, 2024

Even if you do not agree with the values of the (prior) probabilities, you can choose your ov and replicate the process rootclaim uses. It is far from "shoddy" and research on Bayesian methods modelling these kinds of events is highly valuable.

REPLY (1)

Tobias Schneider Mar 28, 2024

I have nothing against Bayesian reasoning. Use it all the time. But the priors, the "evidence", the weights attached were so out-there, I just assumed arguing with them like arguing with Alex Jones who has taken a stats course. Unfortunately statistical methods don't cure brain worms (and in fact might even provide them with a rationalis veneer).

REPLY (1)

The Author Mar 28, 2024

Can we then Aumann agree that: the Rootclaim method with better priors and handling of evidence would not necessarily be shoddy, the current analysis is or n be, and the method itself remains valuable enough that it would be good if they continued to work on it (in Scott's framework, (2)).

REPLY (1)

Leo Abstract Mar 28, 2024

With good-enough priors, yes it would be good enough (by definition). But h good would those have to be? Is it possible to have priors good enough that t method isn't just laundering assumptions through some fancy statistics, as meta-analyses or financial projections or suchwhat often are? I'd answer the questions as "at least 10x better than we have" and "not really, no."

REPLY (1)

The Author Mar 28, 2024

Strange place to comment, I was aiming for agreement with Tobias.

REPLY

Randomstringofcharacters Mar 28, 2024

How so?

REPLY

Saar Wilf Mar 28, 2024

You happened to laugh at an analysis where we turned out to be spot on. There is no now c verified video evidence of the opposition carrying out the Ghouta sarin attack.

https://blog.rootclaim.com/new-evidence-2013-sarin-attack-in-ghouta-syria/

REPLY (1) S

JoshuaE Mar 28, 2024

If you are going to claim your analysis is spot on, please link to a credible independent source. Otherwise this comes across as we believe this unlikely thing and used our analysis you find shoddy to conclude we were right so you should not consider our analysis shoddy.

REPLY (1) S

Saar Wilf Mar 28, 2024

Follow the link. It describes external forensic work which you can verify yourself.

REPLY (1) S

JoshuaE Mar 28, 2024

I did follow the link. My point is if you were vindicated I would expect to be ab read about it in an independent source. Otherwise I need to look at all of you sources and see both which are in support of your claim and which you are claiming support you and look at the evidence in detail. Lab leak as a plausib theory has been printed in the NYTimes since at least 2021, has any mainstre newspaper published anything about this?

REPLY (2) S

Xpym Mar 28, 2024

How often do mainstream newspapers publish anything about Ukrainian misdeeds in the war? Probably something like 1/1000 ratio compared to Russian ones, whereas a reasonable prior of their prevalence would be li 1/10 at best. Similar sort of bias is probably at play with Syria.

REPLY (1) S

JoshuaE Mar 30, 2024

I'm not discussing the ratio of coverage but here is an article from le monde that is easily searchable that backs up the claim that Ukrani committed war crimes.

https://www.lemonde.fr/en/international/article/2022/04/09/ukraine-military-accused-of-war-crimes-against-russian-troop_5980121_4.html. If there was anything that resembled reasor evidence, it would get some coverage the trouble is Saar has motiv reasoning that is not convincing without external collaboration. Otherwise the best you can claim is that there is some evidence tha the syrian weapons were a false flag and again it's not your (Saar's) method that proves the analysis is spot on, its how strong you trust conclusions from the grayzone/wikileaks that the appearance of impopriety means the conclusions are false.

REPLY S

Saar Wilf Mar 29, 2024

Sorry, I thought it was obvious. These outlets have no incentive to publis such findings. Sadly these are the kinds of things you need to verify yourself.

REPLY S

Philipp Markolin, PhD Mar 28, 2024

Agreed.

REPLY S

Mark Neyer Mar 28, 2024

How do you account for “one of the most powerful political actors in the world which has a histc of controlling information flows runs the area in question and can be manipulating what evidenc available?”

I get that you have to work with the evidence you have. When the evidence you have is passing through a filter that has pretty strong incentive to hide its own wrongdoing (if that were to exist) how does that fit into your analysis?

Did they assess a probability that, ie, certain research records where the researchers tried that ~exact~ furin site were destroyed by the CCP? Wouldn’t that just be a “naked prior”?

REPLY (3) S

Randomstringofcharacters Mar 28, 2024

The CCP were trying to cover up zoonotic origin as well at first, with arresting Dr Li and cleaning up the site. Because it was seen as politically embarrassing, as China said it had fi wet market issues after SARS. So that applies on both sides.

(The Chinese official position now is to heavily imply it came from abroad)

Also a question of whether, in the lab leak hypothesis, they'd have been able to cover up la amounts of evidence effectively, given how inept the initial covid response was. And officia different levels acting at cross purposes to cover their asses

REPLY (1)

S

Simon stats Mar 28, 2024

This is not true. George Gao said in January 2020 that it was from the wet market. Afte leading sampling there and the case search, he changed his mind in summer 2020

If you look at what usually happens with research-related accidents, they are usually shrouded by obscurity for years. This is likely even more true in China. Compare e.g. tl 1977 Russian flu, sverdlovsk anthrax leak, various SARS leaks. In some cases finding o that an accident occurred can take decades.

REPLY (2)

S

Ch Hi Mar 28, 2024

I think you're expecting a government to act in a unified way. I'm sure if you searc you could find statements supporting every side. And I *have* encountered (repo of) official Chinese statements claiming it came from abroad. So that is one of the positions that they are (or were) taking. I've also encountered denials that COVID exists (e.g. by the mayor of Wuhan), though that was early in the epidemic.

REPLY (1)

S

Simon stats Mar 28, 2024

I was responding to the claim that the government was trying to cover up the market origin in early 2020. This is inconsistent with the head of the China Cl saying it came from the wet market in January 2020.

Obviously, all of their incentives are to cover up a potential lab accident.

REPLY (1)

S

Ch Hi Mar 28, 2024

The problem is that everybody has incentives to cover up, whether or nc they are at fault. If someone things something MIGHT be read as placinc blame on them, they'll try to suppress it. This makes things quite murky.

FWIW, I believe that if the disease originated (as a human disease) in, sa New York, but didn't get the last mutation that it needed to become pandemic until someone in Wuhan caught it, we'd see largely the same response. You can't judge things by "who's trying to cover things up".

REPLY S

everythingism Mar 29, 2024

But you have to ask why George Gao would change his view to denying the wet market origin if it came from the lab. That's a strange thing to do if China's intenti was to cover up a lab leak. What these statements from China and the various lab narratives/hypotheses have ultimately done is leave the origin unresolved within tl domain of global public opinion, an outcome that may not be the worst from China POV. They're able to avoid officially taking the blame for the pandemic and can control the narrative domestically to say the virus came from outside China.

REPLY S

beleester Mar 28, 2024

That would only work if the research was completed but not published before COVID happened. If someone was studying furin cleavage sites in 2018, they would want to publis paper saying "hey, we tried cleavage sites X, Y, and Z and only X worked" and China would have no reason to think that should be kept secret because COVID hasn't happened yet.

REPLY S

Alex Zavoluk Mar 28, 2024

This is discussed in the debates and in Scott's summary. It is a common theme with conspi theories (term used literally) to allege that the conspiracy is both capable of and motivated make the evidence come out to whatever happens to be the case. However, just because y can imagine a hypothetical in which this might happen, doesn't mean it's remotely likely, especially when characteristics like who is involved and how competent they are change fr argument to argument. China is not omnipotent and frankly probably doesn't care much ab lab leak vs zoonosis. Let's consider the epidemological data pointing to the market, which i often alleged to be tainted.

1. The Chinese government first tried to cover up the existence of the pandemic at all, and failed. Also, as Peter points out, if it started much earlier, you would have many times more deaths, which is even harder to cover up.

2. They also tried to cover up (or at least, were incompetent) the wet market, by claiming animal testing showed no results, but then it turned out they hadn't actually tested live anir (roughly, I might be getting details wrong here).

3. Now they claim the virus started in the US; if they were trying to fake evidence of this, th would have had cases cluster near an airport or hotel or something, not a market that is aln exclusively visited by residents.

4. There's multiple different lines of evidence that point to the market, which makes it increasingly unlikely that China could manipulate all of them in a way that's undetectable.

5. If you don't believe the market data, why do you believe the pandemic started in Wuhan all? Couldn't China be trying to hide, like a secret bioweapon plant in a different city? This i: possible, but of course there's no evidence for it.

REPLY (2) S

Mark Neyer Mar 28, 2024

Maybe you can help me understand how this would fit into a Bayesian framework, but here's how I'd think of it:

Suppose someone did indeed to a research and create that exact strain of virus that w ostensibly "not what you'd do" if you were doing research. This would be pretty direct evidence overwhelmingly in favor of the lab, right?

Yet if that evidence existed it would be far easier to track down and destroy, precisely because someone would know "holy crap I did this" and that specific person would ha very strong incentive for that evidence not to go public.

I don't believe that "China make the evidence go however it wants", I just don't see hov account for "people who knew they caused this could destroy that evidence far more easily than we could find it, and would have a very strong incentive to do so."

REPLY (1) S

Alex Zavoluk Mar 28, 2024

> Suppose someone did indeed to a research and create that exact strain of virus was ostensibly "not what you'd do" if you were doing research. This would be pre direct evidence overwhelmingly in favor of the lab, right?

I don't follow what you're saying here. It looks like "Suppose it was created in a lal that would be evidence in favor of the lab." Can you rephrase? Is there a statemer missing here?

> Yet if that evidence existed it would be far easier to track down and destroy, precisely because someone would know "holy crap I did this" and that specific pe would have a very strong incentive for that evidence not to go public.

I agree that *if* some lab workers knew they had created covid, and knew how the had done it, they probably would and probably could destroy at least some evider It's not at all clear to me how well this would actually work (i.e. would any evidenc remain? would there be evidence of this sort of cover up?) Do you have more specifics?

(Even in the case of lab leak, there might not have been any such cover up--I have seen the theory that the leak was totally accidental and came directly from sampl so they might not have even realized at first even if they were to blame, and WIV workers were behaving pretty normally through December at least. Conversely, ev in the case of zoonosis, panicked workers could have engaged in some sort of co up if they had some vague idea they might be considered responsible. So FWIW I don't think that either "there's some evidence of a cover up at the lab" or "there's evidence of such a cover up" would be definitive evidence either way.)

> I just don't see how to account for "people who knew they caused this could destroy that evidence far more easily than we could find it, and would have a very strong incentive to do so."

Criminals try to hide evidence all the time, but still get convicted (or convinced to plea deals). It's actually really hard to successfully hide, destroy, or fake all of the evidence for something this big, and also do the same for the cover-up, and for th cover-up, etc. There were whistle blowers for previous outbreaks, including lab le of Sars 1, as well as for Sars 2, from the hospitals, in spite of Chinese crackdown.

You can certainly put a cap on how confident you are in a theory--it's not like the judges, or Miller, or Scott, or me, or the superforecasters, are >99.9% confident ir zoonosis. Seems like mostly in the 80-95% range. But it's not correct to just throv the evidence out.

REPLY (1) S

Mark Neyer Mar 28, 2024

I agree with most (all?) of what you're saying but still don't understand something. I couldn't glean this from the article.

How did "people have an incentive and ability to destroy evidence" influence computed odds here?

Yes i understand that destroying evidence would likely happen either way (du panic, not wanting to be blamed, not knowing the true cause) and that it can hard to totally cover something up and that people often are convincted desp trying to cover up evidence, etc.

I’m not questioning the outcome of the debate.

I’m just trying to understand how “people had an incentive to destroy evidenc influenced the computed odds. It seems like the conclusion is “that doesn’t matter because there are multiple trails of evidence that point to the zoonotic case”, but I can’t tell if this is “it shifts odds by 5%” or “we just didn’t comput that because it doesn’t seem necessary” or what.

REPLY (1) S

Alex Zavoluk Mar 28, 2024

> How did “people have an incentive and ability to destroy evidence” influence the computed odds here?

Miller spends a fair amount of time in the debates on why the early case data is reliable enough, in ways that can't easily be faked, as well as reas to believe that China is neither capable of nor willing to fake the data in a way that explains the evidence. So I think the conclusion is "that's theoretically true but it seems unlikely to explain anything."

I think the highest level answer is "everyone rounded the BF used for ea piece of evidence down, based on how likely they thought the evidence be wrong." If you want the specific numbers the judges used, you would have to dig into their reports, or perhaps ask them directly. As far as I ca recall, I don't think Rootclaim really dove into this point, but rather just u it at a very high level ("those data could be wrong, so the evidence provi is weak") so the judges may have just taken it has most likely that the da are accurate.

Keep in mind that the data merely being limited (which is known to be th case) doesn't necessarily tell you much, because you have to explain wh the cases cluster at the *market* specifically. If want to claim the pande started at the lab, it's not enough to say that maybe there are cases mis because it would still be a hell of a coincidence that the cases we do kn about are all at the one place in the city most likely to be the origin of a zoonotic pandemic.

REPLY (1) S

Mark Neyer Mar 28, 2024

Thank you! This seems to answer my question. I appreciate your patience here, it's a real service.

REPLY S

beowulf888 Mar 31, 2024 *Edited*

A few minor niggles...

> 1. The Chinese government first tried to cover up the existence of the pandemic...

It's a bit more nuanced than that. When people started coming down with mysterious cases of a pneumonia-like disease during the first weeks of December, the local hospi MDs and administrators alerted the health authorities of Wuhan, and they in turn notifi the Mayor of Wuhan and the Governor of Hubei. The mayor and the governor tried to h it up because a big party convention was scheduled to be held in Wuhan (early in Jan IIRC). The mayor and governor of Hubei Province decided to go forward with the conference. They felt a small outbreak of some unknown pathogen would reflect poorl their administration, and they didn't alert the national authorities to the outbreak—unti people were crowding the emergency wards, and they couldn't cover up all the hospitalized patients. (This would have been around the third week of December wher they couldn't cover it up anymore—although it didn't reach the Western media until the first week of January.)

Local dithering and silence wasted two critical weeks, but some of the hospital MDs notified the folks across the river at the WIV, and they were able to alert the National Health Commission of the PRC that they suspected they had a SARS-like outbreak on hands. Not long after the MDs were beaten and arrested by local Gong An Bu, probabl the behest of the local authorities who were embarrassed about the outbreak happeni on their watch (again, it's unclear who directed them to do this, but the Mayor and the Governor had the motive).

The national health authorities took the warning seriously, and they informed Chairmar that the situation was potentially quite serious. He immediately fired the Mayor (and th the Governor) for incompetence and he sent in a hand-picked troubleshooter to handl the situation. (I wasn't able to find out what happened to the mayor and governor. I do know if they were allowed to fade into private life or were maximally disciplined.) Once National-level health authorities assessed the situation (and it took them about a week they alerted the WHO at the end of December, and the Chinese started sharing data w the WHO, the ECDC and the CDC. The local coverup wasted two critical weeks, but it w at the local level, and not at the national level.

> 2. They also tried to cover up (or at least, were incompetent) the wet market...

Due to the case cluster, the Chinese national health authorities immediately focused o the wet markets in that district. SARS1 and some Avian Flu outbreaks had started at we markets, so the wet markets within the initial case cluster were immediately suspect. A once it was clear that they had something dangerous on their hands, public health authorities immediately shut down the wet markets in Wuhan (I believe they shut them down all over the city). The animals were disposed of, and the market stalls were disinfected. From a public health perspective, this was a wise decision—but from a dat gathering perspective, they lost the chance to test the animals when they incinerated carcasses. Again, I may be misremembering some of the details but they invited the W to visit the site in mid-January. Either the Chinese or the Chinese with the WHO collec swabs of DNA and RNA from the stalls. So the disinfection happened after they collect the DNA and RNA. The Chinese sequenced the virus and they sent the genome to the WHO, to ECDC, and the CDC on January 15th. It wasn't until March when the CIA direc and Trump started accusing the Chinese of covering up a lab leak that the Chinese stopped cooperating with us. Then naturally, they blamed us for creating the virus.

It's important to note that wild animal farming is a 7 billion dollar business in Hubei, an Wuhan is the second only to Guangzhou as a transshipment center for the animals anc their byproducts. It's a big business, and the authorities don't want to see it shut dowr because it lines their own pockets. So their impulse is to cover up instead of disclose because of all of the negative attention they've garnered worldwide.

REPLY S

Torches Together Mar 28, 2024

I enjoyed this piece - I tried to follow this story in 2020 and have always leaned more zoonosis tl lab leak (I believe I said 90-10 publicly in 2021, probably closer to 95-5 now).

Despite this, I've often been frustrated and annoyed by mainstream virologists who dismiss LL v poor reasoning, and almost always find myself defending (more reasonable) LL proponents.

I guess that this annoyance has led lots of people in rationalist-adjacent circles to lean further L than they should have. I wonder if this is a common rationalist failure mode we need to update c

REPLY (5) S

The Author Mar 28, 2024

Without caring about which one is actually true, we can look at the costs of the beliefs of tl were true or false. If LL is true, and you fight against some sort of conspiracy, that is some positive U. If LL is false and you fight against an imaginary conspiracy, that will worst case cause skepticism and more investigation. Believing that it's zoonosis means you will be

enforcing a conspiracy if you are wrong, which is negative U. Otherwise slightly positive U t
most people, but less than what you'd get if you were right and fighting a conspiracy. Elieze
says one should believe that which is true, but actually this sort of analysis of costs and gai
of being wrong / right is very useful, because it can help to navigate the unvertainty of wha
you should believe in.

REPLY (6) S

Randomstringofcharacters Mar 28, 2024

If we want to go Hansonian about it, saying you think lab leak is at least plausible signa
sophistication and open mindedness

REPLY (1) S

The Author Mar 28, 2024

I did not take signaling that precisely into account but this is absolutely true and
might have some wider implications (I did not have solid concrete examples at tim
writing).

REPLY (1) S

polscistoic Mar 31, 2024

It certainly is an important point. Every statement you make about the probab
of something is at the same time a signal you send about what kind of persor
you want to be perceived as, and/or want to perceive yourself as.

This is unavoidable.

REPLY S

Ch Hi Mar 28, 2024

You're leaving out the utility of finding another reason to blame some group you dislike
feel this is a major driver of many human beliefs.

REPLY S

quiet_NaN Mar 28, 2024

Unless you go all Pascal's wager (which is obviously bullshit), your beliefs don't have
immediate consequences in the world.

In the brains-as-Bayes-engines model, beliefs are probabilistic in nature.

What you do about your beliefs is then another question subject to cost/benefit analys
regarding your utility function. You might want to state your true beliefs, exaggerate th

stay quiet, pay lip service to the opposite side or even do your very best to argue agai
your own beliefs.

There are some corner cases where you might chose to stay strategically ignorant due
direct effects of your beliefs (e.g. infohazards, unconditionally knowing about your top
health abnormalities might encourage hypochondria in all the cases where you don't h
serious health problems, or psychiatric disorders being at least somewhat memetically
infective in populations), but for the most part the Litany of Tarski holds: if X is the cas
then you want to believe X.

REPLY (1)

smopecakes Mar 28, 2024

I think Pascal's Wager is true. Probabilistically speaking, life is meaningful and consequential enough to treat it as such

If I had to give a number on a test I might say that our life is 60% likely to be materialistically determined. To strengthen the point, we'll go with 99%

Now multiply the likelihood by the consequences. Again, to strengthen the point, I
say it's probably not very consequential to both exist in a 'meaningful' and non-random universe and correctly believe that it is. So 99% random or probabilistic existence x 0 consequences of being correct or incorrect about that. And 1% non
random x .1 consequences of it being true. Point one units of consequence weigh
against 0 seems very noticeable in an absolute sense

With my own weights of likelihood and suspicion that human beliefs and actions c
be wave like, propagating and amplifying despite small discrete effects, then my t
is that I am compelled to act as if my life and beliefs are potentially consequential
subject to free will

REPLY

Yug Gnirob Mar 28, 2024

>If LL is false and you fight against an imaginary conspiracy, that will worst case cause
skepticism and more investigation.

The worst case of convincing people that someone is out to get them is that they deci
to preemptively attack those people.

REPLY

Kenny Easwaran Mar 28, 2024

It's a little weird to cite Eliezer as the person who says one should believe that which is true! This really seems like something that should be much more fundamental than on guy who's famous for an unrelated issue (AI safety).

REPLY (1)

MicaiahC Mar 28, 2024

He wrote less wrong and also a screed on what "the simple truth" is, as well as "H Potter and the Methods of Rationality", not "Harry Potter and the sealing of horcruxes" or similar.

I don't think you can say he has *no* claim on being the truth guy, or he isn't famo for wanting to believe true things.

REPLY (2)

Peasy Mar 29, 2024

Thank goodness he came along in the early 21st century and upended thous of years of philosophic practice by inventing the idea that people should beli true things to be true!

REPLY (1)

Concavenator Mar 30, 2024

I mean, that might not be a novel idea, but it's not really universal either. You'll find plenty of people insisting that sometimes believing or teachin false things -- even tricking yourself into believing things you don't know be true -- is a good idea. (Plato and Pascal being two famous examples.

REPLY

The Ancient Geek Mar 30, 2024

To be the Truth Guy, he doesn't have to say it, he has to be the first or the be some way. A lot of the problem is that EY's fans tend to be naive about the history of thought, so it all seems new to them. It's not even the first iteration out-of-the-mainsteam philosophy that claims to answer everything.

The thing called "rationalism" has little to do with rationalism as the term is understood in mainstream philosophy. It is better understood as part of a pat where a succession of small, insular groups engage in a very self confident s of amateur philosophy. Unlike other cult-like groups, they claim to have a str orientation to science, logic and reason(and to be able to implement them mu better everyone else).

Examples include Yudkowsky's "rationalism", Critical Rationalism (Karl Poppe philosophy, articularly David Deutsch's version), Ayn Rand's Objectivixm and Korzybski's General Semantics. There are often several such movements overlapping each other, some are reactions to others, and some borrow idea: from others.

In more detail, This kind of Thing is characterised by:-

• Being science-orientated , but having much more specific claims than "scie good"

* Being strongly to stridently irreligious and anti religious. Also opposed to N Age thinking and continental pilosophy.

• Being largely outside of mainstream academia etc

• Being an insular group that mostly talk to each other

• Having difficulty in communicating with outsiders , in any case, because the own theories are expressed in a novel jargon.

• Centering on a charismatic leader, with a set of mandatory writings

• Having an immodest epistemology..which claims to be able to solve just ab any problem..

• ..which is based on a small number of Weird Tricks. Nothing requires the sa level of deep study as mainstream academia, and the key ideas are often brie slogans.

REPLY (2)

S

MicaiahC Mar 30, 2024

> To be the Truth Guy, he doesn't have to say it, he has to be the first or best in some way.

I used lower case truth guy, clearly in reference to him becoming well kn in some respect for this. I use it in the same way that I would call a somewhat competent local mechanic the "car guy" or I can call Bill Nye science guy" without claiming he's the best person in the world at scien or that he invented science.

Or that I was replying to the statement

> It's a little weird to cite Eliezer as the person who says one should beli that which is true!

Which, like, do you believe that you should only cite the things from the formulation or inventor? Maybe start auditing Kenny about what other dastardly crimes against citation he has committed would be better.

I feel it's not a coincidence that this type of misreading is common to, as would say, "haters",whom all seem to share the following traits, excuse r they are Characterized by:

> Perceive any group seeking improvement to be automatically suspect

>Incapable of reading even slightly positive sentiments about their targe and stay silent

>seem to predictably misread incredibly common use of language, seeir "good vibes" and assuming it means GOAT or inventor being a common example

>Incapable of putting forth or discussing any object level example, preferring instead to talk in innuendos about how there's a large populat of informed disagreement, yet fails to link or write something. If an exam is provided, it is full of guilt by association and sneering arrogance and li else, and they converge on it like little flies, congratulating each other ak how many good points have been made.

> does not seem to be aware of their own constant isolated demand of rigors, or that many of their objections can be answered with less than fi minutes of thought

> become obnoxiously self congratulatory about how they used middlinç low wit to dunk on their targets

You see, listing a bunch of negative traits is a great way to be truth seek If someone tries to engage, it's because they are insecure and you, yes y personally are correct, no matter how obviously derailing or irrelevant th list is. If you can play to an audience, that's all that matters. :-)

Well, you know, one can choose the quality of their friends, but not their enemies. There are ways to choose the quality of your enemies, but the world is not ready to hear them.

REPLY S

polscistoic Mar 31, 2024 *Edited*

I am puzzled that you include Karl Popper here, his view on the demarca criterion between science and non-science (i.e. that a hypothesis shoulc

least in principle be falsifiable) is very mainstream among scientists. (I d
not know Deutsch' interpretation, though.)

REPLY (1)

S

Vampyricon Aug 8, 2024

IME everything after Popper is basically ignored by scientists. Ironic
only a very tiny minority of scientists actually follow falsificationist practices.

REPLY

S

Arie Mar 30, 2024

If you believe in malicious plots that aren't real you reduce social trust which is one of
main predictors of sociatal succes. Messing with that has huge potential negative U.

REPLY

S

ardavei Mar 28, 2024

I think this is definitely a common failure mode in general, and I observe it way more often i
rationalist circles.

Personally, I experienced the exact same thing with the mask and airborne debates. I still g
emotional reaction every time I see someone wearing a cloth mask.

REPLY (1)

S

Ch Hi Mar 28, 2024

FWIW, cloth masks are a major help in many environments. They slow down the speed
the air ejected by someone who is infected, so it tends to remain near them. They also
absorb moisture, so any wet globules ejected are likely to be absorbed.

Note: Almost all masks are more effective at preventing an infected person from sprea
the disease than they are at preventing someone from catching it.

REPLY (1)

S

Viliam Apr 2, 2024

I fully agree with you, but the standard response is that (1) wearing masks is "dehumanizing", (2) we will all catch the diseases anyway, sooner or later, and (3) many people wear the masks incorrectly anyway. Therefore, the proper way to sig
sophisticated thinking is to refuse wearing the masks. Especially when the cost of
doing so will mostly be paid by the others.

REPLY S

Randomstringofcharacters Mar 28, 2024

Yeah, I think. There's a common failure mode of overcorrection from "don't blindly follow th mainstream" to believing the most popular contrarian option by default.

You see a similar thing in the left with people over correcting from "America is always right" supporting genocidal dictators as long as they oppose the USA.

REPLY (2) S

Tophattingson Mar 28, 2024

As Scott described in the article, Lab Leak is the "mainstream" view among the genera public, even if it's contrarian in academia and prestige / elite / media / whatever you wa to call it circles.

REPLY (1) S

Biff Wiss Mar 28, 2024

JFK conspiracies are also the "mainstream" view. I think it's possible for "mainstre to be a slight "term of art" even if the raw survey data says otherwise.

REPLY (1) S

Xpym Mar 28, 2024

Well, there is a big Oscar-winning Hollywood movie heavily implying that the official JFK narrative isn't the whole story, so calling that view mainstream se reasonable in any sense.

REPLY S

Level 50 Lapras Jun 20, 2024 *Edited*

> You see a similar thing in the left with people over correcting from "America is alway right" to supporting genocidal dictators as long as they oppose the USA.

Ironically, nowadays the right tends to do the same thing, due to associating the US wi Evil Wokeists. It really is a trip, seeing politics flip to basically the exact opposite of 20 years ago.

REPLY S

Minus Mar 28, 2024

Well there's the question of how far this was an overcorrection. It may very well be that it w correct to put (as a random number) 80% on it given your state of information & willingnes investigate & priors, while 90% was too much.

(I think it is easy for social-posts, like notable blogposts, pointing out that some belief of th community was likely wrong to end up having the same problem of overcorrecting to "don' disagree" rather than "what degree did we do worse than we should have")

REPLY (1)

Kenny Easwaran Mar 28, 2024

I think that there's almost no difference between 80% and 90%. Even if one believes i objectively correct priors, someone who was off by this much is basically as right as possible.

REPLY (2)

Minus Mar 29, 2024

I was using them as random example numbers, the same logic holds for 5% vs 30

REPLY

polscistoic Mar 31, 2024

True.

And there is a danger if offering more precise probability estimates, in particular v offering predictions about the future (and even if you offer them only to yourself): danger that you may convince yourself that you have more precise knowledge tha you actually have, or can possibly ever have.

REPLY

Kenny Easwaran Mar 28, 2024

> I wonder if this is a common rationalist failure mode we need to update on.

Absolutely. I sometimes feel like a very large fraction of comments on any topic on this blog reaction formation by rationalists who are annoyed at something that MSNBC watchers are saying and have talked themselves into the diametric opposite to rationalize their annoyanc

REPLY (2)

Peasy Mar 29, 2024

It's often worse than that--it's rationalists who are annoyed at a caricature of what MS watchers are saying, and have talked themselves into the diametric opposite of that caricature.

REPLY (1)

Kenny Easwaran Mar 29, 2024

True!

REPLY

Level 50 Lapras Jun 20, 2024 *Edited*

Nate Silver justifies his belief in lab leak pretty much entirely with frustration with the mainstream media's attempts at narrative control in 2020 (nevermind that the mainstr media has been breathlessly *promoting* lab leak ever since). It really is sad to see thi from someone who should know better.

REPLY

Simon stats Mar 28, 2024

I don't think this grapples with the problems the Mr Chen case poses for the zoonosis argument Both sides in the debate agree that Mr Chen contracted symptoms on 16th Dec. This suggests was exposed around 10th Dec. He thinks he was exposed on the train or in hospital on 8th Dec.

This raises two problems:

1. Mr Chen lived 30km away from the Huanan market on the south of the river, and never went anywhere near it in December, according to his interviews. His case was only recorded because happened to be transferred to the top tier Wuhan Central Hospital, which is a sentinel hospital f respiratory outbreaks in Wuhan. It is in the north of the river from his local hospital because his relative happened to work there. This is evidence of a geographical bias in he case search. How many other cases were missed on the south of the river because they happened not to visit top hospitals, which were largely near the market.

Geographic bias in the case search is also confirmed by:

- Literal statements to the effect that 'we focused too much on the market' by the man in charg the case search https://www.bbc.co.uk/sounds/play/m001ng7c

- A mountain of evidence in official Chinese documents of a market-bias in the case search. https://journals.asm.org/doi/full/10.1128/mbio.00313-23

- This short statistical argument -https://academic.oup.com/jrsssa/advance-article-abstract/doi/10.1093/jrsssa/qnae021/7632556?redirectedFrom=fulltext&login=false Cases with links to the market were closer to the market than linked cases. This is evidence FOR ascertainn bias in the case search.

2. Given that there was a confirmed case 30km from the market around 10th of December, this suggests widespread community transmission across Wuhan in early December and late Noven
i.e. not that there was a zoonosis in early Dec. The first confirmed case was on 10th/11th Dec (e
for that case, it is unclear whether the first case was market linked). There was no published retrospective case search. The first case was initially thought to be 1st Dec. Usually if you do contact tracing and a case search, the first official case goes backward in time. In this case, it w
forward in time to 10th Dec.

REPLY (3)

Ch Hi Mar 28, 2024

As I wrote in an earlier post, I think it was circulating in rural China long before it was officia
detected. Catching it on the train fits in with this quite well. My main hypothesis is that a "merchant" carrying produce (of some sort) to the wet market was the source of that epicenter. (That could actually be widened a bit, as all that is required is that someone infe
in a rural village for some reason visited the wet market area for awhile.)

Remember that prior to special tests being developed, COVID was being diagnosed as "atypical pneumonia", at least in a Washington (I presume state rather than DC) hospital.

REPLY (1)

Jeffrey Soreff Mar 28, 2024

>Remember that prior to special tests being developed, COVID was being diagnosed a
"atypical pneumonia", at least in a Washington (I presume state rather than DC) hospit

Good point!

Personally, I'm somewhat skeptical about the "geographic focus" evidence because th
so very many asymptomatic cases. Even if there had been consistent diagnosis from patient 0 onwards, we would have only been seeing the tip of the iceberg.

(I have to admit, I haven't been following LL vs. Z terribly closely.)

REPLY

Ethan Mar 28, 2024

It's common for COVID to have an incubation period of only a day or two, so the fact that h
developed symptoms on December 16th doesn't exclude that he was exposed on Decemb
or 15th.

REPLY (1)

Simon stats Mar 28, 2024

Rai et al (2022), a meta-analysis of the covid incubation period for patients infected by early variants from January to March 2020, found that the average time from exposure symptom onset is around 5.7 days, with a 95% confidence interval of 5.2 to 6.3 days. suggests he was likely infected around the 10th December.

https://link.springer.com/article/10.1007/s10389-021-01478-1

REPLY (1)

Ethan Mar 28, 2024

I think you're misreading the paper. The 95% confidence interval means that there 95% chance that the mean incubation period is between 5.2 and 6.3 days. It does say anything about how likely a 1-2 day incubation period is in an individual.

I haven't read the paper (I only read the abstract), so perhaps there's information there about how common a 1-2 day incubation period is.

REPLY (1)

Simon stats Mar 28, 2024

Ah yes, I think you're right. In either case, in expectation, he was exposed arc 10th Dec, and thinks he was exposed on the 8th

REPLY (1)

Freedom Mar 28, 2024

Does when he thinks he was exposed matter at all? Why should we assig any weight to his speculation? He had symptoms in the hospital on Dec which seems very consistent with contracting COVID in the hospital a fe days before that.

REPLY (1)

Simon stats Mar 28, 2024

I agree it is consistent with that, I am just saying that the average tir to symptom onset is 6 days. he could also have been exposed ten c earlier, per this logic, which is a further update from zoonosis and tl two effects cancel out, no?

REPLY (2)

Kenny Easwaran Mar 28, 2024

If there was any substantial reason to believe his exposure was days earlier, that would give a substantial reason to worry abou

hypotheses that said there was no spreading virus 10 days earl
But the fact that (say) 10% of people take 10 days to develop symptoms doesn't mean that there's a 10% chance that somec who developed symptoms on Dec. 16 was exposed on Dec. 6. can't just invert probabilities like that.

As long as there's a 10% chance of someone developing symptoms in a single day, this is at most 10-to-1 odds evidence against a hypothesis that says he was infected in the hospital (surely less than that, because the non-hospital infection hypothesis didn't give 100% likelihood for symptoms first emer on Dec. 16).

REPLY (1) S

Simon stats Mar 29, 2024

See below comment about this illustrating bias in the case search

REPLY S

Scott Alexander Mar 28, 2024

Author

Aren't we selecting him as a person who got symptoms surpris early? Supposing that someone infects many people on the 14t and we choose the one who develops symptoms earliest to be surprised by, the person we're surprised by will show symptom December 16th.

REPLY (2) S

Simon stats Mar 29, 2024

The point is that Mr Chen was only tagged as a case for idiosyncratic reasons. If there were non-biased case searc then I agree it wouldn't be much evidence. But he was onl counted as a case because he transferred to a sentinel hospital north of the river for idiosyncratic reasons. This w not have happened for the other people with covid in Jian (30km from the market) at that time. This suggests that th were lots of cases (not just mr chen) probably in late November far from the market. Does that make sense?

REPLY

Aristophanes Apr 2, 2024

What the selection mechanism is really depends on what generating procedure you are hypothesizing.

A) Suppose Mr Chen (and 100 others) are "successfully" exposed to Covid while in hospital c. Dec 14. Some of thos people will have short incubation periods, some will have l ones. By chance, Mr Chen happens to be the one with the shortest window, has symptoms by Dec 16. Someone goe "woah, the average incubation to symptoms is 5-6 days, s probably caught it pre-hospital", on Dec 10 on average, be the first known wet market case. This would be a mistake reason we're noticing Mr Chen in this story is we're select on the fact he got symptoms quickly. But in that case, you should see all his exposed hospital cohort come down wit symptoms over the following week. (Note that if the hypothesis is that Mr Chen is the only / one of the few tha exposed in hospital at this time / by the same source, then haven't actually strongly selected him on "got infected quickly", and his early symptom date returns to being a relative surprise - and thus evidence against "infected in hospital" vs "infected pre-hospital". Not a 100-1 factor - q symptom onset isn't that rare. But a 5-1 or a 10-1 factor is possible).

B) Suppose instead the hypothesis is (per your nearby comment, paraphrasing), "the first known case - the shrin vendor who got symptoms on Dec 11, probably wasn't act the first human case, she could easily be anywhere in generation 2-4+, lots of cases are undetected, one of tho chains infects Mr Chen on the train on the 8th/10th/whate then sure, his early symptom date (Dec 16) is easy to expl away (i.e. we don't need to appeal to quick hospital infecti But the way we explain it away is "potentially lots of undetected non-market transmission prior to the first *detected* market case". In this 'world', there's no/minima selection into noticing (in the data, not by us) Mr Chen jus because he got sick quickly (in fact, we would have only v

weak evidence he got sick quickly, but in the same manne
is true for any early case detection: if a + b is low, some of
accrues to our estimate of E(a) and some to E(b)). Rather
was captured in the data for the idiosyncratic reasons that
was transferred to a specific hospital well away from wher
lived. But then, this implies there are likely many other ear
unlinked cases like Mr Chen that never made the official d
counts because - unlike Mr Chen - they did not share his
idiosyncratic hospital transfer. If we only captured Mr Che
the data due to 1 in a 100 luck, there were probably 100 ot
like him. Which then is strong evidence of geographic/trav
history bias in detection of early cases. (That the search w
geographically/travel history biased seems very clear - bo
the primary sources that list linkage to the wet market as a
necessary requirement through mid January 2020, and in
fact that epidemiologically unlinked cases in the early data
physically closer to the wet market than linked cases. This
smoking gun for non-random search. See links provided b
others above).

So in some sense, Mr Chen only raises the probability of tw
things that were already quite probable.

(i) the earliest infections were never captured in the data;
first case is probably several generations in.

(ii) the early data was biased in searching for cases near/li
to the wet market

(Note that both of these things follow - in the sense of rais
probability - from Mr Chen's case even if, by a huge fluke,
was the only infection (in his generation) in the
unlinked/undetected transmission chain he was a part of.
Whether there was 0 others, or whether there was 100 pe
like him is relatively immaterial - it's just ex post chance.
However many infections did exist, they are unlikely to hav
been detected).

But both (i) and (ii) are bad for the wet market hypothesis.
we knew that we had detected the location of the first infe
person, "it happened at the wet market" would have a larg
odds ratio for the zoonotic hypothesis. But if we merely kr

"well, lots of generation 3-4-5 cases happened at the wet market", the relevant question is Pr(lots of gen 3-4-5 case wet market| zoonotic) / Pr(lots of gen 3-4-5 cases at the w market|lab leak). But the characteristic that makes a wet market have a high chance of hosting spread for gen 3-4- cases has ~nothing to do with the wild animals present, ar instead loads very heavily on "how good a location is this f superspreading" * "how likely is it that authorities will notic get suspicious quickly of a linked outbreak at this location (due to the history of SARS CoV 1 etc)." But this condition probability differs *much* less between zoonotic and LL, a becomes increasingly weak evidence the bigger a phenom (ii) is.

As a simple example, consider the Omicron outbreak in Australia, where testing was very good at the start (and w know the first cases had to be imported from overseas). T first imported case was discovered in Sydney on Novembe 27. The first community transmission was discovered on December 3. On Dec 8, 680 people "checked in" to a nightclub about 2 hours north of Sydney. It was a super-spreader event, 295 of them tested positive.

https://www.abc.net.au/news/2021-12-15/newcastle-omic outbreak-largest-covid-superspreader-event/100699092

https://www.newcastleherald.com.au/story/7634098/study shows-how-two-men-infected-hundreds-at-argyle-house

If testing/detection had been less widespread (as is inevit early on in an outbreak, e.g. in Wuhan), these nightclub ca would be the first detected. But this tells us much more ab what sort of locations are good for superspreading, and n whether new viruses are likely brought into existence by licentious dancing.

REPLY S

Scott Alexander Mar 28, 2024 *Edited*

Author

Not sure why this is mysterious.

The first officially confirmed case, the shrimp vendor, got sick December 11. So she was probably infected December 5. She probably wasn't the first human case, because she did interact with the raccoon-dogs; more likely a raccoon-dog vendor got (quietly) sick first an spread it to some other vendors. It's not even clear she was in the second batch of cases - 5-10% of COVID gets detected, probably less when it was first starting, so she could be anywhere from one to four transmissions away from human Patient Zero. So human Patient Zero probably got sick in late November or early December.

Let's say Patient Zero got COVID November 27 (with high error bars). He infects other peo at his workplace (the market), and some of those people infect still other people at the mar or other people elsewhere (for example when they're on the train home from work). COVID case numbers double every 3.5 days, so by 12/11 when the shrimp vendor starts feeling sic there are somewhere around 16 cases (with high error bars - and maybe more if there was second spillover for Lineage A, but ignore that for now). Any of those people could have passed it to Mr. Chen on the train on the 8th/10th/11th/whenever.

REPLY (1)

S

Simon stats Mar 29, 2024

See above comment re this illustrating bias in the case search.

REPLY

S

Simon stats Mar 28, 2024

My final comment. It is important to consider how different SARS-CoV-2 is to other zoonoses. I a challenge for zoonosis proponents to find me a zoonosis with all of the following features

- Spillover occurred after 2000 when sequencing became much cheaper

- There were more than a hundred human cases

- There are zero infected animals.

This characterises SARS-CoV-2, but no other zoonosis meets these criteria. Why?

REPLY (3)

S

Ethan Mar 28, 2024

What evidence do you have that there are zero infected animals? I only see evidence that n of the animals that have been tested were infected with an essentially-genetically-identical pathogen, but there are many animals that were not tested. (I assume the claim you're mak is about essentially-genetically-identical pathogens, since there are plenty of samples fron animals of coronaviruses closely related to SARS-CoV-2.)

I think there are also some viral hemorrhagic fevers that haven't had their animal reservoir identified yet, though I'm not very familiar with this, and I don't know if any of them have ha over 100 human cases.

REPLY (1) S

Simon stats Mar 28, 2024

Yes I am saying that they haven't found an infected animal in their testing efforts, whic were very extensive, and included testing of species sold at the market in Hubei and across China. Usually, for a zoonosis with more than 100 human cases, they find nume infected animals and the first people infected mostly interacted with those animals. Th did not happen for SARS-CoV-2.

This is true for: SARS-1, MERS, H7N9 avian flu, 2018-2021 Langya Henipavirus outbre China, NIPAH, and all other zoonotic coronavirus spillovers with more than 100 human cases.

REPLY (2) S

Freedom Mar 28, 2024

I guess you mean at the initial detection? Since obviously many (many) infected animals have been found subsequently.

REPLY (1) S

Simon stats Mar 29, 2024

yep i mean infected with a progenitor

REPLY S

everythingism Mar 29, 2024

Some important context here is that according to the WHO, China also claims to h found zero infected animals total, during the course of the entire pandemic, even COVID swept across the country infecting (most likely) hundreds of millions of people.

It seems wildly implausible that there are really zero animal infections (including tl passed on by humans) in a region as large as the entire continent of Europe, that has millions of susceptible species being raised on farms. More likely China is eith failing to identify infected animals or failing to report this information to the world.

REPLY S

halvorz Mar 28, 2024

2013-16 West African Ebola outbreak; almost 30k cases, no animal intermediate. This is probably the most notable zoonotic episode of the last 20 years apart from SARS-2, I'm surprised you missed it. There are also 7 other Ebola outbreaks that match your criteria. We been studying Ebola for over 40 years and have yet to determine the animal reservoir. It too 20 years to identify the reservoir for HIV-1's progenitor. Sometimes finding the reservoir is easy, sometimes it's hard. Typically it is easy when you have lots of cases and the virus is n very efficient at human-to-human transmission, because that necessitates lots of separate zoonotic events, which necessitates lots of infected animals. For something that spreads fa (i.e., the kind of virus likely to start a pandemic), you don't need a big reservoir, so you have smaller target. For example, we *did* find the reservoir for the 2009 flu pandemic, but it to years: https://elifesciences.org/articles/16777

REPLY (1)

Simon stats Mar 28, 2024

I hadn't actually looked into ebola yet, so I take it back

I thought the ebola host was known to be bats?

also hadn't looked at the swine flu one, I was overconfident there.

I agree re HIV, but that was before 2000 and sequencing was much more expensive.

REPLY (1)

halvorz Mar 28, 2024

There is strongly suggestive evidence for bats -Marburg virus has been isolated f bats, we've been able to identify antibodies in various bat species that react to Eb and in one case we were able to isolate a short bit of RNA that was similar in sequence to Ebola, but nobody's been able to isolate the virus, nor is it clear how might get to humans from bats.

HIV reservoir was confirmed to be chimps in the mid-2000s, well after sequencin technology was cheap enough to do it. Main limiting factor was just that many chi populations don't have it, plus having to go into the jungle and track chimp troops isolate fragile RNA from their poop...not at all trivial to do.

REPLY

Scott Alexander Mar 28, 2024

Author

Doesn't SARS1 meet these criteria?

Wikipedia says they found "SARS-like" viruses in civets, which made them think civets wer
the natural reservoir, but I don't think they ever caught SARS itself in the wild.

REPLY (1)

S

Simon stats Mar 29, 2024

There's different types of evidence for infected civets, some of which comes from higl seroprevalence among civet traders.

In May 2003, Guan et al (2003) identified SARS-CoV-like virus in animals in a live-anim market in Shenzhen, Guangdong Province, China. Guan et al (2003) also tested for antibodies among workers in the market. They note that "8 out of 20 (40%) of the wilc animal traders and 3 of 15 (20%) of those who slaughter these animals had evidence c antibody, only 1 (5%) of 20 vegetable traders was seropositive." This suggests that the majority of the infections of the 11 people with close contact with animals were zoonot

Among 508 animal traders, 66 (13%) tested positive for IgG antibody to SARS associa coronavirus by ELISA, while the control groups including hospital workers, Guangdong CDC workers, and healthy adults at clinic had an antibody prevalence of 1–3%. Among animal traders, the highest prevalence of antibody was found among those who tradec primarily masked palm civets (72.7%), wild boars (57.1%), muntjac deer (56.3%), hares (46.2%), and pheasant (33.3%). Those for cat, other fowl, and snake were 18.6%, 12% and 9.2%, respectively.

We have no such evidence for covid. As I have noted, none of the cases in the market worked at the animal stalls; there is no evidence that they had close contact with anim

REPLY (1)

S

everythingism Mar 29, 2024

That's an interesting result and something similar was later found for wildlife work in Myanmar and Laos -- antibodies to SARS viruses but the workers themselves having no memory of being sick. However one reason we have no such evidence f COVID is because this kind of testing for workers at the Huanan Market was never released by China. There also is an article stating that one of the first to get sick a the market interacted with animals, although it's unclear to me whether this perso was included in the WHO report or not. The first known case at the market, the shrimp vendor, has said she thinks she may have been infected by sharing a toilet with the wild game traders.

REPLY

S

Richard Weinberg Mar 28, 2024

Thanks for your deep and well-reasoned dive into a truly important topic. For me, your most important take-home messages were a) Bayesian reasoning has serious practical limits; b) we n to take the potential dangers of viral lab leaks seriously; and c) we need to take the potential dangers of natural zoonotic spread very seriously. One point I haven't seen mentioned is that th Earth harbors roughly 10**27 virus particles.

REPLY (3)

Doug S. Mar 28, 2024 *Edited*

I'd guess that most of them are bacteriophages...

REPLY (1)

Richard Weinberg Mar 28, 2024

Probably, and most are marine. But it's a big number even if you cut it x 1,000. It's a bi number even if you cut it x 1,000,000. It's a big number even if you...

REPLY (1)

Jeffrey Soreff Mar 28, 2024

Well, you _could_ think of it as "just" about one kilomole or so... :-)

REPLY

Michael Mar 28, 2024

It's also interesting how b and c support somewhat opposite reactions to gain-of-function research. If zoonosis is the bigger threat, we'd want to learn more about how this happens.

REPLY (1)

Richard Weinberg Mar 29, 2024

good point. I'd say that on the one hand there probably *should* be more gain-of-func research, but it needs to be performed far more cautiously. ALL such work on virus sh be performed in BSL-3 conditions, using only virions that have been genetically cripple be intrinsically non-virulent. I guess the question is whether the scientific "community can ensure that some jerk somewhere doesn't do something really stupid. Unfortunate we tend to underrate the raw power of Stupid.

REPLY

polscistoic Mar 31, 2024

a) is particularly important. What this blog post has shown beyond reasonable doubt is that "real" Bayesianism is not for amateurs.

A little knowledge is a dangerous thing. Better to only teach students who will be tasked to make difficult decisions in their professional lives (such as medical doctors, psychologists, investigators, judges and child protection officers) awareness of cognitive biases & where heuristics may go wrong, and NOT to also teach them formal Bayesian analysis - as they ar unlikely to have the time & information & sufficiently advanced methodological skills to do i properly (plus be aware of all the pitfalls).

REPLY S

NoRandomWalk Mar 28, 2024

My rule of thumb for something like this is 'whose evidence stands up to scrutiny'. I am not sma enough to understand any of this stuff, but I vaguely knew about the evidence beforehand from podcasts and social media. And then Peter just explained it all away effectively without significa pushback. This never happens in UFO debates. Or political conspiracies. Someone spends a bil hours researching sounds smart makes 30000 claims and then you pick 5 at random and 4 are nonsense and 1 is true or misleading or misinterpreted. I notice that my plausible theories of lab after hearing Peters argument about the implausability are *not* those advanced by rootclaim a so I assume they are obviously wrong I just don't know why but I'm confident Peter could convir me instantly. Rootclaim's inputs fell apart under scrutiny', which makes me doubt his statistical analysis. Garbage in garbage out. Note my real confidence about who is right is super low becau am dumb. But the debate definitely moved me away from lab leak 70-30

REPLY (1) S

Saar Wilf Mar 28, 2024

You are conflating the spoken debate part, where Peter had far superior memorized knowledge, to the written parts, where there is basically no zoonosis evidence that survive scrutiny. We'll elaborate in a post soon.

REPLY (4) S

NoRandomWalk Mar 28, 2024

I look forward to reading it, and I have in anticipation 'rolled back' my change in beliefs until I do!

REPLY S

Theodric Mar 28, 2024

“Okay we got our asses kicked in the mutually agreed format *that we ourselves proposed*, but now after the fact we want to re-litigate in a format that plays more to c advantages” *really* feels like shifting the goalposts in an unfair way.

If you take that route I think it would only be fair to give Peter access to the same resources you have in doubling down on the research and time to refute your refutatio

REPLY (2)

Saar Wilf Mar 28, 2024

i was referring to the written parts of the debate. anything that didn't require real responses.

REPLY

MicaiahC Mar 28, 2024

I agree it should reflect poorly on Saar that he didn't realize that sophisticated debaters are in fact a huge problem (see: https://web.archive.org/web/20190129225836/http://commonsenseatheism.com/?p=1437 and to be clear I don't begrudge him for not knowing about a link rotted p from 2018 by a niche atheism blogger).

But I feel you should be engaging in the object level too, and realize that a spoken debate in fact is not the best at eliciting truth, for usual status reasons as well as t reasons Saar cited:

1.Debate importance is rated via time spent on a topic, not its actual importance, Scott's point about the UFO detail debater.

2. It mostly explores how many layers of point-refutation each person has at the moment of debate, rather than what the reflectively endorsed endpoint would be. would rather people take the time to decide how much each point matters to their main point, be explicit about this ahead of time and then decide if hashing it out is good use of time. This is difficult to do in real time format.

3. If it is indeed true that Saar has found the ~one epistemology~, it seems much better to agree upon that frame (but not its conclusions) beforehand. This is not ju to advantage it, but if it turns out to be as dismal as debate afterwards, well, that should give us a pretty good idea how well it generalizes.

That said, you are correct that retooling the die is a social faux pas, and that we should downrate the second round result.

I think if you are as confident on lab leak as Peter is, you should be rubbing your hands together, salivating that Saar is going to eat crow in HD this time.. Maybe n by Peter in particular, because Peter sick of this and won.

I think Saar should state up front that he commits to doing only this one round of welching / sore losering, and have it framed for everyone to see after, if he wants

dodge the "after wit who keeps coming up with reasons why he is right 3 weeks a
losing the argument" accusations.

REPLY (2)

Randomstringofcharacters Mar 28, 2024

Seems more like he was fine with the format when he thought he was the mo
sophisticated interlocutor, and changed his mind afterwards when he lost.

REPLY (1)

MicaiahC Mar 28, 2024

Yes, this was included in my initial assessment of "penalty for not understanding that debates suck". I don't know why people keep thinkin it's good to debate! Judges predictably flop the other way they started a the beginning, we know about the existence of truth independent debat This is an opinion I've had before I even heard it was a debate, and to my this mostly looks like every other case where an amateur debater loses, had a better case (not that I think rootclaim had a better case, but rootcl at least pattern matches to non sour grape, honest dismay at debates sucking).

I don't think sophistication is a good way of tracking truth, because sophistication is often displaying that you understand deep nuances and details, regardless of how it connects to the bigger picture. So anything tracks sophistication tracks more of your ability to fight on specific sub topics, rather than subtopic weighted by centrality to claim.

I think I can be convinced away from this position if Saar goes on to repe this similar behavior, or if he reflects and decides he was wrong after all. if Saar wasn't willing to additionally show his reasoning, or welches on th actual bet.

REPLY

sturo Mar 31, 2024

This is not a niche atheism blogger and the post is not from 2018.

REPLY

Radu Floricica Mar 28, 2024

This post has actually left me with a deep curiosity on Rootclaim itself. To be honest, rather than the Nth post on the topic of lableak I'd very much rather read a primer on h to use the method.

REPLY

NoRandomWalk Apr 5, 2024

I read your reply and found it quite persuasive, and am tentatively back in the lab leak camp

REPLY

Hilarius Bookbinder Mar 28, 2024

A lesson Scott draws is that Bayes is effectively impossible to apply in complex real-world situations, and that ultimately we are forced to rely on our intuitive reasoning. We'll just throw a math in at the end to make sure we aren't making some clear and obvious blunder. I suggest thi means the true epigraph for ACX should not be "P(A|B) = [P(A)*P(B|A)]/P(B), all the rest is commentary," but rather "commentary, and all the rest is P(A|B) = [P(A)*P(B|A)]/P(B)."

REPLY (2)

ultimaniacy Mar 28, 2024

Well, I would argue that Hillel's standing-on-one-foot summary of the Torah was similarly t simple to apply in reality and largely just ends up being a rationalization for our intuitive aesthetic judgements, so I'd say that just makes the epigraph all the more fitting.

REPLY

polscistoic Mar 31, 2024 *Edited*

My main take-home point also.

Like cost-benefit analysis that is not done by a very skilled and aware-of-own-biases practitioner, hard/real Bayesianism seems to create a danger that uncertainties are hidden the priors & underlying assumptions & the empirical material used for the calculations, rath than appearing openly for all to see.

REPLY

TGGP Mar 28, 2024

> When psychoanalysts claim their therapies work, they don't mean that someone who just reac two page "What Is Psychoanalysis?" pamphlet can do good therapy. They mean that someone v spent ten years training under someone who spent ten years training and so on in a lineage bac Freud can do good therapy.

And Robyn Dawes in House of Cards showed the evidence doesn't support the idea that trainin
makes them any better.

REPLY (2)

TGGP Mar 28, 2024

I had forgotten that over a decade ago I'd written about Dawes vs the concept of "metis" among professionals here:

https://entitledtoanopinion.wordpress.com/2010/11/28/robyn-dawes-robin-hanson-as-antidotes-to-james-scott/

And a follow-up here:

https://entitledtoanopinion.wordpress.com/2011/01/26/bogus-expertise-as-weapon-not-of-weak/

REPLY

polscistoic Mar 31, 2024 *Edited*

Jeremy Howick (The philosophy of evidence-based medicine) cites the derogatory acronyr GOBSAT related to the problems with "Metis": Good Old Boys Sat Around A Table (and agr on diagnosis & treatment).

REPLY

Slaw Mar 28, 2024

"One read out of 200,000,000 is completely statistically insignificant," said Pond. "It really had r SARS-CoV-2. There is no evidence based on genetic analysis there was SARS-CoV-2 in that sample. One read out of 200,000,000 — it could have been a low level of trace contamination."

What's more, as Bloom's preprint reports, Q61 was the only swab above a certain threshold for raccoon dog genetic material that contained any SARS-CoV-2 RNA at all: "13 of the 14 samples at least 20% of their chordate mitochondrial material from raccoon dogs contain no SARS-CoV- reads, and the other sample [swab Q61] contains just 1 of ~200,000,000 million reads mapping SARS-CoV-2." When Bloom plotted the quantity of animal genetic material found in the swabs w their SARS-CoV-2 RNA content, he determined that there was in fact a negative correlation between the abundance of SARS-CoV-2 and genetic material from raccoon dogs in the swabs.

REPLY

Elle Mar 28, 2024 *Edited*

Wow this was a great read.

But also.... What happened to discussion of "NIH-funded gain-of-function research" by Peter Daczak and EcoHealth alliance? Is this the "DEFUSE" grant I'm question? It makes not mention o University of North Carolina - that's new to me, though of course I didn't delve deep into the gra

https://www.science.org/content/article/nih-restarts-bat-virus-grant-suspended-3-years-ago-trump.

I also wish it would be possible to go and collect all the Twitter posts on this topic from when th situation was unfolding, before and after the pandemic left China, including all sorts of claims: spread through plumbing, claims of corona virus circulating months (!) before the official first ca etc. -- do an analysis of how much is real, how much is fake. For example I remember following closely everything I could read and don't remember racoon-dogs ever being mentioned, but pangolins were.

REPLY (1)

S

Kenny Easwaran Mar 28, 2024

Pangolins were mentioned in March 2020 as a hypothesis. Raccoon dogs became the more prominent hypothesis by early 2021, when the WHO investigation got samples from the ma Obviously, there are severe limitations on what you can conclude with any confidence from year-old samples, but they're not as much as the limitations on what you can conclude with any samples.

REPLY

S

hwold Mar 28, 2024 *Edited*

Can someone explain to me Peter's argument "doubling rates at the wet market points at zoono rather than a superspreader event" ?

Intuitively, the two scenario (zoonosis / superpreader event) looks the same to me : an infected worker/racoon dog) (goes at/is brought to) the market and infects market participants. I expect same doubling rates in both cases.

REPLY (7)

S

Tatu Ahponen Mar 28, 2024

A least insofar as I remember, "superspreader" was typically used in the public Covid deba to describe cases where the disease vector, for whatever reason, would spread the disease particularly many new targets. Ie. if a typical Covid patient ended up spreading it to two oth (I can't remember the actual numbers), a superspreader would spread it to twenty.

REPLY

S

Deadpan Troglodytes Mar 28, 2024

Yeah, that argument was weak and I would have liked to see it either elaborated or conclusi
refuted - it's a high-load-bearing part of the zoonosis argument.

Early research showed that the original COVID variant spread bimodally: most of the spreac (~80%?) was caused by a small number of vectors. How is the evidence incompatible with idea that a WIV employee seeded infections that developed over the course of the followin week?

Furthermore, even if that model isn't correct, what plausibility advantage does "infected raccoon-dog" have over "WIV employee infected a HSM vendor who then became an vecto (asymptomatic or not), infecting multiple people over the course of a week or two"?

REPLY (3) S

Theodric Mar 28, 2024

The plausibility advantage of "infected raccoon dog" is that there is a good reason for racoon dog to infect someone (or someones) at the wet market, but nowhere else - raccoon dogs don't take crowded public transport to and from their office job and thei meat shopping.

For the WIV employee, there is an extra layer of coincidence that needs to occur for th to only infect someone else (or at least, only infect someone else in a way that "catche on" and spreads) at a wet market that isn't particularly close to the WIV itself.

Basically, conditional on the probability of the wet market being the epicenter of the subsequent epidemic, the probability is higher that this is because it originated there, originating at the WIV and being carried to the market, not creating other outbreaks elsewhere in Wuhan.

REPLY (1) S

Deadpan Troglodytes Mar 28, 2024 *Edited*

That's fair: I agree that the "infected a vendor" model deserves a penalty relative "infected raccoon-dog" (all else being equal). A small penalty for the extra layer c coincidence, and a somewhat larger one raised by the question of "why HSM and anywhere else?" (I don't think zoonotic-origin proponents have convincingly refut the possibility that ascertainment bias undermines key parts of their case, though grant that my priors have been shifting in that direction.)

REPLY (1) S

JoshuaE Mar 28, 2024

Peter's point is that it can't be spreading too much more widely in early December because otherwise when testing becomes widespread you would expect a much higher rate of infection and death therefore we should not consider the HSM initial location to be biased (e.g. there is strong evidence tl the first and most subsequent early Human-to-Human infections occurred at HSM). I think Peter (and the Judge's explanations) was extremely convincing the probability of zoonotic origin is much more likely than an infected lab wor traveling to HSM and starting the pandemic. In addition Peter did a good job reducing my opinion of the likelihood that Sars-Cov-2 is caused by Gain of Function (or any other human manipulation).

REPLY S

REF Mar 28, 2024

Peter did say (something along the lines of) that if you want to assume that a WIV employee left work and went to the wet market then went home and quarantined for a week you are welcome to. It isn't as if he ignored the scenario. I think the point is, if yo know you are infected, why go to the wet market at all and if you don't know you are infected then why are you quarantining...

REPLY (1) S

MicaiahC Mar 28, 2024

Isn't the most transmissible part of the disease right before symptoms appear? Wouldn't that exactly match this behavior, and there's no mystery?

I would be substantially wrong here if minimal virus shedding happens before symptom onset.

REPLY (1) S

REF Mar 28, 2024

Yup, if you want to come up with contrived scenarios, you can indeed make t data fit any hypothesis.

REPLY (1) S

MicaiahC Mar 28, 2024 *Edited*

https://www.goodrx.com/health-topic/infections/viral-shedding-and-hov prevent-the-spread-of-infection

Seems to think that this is true, and cites https://www.bmj.com/content/376/bmj.o89#:~:text=Previous%20studies 0suggest%20that%20the,on%20or%20before%20symptom%20onset

Saying that virus transmission starts two days before symptom onset.

I had the data first, before generating the hypothesis. I wasn't sure that I remembered correctly.

Do you think my scenario is still contrived? Or that I said this as a post h justification?

REPLY

hwold Mar 29, 2024

> Early research showed that the original COVID variant spread bimodally

I think that's a very interesting point ; can you find the paper ? I failed to find it.

REPLY (1)

Aristophanes Mar 29, 2024

The links I sent to you elsewhere should cover this.

REPLY

Simon stats Mar 28, 2024

To me, this argument makes no sense. The market was either: (a) a zoonosis plus a superspreader, or (b) only a superspreader. All of the confirmed cases at the market were f human-to-human transmission. None of the confirmed cases were wildlife sellers (unlike SARS). The confirmed cases were spread across the large area of the whole market.

REPLY (2)

Theodric Mar 28, 2024

Your b) is actually "a lab leak origin outside of the market, that only becomes a superspreader at the market". If the origin is not in the market itself, you need to accou for the probability that an infected superspreader from the lab (or infected by someon from the lab once or twice removed) infected a substantial number of people at the market, but not apparently anywhere else.

REPLY (3)

Simon stats Mar 28, 2024

yes agreed

REPLY (1)

Simon stats Mar 28, 2024

well actually unless cases were missed because of ascertainment bias in the case search

REPLY (2)

S

Theodric Mar 28, 2024

Genuinely asking: But do the subsequent case rates actually support tha Sure, initially all the focus was on the wet market, but if there were other simultaneous epicenters surely that would be ascertainable after the fac

And lab leak proponents would be highly motivated to find such data. Instead they (at least Saar does) seem to mostly concede that the wet market was the source of the first significant outbreak, plus or minus so low level earlier circulation in weakly sourced or anecdotal data.

REPLY (1)

S

Simon stats Mar 28, 2024 *Edited*

To clarify, my initial comment was just making the logical point that market was a superspreader (and maybe also a site of zoonosis), n that it was necessarily the *only* superspreader.

I don't see why you think other epicentres would be ascertainable a the fact? There was a retrospective case search that ended in mid F 2020. There was a market bias in play for around half that time and case count actually fell for part of the search. In my other comment, I highlight evidence for ascertainment bias. for example, you read this report about Zhongnan hospital, a hospital next to the WIV BSL-2 and BSL-3:

"The South China Seafood Market [link requirement] alone ruled ou most of our suspected cases." Yang Jiong said that Zhongnan Hosp is 10 or 20 kilometers away from the South China Seafood Market. M of the suspected cases received in the early stage did not have a history of exposure to the South China Seafood Market, but they ha obvious clinical manifestations of unexplained viral pneumonia. However, the district experts refused to identify them as unexplaine pneumonia after consultation. They could not be referred to Jinyint Hospital, and samples could no longer be sent for genetic sequenci

https://docs.google.com/document/d/1mK2CFD8IeSYrT26BLVRuiJS K-Q8-5K71byJUsmbA/edit

Because many cases in this area were not market-linked, they coulc be confirmed at Jinyintan (the premier infectious disease hospital), there was limited opportunity to identify clusters in this area.

Similarly, Mr Chen was only registered as a case because he happe to be transferred to a sentinel hospital north of the river. This sugge that other cases where Mr Chen lived (35km from the market) woul just never have been found.

REPLY (1)

Theodric Mar 28, 2024

It's frustrating that everyone here is taking "it was a superspreader" without actually addressing the question of whether it was, which is the actual crux of the argument.

If there really is good evidence that the market was not the ma origin of the outbreak, or at least that it's likely other cases we spreading equally quickly from other sites, I would agree that t reduces the probability of zoonotic origin at the market. Which why it's weird to me that Saar essentially concedes this and tri explain it rather than attacking the prior that the market was actually the epicenter in the first place.

REPLY (1)

Simon stats Mar 29, 2024 *Edited*

I think we might be talking past each other. I was assumin that zoonosis people also think it was a superspreader eve which I am understanding as an (indoor) location with a substantial number of confirmed cases from human to hu transmission. Everyone on both sides agrees that the confirmed cases at the market were from human to huma transmission. Are you defining superspreader in a differen way?

REPLY (1)

Theodric Mar 29, 2024

Peter explicitly claims that it was not a superspreader Scott covers in his summary - his argument is that th human to human transmission at the market was at a “normal” or average rate compared to typical Covid transmission seen later in the epidemic.

A “superspreader” event is not merely a single locatic with multiple cases associated with it, it’s a place whe an individual or individuals infect way more than the u number of other individuals (I believe the usual rate is that each case of Covid can be expected to produce 2 additional cases, where a superspreader might infe say 10 other people, making the growth rate of cases much faster).

If the market was a superspreader event, that could h explain why it was the apparent origin of the outbreak even though the actual first infections occurred elsewhere. If it was a normal transmission event, it ma it more likely that a vendor at the wet market really wa “patient zero”.

REPLY (1)

S

Simon stats Mar 29, 2024

I see that makes sense

REPLY

S

Kenny Easwaran Mar 28, 2024

You have to assume not just early cases missed because of ascertainme bias, but also some explanation for why there aren't five times as many cases a couple weeks later if it had been spreading a week longer than hypothesized.

REPLY (2)

S

Deadpan Troglodytes Mar 29, 2024 *Edited*

Before kvetching, I want to concede that this is a good argument, a it's been a big part of why I've updated away from assuming ascertainment bias.

However, I have never seen anyone substantiate its premises. I thinl that's important because there are two related areas of uncertainty underlying the point: first, that we got sufficiently accurate epidemiological data out of China during that early period and seco that we know what the R0 was at the time well enough to make the judgement.

Beyond that, the epidemiological dynamics of COVID were weird: m "punctuated equilibrium" than steady spread. Admittedly, my understanding of that is mainly rooted in subsequent history, when were in effect and people were on guard, but this is not an uncomm model for how diseases spread.

Maybe Peter addresses these issues in the debate, but I have yet tc anyone do so to my satisfaction, so my position remains something "there's only a 50% chance we have information enough accurate t make these kinds of judgements."

(Edit - reading down thread, I see now that Peter linked to his analy: of the spread, which I'll read as soon as I have time.)

REPLY S

Simon stats Mar 29, 2024

So, am I right in thinking you don't think there was ascertainment bi the case search? I'm not sure I understand. In that quote above, yoι have >10 hospitalized cases at Zhongnan that don't get registered because they didn't have a market link. Assuming there are ~10x tot cases, that's already 100 cases you don't see for market biased reasons, (in an area around the WIV)

REPLY S

John Schilling Mar 29, 2024

Whether Patient Zero is a careless WIV lab tech who is also a wet market custome an unlucky wet market vendor, they both go home at the end of the day, probably family, and very probably a normal life where they go out and do things at least ur they get sick. So in either case, the pandemic starts with A: a superspreader even the wet market and B: an infected guy going around spreading the disease the ordinary non-superspreader way. I'm not seeing how you can use observed doubl rates or anything like them to distinguish between the two.

If there were confirmed cases predating the market outbreak, those would give us great insight into what happened, but AFIK there aren't. The fact that there aren't tells us that Patient Zero infected very few people outside of the market and they just thought it was a cold.

REPLY S

Hafizh Afkar Makmur Oct 20, 2024

I think it'd be plausible if an infected from lab is only able to infect exactly one oth person in the market (his/her spouse?), and this person is a super spreader. Matcl what I know about the bimodal distribution. I'm still leaning on zoonosis, but my ra of 30% for lab leak is still too big to ignore.

REPLY S

REF Mar 28, 2024

Peter seemed to have adequately debunked, "superspreader at market," based on gro rates and start dates. Why do you ignore this here?

REPLY (1) S

Simon stats Mar 29, 2024

See above comment re talking past one another.

REPLY S

Mr. Doolittle Mar 28, 2024

I feel like he's leaning very heavily on the average spread, but that's essentially his argumei That if early COVID spread like we know it did later, then the size of the spread would have been much larger if it was earlier, which Saar's argument requires.

Given small numbers at outset, I don't think he can actually rule out low and intermittent sp for a few days maybe weeks before catching in a more randomized way. If someone was infected early and then only spread to one or two other people and they only spread to one other person, it may have delayed the virus from spreading as we saw it later. Once it hits a certain threshold that seems very unlikely, but below that threshold we really can't say that

REPLY (2) S

Aristophanes Mar 28, 2024

Indeed. There is a paper (can dig it up if you want) showing that (iirc) the model numb people to infect (if you're infected) is 0 and the next most common is 1. Many many ch either die out completely or take several cycles to explode.

REPLY (1) S

hwold Mar 29, 2024 *Edited*

Yes, I'm very interested if you can find that paper. It could help shed light on the t question "on average, how much time do we expect to see between the first case the first superspreader event ?"

One key point is that Peter insist on "very little time because exponentially doublir rate", which I believe is wrong.

REPLY (1)

S

Aristophanes Mar 29, 2024

Yes, I agree - I think the uncertainty about the start date is far larger than he' implying from "we know the doubling rate, so therefore it's tightly bounded". you know the exact doubling rate you can talk about aggregate dynamics usi this math when the number of infected is large, but when the outbreak is sma size the variance can be very large.

Now onto the specifics - COVID-19 transmission is characterised by a large degree of overdispersion. This means most people infect very few others, an the vast majority of transmission is driven by a small fraction of the infected. Multiple studies from 2020 found that over half of infected people never infe anyone themselves. Here are some links:

a) https://www.nature.com/articles/s41591-020-1092-0 - See Figure 3

b) https://pubmed.ncbi.nlm.nih.gov/32685698/ - See Figure 2

c) https://www.nature.com/articles/s41467-020-20235-8 - "The remainder (57%) did not lead to a transmission event"

d) https://www.nature.com/articles/s41598-020-79352-5 - See Figure 2

REPLY

S

Theodric Mar 28, 2024

Sure, but why would the virus only "break out" at the market? I don't think the claim is these early cases were some less virulent strain (the A and B types are both pretty infectious). So if it was spreading at a low rate throughout Wuhan, there are a *lot* of potential breakout sites where it didn't break out, or create multiple epicenters at roug the same time as the wet market.

Which isn't totally implausible! But it's probably less likely than "no really, the first case were spread in the wet market and not somewhere else".

REPLY (2)

S

Mr. Doolittle Mar 28, 2024

That's the other part of Peter's argument that I didn't like, the almost slavish follo of "less likely = wrong" when at any point something unusual could very well have happened. That seems to be Scott's point at the end, where we know for sure tha least one of two very unusual things actually did happen.

I don't know if I have enough knowledge to say that it is "probably less likely" or n but that's not the final determiner of truth.

REPLY (1)

Theodric Mar 28, 2024

This strikes me as a weird argument to defend someone whose whole projec rigidly following Bayesian probability. "Probably less likely" IS the final detern of (as-close-as-you-can-get-to) truth.

REPLY (1)

Mr. Doolittle Apr 2, 2024

I get his point, but I simply disagree with it. If that means I disagree with Bayesian reasoning then I'm okay with that too.

REPLY (1)

Theodric Apr 2, 2024

I think you miss my point: slavish devotion to Bayesianism isn't Pete position, but Rootclaim's.

If you reject Peter's argument on non-Bayesian grounds, then Peter lost the battle, but Rootclaim loses the war.

REPLY

Aristophanes Mar 29, 2024

Something as high as like 60-70% of people infected with Covid appear to not inf anyone (in a seronaive environment). It might even be more skewed than that onc you account for under detection of asymptomatic cases to begin with (I'm guessi the never symptotic don't spread it much)!

This makes chains that die out, or chains that barely survive for a few generations then explode completely plausible.

REPLY

Theodric Mar 28, 2024

I understand “superspreader” to mean something like “an infected individual who sheds wa
more virus than usual in a confined environment creates a localized, discrete spike in infect
rates”. Or in any case “a discrete event with a much higher rate of infection than the base ra
expected by the observed growth rate of the epidemic”.

Peter is saying the wet market doesn’t look like that, it looks like a “normal” rate of infectior
and growth in cases.

Why does that matter? The two sides of the argument both need to explain why the epiden
was first detected and seemed to spread from the Wuhan market, rather than somewhere e

Saar’s claim is that basically Covid leaked from the lab sometime prior to the wet market outbreak, and was circulating around at a low rate, but a “superspreader” event in the wet market made it the epicenter of subsequent infections. This theory takes a major hit if the spread in the wet market was not atypical, because in that case the market should just be c
of many similar outbreak sites started by the various infected folks circulating around, rathe
than a clear epicenter.

Peter claims that the wet market was the actual origin of the zoonotic transmission. It looks
it all spread from there because it *did* all spread from there. The fact that there was nothi
“super” about the spread at the market supports this theory - wet market infections don’t r
to swamp other potential epicenters because there are no other epicenters.

REPLY (1) S

REF Mar 28, 2024

No, superspreader is, "went to my own wedding with COVID and kissed all 200 guests

REPLY (1) S

Theodric Mar 28, 2024

Either way, it comes down to one person infecting way more than the average 0-2
people per case, right?

REPLY (1) S

REF Mar 28, 2024

It does. But it is far more likely to be situational then something anomalous al
viral shedding.

REPLY S

Alex Zavoluk Mar 28, 2024

> I expect the same doubling rates in both cases.

Yes, this does make sense, which is the point. Such a scenario is a priori unlikely for lab lea since it requires a lab worker to go to the market and infect someone there, but infect no o anywhere else, either in the lab or in between, since there are no other early clusters. To m this hypothesis seem more plausible, Rootclaim asserted that the market provided very go conditions for much faster growth of Covid, possibly even a superspreader event (i.e. a cas where a single person infects a dozen or several dozen people all at once, greatly in excess the average number of secondary infections, which for early covid is less than 2. This woul explain why the market dominates the early cases, but as Peter points out, these claims ha no evidence and are contradicted by available data. If the spreading rate is the same inside market as everywhere else, then the strong clustering of early cases to the market is very g evidence in favor of the pandemic starting there (and thus, in favor of zoonosis).

Bu the upshot is that this argument is specifically a response to Rootclaim's argument, so i don't believe the latter, then this part would be irrelevant for you.

REPLY (1) S

Theodric Mar 28, 2024

This is a better summary than I was able to put together.

I think Rootclaim needs to defend their assertion of a superspreader event at the mark because otherwise Peter seems to have convincingly shown that it wasn't, and if it wa it's pretty good evidence in favor of the wet market theory.

REPLY (1) S

Alex Zavoluk Mar 28, 2024

I don't think they're going to do that, and from Saar's point of view they don't real need to. This is his whole shtick with "both sides shouldn't have high BF" and Sc analogy with the Muslim UFO spotter. Maybe it wasn't a superspreader event at t market; that was just one possible explanation. But this example (allegedly) show that such explanations are not that rare or hard to find, and so this evidence must have a low Bayes Factor.

REPLY (2) S

Theodric Mar 28, 2024

In something like this, you might end up with weirdly extreme probabilities si because global pandemics are rare, and will necessarily be the result of a ch of individually unlikely events occurring. I know in theory all these probabilitie are supposed to be "conditional on Covid actually happening" but that's easi said than done.

I get the Muslim UFO Spotter analogy, but I still feel like Saar is a bit dodgy h

Your model should account for all possibilities even if there is just a "what if t did happen, but not in one of the specific ways I already quantified" uncertai term.

If your conclusion is correct, but for a reason your model didn't even anticipa you just got lucky.

As it is, it looks like Rootclaim simply didn't consider the possibility that the w market wasn't a superspreader, which raises the obvious question "what else they miss".

REPLY (1)

Alex Zavoluk Mar 28, 2024

> In something like this, you might end up with weirdly extreme probabili simply because global pandemics are rare, and will necessarily be the re of a chain of individually unlikely events occurring.

I think this is likely to be true.

> I still feel like Saar is a bit dodgy here.

Oh, I agree that Saar's argument is extremely dodgy. His argument relies the Bayes Factors that he produces being both high and extremely robu that is not the case, it totally falls apart. I was just explaining why I don't think Saar is likely to further support the superspreader claim, or care ab it.

REPLY (1)

Comment removed Mar 28, 2024

Comment removed

REPLY (1)

Alex Zavoluk Mar 28, 2024

I believe they presented some arguments as to why we would expect it to be a superspreader location a priori and other plac would not be (e.g. unique combination of traffic, poor ventilatic and other factors) as well as some examples of market superspreading events elsewhere. It was all just very indirect a speculative compared to what Peter showed.

REPLY (1) S

Comment removed Mar 28, 2024

Comment removed

REPLY (1) S

Alex Zavoluk Mar 28, 2024

I don't recall exactly what Rootclaim argued. I think th A) cherry picked what factors were relevant, and B) fa to quantify any of the effects they identified.

REPLY S

Kenny Easwaran Mar 28, 2024

I'm confused about the "both sides shouldn't have high Bayes factor" claim. plenty of ordinary claims, I can very easily come up with extremely high Baye factor pieces of evidence on both sides.

Like consider the question of whether I was born in India or in North America last name is a very high Bayes factor piece of evidence in favor of the India hypothesis, and my first name is a very high Bayes factor piece of evidence i favor of the North America hypothesis. I haven't thought of any evidence I ca provide that has a high Bayes factor in favor of the hypothesis that I was borr South America over the North America hypothesis (the fact that I currently liv Southern California is probably single digit Bayes factor evidence in favor of South America over India).

In this case, there's very good explanation available for the anomalous evider (my father was born in India), and not every false hypothesis has an equal amount of high Bayes factor evidence.

But if you do a very serious search for anomalies related to something, you're likely to find a few anomalies in favor of some particular false hypothesis, eve most false hypotheses don't have any in their favor. There's a reason we're debating whether it was from the wet market or from the virology lab, and no considering whether it spread from pet dogs, or was planted by the CIA, or c from bats living in someone's attic.

REPLY (1)

Alex Zavoluk Mar 28, 2024

I believe what Rootclaim argues is that if you have a high BF on conflictir hypotheses, then one of them is either wrong or p-hacked. Your name example is a good one, though, since each piece of evidence is so simila it's hard to allege p-hacking for either side. I wonder how strong each of those pieces of evidence actually is? Like "Kenny" being 1/1000 in favor NA and "Easwaran" being 1/5000 in favor of India? I have no idea what tl numbers actually are.

And of course, unlikely events can still happen. For most people, their fir and last names probably don't provide such a high level of disagreemen think this has to be the case, right? If lots of American-born people had Indian last names, then the Bayes Factor for "Indian last name" would by definition not be very large? I'm just thinking about this quickly and not doing math.

> But if you do a very serious search for anomalies related to something you're likely to find a few anomalies in favor of some particular false hypothesis, even if most false hypotheses don't have any in their favor.

Yes, I think this is related to Rootclaim's assertion of p-hacking. But I thi this is also an easy failure mode, which several people have pointed out prevalent in the online lab leak community, of basically going on fishing expeditions with a high number of degrees of freedom and not accountii for the number of opportunities for an anomaly.

REPLY (1)

Kenny Easwaran Mar 28, 2024

> I have no idea what these numbers actually are.

I also have no idea - I could imagine them being anywhere from a fa of 10 to a factor of 1,000,000!

In any case, I think that it's a bit awkward to talk about "p-hacking" when we're trying to do a Bayesian analysis. We know that most of evidence people come up with is going to be gathered in a highly nc random way!

I'm a huge proponent of Bayesianism, but I don't see much prospec explicit calculations of this sort to be that useful in this kind of form at least in part for this reason.

REPLY

tgof137 Mar 28, 2024

For the exponential math argument and its relevance, start watching here:

https://www.youtube.com/watch?v=d1dbfoK8nSE&t=870s

It shows that the market outbreak can't just be an artifact of ascertainment bias because tl weren't enough cases at that time in the outbreak, the size of the market outbreak in early-December is roughly what you'd expect to find in all of Wuhan.

So it can still be a lab leak, it just needs to go straight from the lab to the market and nowhe else (with both lineages). That's very unlikely and needs to be accounted for in any sensibl bayesian analysis.

If you think that the exponential model is too simplistic to make that point, I've also done m detailed epidemic simulations to demonstrate a similar thing:

https://twitter.com/tgof137/status/1772417277670871113

REPLY

Elle Griffin Mar 28, 2024

Wow, this was incredible.

REPLY

Omer Mar 28, 2024

Is it really such a coincidence that the outbreak occurred naturally nearby the lab? It's reasonab guess, I don't know) that the lab was strategically placed there in the first place due to the area' wildlife, which host many animals relevant to their virology research. If so, the alleged coinciden significantly fades.

REPLY (2)

Comment removed Mar 28, 2024

Comment removed

REPLY (2)

Charles Pye Mar 28, 2024

My theory is that the local government of Wuhan knew they were at risk, because so n
rural farmers in the area were bringing wildlife into the wet market. So they funded the
WIV. As you say, a big research institute kind of has to be near a city. It's not going to b
Bruce Wayne living in a bat cave.

I wish I knew more about the internal politics of China though.

REPLY (1) S

JoshuaE Mar 28, 2024

It's somewhere between a pandemic started in San Diego and you discovered Sar
Diego had a relevant lab (8th largest city in the country) and Los Angeles county l
a pandemic start in downtown and there's a relevant lab in Pasadena (based on si
of Wuhan vs American cities).

REPLY (1) S

Charles Pye Mar 28, 2024

I don't think either one is really a fair comparison. Wuhan has *11 million peop
so it's roughly the size of NYC and LA combined. But it's also in an area wher
there are still a lot of desparately poor people living in the country and huntin
whatever animals they can find as game meat, on a scale which just doesn't
in the US. "Anything that walks, flies, swims, or craws..."
(https://www.quora.com/Do-the-Chinese-eat-everything-that-walks-swims-f
and-crawls)

REPLY (1) S

JoshuaE Mar 28, 2024 *Edited*

I was using LA county (although thinking of metro area although the two
not super diffrerent in the case of LA) not LA City (LA metro area has 12
million people). My assumption is a pandemic starting in LA would most
likely be caused by travelers so it wasn't meant to be exactly the same b
rather it's not surprising for there to be a city/metro area with both a lab
a pandemic. When I was picking US cities (metro areas) my though it tha
many people had not heard of Wuhan until 2020 but it's not some podur
village.

REPLY S

Omer Mar 28, 2024

This answer misses the point. Does it make sense to build a lab tasked with collecting samples for coronaviruses research in the London, in the same way it that it does in Wuhan?

REPLY (1)

tg56 Mar 28, 2024

Not sure, but isn't agreed by all in the argument that the other top lab in the worlc researching Coronaviruses was in North Carolina?

REPLY (1)

Omer Mar 28, 2024

From the post: "Peter: As mentioned earlier, the DEFUSE grant was rejected. Further, the grant said that the Wuhan Institute of Virology was responsible fo finding the viruses, and the University of North Carolina would do all the gain function research.".

I don't know what exactly "responsible for finding the viruses" operatively means, but it might mean that their location is indeed non-random, and that t was their "competitive edge" so to speak.

REPLY (1)

Kenny Easwaran Mar 28, 2024

It's "non-random" in the same sense that Miami being a major banking h for Latin America is non-random. It's not especially close to the places t it's serving, but it is a natural location to be an intermediary between the places that it's serving and the broader world of finance or biological research.

REPLY

John Schilling Mar 29, 2024

I don't guess, I've looked into it, and no. The lab wasn't placed in Wuhan because of the ba pangolins or racoon dogs or whatever live near Wuhan. First off, the bats we care about live away from Wuhan, and any pangolins or raccoon dogs we care about have to have lived nea the bats. Second, it's the "Wuhan Institute of Virology", not the "Wuhan Institute of Coronavirology" or "Wuhan institute of Zoonotic Virology"; it was not initially meant to specialize in any particular sort of virus that would call for any particular location.

Probably the WIV is in Wuhan because when it came time to found a big Institute of Virolog the next guy on the CCP's "we owe this guy a favor, let's build a big infrastructure project ir district" list was from Wuhan.

REPLY S

Neeraj Krishnan Mar 28, 2024

Thank you dashing young blogger!

REPLY S

Metacelsus Mar 28, 2024

I think the "lab leak" side of the debate focused too much on the probability that SARS-CoV-2 w engineered, and didn't properly account for the scenario that SARS-CoV-2 was a natural virus collected by virologists that accidentally leaked. In my opinion the latter scenario is more likely t the former.

REPLY (2) S

Scott Alexander Mar 28, 2024

Author

Tell me more. My impression was that one of the strongest pieces of evidence for lab leak i the weird furin cleavage site. If you say it wasn't engineered, you lose that evidence - what you gain in exchange?

REPLY (6) S

Comment removed Mar 28, 2024

Comment removed

REPLY (1) S

Kenny Easwaran Mar 28, 2024

The weird furin cleavage site isn't evidence for that. It's no more likely under that hypothesis than under the pure zoonotic hypothesis.

REPLY (1) S

Comment removed Mar 28, 2024

Comment removed

REPLY (1) S

Kenny Easwaran Mar 29, 2024

No. CRISPR creates cuts and introductions at planned sites and with planned new sequences. Having something that no one would have plan is not particularly likely.

REPLY S

Philipp Markolin, PhD Mar 28, 2024

It is 2024, there is 0% doubt that the Furin cleavage site is natural and was not introdu by an engineer. Long list of reasons include (not comprehensive):

-All previous FCS created by researchers (for example here in 2006, or here in 2009, a published years before the "secret" DEFUSE proposal suggesting similar work got play up as a smoking gun) have avoided using an "insertion", and rather chose to "substitu rewrite-existing nucleotides in this position to change the encoded amino acids to cre an FCS. This is necessary and basic biology because the addition of new amino acids any spike protein will likely disrupt its structure and break the whole thing.

- Curiously, an additional, unnecessary proline residue is part of the insertion. Proline i not part of any FCS motif, however, the amino acid is considered a helix-breaker; something that will mess up your protein structure with a high probability. Why risk introducing that?

- Even more curious, the whole sequence insertion is "out-of-frame", meaning the nucleotides were not introduced neatly but rather staggered for no reason, again something that scientists would not do ordinarily (or at all)

- However, the FCS insertion is right next to an RNA sequence motif prone for copy-ch recombination, arguing for this natural mechanism bringing it about (and copy-choice recombination is reading-frame independent and favors out-of-frame insertions 2:1)

- The GC content of the insertion is 83%, very unusual for artificial introductions

- The coded amino acids themselves make an odd, non-canonical, and weak recogniti motif (RRXR) for furin-like proteases; a motif that might ordinarily not work compared strong motifs such as RRSRR (which would usually be used by virologists).

- The FCS cleavage at S1/S2 destabilizes the original (Wuhan-Hu-1) spike protein, whi only got compensated partially when the D614G mutation emerged quickly in the early pandemic, arguing for a recent acquisition, not design or cell culture

- Furthermore, potentially because the FCS in the original SARS-COV-2 is not that stal it is lost very quickly in Vero cells and other typical cell culture systems. Selection pressures seem to run against it in these research-associated amplification systems; s how exactly did an FCS-containing virus that first has to be amplified in cell culture be it can infect a researcher escape from a lab and do so with an FCS?

But these alone is not proof. Who knows what weird and arcane experiments those Chinese researchers would cook up in a lab, right? (Some lab leak truthers with no molecular biology experience would say. For actual genetic engineers, the situation is already clear as day) Fortunately, we do not need to rely on experience or speculations about any motivations or experimental procedures Chinese scientists might have follo

This is because nature just has more secrets hidden in the FCS that humans could not have come up with in the first place:

After intensive mechanistic studies of the FCS in SARS-CoV-2, researchers discoverec hitherto unknown synergistic interplay between the odd, suboptimal FCS and the gene backbone of the virus (which we by now know is natural because of recombination).

- We have learned that pre-cleavage of SARS-CoV-2 S1/S2 by furin is what boost TMPRSS2-mediated cell fusion and thus impacts cellular tropism and transmission dynamics (ergo infectivity)

- But it turns out, that was not the full mechanistic story. In nature, no genetic element acts alone

- The viral QTQTN amino acid motif is an uncommon natural sequence in several sarbecovirus spike proteins directly upstream of the FCS sequence

- In cell culture experiments, QTQTN is not stable and repeatedly lost, similar to the FC Loss results in impaired viral replication.

- QTQTN determines how tightly the loop harboring the FCS is bound to the spike prot thus it regulates spike protein stability and shapes how well the Golgi-bound furin-like proteases have access to recognize the FCS motif and pre-cleave at S1/S2

- The QTQTN motif is also glycosylated and loss of that glycosylation impairs viral replication as well.

- Researchers showed beautifully that there is an intricate functional interplay betweer FCS, the loop length, and glycosylation: These separate but co-dependent elements w together synergistically and all elements are ultimately required for efficient FCS pre-cleavage (and with it, impact viral replication and pathogenesis). In nature, no genetic element acts alone.

Co-dependency and synergy between genetic elements are hallmarks of evolutionary selection; these type of complex interactions are extremely hard to design for even wi perfect knowledge of all structural components, which nobody had in 2019. In the cas SARS-CoV-2's FCS, there is no conceivable way how engineers could have designed t observed- but hitherto undiscovered- mechanistic synergy purposefully.

Take this together with all the other unexplainable design choices listed above; and yo
have your answer. The polybasic cleavage motif recognized by furin was not artificially
inserted

REPLY (3)

Comment removed Mar 28, 2024

Comment removed

REPLY (1)

Philipp Markolin, PhD Mar 28, 2024

Many technical reasons. Maybe the simplest: Viruses adapt to their host syst
The original SARS-CoV-2 could not infect mice. Even with mice carrying a hu
copy of the ACE2 receptor, the virus shows none of the characteristic
adaptations that it acquired once it learned to infect mice (alpha, delta varien
that one would have expected to see earlier if it has ever been passaged in su
a system. The real problem is actually transmission, but that is more complex
quickly write out here.

REPLY

Simon stats Mar 28, 2024

Isn't the out of frame point relative to the backbone used? On the lab leak hypoth
the backbone used is unknown, so we don't know the insert is out of frame

REPLY (2)

Philipp Markolin, PhD Mar 28, 2024

Nope. Out of frame with respect to the used reading frame for the whole spik
and that is based on the coding amino acids that barely differ between the
various spike proteins because they are functionally conserved

REPLY (1)

Simon stats Mar 28, 2024

I see, thanks. Is it possible that there might be differences between the
spikes of some virus that they had that we don't know about that would
make the insert in-frame?

REPLY (1)

Philipp Markolin, PhD Mar 28, 2024

Short answer no, longer answer: still no, but never say never; but we would need some really strong evidence to suggest something like being possible

REPLY

tgof137 Mar 29, 2024

Yes, it's relative to whatever nucleotides are there already. Though I tried alig it to every known similar virus and it was out of frame for all of them.

In theory, the leading Serine could have been spelled with different codons, i whatever virus the lab started from. There are 6 codon choices for serine, on one would work, so I called this a 1 in 6 chance at the debate. Probably a cru way to do it, but Rootclaim wanted to quantify everything.

REPLY (1)

Simon stats Mar 29, 2024

So then isn't it wrong to say that the insert was definitely out of frame? I correct to say that if there were an unknown virus that the WIV following workflow outlined in DEFUSE, the insert could have been out of frame bu could not have been. From a Bayesian point of view, one has to weigh th against the possibility that the furin cleavage site emerged naturally in a farm in Hubei. The 850 sars-like coronavirus don't have a furin cleavage site, suggesting it is selected against in nature.

One also has to think about the timing of the acquisition when weighing probabilities. The furin cleavage site emerged the year after the WIV was a proposal to insert the site. The mutations in the virus also suggest that furin cleavage site emerged recently around Oct/Nov. Choe and Farzan (2021) argue that "The notable emergence of D614G suggests that the acquisition of a destabilizing furin site was a recent event". So, zoonosis to posit that the furin cleavage site emerged only once naturally, was ne found in nature despite them looking, and caused approx 1-2 spillovers i Wuhan city.

REPLY (1)

tgof137 Mar 29, 2024 *Edited*

Yes. I dealt with it in a probabilistic manner by assuming the original virus could have different codons.

I suppose what you really want is the ratio, i.e. 1/3rd of natural inser would be in frame, but let's assume closer to 100% of lab inserts wo be in frame, so you could change my numbers that way. But I was already being generous by assuming that the natural viruses have e priors for all codons at that spot, when it's actually out of frame for every one we know about.

The presence of the furin cleavage site itself is dealt with separately the probabilities. To some degree that may just be included in your priors -- i.e. the judges both assumed a serious coronavirus pander on the order of 1 in 100 years, it wouldn't be serious without the furi cleavage site, so it's double counting to include that.

Saar used a factor of 2, that the lab pandemic is twice as likely as th natural pandemic to have such a site. That's probably double count but maybe not, since his pandemic prior was 1 in 20 years. I could s going up to a factor of 5 here since we don't know exactly how ofte the serious coronavirus pandemics are.

Then the last question is do you get separate odds for it being an "inserted FCS" vs just an FCS. This is the real thing Saar thinks is important, he gave it a factor of 50 at the debate but later said he privately thinks it's 1 in a billion, or something like that. He thinks tha the lab would totally put a 12 NT insert into a coronavirus but nature could never do that.

Strangely, he actually has that backwards -- no lab in history has ev inserted 12 NT's to make a cleavage site. They typically mutate exis amino acids, rather than inserting. On the other hand, nature has sh that it repeatedly does 12 NT insertions at random spots, including sometimes at or near the FCS. So the odds here probably actually le towards nature, if you understand how virologists normally work (Sa does not).

It's not the least bit clear that the DEFUSE grant suggests to "insert site" -- that's a highly motivated reasoning that few virologists agre with.

Finally, the fact that this is a sarbecovirus with a furin cleavage site kind of weird -- it's never been observed in such a bat virus. But tho are enteric viruses, not respiratory viruses, so the FCS is selected against. In an intermediate host, it's quite likely that this feature wou be selected for. We just don't know of many sarbecoviruses which h

spilled over into something else... I think that's just SARS1, SARS2, a
a couple pangolin viruses? Maybe there are others I don't know abo

So it's kind of a 0 divided by 0 situation, going off prior odds. 0 out
labs have done anything like this. 0 out of 3 natural viruses were jus
like this.

Very hard to look at that situation and confidently assign some stro
bayes factor for the lab.

REPLY (1)

Simon stats Mar 30, 2024

Thanks for the very detailed comment, it's very helpful. I was m
thinking of Markolin's claim that the FCS definitely wasn't inser
which seems wrong.

How do you interpret what DEFUSE says their plan is with resp
to the FCS? Did you see those recent FOIA documents where t
have an arrow with FCS pointing to the S1/S2 junction, which is
where the FCS is in sars2?

Isn't one part of the update from DEFUSE the timing? They put
this research proposal in 2018. The WIV has a history of copyir
leading edge work done by Baric but at lower biosafety. The ne
year, something broadly in-keeping with that workflow emerge
Wuhan. Whereas the odds on a coronavirus emerging naturally
with an FCS are just spread out over 100 years.

Regarding the natural insertion of the FCS, what do you make o
the point that this requires intense transmission among the
intermediate host? This is what happens with avian influenza to
make it highly pathogenic - spread among thousands of birds
leads to acquisition of a cleavage site. But then on the natural
scenario, (1) the infected hosts should be easy to find; (2) we
should expect many more than 1-2 spillovers.

REPLY

Arbituram Mar 28, 2024

Thank you for this thorough explanation of that particular point! I hadn't followed i
properly before.

REPLY

Écorché Mar 28, 2024

The existence of the furin site means relatively little because no FCS implies no pande and we wouldn't be asking the question. The strongest evidence for engineering is hov closely the virus matches the DEFUSE proposal.

Weissman does devote some space to what he calls the ZL hypothesis and finds that e if there was a purely zoonotic spillover, it probably made it to Wuhan via research.

REPLY (1) S

tgof137 Mar 29, 2024

Good point -- it's easy to double count in a bayesian analysis, so if you start with odds for a serious natural pandemic and a serious lab leak pandemic, you probab can't go on to include things like "the virus has a furin cleavage site" or "the virus binds to human ACE2". (Rootclaim did both in their analysis)

REPLY (1) S

Écorché Mar 29, 2024

Weissman gets this right (and may have been the first to explain it). That see more relevant to me than your victory over Rootclaim.

REPLY S

Notmy Realname Mar 28, 2024 *Edited*

I lean towards lab-leak and am skeptical early Covid would have developed in nature.

Beyond that, what I don't think is addressed, I think it's significantly more likely that a pandemic coronavirus that pops up in the one city in East Asia with the biggest coronavirus labs, rather than any of the other hundreds or thousands of cities with wet markets, would be an entirely natural virus that made its way there due to improper sa precautions by the lab. The lab collected coronaviruses, was full of both resident and visiting researchers who work with coronaviruses, and was known for having lax safety standards.

Even if Covid wasn't engineered in the lab (I think it was), I think it's more likely that it would be a natural virus that appeared in the wet market by way of proximity to the lab than by way of an infected wild animal brought directly to the wet market from the wild

This would be essentially a zoonotic lab leak but I would award it to team lab leak.

REPLY (2) S

Kenny Easwaran Mar 28, 2024

I don't think there are a billion people that live in cities as prone to wet lab spillove as Wuhan is. Given that 11 million of those people live in Wuhan, this is at most a factor of 100.

REPLY (1)

S

Aristophanes Apr 2, 2024

Weissman is his blog post (linked by others - https://michaelweissman.substack.com/p/an-inconvenient-probability-v57) makes a pretty substantial seeming point that Wuhan actually had a very low share of the wild animal consumption trade. So Pr(Wuhan|Zoonotic, wet marl <<< 0.01.

REPLY

S

Shawn Apr 16

I'm replying two years late but there's zero shot COVID was engineered.

REPLY

S

Juanita del Valle Mar 29, 2024

Scott, I don’t understand understand your question. The only reason the lab leak is considered a possibility at all is the improbability of a coronavirus pandemic starting ri next to one of very few global coronavirus research centers (indeed a facility which at time had sampled and sequenced the closest-then-known relative to COVID-19).

Additional claims about furin cleavage sites may support certain scenarios within the l leak umbrella, but their strength or weakness does not detract from that starting improbability.

REPLY

S

Metacelsus Mar 30, 2024

What you gain in exchange is that you no longer need to assume that the Wuhan virologists were actually doing any engineering of their viruses.

And the furin cleavage site doesn't look engineered to me, someone who wanted to engineer a furin cleavage site would use different amino acids (as the debater mentior

REPLY

S

Anna Rita Mar 28, 2024

I'm guessing that Wilf didn't argue this theory for two reasons:

1) He views it as much less likely than a leaked gain-of-function experiment. Rootclaim's website shows a probability of 89% of gain of function escape, but only 3.2% for a zoonosi collection accident.

2) According to their public discussion, it was explicitly out of scope of the debate:

Peter Miller:

>Would a zoonotic collection accident be considered a lab leak or zoonosis, for the judging this debate?

Saar Wilf:

>The judges will have to decide which of the two hypotheses is more likely: GoF vs unassis zoonosis. So it's possible they are both wrong, and there will still be a winner.

https://www.astralcodexten.com/p/contra-kavanaugh-on-fideism/comment/12792183

REPLY (1) S

Kenny Easwaran Mar 28, 2024

That's really interesting! I had convinced myself that zoonotic collection accident was main plausible lab leak hypothesis, though gain-of-function leak is also relevant.

It is unfortunate that there's not much discussion of the three hypotheses explicitly - everyone seems to present it as a binary choice (though they sometimes differ betwee which two options are presented).

I don't know of any fourth hypothesis that is meaningfully supported by anyone.

REPLY (2) S

Anna Rita Mar 28, 2024

To be honest, I find the zoonotic collection lab leak theory to be the least interesti variant of the lab leak theory. If all that a Wuhan researcher did was move an exist virus into a populated area, then they did nothing for the virus that could not have been done by a miner who stepped in bat shit.

REPLY (2) S

Kenny Easwaran Mar 29, 2024

Right. But the lab is continually going out and looking for new caves full of unsampled viruses, while miners look for a site with minerals and keep workir there.

REPLY S

Juanita del Valle Mar 29, 2024

If the pandemic was caused by 'zoonotic collection', there is every probabilit pandemic might otherwise have never happened. Scientists take their sampl from very remote locations, and deliberately get close to bats in a way that m miners would be averse to.

This is intuitive, because no pandemics have been caused by miners, despite there being many more miners underground at any moment than virologists.

In other words, it would be a massive deal. And going forward it would have h implications for what precautions scientists should take when sampling (if th permitted to sample at all).

REPLY S

tgof137 Mar 29, 2024

I don't think it's likely for bat virus sampling to produce a pandemic. In the whole history of the WIV, collecting 20,000+ samples, they only ever managed to cultur sarbecoviruses. Usually it's just enough RNA to get a sequence but not so much t you can get yourself sick.

There are also signs that the starting virus was already adapted to a respiratory system, whereas the bat viruses are enteric viruses. The furin cleavage site is the most notable such feature, but there are other smaller ones. So it was in some species other than a bat, before it jumped into humans.

I suppose if researchers went out sampling raccoon dogs or some other animals a a researcher got sick, that could do it. But I don't see why the hypothetical scenar researchers going out and doing that should be any more likely than zoonosis at t market, where we know for sure they were importing lots of exotic animals.

REPLY (2) S

Kenny Easwaran Mar 29, 2024

The main thought I have is that while there's a lot of people gathering wild animals for wet markets, the relatively few lab workers going out to gather wi animals are specifically looking for viruses, and will thus be picking up disproportionately many (whether it's overall greater or equal or less is not obvious).

REPLY (1) S

Simon stats Mar 29, 2024

Another way to look at it is that the only known traffic of sarbecoviruses from the location of all of the closest known relatives to covid was via science, not via the animal trade.

REPLY (1)

S

Juanita del Valle Mar 29, 2024

Exactly. The reason to at least consider 'zoonotic collection' is that know researchers were going and continue to go collect samples fr remote locations where the closest known relatives to coronavirus e Only one cross-species jump is required.

In contrast, all animal trade scenarios currently involve multiple cros species leaps between unknown animals with essentially no idea as how, when, or where that occurred.

REPLY

S

Juanita del Valle Mar 29, 2024

I don't understand why you think it's impossible for an enteric virus in bats to evolve into a respiratory virus in bats?

REPLY

S

tailcalled Mar 28, 2024

The issues with extreme odds, focusing on the big picture, and eating one's losses are all relate the same phenomenon: The high complexity of the underlying processes.

Reality is not generated by a simple "zoonosis" factor competing against a "lab factor", as one would think if one naively took the simplest Bayesian models seriously. Rather there was a wet market and a lab and a bunch of people all doing a bunch of different things, etc.. Way too com to explain. (Though because of funky math where we condition on the fact that there was a pandemic, we also can't ignore that there is a competition between these two factors; they just aren't the generators of reality, and therefore not of the evidence either.)

It is extremely common to have pieces of evidence with extremely low priors, because these pie of evidence contain a lot of information about the specifics of these underlying processes, and probability drops exponentially with information. Notably, contra focusing on the big picture, accurately integrating this evidence requires building a narrow, specific picture of what happen because otherwise you don't really know how to evaluate the interactions between the probabili when conditioning on multiple things.

As you build these complex, low-probability models, that means you end up eating a bunch of losses, in rough proportion to the evidence you add to your model to account for the complexity your resulting model of what happened doesn't have very low prior probability, you're not thinki enough detail.

REPLY (2)

S

tailcalled Mar 28, 2024

Important link: https://www.lesswrong.com/posts/JD7fwtRQ27yc8NoqS/strong-evidence-is common

REPLY (1)

S

polscistoic Mar 31, 2024

Thanks for the comment and the link. Useful when warning bright students not to be t much born-again Bayesians. (Beware of the young convert.)

REPLY

S

Kenny Easwaran Mar 28, 2024

Yeah, I was very unclear what's going on with the claim that it should be unlikely to find hig Bayes factor evidence on multiple sides of a debate.

It's certainly unlikely that any particular high Bayes factor evidence exists for a false hypoth - but many high Bayes factor pieces of evidence for the *true* hypothesis are *also* very unlikely!

But if a lot of people are digging very hard, they're going to find a lot of unlikely things. Anc there are some false hypotheses that are in some sense "close to true", you'll probably finc few things that strongly support them.

REPLY (1)

S

tailcalled Mar 29, 2024

I'm somewhat unsure about the math for Bayes factors per se. Assuming you're doing Bayesian updating correctly on a good model, you should be well-calibrated, so you shouldn't be updating outside the 0.1% to 99.9% range until you actually have a ~final conclusion. This sure seems like it means that strong Bayes factors should only line up the true hypothesis.

I think the way one gets extreme Bayes factors is model misspecification. Like both hypotheses often need to eat the odds for the information contained in a piece of evidence, as they both need to be updated to give high probability to that information, the low odds for the information doesn't directly give strong Bayes factors. But then th models should be ready to predict similar information in the future, and if they confide disagree on what "similar information" means, that gives you the crux for what settles debate.

But if one of the models get updated wrongly, for instance by not adding a factor that explains the information, then when similar information comes around, the model that updated wrongly gets forced to eat this information too, which might turn into an extre update against it.

REPLY

JBG Mar 28, 2024

Did either side get into the US intelligence assessments at all?

It seems to me that by far the strongest argument for lab leak is the US government conclusion that effect. These agencies have an awful lot of raw data that no one else does, and the resourc to put a lot of analytical effort into the problem. Obviously, they're fallible but they have access the same scientific data as everyone else plus whatever kinds of useful classified raw data the U government has scooped up over the years and they employ experts on stuff. That's enough fo pretty decent presumption in their favor, especially when so much of the debate centers on dat from Wuhan that could easily have been manipulated.

I'd find the zoonotic origins argument a lot stronger if someone from that side could explain wha the FBI and DOE may have seen that led them to get it "wrong" based on some premise other th incompetence.

REPLY (5)

Mr. Doolittle Mar 28, 2024

I also found it odd that official government positions were not discussed. Even if one side c dismiss them, you'd think they'd still talk about it?

Of course, independent researches without access can't talk about classified data they do have, and that makes the argument an appeal to authority instead of something that can be modeled and debated.

REPLY (1)

JBG Mar 28, 2024

You can absolutely talk about classified data you don't have -- historians and journalis do it all the time and learning to read through redactions is a fundamental skill in those fields. There are still valid inferences to be drawn even if you can't see the underlying material; you read around the redactions and see what could be under there and what could mean.

In this case, for example, the fact that different intelligence agencies disagree with eac other indicates there is not some kind of classified smoking gun. Like, if there were tap of the scientists saying "oh shit we did this" intercepted by the NSA, then all the agenc would line up behind that.

We also know that the FBI and DOE both think a lab leak is the explanation, but accord to the declassified report, they reached that conclusion for *different* reasons than or another. We also know that the DOE changed its position between 2021 and 2023 bas on some kind of new intelligence received in that period, whereas the FBI thought it wa lab leak all along. Those are some very interesting data points.

You can also consider what the intelligence community is geared towards understandi and what kind of information they could plausibly have. These are not public health agencies. They're geared, more than anything, towards gathering and analyzing information about the Chinese government and military. So, they real comparative advantages in understanding the implications of things like Chinese cover-ups or disinformation but no comparative advantage in a debate about furin cleavage sites or whatnot. That's also important and gives us some sense of what could fill in the blanks

I haven't done the work myself to work through this (and was hoping someone else wo because I'd be very interested) but there are also plenty of bread crumbs in what has l publicly released that you can start following.

The most important exercise, though, is just to try to put yourself in their shoes and th "What could they know that would lead to conclusion X" and especially "What could th know that would *incorrectly* lead to conclusion X." If you can't give a compelling ans to that second one, then you have to rethink your own confidence level.

REPLY (1) S

Kenneth Almquist Apr 5, 2024

Those are fair points, but the problem is made more complicated because in addi to not having access to the evidence, we also don’t know in any detail what the conclusions were. The DNI says that the FBI and DOE both “assess that a laborato associated incident was the most likely cause of the first human infection with SA CoV-2.” That could mean they think that a researcher collected the SARS-CoV-2 i the field and become infected by it. It could also mean that they think SARS-CoV- was genetically engineered in a lab and then transmitted to humans. Those are ra different scenarios.

REPLY S

quiet_NaN Mar 28, 2024

If you are the US president, you can perhaps trust what the intelligence services will tell yo

If you are the public, you can not trust them at all.

FFS, we are talking about groups who have worked hard to stage coups and went out of the way to torture prisoners in gitmo. Psyops are one of their tools. Do you really believe they would draw the line at lying to the public?

Before the NATO warning about Putin became true, my general stance was 'the tell for lying public statements from the intelligence community is that their lips are moving'. I still refuse update on them, even if they tell the truth, they don't want you to believe that because that because it is true (simulacrum level 1) but because it is advantageous for them if you believ (level 2).

Yes, I am old enough to remember Powell trying to convince the UN security council with hi 'intelligence' about Saddam's weapons of mass destruction.

REPLY (1) S

Freedom Mar 28, 2024

You don't think Powell believed what he was saying? Is Powell even an intelligence guy

REPLY S

Tatu Ahponen Mar 28, 2024

Wouldn't one simple reason be that US is in a cold war with China, and in such a situation intelligence agencies might be predisposed to prod people towards explanations that place China in a worse light than other poential explanations?

REPLY S

Kenny Easwaran Mar 28, 2024

Weren't there several intelligence agencies that reported their overall conclusion was mode support one way, and several that reported their overall conclusion was moderate support t other way, and none that reported their overall conclusion was strong support for intelligen design?

REPLY (1) S

JBG Mar 28, 2024

The FBI assessed at medium confidence that COVID was a lab leak. No one else got al
low confidence.

The Department of Energy (really meaning the US national labs) assessed at low confidence that COVID was a lab leak.

The National Intelligence Council and 4 unnamed agencies assessed at low confidenc
that COVID was zoonotic.

The CIA and one other unnamed agency didn't reach a conclusion one way or the othe

REPLY S

everythingism Mar 29, 2024

While it seems reasonable to assume government agencies might have better evidence tha
the public, what's actually happened on this topic has been people in the government push
stories that end up being false or exaggerated.

These have included claims such as scientists at the lab being the real first cluster, that SA
was part of a bioweapons program, that there's evidence a safety incident happened at the
in late 2019, that a vaccine scientist was executed by being thrown off the roof of the lab, tl
the virus was already spreading at the Wuhan Military Games. All of these are from US government sources and all of them appear to be false, exaggerated or lacking evidence. Exactly what leads high-level national security officials to push things like this without gooc
evidence isn't totally clear -- maybe it's the intersection of US-China geopolitical rivalry wi
complex, unknown topic -- but all of this suggests the FBI and DOE's conclusions might ne
to be taken with a grain of salt as well.

We also know that neither of their assessments changed the views of any other agencies, which seems surprising if they really had strong evidence. IMO it's more likely their assessments simply represent the view of particular analysts at these agencies, who came
with their own theories about what happened based on limited evidence like everyone else.

REPLY S

Jon Mar 28, 2024

If SARS-Cov-2 did not come from WIV then presumably WIV and the Chinese government woul
have had a very powerful motivation to open their records to outside investigators and make all their people available for interviews. Instead they clammed up. Why?

REPLY (4) S

Thomas Kehrenberg Mar 28, 2024

The problem is that the Chinese government is also unhappy with zoonosis, because it's st somehow embarrassing. The narrative pushed in China now seems to be that the virus did originate in China but was brought in to Wuhan in frozen food. This actually happened seve times after China's zero-covid strategy started so it's a convenient theory to push the blam on.

REPLY (1)

Jon Mar 28, 2024

Ok, but that still leaves them with a strong incentive to open the WIV records if they ar consistent with the argument that WIV was not the origin of the virus.

REPLY (1)

Jeffrey Soreff Mar 28, 2024 *Edited*

At this point in time, anyway, would anyone believe opened WIV records that apparently exonerated them? The CCP would have had ample time to cook the bc (I doubt that I would trust _any_ government's claims in an analogous situation.)

REPLY

Kenny Easwaran Mar 28, 2024

All the usual reasons that most virology institutes don't generally open up their records to outside investigators. Plus possibly whatever not-totally-kosher things they might have bee doing there.

REPLY

Zach Mar 30, 2024

These kinds of arguments were used to devastating effect in the run up to the 2003 Iraq W Saddam Hussein refused to allow inspectors to see his chemical/biological/nuclear weapon facilities. The only plausible reason for this was that he had secret weapons that he wanted hide from the West.

As it turned out, he did not have secret weapons.

Instead, he wanted to appear like he had secret weapons to scare off Iran, a large hostile neighbor which he had fought with repeatedly in the past. A totally rational reason to want appear to have secret weapons of mass destruction which did not occur to the American government before they made the fateful decision to invade.

Trying to ascertain the intentions of your enemies is something humans are particularly bac I wouldn't use it as a basis for making decisions, if it can be avoided.

REPLY (1)

polscistoic Mar 31, 2024 *Edited*

It cannot be avoided.

...if you could avoid ascerting the intentions of other people (not only your enemies) in your everyday life (rather than only believing you do not have to do this), you could no walk down any street without being on constant alert and armed to the teeth.

...Because then you could not ascertain if the passers-by are unlikely to have the inten to suddenly and unprovokedly attack you.

Ascerting the intentions of other people, at least in the limited sense of being reasonal sure they have not the intention to suddenly attack you, is nothing less than THE fundamental feature that makes in possible for humans to live in societies consisting a of non-kin. It is the founding stone of sociology.

REPLY

everythingism Mar 31, 2024

The Chinese government's default reaction is to cover up anything remotely controversial, for stupid or irrational reasons, so failing to open one of their high-profile labs to foreign investigators is not really that surprising. They did allow the WHO to go there and interview scientists for several hours about their work, and Zhengli Shi gave an interview to Science where she answered questions about specific lab leak allegations. From China's POV this probably represents more openness than usual (by comparison, the WHO did not get the chance to talk to animal traders at the market).

But ultimately the CCP cares much more about domestic public opinion than international. within China itself, the widespread belief is that the virus came from overseas, possibly fror US lab.

REPLY

Sam B Mar 28, 2024

I hadn't heard of Rootclaim before this, but that first screenshot which puts the odds of widespr fraud in the 2020 election at 8.7% tells me all I need to know -- that is orders of magnitude too l as anyone who understands how us elections are run would understand. Seems like there's something about this method, at least as practiced by Saar, that produces excessive probabilitie out of consensus or malicious explanations for things.

REPLY (1)

Saar Wilf Mar 28, 2024

This is a known bias we have. We have several safeguards against reaching extreme probabilities, and it often causes us to overestimate low probabilities. We have improved th mechanism since, and would probably get a lower number for election fraud today.

REPLY (1)

Sam B Mar 28, 2024

I'm sure this is not a new argument to you, but if you admittedly exaggerate a say .000 chance to .05 why not .05 to .5? And a number of exaggerated small risks will add up t very high probability quite quickly.

REPLY (1)

Saar Wilf Mar 28, 2024 *Edited*

because this mechanism specifically pushes posterior probabilities to the middle

REPLY (1)

Caleb Benson May 11

But if people took the odds like this as the definitively true odds, this push to middle would risk overplaying lots of catastrophic risk. Is there is a single dig percentage chance of say 9/11 a nuke in a major city or related manmade massacre no one would live in cities. People correctly "know" that this isn't tl case and the push to the middle approach if incorporated would impose unfathomable costs.

REPLY (1)

Saar Wilf May 11

indeed. this is a known weakness. It is gradually reduced as we gain confidence in our process.

REPLY (1)

Caleb Benson May 25

But using what framework? How aren't these just empty numbers? I a certain point numbers turn to zero

REPLY

Shaked Koplewitz Mar 28, 2024

On the subject of this showing up baysian analysis, since viewers all tried to track it and got different results anyway:

Quant trading also works like this - it's basically using baysian inference for the stock market - a also ends up with a lot of big discrepancies and requiring a lot of Metis to get right. But it's still consistently significantly outperformed (in terms of real money earned in the real market) over traditional non-quanty trading. To the degree that baysian analysis does work in real life, this is I'd expect it to look like.

REPLY (1)

MicaiahC Mar 28, 2024

One thing that I'm confused about is that I often see the following two claims being made:

1. Quant traders / financial people in general are one of the most calibrated and truth tracki professions.

2. Basically all traders, after accounting for fees, risk, and long run performance, underperf the index.

Which seems contradictory. However, there are obvious ways in which these claims can be both true.

1. We're talking about different populations

A. People talking online are highly selected by the competence of their peers, so all the anecdotes online are selecting for say, the top1% of traders.

B. There are two criteria for success, marketing and actual financial talent. Funds that are mostly good at selling a story to retail investors "drag down" the average, but remain solve because their customers do not understand "underperforming the index". Calibrated peopl come from the latter, or produce models for the former that are then organizationally neute in some way.

C. This is just talking about two time periods. Kahneman released his analysis something lik 20 years ago, and most of what I read about this was from that timeframe. 20 years is a ver long time for a bunch of intelligent people to incorporate new information and develop goo norms around them.

2. The second situation is that one of the premises above is flawed.

A. Individual models do perform better or worse, but they do so for significantly short perio of time, or the models themselves are calibrated, but the financial EVs aren't, so you end u with something like a black schols situation, where most calibrated models do well, then someone accidentally has a programming error flipping a sign, or something which wipes o lot of value, and these unlikely incidents form a balancing factor for performance. So calibration does not necessarily track financial performance.

D. The "fees" part of the "outperforming the index fund" is where all the magic happens: th individual trader or firm wins by having good models, but they aren't good enough for the r investor. So calibration does track financial performance, just not enough.

Hmm, maybe ai should talk to finance people more.

REPLY (3)

S

1123581321 Mar 28, 2024

"All traders", taken literally, are the index, and therefore must underperform due to vari frictions.

REPLY (1)

S

MicaiahC Mar 28, 2024

That's not it, I don't think, since the index is a specific algorithm weighting everytl and "all traders" here specifically mean "like 200 firms that Kahneman looked at i Wall Street" or something like that.

I'm not sure if that's definitionally true or false, it is the case that every trade has a buyer and a seller, but I have an intuition that at least some movement of the stoc market reflects "real underlying wealth" whatever that means, and that it would b possible to have economic growth such that indexes in fact do not perform well. Tricky, tricky, I feel like my intuitive model is wrong in some way, but I don't understand what.

REPLY (1)

S

1123581321 Mar 28, 2024 *Edited*

If Kahneman looked at a representative sample (and the guy was meticulous, expect him to had done that), then their performance should closely match tl of "all traders". The sense in which "all traders are the markets" is kind of definitionally true - "the market" is just the sum of all trading transactions. Sc "market going up" means transactions overtime happen at higher and higher prices - the reasons for the uplift are not important for this specific understanding. Once we know the traders make up the market it becomes cl that they - collectively - underperform by the amount of "friction", i.e., costs transactions.

There's a subtle but important complication though: what we call "the marke usually refers to some kind of a market index, e.g., Wilshire 5000 or some su

These indexes attempt to represent the market, but are not "the market", and therefore have their own deviations, or "tracking errors". So in reality, it is possible for anomalies to occur where, for a period of time, it would look like traders" outperform "the market". When in reality what this means is "a representative sample of traders" outperformed "a market index".

REPLY (1)

MicaiahC Mar 28, 2024

No, Kahneman was specifically looking at the trading activity of Wall Stre firms, which isn't representative of all trades. Even then I don't think this would be definitionally true: for example, you could have one super suck firm, who trades in the maximally ignorant way, and has deep pockets su that they don't stop trading. You'd get the result that most firms do outperform the market. I think you have to do the hard work of actually understanding how both statements came to be, sorry.

REPLY (1)

1123581321 Mar 28, 2024

Oh, sure, "most" is not "all", it's entirely possible for there to be out with outsized impact. In fact, we know this has happened - most famously in 1998 with the LTCM collapse.

REPLY

Shaked Koplewitz Mar 28, 2024

Another piece is that hedge funds are generally more consistent than index funds (eve they match them on over), and for a lot of people uncorrelated returns are more valuat (see e.g. herre https://capitalgains.thediff.co/p/naivecomparisons)

REPLY (1)

Shaked Koplewitz Mar 28, 2024

Also note that Quant traders vs finance people in general aren't exactly the same groups (the quant heavy places have generally done better over the last two deca or so)

REPLY (1)

MicaiahC Mar 28, 2024

Yeah, this looks like the "Micaiah is unreasonably conflating two groups, quai and non quants into one, and people are predictably talking about quants wh he hears about it online" and "outperforming index, isn't the only way to be tı seeking, turns out having a model that doesn't rely in default economic grow and still earns money has to be more accurate since you don't have that as a crutch".

Feeling pretty not confused now. Whooooo.

REPLY S

Majromax Mar 29, 2024

There's one more potential point of explanation: exploitable mispricings are finite. Trac itself moves the market, so if you have a better model than the market consensus the absolute amount of money you can gain from trading upon it is still quite limited. Large funds – ones open to new investors and large enough to move the statistics of "beatin the market" – are too large to meaningfully exploit these opportunities.

Hence, you see hedge funds and other large investment funds (pensions, for example) look at less liquid markets like private companies and privately-placed loans. In these markets, there's less public trading and thus less market consensus (thus more potent for mispricing) and without public quotes individual trading is less likely to move the average price for everyone else. If I'm a hedge fund and I know I can give some set of private companies loans at 0.1% less than my competitors, issuing that first loan isn't a immediate, obvious signal to my competitors that they have something wrong.

REPLY (1) S

MicaiahC Mar 29, 2024

This doesn't roll off the tongue at all when I reply to "If you're so rational why ain't rich"!!

Thanks. That might also explain another flaw in my thinking about the study of be market index returns: if the firm's base rate isn't market returns but some differer distribution, yeah, sure it's technically bad to give them your money, but that does mean they're inaccurate. They're playing a potentially harder game.

REPLY S

Rachael Mar 28, 2024

"1/10,000 Wuhan citizens work at the wet market. So if a lab leak was going to show up somewh random, the wet market was a 1/10,000 chance."

Why are we only counting the people who *worked* there? Wouldn't the relevant number be all people who visited there (a much higher proportion of the population)?

REPLY (1)

Saar Wilf Mar 28, 2024

The exact claim was about the index case being a market worker. That's generally OK. The main mistake is underestimating all the factors that would make that happen even if there v no spillover there.

REPLY

Adam Mar 28, 2024

First image: "The election was no different from previous elections" is not inconsistent with the claim that there was widespread fraud. As stated, none of the options are mutually exclusive.

(this is true regardless of the actual answer to the question)

REPLY

Ch Hi Mar 28, 2024

I disagree with some of the arguments. I think COVID originally started spreading in rural village didn't spread fast, because there wasn't that much unventilated space, and it wasn't well adapt to humans. And it wasn't detected, because until special tests were developed, COVID just look like "atypical pneumonia". That is where it adapted to humans. Then it was carried into the wet market by a vendor. So it did spend some time learning how to spread in humans, but it wasn't b gain-of-function research.

REPLY (1)

Mr. Doolittle Mar 28, 2024

How does this theory address the fact that the nearest comparable virus was from a cave 1 miles from Wuhan?

REPLY (2)

Ch Hi Mar 28, 2024

It doesn't directly address it. But if I am correct, our knowledge of what viruses are circulating where is quite patchy and incomplete. I've heard that in some cases of xenonotic diseases it's taken multiple decades to track down the "original source"...wi nobody trying to make excuses or prove that it isn't their fault.

Also, if the virus was originally a bat virus, bats fly, and some are migratory. So perhap they just haven't been looking in the proper caves to find a closer example.

FWIW, I don't think the evidence is sufficient to invest ANY of the theories assigning bl with much belief. People can't even agree on what was the most probable non-human host. (Why not some weasel...we know it spread well among mink.)

REPLY (1)

S

everythingism Mar 31, 2024

The main challenge to this idea would probably be the fact that the pandemic see to be descended from the strain of the virus that emerged in Wuhan, with scientis arguing it traces back in terms of timing to late November, right around the time tl Huanan Market outbreak would have begun. This is why the scenario of spillover a the market seems to make sense, especially given there were multiple potential h species sold and slaughtered there. As it happens one of the stores at the market a website that was archived on the Wayback Machine and you can get an idea of t range of species that were being sold:

https://web.archive.org/web/20190707014745/http://whdaz.com/

Some of the animals were being imported from the part of China where similar vir circulate in bats, while others are the same species found infected during SARS-C 1.

But all of that could be wrong of course. Apparently there was a wave of influenza spreading through Hubei province right before the pandemic began so you could speculate that this masked smaller-scale rural outbreaks. There's never been retrospective case-finding outside of Wuhan that's been publicly released. There' report in a Hubei paper that markets in Enshi, where civets and raccoon dogs solc Wuhan were raised, were shut down on December 23, 2019 -- a week before the Huanan Market in Wuhan was closed.

https://www.independent.co.uk/asia/china/covid-coronavirus-bats-caves-hubei-b1940443.html

China has also failed to release testing of blood banks outside of Wuhan, and test of the animal trade in general has been limited. Officially the government still clair that zero animals within their borders have tested positive at any point during the pandemic and there have been no outbreaks on fur farms in spite of tens of millio susceptible species being raised there. So it's difficult to draw conclusions but th genetic history seems to point to the outbreak in Wuhan being caused by spillove humans either at the market itself, or at a closely linked location such as a farm th supplied the market.

REPLY

S

Kenny Easwaran Mar 28, 2024

Is 1000 miles nearer or farther than you expect for a virus that is separated by several years or decades of evolution from the virus we're talking about?

REPLY (1)

Mr. Doolittle Apr 2, 2024

According to Saar's argument, which Peter did not refute, the nearest example we found was located 1,000 miles away. If there are no intermediaries between that sample and what we later saw in COVID, then that seems to increase the chances this was a modified strain. If there are intermediaries, then we would need to know where those were located in order to determine how their descendants ended up Wuhan.

REPLY

Nick Collins Mar 28, 2024 *Edited*

Whichever debate participant is most well-versed is most likely to have thoroughly understood t topic and thus is most likely to have a correct intuition on the subject. Thus, we should just deci on these issues based on our subjective intuition about which debater seems to be most knowledgeable and rational, and not reason through the content of the debate except to fact-ch the debaters and ensure their reasoning follows so they can't get away with intelligent-sounding bullshit. For example, instead of checking every single fact, just check a simple random sample them. This approach may seem objectionable to some people, but it's probably what we're all subconsciously doing anyways, and it's probably approximately correct.

REPLY (1)

eleventhkey Mar 28, 2024

>Whichever debate participant is most well-versed is most likely to have thoroughly understood the topic and thus is most likely to have a correct intuition on the subject.

I just don't think this is true. "Crank that goes down an obsessive rabbit hole researching conspiracies" is an archetype that will tend to end up (a) having a lot of domain knowledge and (b) being completely incorrect.

Like, I'm pretty sure if you took a flat-earth nut and put them in a debate against a random person chosen off the street, the flat-Earther is going to be far more well-versed. I've spok flat-Earthers! They love getting into minutia about discrepancies in NASA image processing and flaws with Newtonian physics, and Ancient Greek historiography, and so on. The avera person on the street is going to be a lot less knowledgeable on all of these things!

Being "well-versed" just selects for obsessiveness, not correctness.

REPLY (1)

S

Kenny Easwaran Mar 28, 2024

That was exactly the example I was going to come up with. I do enjoy learning from fla earthers, because they dig up interesting anomalies, even though I think they've fundamentally learned the wrong lesson from them.

(To be fair, the same is probably true for most real experts for at least some of the thin that they have real expertise on.)

REPLY

S

Nikita Sokolsky Mar 28, 2024

> this disease that came out of nowhere and ruined all of our lives for a few years

I wish we could have a debate on what the proper response to COVID should have been. In theo governments refused to do much other than developing vaccines, COVID would've had a significantly smaller impact on our lives. But also this would've also had significant costs to mar people in the form of losing their elderly relatives and more people would've had long COVID.

I wish we could have a Rootclaim debate on what the Western countries should've done in late February/early March as a response to the virus. I think there's a large degree of disagreement about this question to this very day and it's definitely not been discussed enough in retrospecti

REPLY (2)

S

ProfGerm Mar 28, 2024

Asking the real questions! Lab leak vs zoonosis strikes me as mostly a culture war question more than anything meaningful, and serves a distraction from these brass tacks of: okay, whatever, what do we do *next time*? I fear we'll repeat the 2020 psychosis instead of havi better response.

I recognize some people are working on this- biosecurity has become a larger EA cause, as example- but I suspect that the "people at the top" will be just as ineffectual, inattentive, ar political as last time.

REPLY (1)

S

Mark Roulo Mar 28, 2024

With regards to the culture war aspect, I collected a (possibly relevant) list of headline the lab leak hypothesis over time. My intent was to try to see how normal people migh the news over time on the lab leak hypothesis so I pretty much just focused on headlin

and restricted it to fairly mainstream sources (I'm sure it would look different if I includ
for example, Daily Mail).

If we see this sort of thing the "next time" then I expect we will see similar belief splits.

Part of the problem is that I don't think most people are very good at treating headline "this might be true ... but possibly not" instead seeing things as "the experts have changed their minds ... again"

http://mistybeach.com/mark/LabLeak.html

I don't have any concrete suggestions.

REPLY S

Tophattingson Mar 28, 2024

There isn't any clear correlation between governments doing stuff and less covid happenin Sweden is the obvious go-to as a country that did little and got same or better results than peers, but Peru is also an example of a country that did a lot and got an unparalleled catastrophe for it.

REPLY (1) S

Rockychug Mar 28, 2024

Sweden got much worse results than Norway for example. I think this comparison is m informative than with countries such as France/UK/Italy

REPLY (1) S

Tophattingson Mar 28, 2024

Why exclusively compare Sweden with another country that had relatively lax restrictions? Regardless, middling short term results and brilliant long-term result (lowest or near-lowest excess deaths in Europe) for the one country that notably rejected all but the most modest of restrictions is hardly a ringing endorsement o restrictions doing anything.

REPLY (1) S

Rockychug Mar 28, 2024

Because they both share similar climate, culture, and density of population. I assume there's also no big difference in term of popularity as a touristic destination between the two. Norway had a stricter policy than Sweden, schc and several business were closed. This was not the case in Sweden.

REPLY (1) S

Tophattingson Mar 28, 2024 *Edited*

How confident are we that climate, culture and population density are ev relevant? Relevant enough that we should ignore Peru's terrible results v restrictions, and Sweden's good results with restrictions, because one neighbour of Sweden slightly outperformed it in the short term but not ir long term? Does the Scandinavian Taiga perform spooky action at a dist by affecting transmission in Stockholm?

Stringency of restrictions and covid deaths looks like a scatter plot with correlation. This is relevant, and can't be excused away with culture.

Regardless, this is a post-hoc epicycle. When lockdowns were brought i they were advertised as necessary everywhere to avert megadeaths. No that some countries didn't have to do them because culture or weather. that rural areas didn't need to because population density. And not that outcome would instead be going from Low Norway deaths to Middling Sweden deaths. If you believe that certain population densities, cultures weathers mean you don't need to lock down, you're already 90% the wa opposing lockdowns, you just differ in the universality of opposing them

I agree Norway had a stricter policy than Sweden. It was, however, still m lax than Europe.

REPLY S

Devaraj Sandberg Mar 28, 2024

Awesome analysis. And really good that you picked up on the personality factor at the end. Kind odd that Rootclaim do deep Bayesian stuff on the arguments, but do no personality screening fo willingness to be objective / fear of losing a debate. The adversarial system is super-Western, I guess, with two dudes fighting their corners,. But in this instance it seems to me like it could be huge factor to swing things.

The comment at the beginning suggesting that it was too bad Putin doesn't have cancer felt a li odd to me, BTW.

REPLY (1) S

Anteros Mar 28, 2024

The snide comment about Putin also grated with me. But I remember Scott's thing about B Laden and Thatcher - if Putin was assassinated (for e.g. by the CIA) what would my reactio be?

REPLY (1)

Devaraj Sandberg Mar 28, 2024

I think the human tendency to try to anthropomorphise our way out of problems can b issue. Bad guy here, bad guy here...

REPLY

William H Stoddard Mar 28, 2024

You know, I looked at the "widespread fraud" one at the top, and it struck me that it didn't even consider that I would think was a much more likely hypothesis for it: That several big city politic machines, acting independently and in parallel, jumped in an applied diverse methods of falsifyi the outcomes, in just those places where such falsification was both (a) readily manageable anc critical to their respective states' electoral votes (giving motivation). It makes me suspect that y can get the results you want from Bayesian methods by carefully curating the initial hypotheses compare. Not very different, perhaps, from C.S. Lewis's trichotomy about Jesus.

REPLY

quiet_NaN Mar 28, 2024

Hot take: Peter clearly failed to convince anyone.

The lab leak odds, in log10 (i.e. orders of magnitude are):

Peter -20.7

Saar 2.7

Eric -3.1

Will -2.5

Scott -1.2

Daniel -1.4

One of these numbers is clearly an outlier. Scott mentions it and calls it "trolling", I would argue it is debating in bad faith. 2e-21 is a ratio which is just silly. For one thing, the gain of function at WiV pathway is not the only pathway towards a lab leak. The WIV could also have released a naturally occurring coronavirus at the wet market. At 2e-21 odds, we would probably have to consider the possibility that the WIV built a time machine and went back in time to infect the we market.

Apart from the pangolin which got retconned out in favor of the raccoon dog, one thing I remem from the earlier discussions was that proponents of the lab leak pointed out that the bat with th virus likely did not come from Wuhan. Given that Saar has not made that argument, is that no lo the case? I imagine that by now we have a much better understanding of naturally occurring bat corona viruses in China.

REPLY (1) S

Mr. Doolittle Mar 28, 2024

Scott included several arguments where both sides agree that the source virus was from ba in a different region than Wuhan.

REPLY S

Philipp Markolin, PhD Mar 28, 2024

Hi Scott,

thank you for mentioning my "preachy and annoying" blog. :P

Can't do much about my eyes rolling backward on this topic, but since you are all about evidenc and probabilities; you might want to keep your eyes peeled for my next 15000-word sci comm article that comes out in two weeks and will bring people to the cutting edge of origin science, a well as put the "gain-of-function research" narrative to bed once and for all.

Promised :)

Cheers

REPLY (3) S

Howie Mar 28, 2024

>preachy, annoying

I just read your piece. He's not wrong. If your goal is a broader audience you may want to reflect on it.

REPLY (2) S

Leo Abstract Mar 28, 2024

There are plenty of people who love tone like this. He's solidly in the middle of his lane and will do fine.

REPLY (1) S

Howie Mar 30, 2024

Sure, the lane is a circlejerk of signaling and dopamine hits from feeling superior. real persuasion or work is being accomplished.

REPLY

S

Philipp Markolin, PhD Mar 28, 2024

Hihi, opinions differ, so do goals.

Maybe my goal is to accurately reflect reality, educate, and have fun writing, rather tha catering to audience demand by constantly upholding the false equivalency spiel of influencers that care more about popularity than science :)

Scotts friendly bothsidism here towards a fanatic multimillionaire is what makes me cringe... guess that is what influencers do to be popular with the masses... "don't mak enemies of powerful people, don't rock the boat, give your audience what it wants fror you, everybody's opinion - no matter how unfounded - has value and must be taken seriously, isn't it that both sides have good points... maybe they might not be entirely correct, but those pesky Zoonati scientists certainly have their fair share of problems. too.." *yawn*

Seen it a hundred times, but a spade remains a spade no matter how many charitable takes are written about how it can also be seen as a fork

REPLY (1)

S

tgof137 Mar 28, 2024

I think your writing is good and your videos are as well.

But I also think that the tone would put off anyone who's on the fence about the is because you lump together lab leak believers with other conspiracy theorists, anc makes it sounds like everyone who's taken in by the theory is just an idiot.

But support for lab leak runs around 60%, and plenty of smart people believe it. T aren't all anti-vaxxers or flat Earthers of UFO cultists.

I got taken in by it for a while, at least enough to say it was 50/50. You even did, yourself, for a briefer time (like 1 month in 2021, or something?)

I will admit that it's a struggle to write an article that simultaneously preserves the of the person who fell for a bad theory and also debunks that theory effectively. S you have to pick some approach there, and yours is just different than mine.

REPLY (1)

S

Philipp Markolin, PhD Mar 28, 2024

Maybe I should have a disclaimer somewhere:

"People often frown when I call the currently advanced "man-made" speculations conspiracy myths, because "how can you stigmatize 2/3 people conspiracy theorists?". So let me quickly clarify: Conspiracy myth is an accur functional term for these sets of ideas and attached worldviews. However, believing a lab leak is likely, or having fallen for one or the other false myth dc not turn ordinary citizens into conspiracy theorists, nor does it make them gullible or stupid, just incorrect. We all hold wrong, uninformed, ignorant, or shortsighted beliefs on many issues, make mistakes, or just get unlucky with information diet. Yet we are usually able to update our beliefs and correct co by ourselves when such opportunities present themselves, while conspiracy theorists usually can not. The term "conspiracy theorist" should-in my opinic be reserved for the accurate description of participants and activists in close knit conspiratorial communities that are hyper-engaged and trapped in these encompassing worldviews, even victims of it. They need help and deprogramming; they do not deserve our attention, but also not our scorn. A ire goes towards those who wield and amplify conspiracy myths as a tool for popularity, persuasion, profit, or power. Self-serving influencers, populist commentators, unethical journalists, calculating politicians, fanatic activists, power brokers, media manipulators, conflict entrepreneurs, contrarian grifter and their ilk would probably fall into that group. What they have in common is that they are a type of social elite with an outsized impact on the discourse tl they poison for everybody else."

REPLY (4)

tgof137 Mar 28, 2024

Yeah, it's a hard balance to strike. I loathe a few of the people that professionally lie and manipulate the public on this issue and a few othe (like vaccines or climate change).

But I have no ill will towards the average person that just thinks it's a lab because it's all really confusing and there are more newspaper articles about lab leak than about zoonosis. To them, it's more like:

"I've been there! I was there! I'm trying to save you a year of your life so don't have to spend as much time as I did, sorting it all out."

REPLY (1)

Philipp Markolin, PhD Mar 28, 2024

Yeah, no disagreement here.

REPLY S

Moon Moth Mar 28, 2024

That, in its entirety, would be a better title for your article.

REPLY S

Simon stats Mar 29, 2024

In the expert poll, one fifth of respondents thought a lab leak more likely The median expert puts it at 10%. This is, by definition, not a conspiracy theory

REPLY S

Howie Mar 30, 2024

>However, believing a lab leak is likely, or having fallen for one or the oth false myth does not turn ordinary citizens into conspiracy theorists, nor does it make them gullible or stupid, just incorrect.

I agree. So why take such a derisive and condescending tone? There is r work to be done to dispel the lab leak theory. You seem to be more interested in signaling than actually changing people's minds. Because y writing is laundry list of things not to do when trying to persuade.

REPLY S

Leo Abstract Mar 28, 2024

I had to ctrl-f to find his mention of you after stumbling over your comment here, which sto out visually due to the annoying use of emoticons.

Does your blog contain a post about how you so fluently acquired journalism-speak? Did yo really start out in 'science' without a journalism degree? Can you really write 15,000 words that breathless, snarky tone on a regular basis? While you'll clearly never write for the NYT, is more likely due to discrimination than lack of ability. You seem lke you'd be a great fit.

REPLY (1) S

Philipp Markolin, PhD Mar 28, 2024

Is that a backhanded compliment or a diss? xD

REPLY (2)

1123581321 Mar 28, 2024

FWIW I read it as a compliment.

REPLY (1)

Philipp Markolin, PhD Mar 28, 2024

Yeah, I am about 50/50 on this. Maybe we should set up a rootclaim debate t figure it out

REPLY

Leo Abstract Mar 28, 2024 *Edited*

Mostly a compliment. You're in a comment section famously inimical to the NYT, b we'd all agree there's a hierarchy to journos. That is, while they're all bad for socie only most of them are bad at their jobs. Some are talented and competent. The di the implication that if it weren't for anti-white, anti-male bias you'd already be a b deal at the NYT. I have no idea if I'm joking when I say that part.

*edit: were[n't]

REPLY (1)

Leo Abstract Mar 28, 2024

Separately, I'm bothering to write the compliment because I work hard at goo mental hygiene. I hope never to lose the ability to give credit where it is due. annoying emoticons and snarky tone probably activated some prey-drive-adjacent subroutine in my brain that looks for easy dunks, but you don't prov any.

REPLY

Scott Alexander Mar 29, 2024

Author

I enjoyed your blog post and am grateful you chose to cover this very interesting event. I dr from your post when writing mine and hope I gave you enough credit.

But yeah, I'm pretty jaded about the skeptic community, for reasons I've talked about at https://www.astralcodexten.com/p/contra-kavanaugh-on-fideism and

https://slatestarcodex.com/2014/04/15/the-cowpox-of-doubt/, and fairly or not your blog p
reminded me of them.

I don't know if you're joking when you say you'll put the gain-of-function research narrative
bed once and for all, but I look forward to reading what you've got to say.

REPLY (2)

Philipp Markolin, PhD Mar 29, 2024

Tongue-in-cheek of course, my whole point is that the lab leak theory is dead, but the lableak myth will never die for factors that go beyond scientific uncertainty.

No matter what I write, media manipulators will not stop pushing a false narrative.

However, for those with an open mind and a real interest, yes, the next article will redu
scientific uncertainty to a point where no "gain-of-function" scenario remains possible

Looking forward to reading your comment then!

cheers

REPLY (1)

Xpym Mar 29, 2024

Huh, for me the real "gain-of-function narrative" is that regardless of whether it's responsible for COVID in particular, it might very well become responsible for the pandemic, possibly an even worse one, and whatever dubious benefits this resea
brings can't pass any remotely reasonable cost-benefit analysis. I'd be much mor
interested to see a discussion of that.

REPLY (1)

Philipp Markolin, PhD Mar 29, 2024

Will be extensively covered.

But tl:dr is: GoF is not a precise term, but has both risks as well as benefits th
are outsized. Way forward is that cost:benefit needs to be assessed based o
individual experiments and by experts who understand what they are talking about.

REPLY

Philipp Markolin, PhD Apr 11, 2024

Hi Scott, here is as promised, the investigation that will put the "gain-of-function virus narrative to bed. Enjoy and let me know what you think. https://www.protagonist-science.com/p/treacherous-ancestry

REPLY

A1987dM Mar 28, 2024

> Fourth, for the first time it made me see the coronavirus as one of God's biggest and funniest jokes.

The punchline is "hindsight is always 2020"

REPLY (2)

George H. Mar 28, 2024

Grin, thanks.

REPLY

Vampyricon Aug 8, 2024

Made me laugh!

REPLY

E Dincer Mar 28, 2024

I was pretty sure about lab leak, now I'm 50/50. Normally I should be frustrated to end up 50/50 after reading all this but I'm filled with a feeling of satisfaction. Great job for all involved.

REPLY (2)

Jeff Mar 28, 2024

Similar for me. This nudged me from 85% lab leak to maybe 60%. I do wish the policy conclusions section included some reiteration of the lesson that clearly we need to clamp c on GoF research and generally need a ton of better, independent oversight of labs handling/creating dangerous pathogens.

REPLY

Tristan Mar 29, 2024

I went from 85% lab leak to 80% natural origin. The virology stuff seems pretty convincing.

His talk about the databases doesn't seem to line up with what I remember reading from Al Chan however. I'd be interested to see them debate.

REPLY

Steve Cheung Mar 28, 2024

I am WAY out over my skis here...but it just seems like there is no objective way to actually TEST of your priors in a scenario like this. So you're just giving a finite number (and "false" sense of precision) to what actually remains a messy and murky quandary.

REPLY (1)

Kenny Easwaran Mar 28, 2024

Priors aren't the sort of thing that can be tested. Scott thinks that some priors are better th others, and that there's an objectively best prior out there, but I don't think anyone has ma any better progress on this than "just assume everything is equally likely to begin with, and don't worry too much about what "everything" means". (Maybe that was a bit too snarky.)

REPLY (2)

Steve Cheung Mar 28, 2024

But that's what I mean. These guys calculated posterior probabilities to 6 or 7 decimal places. That's an incredible amount of precision to be claiming when your starting poir an unquantified, unverified, and seemingly unverifiable prior.

So this debate brought out a lot of facts and information I wasn't previously aware of. it was presented by both sides incredibly well. This would be close to a best-case exar of the concept, i would think. And it appears "zoonosis" "won". Yet we are in fact still n closer to knowing who is right. That's a fairly circuitous 15 + hr journey for that kind of payoff.

REPLY (1)

Kenny Easwaran Mar 28, 2024

It depends on what you mean by "right". If you mean "who supports the theory th true", I think we are closer to knowing that. If you mean "whose degree of belief is closer to the objectively correct degree of belief", well I don't think that exists.

REPLY (1)

Steve Cheung Mar 28, 2024

I agree (to the minimal extent I understand this stuff). It seems, as you say, th there is no way to test how accurate your prior (or your belief) actually is.

And when I say "right", I do mean "true" or "objectively correct".

But this is what I'm referring to: without being able to say your prior (or starti belief) is correct (or how objectively relatively true your starting belief might how confident can you be that any reasoning predicated on that starting poir will lead you to the correct place/conclusion? How confident can you be that perfectly constructed and engineered thing will remain standing, when you'v built it on sand?

REPLY (1)

S

Kenny Easwaran Mar 29, 2024

If the conclusion is a probability as well, then there's again nothing it me for that to be "right" or "correct".

What we do have a guarantee of is that if something is true, then pieces evidence that increase your probability for it will be more plentiful, and if false, then pieces of evidence that decrease your probability for it will be more plentiful. No matter what prior you start with, an investigation that continues finding new pieces of relevant evidence will in the limit conver to 100% if the claim is true and to 0% if it is false.

We can't put bounds on that to guarantee that at any particular finite po the investigation you will be particularly close to the truth, but we can guarantee that in the limit you will.

To ask for something much stronger than that is to ask for a guarantee c omniscience.

REPLY (1)

S

Steve Cheung Mar 29, 2024

I agree that the objective is to get closer to knowing the true answe without necessarily reaching a point of certainty. But this debate to exposes some serious limitations to Bayesian probability.

Both sides were well researched and seemingly agreed to most of t facts. However, they had widely disparate priors that affected their interpretation of those facts...which then in turn affected their priors other or subsequent facts. And they ultimately came to wildly and widely disparate results. And there's no objective test of their priors it seems their conclusions are driven by their views of facts which ir turn are driven by their prior biases. This does not give me confider in the objectivity of the process or the accuracy of any result based Bayes.

REPLY (2)

S

Kenny Easwaran Mar 29, 2024

The issues you mention are inescapable features of investigati conducted by finite beings without any infallible methods. Alternatives might be able to guarantee agreement between investigators, but can't guarantee convergence to the truth.

Some people like Scott think that bayesianism can be modified way that makes it objective. I don't think that objectivity is possible, though accuracy in the limit is.

REPLY (2) S

Steve Cheung Mar 29, 2024

Agree. In my field we traditionally use frequentist stats. Bu do apply Bayes implicitly. There is some movement to doir more scholarly stuff using Bayesian methods. Steep learni curve for me. But it's good to see some of the practical limitations as illustrated here.

REPLY S

polscistoic Mar 31, 2024 *Edited*

Not at all well-versed in "real" Bayesian analysis, so mayb this question is off the mark - but why not do some forms sensitivity tests, to see how robust the conclusion is for changing parameters? ? Choose different priors, see wher that leads, or use different follow-up assumptions, like in c benefit analysis?

REPLY S

everythingism Mar 31, 2024

"This does not give me confidence in the objectivity of the pro or the accuracy of any result based on Bayes."

But clearly in some cases the priors can be based on good dat agree that when confronting such a complex, unknown questic seems absurd to think you can can get a convincing answer through likelihood analysis alone. It might be more useful as a of deconstructing how you arrived at the view you hold, and se whether any of it is based on wrong assumptions.

REPLY S

Hafizh Afkar Makmur Oct 20, 2024

I think it's not that the priors are inaccurate (which they always are), but the margins a too big. So big that that it makes both conclusions likely if you choose either end of th margin. I usually take that as a sign that whatever I'll be thinking is useless without reducing that margin.

REPLY

paul bali Mar 28, 2024

i "like" the Cosmic Joker hypothesis, meaning it's the most *interesting* one to me. i also like th idea of the Covid virus as a virus in a superposition, taking inspiration from Scott Aaronson's comment in a podcast that "There is a serious prospect that in our lifetimes people will be able create superposition States of, let's say, a virus". (dunc tank podcast, July 11, 2020, with minor editing from his improv oral) in this krazy view, the two states of the virus are: (i) being from the Wuhan virology lab: and (ii) being from the wet market. the evidence never resolves decisively t one of them. to put it another way, a meta-version of the lab leak hypothesis is true: the Lab is s Schrodinger's Box, and the Covid virus escaped while still in its superpositional state. it moves though our world leaving a neat two-path evidential trace.

https://philarchive.org/rec/BALCDT-5

REPLY

George H. Mar 28, 2024 *Edited*

Very nice, thanks for watching, and reporting. I want to make an argument for agnosticism. It se like we are all inclined to think we must have an opinion on almost everything. What is wrong wit saying, "I don't know"? Being trained in physics, I just want more good data before I try and mak any sort of decision. Eventually I hope we will hear news of some intermediate animal host, or ne from some researcher that says, yeah we were doing GOF on those viruses. Until then I don't se how the odds (for me) can be anything other than 50/50. A sign of my almost complete ignoran and an unwillingness to give anymore credit to one sides 'experts' vs the others. I think the best path for humanity is to assume both things are true. And try to make both forms of possible viru mutation less likely to happen in the future.

Oh, and why are there no error bars in all of this analysis. I'm thinking the error bars should be h

*I read a lot of stuff about this in 2020, said then we need more good data, and stopped spendi time thinking about it.

REPLY (1)

Comment removed Mar 30, 2024

Comment removed

REPLY (1)

S

polscistoic Mar 31, 2024

You are saying it as a joke, but it is a very important point. Why do we care if the probability of someting is this or that? We often care because the probability we assigı a signal to others about who we regard ourselves to be, including who we regard as friends and who we regard as enemies.

...and even if we do not care, we overlook that others will interpret our assigned proba this way at our own peril.

REPLY

S

Mr. Doolittle Mar 28, 2024

Thanks for the write up Scott! This is definitely very interesting.

I find myself walking away with a couple of thoughts. First, about the debate itself:

1) I started out pretty strongly leaning towards but far from certain about Lab Leak being correc lot of this I think stems from finding out that the anti-lab-leak side was a literal conspiracy that h a lot of institutional support to try to shut down the debate. Even those shouting down the theoı secretly thought it might be true (at least early on)!

2) Likely related, I could feel learned helplessness setting in while reading the back and forth descriptions of the debate. With no priors on the believability of either participant, and seeing tl both were making very strong claims that could reasonably answer their opponents, I want to mentally shut down and hold my existing perspective.

3) I end the debate slightly closer to zoonosis/no opinion, but still leaning towards lab leak.

Meta-discussion I feel much worse about Bayesian reasoning, especially the hard version with n Peter clearly has a number that's beyond implausible, but the error bars on all of the numbers s unworkably large. You could adjust all of the factors up and down by relatively small amounts ar come up with any interpretation.

Even more, sometimes really really implausible things happen. As Scott notes here, something r implausible definitely did happen! In fact, across the world, implausible things happen quite ofte People die of incredibly unlikely events all the time - an average of about 7 kids per year in the U have drowned in buckets since 1984. If you had never heard that number, what probability woul you put on the chances of a child drowning in a bucket? There are 22.4 million kids 0-5 in the U right now, with seven per year drowning in a bucket, which suggests about 0.0000003125 chan of a child drowning in a bucket in a given year. No matter how you do the math, the chances are very truly low. But it happens seven times a year.

How would we model COVID's origins if something incredibly unlikely happened? It's not imposs that it originated in neither the Wuhan Institute nor the wet market. In the realm of possible, it m not even have started in Wuhan. Saying that something is *more* or *less* likely is all well and g but reality happens in exactly one way, which can be the very rare event. I think 1/10,000 is inco for the likelihood here, but sometimes 1/10,000 things happen. Just because we say something unlikely doesn't mean it's untrue. Saar implicitly believes that the wet market theory is ~6% likel be true. 6% is far from ruling something out, even if he's arguing as if it's closer to 99.99% the other way.

REPLY

S

ConnGator Mar 28, 2024

Interesting debate, well covered.

I updated my lab leak confidence from 99% to 90%. It swayed but did not convince me due to t somewhat sketchy logic about wet market probabilities.

REPLY

S

WoolyAI Mar 28, 2024

#1 I really enjoyed reading this.

#2 My beliefs shifted from pretty confident in lab leak to 50-50

#3 Reading through your summary, I often got this horrific feeling of college debate and droppir points. College debate has this concept of dropped arguments, where the worst thing you can c not respond to a specific argument your opponent makes, because they automatically win that argument. That's what this felt like, I'm not actually convinced by Peter's overall story, I'm not e sure there's a coherent story there, but Peter kept coming up with specific situations where Saa was wrong and that convinced me.

I'm reading back through the lineages thing and, just reading through, Peters positive argument why the older lineage started spreading later is pretty handwavy. Compare that to the next secti where he goes through why the intermediates are a programming error, he's on point. But I com back to the core question of why the older lineage started spreading later than the new lineage I don't have a great answer to the main point, and I think Saar's is better here, but Peter domina on the subpoint.

I'm just scared of focusing too much on the subpoint and irrationally weighing that too heavily. I dunno? For people who watched it, did you get "debate bro"/Destiny vibes from Peter? That's w this kinda smells like, I notice intense debates should work but I could also watch 50 debate streams and at the end of it be dumber than I started and I felt this way a lot in college debate a well.

REPLY (2)

Tatu Ahponen Mar 28, 2024

Mine shifted from 60-40 lab leak to 75-25 zoonosis.

REPLY (1)

Arbituram Mar 28, 2024

Incidentally, my probabilities were and now are identical to yours. Also, Peter is astonishing, that is a wild amount of effort put into something.

REPLY

Koken Mar 28, 2024

This is probably not the important issue here, but as a former university debater I feel compelled to note that the 'dropped arguments' thing is specific to certain formats of competitive debating and far from universal.

REPLY

TheZeroWave Mar 28, 2024

This was a wonderful read and personally seems like a great experiment. A bunch of credit goes Saar and Rootclaim for courtesy and making this public and educating all of us, whatever positic you hold.

REPLY

Alex Zavoluk Mar 28, 2024

> along with their $100,000 table stakes

I thought the judges fees came out of the stakes, rather than being in addition?

REPLY (1)

tgof137 Mar 28, 2024

I had us both bet 105k so that someone could still say they won 100k.

REPLY

Nathan Cook Mar 28, 2024

An insanely good debate in many ways, and congratulations to Scott on a most excellent summa This reads like a communication from an alternate universe. Seriously, since when are there 15-l highly structured debates at the object level with this level of intelligence brought to bear? That does not happen in this world, except it just did.

Despite the outcome not being what Saar wanted, he has ably promoted Rootclaim as a useful t I want a lot more of whatever this is.

Despite Peter being one of the smartest people I've ever encountered online, arriving at an odds ratio of ~10^21 against lab leak is prima facie evidence that he has not correctly employed Baye reasoning. That sort of ratio is where you start getting "simultaneous miracles". I estimate that t chances of a *natural* zoonotic event, meaning one in which human action was not a causal fac infecting its first patient *inside the virology lab*, as being *considerably* more likely than 10^2 against. I possibly could estimate the chances that I, personally, created COVID, had it released the Huanan Seafood Market, and suppressed my memory of doing so, as 10^21 against. (By this mean if I sat down and calculated how likely such an event was, 10^-21 is an answer I might obt In reality I would throw out any probability that low and group the event with "weird shit happens 10^21 is a sextillion. I can't count that high and neither can Peter.

Not getting answers like 10^21 touches on a real problem with practical Bayesian reasoning, wh is that almost all independent events are only very nearly independent. Clinamina like the Self Memory Wiping Garage Biolab Operator Who Is Me exist in enough possible worlds to start havi an effect if you multiply together enough large Bayes factors. This is far from disastrous to the project of being better Bayesian reasoners but there should be more effort going into developin and teaching the techniques that deal with it. Rootclaim looks like one of the places that might happen.

REPLY (4) S

eleventhkey Mar 28, 2024

>Despite Peter being one of the smartest people I've ever encountered online, arriving at a odds ratio of ~10^21 against lab leak is prima facie evidence that he has not correctly employed Bayesian reasoning.

Agreed. 10^21 is an absurd number. There have only been 10^17 seconds since the Big Bar

REPLY S

Peter Gerdes Mar 28, 2024

TBF I think this just is a matter of approach. Peter is reporting the nominal calculation on th assumption that the theory is correct, the evidence is what it's claimed to be etc etc..

I think it's clear Peter doesn't actually believe those are the betting odds that one should us on the subject.

So I think it's less of an error and more a notational difference.

REPLY

S

Phil H Mar 28, 2024

"Seriously, since when are there 15-hour highly structured debates at the object level with level of intelligence..."

I just wanted to suggest that actually, they happen all the time: in courts of law. And it's instructive to think how that comes about. It requires wigs and centuries of tradition; and re strict rules that get actually get enforced about what you can and can't say; and real consequences at the end.

And there is an obvious legal correlate to this error that Peter was making with his 10^21 cl that poor woman who suffered successive cot deaths and was put in jail for them, because statistics expert multiplied the probabilities of cot deaths together and worked out there wa only a 1 in a million chance she hadn't murdered her own children. She was eventually relea

REPLY (1)

S

polscistoic Mar 31, 2024

I was also thinking of that story related to the dangers of professionals tasked with ma difficult decisions becoming (amateur) Bayesians. Could not find the proper reference though. If you have it, I'd be interested.

REPLY

S

tgof137 Mar 28, 2024 *Edited*

One thing worth noting is that the odds Rootclaim gave at the debate are considerably high their central estimate was 1 in 180,000 for zoonosis and their high end estimate was 1 in 70 million.

To a large extent, I decided to just mock the process and make up even bigger numbers on other-side. (more specifically, I gave fair justifications for many of the numbers, but the ide that you can just multiply them all together is not fair, because you might have picked the factors in a biased way, and in the end the argument should be bounded by whatever confidence you have in your own abilities)

For this write-up, Rootclaim asked Scott to change all their debate numbers to smaller numbers. The furin cleavage site that was 1 in 100 at the debate became 1 in 20, some othe factors just got dropped entirely, etc.

Charitably, I could assume they asked Scott to make the change because they have since become less confident in their arguments.

But, cynically, I might think they did this to make it look like, "they were reasonable, I made numbers that were too large, and the judges were just too stupid to tell the difference so th came to the wrong answer"

REPLY (1)

S

Écorché Mar 28, 2024

"I decided to just mock the process and make up even bigger numbers on the other-si and people here are updating based on the result.

REPLY (1)

S

tgof137 Mar 28, 2024

So do your own bayesian analysis with whatever numbers you find most reasonab you think bayesian analysis is the right way to solve the problem.

REPLY (1)

S

Écorché Mar 28, 2024

I think Michael Weissman's robust Bayesian analysis is dramatically better th anything I or anyone else has come up with on the quantifiable subset of the evidence: https://michaelweissman.substack.com/p/an-inconvenient-probab v57

It is described as "excellent" in the post here but, despite being the most comprehensive and sophisticated analysis to date, its aggregated results are included.

REPLY (1)

S

tgof137 Mar 28, 2024

I asked Weissman yesterday for an updated summarized table but he tol me he's against tables and prefers people read the entire text...

His is a bit like Rootclaim, where he gives the market data zero weight, t lineage A/lineage B data zero weight, and so on.

REPLY (3)

S

Écorché Mar 28, 2024

He explains those choices in detail though. Many people would ben from reading his explanation of why the market data doesn't suppo market spillover.

I know you know all this, but other readers here should note that of three big papers arguing for market origin:

Worobey et al 2022's conclusion is under heavy fire in the statistics journals (Stoyan and Chiu 2024, Weissman 2024).

Pekar et al 2022 already had to correct their code ... and change th text to adopt a more lenient significance threshold. The same guy w found those problems has found further issues that invalidate the conclusion completely, though the authors have not had a chance t respond fully yet. Some ACX readers will be amused to hear that th four errors found in Pekar et al 2022 all had the same sign.

Crits-Christoph et al was debunked before it was even published.

REPLY (1) S

tgof137 Mar 28, 2024 *Edited*

I've read Stoyan and Chiu, as well as Weissman's paper. I think they offer absolutely nothing that debunks the case for zoonos To a large degree, I think the point is just to obfuscate the conversation endlessly, so that people online can say, "well, yo have your paper and we have ours"

I think, in general, many people get confused between "your st is debunked" and "someone raised a counter-argument, which may or may not have been good". For a good explanation of ho this dynamic works, I might recommend:

https://slatestarcodex.com/2014/12/13/debunked-and-well-ref

REPLY (1) S

Écorché Mar 28, 2024

Many people see the market origin papers as strong evide for natural spillover. Weakness in those papers is relevant.

REPLY (1) S

MicaiahC Mar 28, 2024

Weakness is relevant, but is it a central weakness?

If Peter is unconvinced, and you have methodological problems, I think it's more important to explain why y think the methodological points are fatal, rather than on an ambient air of "is weak" or "is strong". Truth flo from the lower level details of reality upwards, not the other way around.

REPLY (1)

Écorché Mar 28, 2024 *Edited*

I don't think it's useful for me to try to reinterpret You'd get a much better picture of the argument clicking the link, searching for the stuff about Pe or Worobey, and following links from there as needed.

[EDIT: I should have made more of an effort to answer this question. I think Pekar et al 2022 mig get retracted. Worobey et al has a bunch of othe work in it, but the geospatial analysis centering c the market looks invalid. There is not enough information in the locations to identify a source.

Weissman's analysis has an interesting historical anecdote: Snow's identification of the cholera sc was not based on spatial proximity.]

REPLY

Michael Weissman Apr 2, 2024

I give A/B zero weight in order to be generous to the zoonotic case. Standing by itself, without the A/B data, without the missing RNA/hostDNA correlations, and without data on Wuhan markets as fraction of China markets, then empirically the HSM cluster would g around a factor of 10 favoring zoo. Putting all those factors togethe tends to weigh against the market case, but a priori markets gave o fraction of the zoo priors, so it's not worth penalizing zoo for the collapse of the market case.

Here's an 11.5 min talk that may help put the JRSSA ascertainment case in context. It's not a side quibble. https://docs.google.com/presentation/d/1y3_KYMnU7YyMjkMvIWcp ZkWTfQ6b2/edit?usp=drive_link&ouid=106543307891005183746&rtpof=true&sd=tr

It plays in ppt if downloaded, but not in google docs.

REPLY

S

Michael Weissman Apr 2, 2024

OK, here's a table, for what little it's worth. Without the explanation not much. Odds are for general zoo (ZW) vs. DEFUSE-style LL.

Priors: some pandemic start in 2019 1/70 ±x10

Sarbeco: 10/1

Wuhan: 80/1 (would be roughly 800/1 for market)

No-host: 3.3/1

Pre-adapt: 2.7/1

Has FCS: 3.3/1

CGGCGG: 7.3/1

Restriction site pattern 70/1

Each of the factors has already been discounted by a hierarchical inclusion of uncertain nuisance parameters. E.g. the FCS factor allo for selection by using only successful non-bat sarbecoviruses for comparison. I include a nuisance parameter for possible irrelevance end up with only the 3.3 factor although there are 27 non-bat speci with SC's detected and zero of them have an FCS.

The final odds are calculated by integrating over a fat-tailed version the highly uncertain priors.

REPLY (1)

S

Michael Weissman Apr 2, 2024

p.s. The restriction enzyme pattern bit only made it in after Em Kopp found the detailed plans in DEFUSE exchanges, allowing estimate of P(pattern|LL). P(pattern|ZW) is estimated in multip independent ways and then hierarchically discounted for uncertainty. The reason this factor (aside from magnitude) is a is that it has nothing to do with selection as a pandemic-cause unlike FCS and pre-adaptation.

REPLY

S

redxaxder Mar 28, 2024

Several of Peter's arguments treat the doubling rate as a real property at low populations. That doesn't seem right.

If patent zero only infects one person, the pandemic is delayed by one doubling period. If they i 4, it arrives one period sooner. What are the relative odds between someone infecting zero, one two, four, or fifty? When there are enough infected this kind of effect averages out and we can j say it doubles at some rate. At small numbers you get swings in infection progress based on ho each person rolls.

Because of that attempts to extrapolate infected populations backward using the doubling rate seem doomed once you get into single digits. The error bars on that kind of estimate should be large. It's noisy.

So I don't think you can confidently say things like this

> The COVID pandemic doubles every 3.5 days. So if the first infection was much earlier - let's s November 11 - we would expect 256x as much COVID as we actually saw

Or this

> Although Lineage A is evolutionarily older, Lineage B started spreading in humans first.

> We know this because Lineage B is more common. Throughout the early pandemic, until the D614G variant drove all other strains extinct, a consistent 2/3 of the cases were B, compared to A. Both strains spread at the same rate, so the best explanation is that B started earlier than A. Since COVID doubles every 3-4 days, probably Lineage B started 3-4 days earlier than Lineage which explains why it's always been twice as many cases.

REPLY (4) S

collin Mar 28, 2024

Yes - the Skagit chorale is a great example of the superspreader dynamics at play. R0 is a p hoc measure for clowns when it's as k-dominated as COVID seems to be. The problem wit "COVID spread in the wet market at exactly its normal spread rate, doubling about once ev 3.5 days. Stop calling the wet market a super-spreader event." is that it's "normal spread ra is an average that is lumping in **the fact that COVID has super-spreader events**.

REPLY (1) S

Comment removed Mar 30, 2024

Comment removed

REPLY (1) S

Aristophanes Apr 2, 2024

I think it sits here unaddressed not because a rebuttal is easy and people are bore but because a rebuttal is hard, and unless rebutted, it seems a fairly devastating point. To elaborate a bit - and to integrate over a few strands of ~independent evidence:

a) We have two lineages, Lineage A appears (with v. high probability) older, and th wet market cases are lineage B (and iirc, no original Lineage A cases at all).

b) We have other relatively early cases that don't have a known market link

c) None of the early wet market cases look remotely like Patient Zero smoking gu (none of the animal trade workers, no infected animals detected at the market etc

d) We see superspreading events at other wet/seafood markets in other cities/countries.

These fit quite well with a story of "the outbreak starts at unknown location, the fi detected cases we see at the wet market are several generations in [something th in the abstract is pretty likely - what are the odds of observing Patient Zero?], a superspreader event happens at the wet market". This story is bad for zoonotic proponents because it doesn't have a high odds ratio of P(.|zoonotic)/P(.|LL)

The response is "well, we can't be missing the first few generations because we k the doubling rate and if were missing the first few generations then by January th would have been far more deaths/cases". And then conclude:

(i) "well Covid can't have started earlier, so there can't be meaningful early non-market spread, so anyone who looks early and is unlinked must have had a really short incubation period", [even though ex ante that's less likely and should be penalised].

(ii) "well, Covid can't have started earlier, so even though Lineage A looks older, s Lineage B is more common, it must have started spreading first"

But these deterministic timing arguments are nonsense because not only is stochasticity potentially important anytime you have small n, the over dispersion c Covid spread is huge so the variance of spread early on is very high. The most common number of people for someone to infect is 0. The second most common The vast majority of spread is driven by a small minority of infected people.

So the argument dismissing the Lineage A vs Lineage B point dies immediately. (Suppose Lineage A has a 1-2 generation head start. What is the probability that a month hence it has more people infected than Lineage B, if both have identical lat spread potential? It won't be far off 50-50, even if you condition on both surviving which we should). You don't even need to penalise Lineage B being smaller as "improbable bad luck, what a coincidence that the more predominant lineage is tl one at the market", because while there is probably lots of room for random luck, most likely reason Lineage B "got lucky" and outgrew A early on is that it (by happenstance) spread at the market while A did not. E.g. if I tell you "Lineage X hi 1-2 generation head start, but the first known Lineage Y case is at the wet market which lineage has larger share a few generations hence", isn't the smart money o

And the point of "the total outbreak would be too big by date T if it started a coup generations earlier" also dies immediately because the 95% CI of outbreak size a 1-2 months given a t0 probably covers at least an order of magnitude.

It seems really bad for Peter's arguments in general that he would consistently mi this obviously wrong argument, to argue against multiple lines of evidence, seemi without realizing that it is wrong.

REPLY

Écorché Mar 28, 2024

I can't find the ref, but there was a paper that looked at changes in reporting criteria during outbreak. In addition to any differences due to the small population, the reporting bias also changed over time.

REPLY

tgof137 Mar 28, 2024

The doubling rate argument is, indeed, oversimplified and invalid in the earliest stages whe you have single/double digits of cases. I have also programmed epidemic simulations that t that into account. Here's a recent write-up:

https://twitter.com/tgof137/status/1772417277670871113

REPLY (1)

redxaxder Mar 28, 2024

How sensitive are the sim outcomes to the choice of distribution here? https://twitter.com/tgof137/status/1772417293181505543

REPLY (1)

tgof137 Mar 28, 2024

I find that for the range of values that I find most plausible, it's hard to support an case that there was a large covid outbreak besides the market.

If I were really motivated to develop a lab leak case, I suppose the approach woul
either be to say that the outbreak in January is much larger than anyone thinks, o
that the growth rate (R0) is slower than anyone thinks. I don't think that changing
dispersion/superspreading rate (k) would impact it as much.

Changing January size is hard because you can bound that by excess deaths and
serology. And you can measure R0 in other cities besides Wuhan, so you can't ma
that so much smaller.

I am, of course, not sufficiently confident in this model to say, "there were exactly
cases in Wuhan, on December 10th". There's a range of values, both because the
parameters aren't exact, because there's a statistical range even within given
parameters, and because the model may not perfectly represent reality.

None of this absolutely disproves a lab leak, either. It just makes it very unlikely
because the lab leak needs to cross town to the market within, say, the first 40 ca
or less. And then both lineages need to show up there, and the clock needs to rev
by chance.

REPLY (2) S

Écorché Mar 28, 2024

Just checking: the simulations you're referring to are your implementation of
et al 2020, which you said was too messy to put up on github?

REPLY (1) S

tgof137 Mar 28, 2024

It's just some code I cooked up from scratch in C++. Some of the input
parameters are based on Hao et al 2020.

I'll post it at some point -- one scientist wants me to publish it, maybe I'l
it that way, maybe I'll just share it otherwise. I'm too busy with some oth
stuff for the next few weeks.

The original motivation was just that I didn't want to trust Pekar's paper
faith, so I wanted to see if the same results just fell naturally out of any
reasonable model. I tried several models and they got similar results.

REPLY (1) S

Écorché Mar 29, 2024

I think the person suggesting you publish it thought that peer revie
would be a good learning experience for you, since you weren't

accepting informal feedback. I'm afraid they weren't suggesting tha
your work was good enough that you should go public without anyb
double-checking it. You didn't seem to understand why people were
telling you to preregister either.

Pekar et al 2022 has serious errors despite having 29 authors and however many peer reviewers. @Nizzaneela has worked in the oper you can see how subtle the bugs are. And you think you can just ha up some C++ and settle the COVID origin question!

Have you thanked Rob Townley yet? If he hadn't taken pity and give you the Hao paper a couple days ago, you'd still be trying to reinver SIR from scratch. You flat out insulted the expert who recommende the Epidemics on Networks book and python code for you.

REPLY S

redxaxder Mar 28, 2024

I was worried about k in particular. The variance in the start time estimate sh mostly come from kurtosis.

REPLY (1) S

tgof137 Mar 28, 2024

The original model I built used a poisson distribution of secondary infect and it still showed a similar result for 2 lineages vs 1. I didn't test that on outbreak timing.

I can do a write up at some point to show how this all varies with each parameter. But I need to get a smaller search space before doing that.

The first thing I was trying to do is get a better answer from a few scient about how to set the mutation rate. The answers I got were highly techn and will take me days of study to sort through. I could just linearly scale some value that I think is right, but I want a better justification than that.

REPLY (1) S

Aristophanes Mar 29, 2024

Isn't Poisson completely wrong? That implies common latent mean across individuals, which is inconsistent with evidence of large over dispersion

REPLY (1) S

tgof137 Mar 29, 2024

Yes. Overdispersed negative binomial model is more appropria
that's what I switched to.

REPLY (1)

Aristophanes Apr 2, 2024

Ok sure, you can run whatever simulation you want, but yc
don't need a simulation to see:

(i) you made a bunch of claims (rejecting various lines of evidence as rejected by the possible start dates) that were reliant on growth ratios being constant ex post in small sample.

(ii) you apparently weren't aware that Covid spread is massively over dispersed (or aware of the implications of t

(iii) with Covid's level of overdisperson, the confidence intervals for statistics like "size of pandemic if started at d t0" or "ratio of number of cases for different lineages base on their start dates t1 and t2" are extremely wide.

REPLY

Jeffrey Soreff Mar 28, 2024

>Several of Peter's arguments treat the doubling rate as a real property at low populations. That doesn't seem right.

True! At small numbers, the fractional scatter is large.

I find it somewhat amusing that in textbooks illustrating radioactive decay and the idea of h lives, they always show curves that go e.g.

1024, 512, 256, ...

not

1024, 512+-sqrt(512), ...

they don't show the randomness of all the individual decays and how that scales.

REPLY

Simon stats Mar 28, 2024

Peter claims that the WIV sampling stopped in 2015. This is not true. See page 60 downwards o this FOIA. https://s3.documentcloud.org/documents/21170561/536974886-gain-of-function-communications-between-ecohealth-alliance-and-niaid.pdf

EcoHealth say this to NIH in 2018:

"As per last year, we will not be subco ntractin g any funds to the intuiti ons in these

countries. All efforts expended in these co untries will be from collaborating

partners and not funded by our award. PI, Co-Investigators or other team

members may con duct short field ttips to assess markets, identify wild life in

them, and anange for shipment of samp les of bats and other high-risk host species

in co untri es that nei ghbor China (Burma, Vietnam, Cambodia, Laos) and that

supply wildlife to the internationa l trad e to China (Tha iland , Malay sia,

Indon esja)."

i.e. they say in 2017, they did sampling in these countries, and all samples would be sent to the ' Weirdly, all of the closest known relatives to SARS-CoV-2 have been found in places the WIV wa actively sampling for more than a decade.

This is not even considering the sampling that the WIV would have done without EcoHealth func

The DEFUSE proposal proposes extensive bat harvesting in Yunnan in 2018. This is also evidenc against them having given up on bat harvesting in 2015.

REPLY (1) S

Simon stats Mar 29, 2024

@Peter, what are your thoughts on this?

REPLY S

Mark Mar 28, 2024

My intuition is that Rootclaim is correct that you should be able to use Bayesian spreadsheets--especially after some adversarial collaboration--and get a format that consistently outperforms intuition and even professionals informally weighing the evidence. It's sort of like how as an ana use fermi-calculations to get much more reasonable estimates of unknowns than just trying to estimate the unknown directly. I think Rootclaim should focus more on testing that.

I'm with you (after reading two long summaries of their debate) that zoological is more probable also that Rootclaim is a worthy project. I wonder if the change in format that Saar should focus c though is more emphasis on the spreadsheet and adversarial collaboration to get both participa using the same spreadsheet. I'd love to see a version of the spreadsheet that two smart opposi

viewpoints people agree is a reasonable setup (without any multi-stage fallacies, etc.) so we ca
see which line items are driving most of their disagreement free of any logical reasoning errors (
at least free of errors that adversarial smart people could identify). And then judges can focus n
on the individual ratios (and any remaining spreadsheet steps that debaters couldn't adversaria
reconcile)

REPLY (1)

Peter Gerdes Mar 28, 2024

The problem is that it's us people don't come pre-equipped with a distinction between prio
and evidence and we can't unlearn facts. Thus, even if in theory it's a good system you car
get reliable data for it because you can't go back and figure out what your prior was. As su
it's very easy to double count or undercount evidence.

It works great in situations with some obvious prior because it's a common situation (do I h
cancer...well I know what the base rate for my demographic factors is and can adjust that b
test results) but most problems aren't like that.

REPLY (1)

Mark Mar 28, 2024

Doesn't normal human reasoning have the same problem but just isn't explicit about it

The hope here though is the adversarial collaboration aspect of it would point out whe
double counting or undercounting is occurring and judges could focus in on those are
as driving the crux of the disagreement.

REPLY (1)

Peter Gerdes Mar 29, 2024

Ofc it does, but the point is that we don't have a reason to think this mechanism solves those kind of failures as it's being promoted to do.

It's like saying: I'm going to solve the political mess we are in by first figuring out w
the government should do and then persuading the voters to support it.

I mean maybe I will but breaking it up into those two steps didn't help, it just gave
names to a way you could imagine making things better.

REPLY (1)

polscistoic Mar 31, 2024

Interesting debate with Mark here. Have you written anything else/longer on t
cognitive & other problems of separating priors and evidence?

REPLY

S

J V Mar 28, 2024

That was really interesting. My bayes argument would be something like "If covid WAS leaked fr the Wuhan research lab, how plausible is that picture of events?" If so I'd expect an average am of "well, we don't know" for scientific research. But each thing that needs to be explained away rapidly adds to "that doesn't sound right". Today I learned that they did do modifications like thi But I also learned that they were usually publishing their research like any academic institution, you need coincidences or cover-up to suppress anything they'd done on COVID specifically.

This includes a lot of things that I don't know. But I still like my reasoning, because I know there' lots of biologists out there who have a better picture than me. And I haven't heard of any reasor biologists to avoid thinking about a lab leak, and don't think it's likely there could be a reason th wouldn't be well-known. (Whereas there ARE reasons for physicists to dislike many worlds, or doctors to dislike "chronic fatigue exists and easy weight loss doesn't") But what I usually hear random biologists is "well, yeah, it could be, but I don't have any particular reason to think it is". so I think that's probably right.

I'm interested in the ways of putting more specific numbers into reasoning but I think you can rapidly get into territory where it's hard to tell if the reasoning is sound. E.g. trying to describe h much of a coincidence each theory is raises questions like, if it was zoontic, how many other pla would have sounded as suspicious as the wuhan research lab? Restricted secret areas? Other k of labs? How unique is the nearby lab, in advance? If it was a leak, how many places would have seemed as suspicious as a wet market? Nowhere else seems very suspicious, but "coming out nowhere" would have seemed completely plausible for zoonotism. Or how much are the lab and pandemic both there because there's convenient reservoirs of coronoviruses nearby, that could I've no idea. I think trying to make these precise leads to "things we didn't know we needed to account for" swamping the analysis.

I ought to read how the probability calculation is actually working. I would be really interested. Although I'm not sure it would help: e.g. I'm more persuaded by my argument of listening to biologists, so any attempt to quantify likelihoods for genetic arguments will be swamped by the chance of something I don't know mattering.

REPLY (1)

S

J V Mar 28, 2024

I should also add, I don't really know how plausible it is that COVID was circulating a few months earlier and just happened to take off at the time we identified. That seems pretty plausible from a general "we can't usually trace outbreaks back to an individual person" expectation. This debate showed some argument why it's plausible it started at the market

I don't understand the biology well enough to really know. But it doesn't really affect how lil think a lab leak is, that's more based on the default in the absence of evidence being spontaneous mutation, not the specific evidence for the market origin.

REPLY

For Lack Of A Better Word Mar 28, 2024

Perhaps this description of the debate wrongly magnifies this, but it struck me as really discredi to the lab leak side that they kept relying on debunked claims. The LL side would bring up thing lime the 90 missing cases, the daily mail guy, or the idea that the market was a superspreader event, and Peter would describe that evidence in greater depth and be able to pinpoint why tha evidence was faulty, which was not rebutted by the LL side. One of the most annoying habits conspiracy-adjacent thinkers online have is to gish-gallop between weak evidence they're only lightly acquainted with, and then not change anything about their thinking when a piece of evide is disproven. I generally find the meta-bayes "what about the odds your argument is secretly ba and mine is secretly good" mostly annoying, but how can you not apply a pretty heavy discount the credibility to a side that is frequently trotting out rather flimsy evidence?

REPLY (2)

Arbituram Mar 28, 2024

Agreed, and this is a significant part of what updated me heavily towards zoonosis. I didn't the impression the lab leak folks updated at all on Peter's comprehensive dismantling of a l number of arguments.

REPLY

Radar Mar 28, 2024

Yes, same.

REPLY

Peter Gerdes Mar 28, 2024

Two big issues with trying to formalize Bayesian reasoning is to properly identify what is already part of your priors and what isn't. Second, using it at scale in a systematic way so that you actu benefit from the correct updating mechanism. After all, the more you just rely on the prior the le makes a difference.

The problem is that we only pay attention to issues once something makes us think there is an interesting question there and often these issues are relatively unique so we have no obvious ch of base rate to use as our prior. And there is the very real danger of pulling in information that is really only a result of the update.

I think distinguishing the prior is one place where AI can sorta help. Maybe not yet, but a great feature with AI is that you can remove data from it's knowledge and see how it would judge thin I'm particularly hopeful for this in politics (this AI predicts your judgement in these political issue 90% of the time and when it doesn't know that Trump supports this policy it predicts you will to

I'm more skeptical about the ability to do this systemically at scale in a way that uses enough interconnections to improve judgement.

REPLY (1) S

polscistoic Mar 31, 2024

Thanks for this add-on to your comment further up, on the problem of differentiating information used to form priors and information used as evidence.

REPLY S

Peter Gerdes Mar 28, 2024

Also a statistical analysis that favors bass sounds awesome!

REPLY (1) S

Kenny Easwaran Mar 28, 2024

Some people talk about mammal origin at the seafood market, some talk about accidental infection from a bat at the virology lab, some people talk about gain-of-function gone wron why does no one consider the hypothesis that it was actually the seafood at the seafood market?!

REPLY (1) S

Peter Gerdes Mar 28, 2024

I'm absolutely convinced the first mammals originated at a seafood market!

REPLY S

Ethan Mar 28, 2024

I think people get caught up about the furin cleavage site mutation because they're looking at tl wrong base rates. To think about the furin cleavage motif, you need to ask (I'm phrasing these questions a bit imprecisely):

- How often are furin cleavage motifs studied in places other than Wuhan? How often are they studied in Wuhan?

- How often are furin cleavage site insertions the cause of viruses learning how to become pandemic? How often are other mutations the cause of viruses learning how to become panden

I don't see people asking those questions. The fact that people don't ask those questions make me suspicious of everyone involved, honestly. These appear to me to be the questions that aris of applying Bayesian reasoning straightforwardly. (I'm also suspicious, as a sidenote, because t question overall is kind of irrelevant: everyone agrees lab leaks can cause pandemics, and every agrees zoonosis can cause pandemics, and everyone agrees zoonosis is more common than lal leaks, so we should just try to prevent both, focusing more on zoonosis than lab leaks.)

REPLY

Peter Gerdes Mar 28, 2024

Also, WTF do people care so much if this particular disease originated from a lab leak.

We have a pretty good grip on the problem of biolab safety (eg the leaks in the UK) and the existence of gain of function research. So whether or not COVID itself happened to be a lab leal isn't reason to update too much on the future probability. Any forward looking policy question shouldn't really depend much on whether COVID was or wasn't a lab leak.

I get the debate is interesting but it's ironic that a claim to be more rational is being tested via a subject that it's irrational to be so concerned about.

REPLY (2)

Taleuntum Mar 28, 2024

If we determine someone to be guilty of something bad regarding this case, we can punish them.

REPLY (1)

Peter Gerdes Mar 29, 2024

But people aren't more morally blameworthy because they were unlucky. I don't think have any reason to believe that the people in Wuhan virology were behaving particular badly so why go looking to punish someone?

If you want to punish people who behave badly we have a great way to do that: look fo evidence of bad behavior and investigate those cases not the ones where things just happened to go badly.

Sure, for some types of behavior that's really the best we can do. The primary evidenc we get that someone might have been shaking their baby is when it goes wrong and t baby ends up hurt/dead (tho still often wasn't shaking). But this is exactly the opposit situation.

We have piles of information about how biosafety labs are run and plenty of cases the behave incredibly dangerously. We know what research gets done because it usually

published etc. So we have plenty of leads about who is behaving in unsafe ways if we a
intent on punishment.

Focusing on the time shit went just seems like hurting someone to give us emotional satisfaction about a sucky thing we experienced and that's awful.

REPLY (1)

Xpym Mar 29, 2024

>I don't think we have any reason to believe that the people in Wuhan virology we behaving particularly badly so why go looking to punish someone?

Well, the talk about doing dangerous research in an insufficiently secure lab does sound pretty bad, and that the US were willing to collaborate with them also isn't good look.

REPLY (1)

Peter Gerdes Mar 30, 2024

A general program to go after people doing what you just described seems reasonable but it doesn't seem we have particularly strong evidence it happe here.

If it shows up fine. But there isn't any reason to particularly focus on this case because bad shit happened. If this was the cause it would be crazy unlikely t was the first time someone was breaking the rules like this.

REPLY

Michael Mar 29, 2024

Most people who care probably have political or emotional reasons, whether it be a desire t blame China, or anger at the US health institutions and distrust of the mainstream media.

But ignoring that, if the risk of pandemic from zoonosis is high and the risk from lab leak is we probably want more gain-of-function research so we can prevent the former.

REPLY

Howie Mar 28, 2024

My favorite post in a long time. Thanks for the recap Scott.

REPLY

Steve Mar 28, 2024

"Either a zoonotic virus crossed over to humans fifteen miles from the biggest coronavirus laboratory in the Eastern Hemisphere. Or a lab leak virus first rose to public attention right near raccoon-dog stall in a wet market."

One thing that strikes me is that there is an asymmetry here. The first part of the above does inc sound like a grand coincidence. It's not like the mutating virus would know that it happened to mutate near a lab. However, the second part is only a grand coincidence if we assume that if the was a lab leak, that it was a) fully accidental, and that b) no one acting in bad faith had the abilit distort data from early testing, etc..

I'm not sure I'm of the opinion that lab leak is thus more likely. But I can't help but think: it sure would be convenient if you were a lab researcher/administrator/related government official and didn't want to be blamed for something you negligently made possible - then it sure is convenie that there is that nice big wet market nearby that perhaps you can shift the blame toward.

My current thoughts: though I didn't sit through the 15 hours - I am now of the opinion that the zoonotic hypothesis is stronger than I had thought before. I am updating to be less confident in lab leak. But I'm probably now to about even odds, fwiw.

REPLY (3) S

Luca Mar 28, 2024

Exactly my thought. I don't understand why this argument wasn't presented or discussed. I know nothing about it, but why would we trust any data or research coming from China on My default prior would be that the ccp goes to extraordinary lengths to fabricate an alterna reality in many other places, and they have reasonably strong incentives to do so here, so wouldn't they? I'm assuming this presents as obviously not the case to the people on your ground otherwise it would have been brought up?

REPLY (1) S

Xpym Mar 29, 2024 *Edited*

Well, the best argument here is that the CCP likely didn't know exactly what, if anythin they needed to fake in the earliest days, so there's no particular reason to suspect the initial data to be (competently, non-obviously) tampered with.

REPLY S

Thegnskald Mar 28, 2024

There are very few coronavirus laboratories, but there are many wet markets. We don't real get to condition on "raccoon-dog stall", but rather the set of all animal stalls such that they plausible carriers such that a zoonotic origin could be posited - so:

Bats

Pangolins

Mustelids

Felines

Dogs

Deer

Rodents

Marsupials

And at this point I'm just tired of listing the animals that COVID-19 or sufficiently close dise have been observed in that somebody could posit a zoonotic origin, given COVID-19.

But it's worse than that, because there's nothing preventing a zoonotic disease from appea in places other than wet markets. That it appeared in a wet market near a particular stall fee like strong evidence because it's so specific, but -any- real world-case would be equally specific - what we actually need to consider is the full reference class of places which migh feel- weirdly specific. Like, it had first appeared in a zoo with an on-site veterinary clinic wh regularly treats wild animals. My local zoo does this, although I can't find statistics on how common this is. Or maybe the zoo just got a shipment of wild animals, which isn't that unus particularly in China. (Sure, none of the shipment of animals tested positive, but did any of raccoon-dogs test positive?) Or maybe it first showed up in a restaurant who had a staff member test positive, and who had recently been at a wet market.

That it first showed up where it did, then, doesn't actually give us very much evidence for c against a zoonotic origin - the reference class for places that intuitively feel weirdly coincidental/specific is huge, because the specificity is post-hoc.

However, that it showed up where it did relative to a coronavirus research laboratory researching not just the disease, but the specific thing that made the disease such a proble is much more unusual; there are only a handful of places in the world which satisfy even a broadened criteria.

You'd have to have quite high priors against a laboratory leak happening at all to arrive at a conclusion, based purely on the location of first observation, to think that it is zoonotic bas on that information. The location is strong evidence in favor of the lab leak hypothesis.

REPLY S

Kenny Easwaran Mar 28, 2024

There's probably under a billion people who shop at wet markets with this sort of crossover potential. Something like 1% of them are in Wuhan.

REPLY (1)

Simon stats Mar 31, 2024

I think you are overstating the numbers here. 38 raccoon dogs were sold per month at markets in Wuhan. So, maybe around 20 raccoon dogs were sold at Huanan in the mor of November 2019. The zoonosis hypothesis outlined by Miller is the following specific claim: the 20 raccoon dogs sold at the Huanan market were infected with covid and caused 8 spillovers, 1-2 of which sustained an ongoing outbreak. This is an extremely small fraction of the relevant wildlife trade in China, <<<1%.

Despite this, there are various points of counter-evidence against the raccoon dogs:

Negative correlation with covid in the environmental sampling, but positive correlation animal viruses the animals were actually infected with.

No raccoon dog ever found infected with the progenitor virus

No raccoon dog ever infected by humans in later outbreaks, despite this happening several times with other animals such as mink.

The raccoon dogs were wild caught in Hubei, not farmed

REPLY

Écorché Mar 28, 2024

Michael Weissman's analysis is by far the most principled and persuasive thing I've read on the topic, it deserves much more discussion than just the link.

REPLY (1)

Écorché Mar 28, 2024 *Edited*

Some examples of things mentioned in Weissman's analysis that don't show up here:

* the initial outbreak at a market proves very little: the initial Beijing superspreader event wa a wet market, and the largest outbreaks in many countries were at markets. [EDIT: actually is mentioned above]

* lab accidents are much more common than you think

* Pekar 2022 looks like it's failing replication: https://pubpeer.com/publications/3FB983CC74C0A93394568A373167CE#12

* spatial statistics experts dismiss the methods of Worobey 2022

* the 2018 DEFUSE proposal really predicted some highly unlikely features of the virus

This was a debate between a smart autodidact and an unprepared multimillionaire, in a forr which prevents easy annotation and rebuttal. It's maddening to watch the debate because Peter gets away with so much. I'm really surprised how much people are updating based oı

REPLY (2)

Radar Mar 28, 2024

Do you think someone reading Scott's summary and someone watching the debate vic would come away with very different impressions? It sounds like you did?

REPLY (2)

Écorché Mar 28, 2024

I think this post goes beyond being a summary of the debate.

REPLY (1)

Radar Mar 28, 2024

In a tendentious way?

REPLY

Hafizh Afkar Makmur Oct 20, 2024

It's actually an interesting issue. I outright skip watching the video (of course) anc even the Scott-written part here, because after several lines I realized I hate deba dialogue format so much. I practically only read Scott's original writings in this art

Even then I'm still updating my prior toward zoonosis though.

REPLY

Aristophanes Apr 2, 2024

I think I basically agree with you on all the substance - I was frustrated by some of Pet questionable claims that were allowed through as fact, and the Weissman post you linl to was very helpful (and does a much better job of discussing what the conditioning se and how that affects what different bits of evidence mean). But beyond everything els agree with you on "why are people updating". If one person is much slicker, and is muc more prepared, and has nice "just so stories" to explain away certain bits of evidence, ***holding everything else constant*** this should reduce how much you update in the favour. A more prepared person is more likely to win a debate on points/style even whe they are wrong. Verbal debates just aren't a very good mechanism of accurate determination of truth, they have a lot of noise/aesthetic factors seem to have great influence on perceptions! You just shouldn't update that much when a priori there's nc reason to think the best arguments on both sides will have been presented / equal scru been applied to claims on both sides. (This of course blows up 100x when you see the slicker person getting away with clearly erroneous claims).

REPLY S

Peter Gerdes Mar 28, 2024

The claim that you shouldn't have high Bayes factors on both sides of a debate is just false.

Every event with sufficient detail involves a ton of unlikely stuff happening. Every time you flip 3 coins you get an exceedingly unlikely list of heads and tails.

Sure, if you understand the data to be the **complete** specification of the even then yes you c get high Bayes factors on both sides because you only get one. But that's not what's going on h They are both cherry picking the aspects of the situation that most favor their hypothesis so of course you get high Bayes factors on both sides.

REPLY S

Universal Set Mar 28, 2024

I don't have anything to add on the COVID origins topic, but I do want to register some (possible disagreement with this bit:

"For example, suppose I win the lottery, I'm told I win the lottery, the lottery company gives me a check, I cash the check, and I become rich. Given that there were 1-in-100-million odds against winning the lottery, the lottery company giving me the check and so on must be at least 100-mil to-1-level evidence - otherwise I should refuse to believe I really won the lottery, even as I enjoy newfound wealth!"

The quip-level counterpoint: You *shouldn't* believe that you won the lottery, because which is more likely -- that I won the lottery and cashed the check and got all this claimed evidence, or t I'm *dreaming* that I won the lottery and cashed the check and all the rest? After all, I've had several dreams where I won the lottery before, and I can get just about any evidence you like in dream (all made up by my brain, of course). Sure, a 10^8:1 update is possible in this case (er, maybe, I'm not sure because it sort of hinges on things I think I know about dreams but which m be false -- after all, I might be dreaming I know them), but the strongest counter hypothesis is tl sort of out of band thing (dreaming, or maybe something like solipsism or a "living in a simulatio or "being in The Matrix" or similar roughly-equivalent thing) rather than something like an elabo prank or a mistake.

Yes, on at least one occasion I have realized that I was dreaming by applying this logic.

This might seem academic or silly -- how do I **really know** that I'm not dreaming my whole lif being continually deceived by Descartes's demon, etc. -- but I'm pretty sure that similar caveat ought to apply to trying to reason about some things that people care about, like "Is there a Goc If you think you are more than 99% sure you know the answer based on Bayesian reasoning, you being overconfident that you (a) understand the various conditional probabilities involved and/o have properly considered all the relevant hypotheses.

REPLY (3)

Comment removed Mar 28, 2024

Comment removed

REPLY (1)

Biff Wiss Mar 29, 2024

> Well, it's pretty obvious that I haven't just been dreaming my whole life because the human brain is simply not powerful enough to simulate the level of consistency that "reality" offers.

But within a dream (or within a dream-within-the-dream, if we accept the hypothesis) generally don't take notice of the inconsistencies.*

So if you were dreaming your whole life, it may well be full of inconsistencies which you simply haven't noticed - but likely will once you wake up.

*Sometimes we perceive noticing the inconsistencies as causing us to wake up (as frequently depicted in movies and television - there's a House M.D. episode that immediately springs to mind, and I vaguely remember the Sopranos using the motif as well). In reality, though, it's the opposite - our conscious mind waking up causes us to notice the dream's inconsistencies.

REPLY

Kenny Easwaran Mar 28, 2024

I think that most people overestimate how confident 99% is, or even 99,999,999-to-1. Tho are both big, but you can reverse the first by observing 7 coin flips all go the same way, anc even the latter just needs 27 coin flips. You're unlikely to find this evidence if the thing you thought was likely is in fact true, but it's actually not all that much evidence you'd need to s to convince you of the a priori unlikely thing.

REPLY (2)

polscistoic Mar 31, 2024

Great example/illustration!

REPLY S

Hafizh Afkar Makmur Oct 20, 2024

On one hand seeing even 7 coin flips all go in one way really should have reversed you belief since it's very unlikely. On the other hand, there's much higher chance that there something that's rigging all those 7 coins to flip that way, in other words they're not independent. Really some belief's priors are so high that one experiment shouldn't be to flip it, because prior of that experiment's validity is lower.

REPLY S

Scott Alexander Mar 29, 2024 *Edited*

Author

I think this is a general case of the principle that anyone in a weird enough situation should think their whole life is a dream or simulation. For example, I think all heads of state and billionaires should think this (though luckily most of them haven't reasoned it through, or el politics would get very weird).

I'm not sure if my situation (famous blogger) is enough to get me in this position; I'm about 50 on this one.

REPLY (2) S

Xpym Mar 29, 2024 *Edited*

>or else politics would get very weird

Is this tongue-in-cheek? I've long thought that we were transplanted into an alternativ clown reality around 2015.

REPLY S

Citizen Penrose Mar 30, 2024

Isn't just being a human after the industrial revolution (and just before the singulairty maybe?) already a very strange situation, rather than being a trilobite during the Camb or something?

REPLY S

Theodric Mar 28, 2024

What’s the Bayesian probability that Scott wrote this entire article to make the pun in the first section heading?

REPLY S

hwold Mar 28, 2024

> So probably we should just accept that the first reported case - a wet market vendor, Decemb
11

So, after searching a bit, the closest article I found is this : https://www.nydailynews.com/2020/03/27/shrimp-vendor-identified-as-possible-coronavirus-patient-zero-leaked-document-says/

A shrimp vendor, not raccoon dogs vendor. This is to me an important piece of information and leaving it out feels very misleading.

REPLY (1)

Kenny Easwaran Mar 28, 2024

Was her stall next to the raccoon dog stall? It looks like most of the early cases were.

REPLY

javiero Mar 28, 2024

I don't want to divert attention from the debate, but since it seems appropriate, I'd like to share post I wrote nearly three years ago proposing an hypothesis that is not exactly lab-leak, though incompatible with it, and not zoonosis (at least as I've seen it stated):

https://medium.com/analytics-vidhya/a-novel-hypothesis-previous-ancestor-virus-ba2ae7b100

Short version:

The propagation of Covid-19 seems to point to certain areas of the world being less affected by disease for no particular reason (no younger population structure, no less obesity, etc...). The m area being Southern China and Northern SEA. This is precisely the area where you would expec zoonosis for a bat coronavirus to happen. Other geographic areas correspond to places where infected people might have first, and in greater numbers, arrived if they had been infected by a previous virus in the aforementioned main area. This and other evidence (see full post) point to possibility that a previous ancestor (to SARSCov2) virus emerged and propagated in that area, before mutating into SARSCoV2.

REPLY

Joshua Hedlund Mar 28, 2024 *Edited*

The competing 1/10,000 claims about the wet market are interesting, but feel too speculative without enough hard evidence for me to know what to do with. Far more salient and intriguing tc are:

Saar's claim that the 12-nucleotide furin site comes out of nowhere...

> COVID - which mostly just resembles BANAL-52 with a few scattered single-point mutations - twelve completely new nucleotides in a row - a fully formed furin cleavage site that came out of nowhere.

vs. Peter's claim that the the furin site doesn't use the expected sequences for such a site and actually uses sequences that wouldn't be expected to work. Plus Peter's claim that WIV didn't h the technology or expertise to insert such a site anyway

> COVID's furin cleavage site is admittedly unusual. But it's unusual in a way that looks natural rather than man-made. Labs don't usually add furin cleavage sites through nucleotide insertion (they usually mutate what's already there).... COVID's furin cleavage site is a mess. When huma are inserting furin cleavage sites into viruses for gain-of-function, the standard practice is RRKR very nice and simple furin cleavage site which works well. COVID uses PRRAR, a bizarre furin cleavage site which no human has ever used before, and which virologists expected to work po They later found that an adjacent part of COVID's genome twisted the protein in an unusual way that allowed PRRAR to be a viable furin cleavage site, but this discovery took a lot of computer power, and was only made after COVID became important. The Wuhan virologists supposedly d gain-of-function research on COVID shouldn't have known this would work. Why didn't they jus the standard RRKR site, which would have worked better? Everyone thinks it works better! Even virus eventually decided it worked better - sometime during the course of the pandemic, it muta away from its weird PRRAR furin cleavage site towards a more normal form.

Peter's claims seem stronger to me, and I didn't see any response to these specific claims in the summary (although I may have missed), but I also can't explain Saar's claim, nor do I have the expertise to assign probabilities to any of this. But these claims seem to me to be the most important and what I would love to see further discussion about.

REPLY (2) S

Thegnskald Mar 28, 2024

I'd put basic internet research on the subject as "very low reliability", so, take this with a gr of salt, but:

Some internet research suggests that the PRRAR furin cleavage site is better-adapted to animals (ferrets in particular) that were used as test subjects at the laboratory in question (poorly adapted to bats and pangolins), and the RRKR cleavage site is better-adapted to humans.

REPLY (1) S

Joshua Hedlund Mar 28, 2024 *Edited*

Interesting, thanks! In which case, Peter's claim could possibly (accounting for salt gra fall under the all-too-common "Simple Absolute Claim About How Humans Always Do

Thing Turns Out To Overlook Real-World Nuance"

REPLY

S

Moon Moth Mar 30, 2024

It seems entirely plausible that the WIV was trying out a new type of furin cleavage site, to examine how it worked. It's evidence against SARS-CoV-2 being a deliberately-created bioweapon, not against it being part of a research project about how animal-borne disease can jump to humans. In my own field I've often copied something new and interesting that I in the wild, just to see how it works and improve my own understanding, and I'm not even a research scientist.

REPLY

S

warty dog Mar 28, 2024

typo: they a found

REPLY

S

Phil H Mar 28, 2024

I just wanted to respond to this point in the conclusion:

"Cool Machiavellian plot you have there, but maybe the fact that you're losing 16%-66% should make you question whether you're really as smart as you think you are...At some point you have start debating!"

The reason that people in the field might be reluctant to accept this point is that they don't, and shouldn't, give a shit what 66% of the American public think. And debating just gives the Ameri public the impression that it has some kind of a right to have a say in what the truth is. If you're virus researcher now, working on how to predict and prevent the next zoonotic infectious epide you really really don't want the public deciding to shut you down or order you to focus on lab lea prevention.

This is obviously very problematic - I'm kinda making the argument for important things to be decided in smoke-filled back rooms. But it's also unavoidable, I think. Science must by its natur difficult and cutting-edge and unknowable to most people; and scientific advances are by defin decided by something other than human opinion - ideally, by what the universe is really like. So there is just an inherent limit on the level to which science can be democratised. Which means t noting which way public opinion blows is unlikely to be very relevant to the people who want to advance science.

That said, I do think that science is getting more democratic, and that that's a good thing. It's ju messy process.

REPLY (2)

S

DavidK74 Mar 28, 2024

"If you're a virus researcher now, working on how to predict and prevent the next zoonotic infectious epidemic, you really really don't want the public deciding to shut you down or orc you to focus on lab leak prevention."

A question in good faith: Is "working on how to predict and prevent the next zoonotic infect epidemic" really what this field is doing? We just had a major infectious epidemic. Did work this field help predict or prevent it?

REPLY (2)

S

Radar Mar 28, 2024

At UC Berkeley in the 1980s (and so presumably lots of other places) there were loads scientists talking about the need for pandemic prevention research. The people who h been doing that kind of work for decades have been working without adequate funding None of these people are surprised this happened. The fact it happened as they predi it would is not evidence for their work being useless.

Millions of people were killed or disabled by polio.... until they weren't. We could make list of many other things like this in the realm of scientific research.

REPLY (1)

S

DavidK74 Mar 28, 2024

Was there any research going on at WIV that was useful in predicting, preventing, combating COVID-19? I'm sure there are lots of smart people spending lots of mo studying viruses there and elsewhere. But what good have they done anyone.

I specifically ask, because it does seem like there's a >0% chance that the resear itself is actually increasing the risk of pandemics.

REPLY (2)

S

Phil H Mar 28, 2024

I feel like the best answer we can give to that is: look at the historical record. there in fact more pandemics now than there used to be, or fewer? I think it's fewer, because many serious diseases like polio have been wiped out in the developed world, and flu has been controlled. You're right about that non-zer probability of science making things worse, but the facts seem to suggest we moving in the right direction.

REPLY S

Radar Mar 28, 2024

There is always a nonzero chance that scientific research will make some thir worse. We have a ton of examples of this. Thalidomide given to pregnant wor causing tons of children with birth defects. There are so many examples.

Do you think vaccines have been not worth it? Antiobiotics? Everyone studyir infectious disease treatment is working with diseases in labs. Would you have them stop?

We have a risk of pandemics built into our ecology on this planet. So far as w know. all the pandemics we've had historically were from nature. If this one wasn't, it was very likely the first one like that. We know up to this point that there have been lab leaks that did cause deaths, you can go read the wikipec page about it. Those deaths are dwarfed by almost any other cause of death can think of.

What would you like to see happen?

REPLY (1) S

Comment removed Mar 28, 2024

Comment removed

REPLY (1) S

Radar Mar 28, 2024

Because I agree with what you're saying, I think I potentially misreac intent of DavidK74 and heard him asking a broader question like wh the use of research that happens in labs that have diseases and infectious agents in them.

I also agree with the need for oversight and accountability for resea absolutely. I just don't want that to happen via general public opinio can't remember but I think the original thread of this discussion was a side comment Scott made about public opinion.

I think it's good to have lots of arguing with expert opinion, including multi-disciplinarily (if that's a word).

REPLY

Phil H Mar 28, 2024

I'm not in this field, so I can't claim any deep knowledge. But yes, clearly some parts o field do do that. I think they did indeed predict more SARS-like outbreaks following SA in 2003, and I'm sure did some work to try to work against them (those models that th kept wheeling out early in the Covid pandemic were created long before it struck). I thi the work on vaccines made a big difference in the Covid epidemic.

I was really making a more general point, that science and technology don't advance through democratic processes. Even the most perfect democratic process couldn't actually help scientists uncover truth more quickly. And real democratic processes are perfect; they are political. If science starts worrying too much about what the public thinks, then it just opens itself up to political meddling.

REPLY (1)

Comment removed Mar 28, 2024 *Edited*

Comment removed

REPLY (1)

Phil H Mar 28, 2024

I guess I'm willing to believe that claim about the development of mRNA vaccines. But I don't think it's relevant to the point I was making?

REPLY (1)

Comment removed Mar 28, 2024

Comment removed

REPLY (1)

Phil H Mar 28, 2024

I think there's two different issues here. My understanding is that a of gain of function research is basically about bioweapons. If that's right, then shining sunlight on virologists isn't going to make any difference at all. Virologists aren't the once who take decisions in bioweapons research. It's the army guys. And they don't accept pul scrutiny.

My point was about what the field of virology should be doing right I assume that their research paths have been affected by Covid. Sc was suggesting that they should get out there and debate the publi make the case that it probably wasn't the lab leak. I think they have good reason not to do that: if they go out and start debating these points, they only risk increased political interference in their researc decisions.

That increased political interference won't make any difference at a gain-of-function research, which will or will not carry on under stric secrecy at the whim of the army guys. But political interference cou make a difference to the work of mainstream virologists, particularl public pressure ends up pushing grant money towards lab leak rese projects. That seems to me by far the most likely political outcome. wouldn't achieve the sunlight that you want (which is mostly achiev through open science publication, anyway); and it could send resea up a bunch of blind alleys.

REPLY (2) S

Comment removed Mar 28, 2024 *Edited*

Comment removed

REPLY (1) S

Phil H Mar 28, 2024

"Do you want a world where 60% of the public are wrong about what scientists are convinced is important truth?"

I mean... I'm intensely relaxed about that possibility. I think you're getting a bit hung up on the messedupness of everything about Covid - which was entirely messed up, I agree - and forgetting the bigger picture.

For example, I'm an atheist, and I guess you are, too. That means that I believe, 100%, fundamentally, that most of th world are just wrong about the the most basic stuff. Is mos the world wrong about the fundamentals of virology? Yes. Unquestionably. Incurably. Most of the world (including me

I'm non-science) will never, can never understand what gc on in virology.

And we don't have to, because the mechanism by which virologists communicate with the world - open science, ar now YouTube - are good enough. They are tried and testeı mechanisms that enable oversight without excessive intervention. Covid may or may not have been a virology c up, but even if it was, there is no alternative model for governance of this field. "Public debate" is not a way for science to work - it's not the way anything works. (Politics political parties and newspapers; the law has courts and h formalised procedures; etc.)

Even though, as you say in another comment, the way the sausage gets made is really ugly, it's always been that way You're throwing out a lot of good sausage because you sa some gristle go into it.

REPLY (1) S

Comment removed Mar 28, 2024

Comment removed

REPLY (1) S

Phil H Mar 28, 2024

But a bunch of your premises just aren't true.

"The gristle may include millions dead and entire economies decimated" - but the whole point of t blog post is that a really really detailed debate concluded that it didn't. Science of course has ki very large numbers in the past - Oppenheimer! - it's been demonstrated right in front of your eyes here that in this case, that wasn't what happenec

"without the scientists in question even deigning join the effort to understand how it happened" - There's no sense in which this is true. They are. ' investigated. They concluded. Their understandi done. It's all online! I understand that you don't a with the conclusion that they reached. But that doesn't mean they didn't do the work.

Moreover, virologists are engaged in an effort to people the answer that they've reached. Watch a news show on this subject, and you'll find virolog educating people. They've written articles and books. Scott is suggesting that the public educa process hasn't been done well enough; but neith he nor you have put forward a good alternative channel or forum for communicating their ideas. I can't think of one.

In fact, this very debate was supposed to be an interesting alternative channel for disseminating information... it worked on me.

“shhh be a good boy” - No one said that.

"smartest people in the world...could distill what mattered out of their great wisdom..." They do. T have. Mainly through official mechanisms like the and WTO, and also through books and articles ar

You're right that it's a nightmare that people still won't believe the scientists when they explain th conclusions. Personally, I can't understand the psychology of those who refuse to listen when th evidence is laid out in front of them.

But seeing as you seem to be one of those peopl can you think of a better way for this communica to happen? What would actually persuade you?

REPLY (2)

Continue thread →

Jeffrey Soreff Mar 29, 2024

>I think there's two different issues here. My understanding is a lot of gain of function research is basically about bioweapons that's right, then shining sunlight on virologists isn't going to m any difference at all. Virologists aren't the once who take decis

in bioweapons research. It's the army guys. And they don't acc
public scrutiny.

I think that this makes the army guys look a bit more secretive
they actually are. After all, some gain of function research is op
published. Admittedly, I haven't got the foggiest idea what _fraction_ of gain of function research is openly published.

In electronics and applied physics, a lot of work is directed tow
military applications, yet is openly published, though, again, I h
no idea what fraction is published and what fraction is kept sec

REPLY S

Radar Mar 28, 2024 *Edited*

I think decisions about what research to fund needs to be made by people who are familiar
the domain of research in question. I don't think we need to think of this as smoke-filled backrooms. I don't want the public deciding which strand of breast cancer research shows most promise. I don't want decisions about research into the rise in colon cancer in young people being made by public opinion about whether genetics, environment, or having gotte
the MMR vaccine is the more important line of research.

REPLY (1) S

Comment removed Mar 28, 2024

Comment removed

REPLY (2) S

Radar Mar 28, 2024

I'm not sure what you're proposing?

REPLY (1) S

Comment removed Mar 28, 2024

Comment removed

REPLY (1) S

Radar Mar 28, 2024

That all makes sense to me.

This is not remotely my field. I hear you speaking about arrogance in a s
field that has questionable value. I can confirm I see this over in my corn
psychology.

REPLY

S

Phil H Mar 28, 2024

This is not right. We can't assume *perfect* benevolence and *perfect* lack of conflict of interest. They're human, after all. But that isn't a minimal assumption. I strawman level assumption. Minimal benevolence and minimal lack of conflict do seem warranted - by, for example, all that paperwork from WIV that people have pored through. That paperwork exists because even in nasty old China, scientists may or may not be interested in creating biological weapons still observe basic sa protocols. That shows minimal benevolence to the rest of us.

REPLY (1)

S

Comment removed Mar 28, 2024

Comment removed

REPLY (1)

S

Phil H Mar 28, 2024

My whole theme here is that information looks different inside a speciali field and outside a specialist field. I'm a teacher, and I "lie" to the childre teach all the time. I do it because they're not yet ready to process the fu complexity of English grammar yet. I would expect virology experts to sa different things to people inside the field than they do to people outside field. So when we observe them doing that, I don't really see it as a "lie". example, it would seem perfectly reasonable to me if they chose to communicate to the media the *most* likely origin story as understood b their field, i.e. wild animal origin; while internally discussing the less likel but more dramatic possibility of a lab leak. That doesn't sound mendaci it sounds like normal professional behaviour.

But I could be wrong about that. Perhaps a significant number of virolog (not all, remember they're not a monolith) behaved badly in the immedia aftermath of the outbreak. Even if they did, that absolutely does not invalidate the whole field of virology, or suggest that virologists need to put under greater public scrutiny. You'll recall that the president of the U at that time suggested drinking bleach. This does not imply that all presidents should henceforth be subject to stronger rules on what they and can't say.

REPLY

S

artifex0 Mar 28, 2024 *Edited*

I'm against using a text format for future long-form debates like this.

I watched this debate an a second monitor while doing other non-cognitively-demanding things working on design projects and playing video games- and I enjoyed it a lot more than the podca and audiobooks I usually use for that purpose. In contrast to those, nothing about the debate fe like dull filler, and yet, unlike most really demanding books, it was possible to miss bits here and there without losing the overall thread of the argument. It's almost a perfect format for content listen to while slightly distracted, and I think that for that reason, debates like this can provide a of value to people beyond just shifting their opinions on the topic.

Granted, a text format would probably result in slightly better arguments- though I don't buy thā Peter ambushing Saar with new claims was a major factor in his loss; the lab leak side had plent time to research claims between debates, and plenty of time to respond later. A book-length wr debate, however, is something I expect almost nobody would be willing to read, since it doesn't provide that extra value as good content to consume while doing other things. It would also requ a lot more effort on the part of the debators, so I'd expect there to be both fewer long-form deb and much less engagement- and I don't think a slight bump in quality would at all make up for tl loss of value.

REPLY (2) S

Radar Mar 28, 2024

Oh interesting argument, I like that. You're saying it's worth losing some potential accuracy outcome on the issue in question in exchange for the public educational value of it?

I don't think very many people are ever going to watch this kind of very long form debate aı so I have a preference for thinking of it as a kind of science project on its own and that maximum accuracy is better for that. And then the NYT or Nature or bloggers or whoever c talk about how this very sophisticated well-run debate with very knowledgeable people happened and they concluded X instead of Y, and they basically do a book review of it.

Oh wow, I guess I really am an elitist, huh?

REPLY (1) S

artifex0 Mar 28, 2024

Education, definitely, but also entertainment- which is a less high-status sort of value, still real value.

Well-produced podcasts and hours-long video essays will often be seen by very large numbers of people, even when they dive pretty deep into technical questions. Dwarke Patel's two-hour-long interview with the Anthropic CEO, for instance, has over two hundred thousand views, and wasn't much less technical than this debate. I also founc bit less engaging- while a good interviewer will challenge an guest's ideas to some de; they'll never do so as thoroughly as an opponent in a debate. Ideas are not only more relevant, but also more interesting when you have good reason to trust that they're no being presented misleadingly- and while in a polite conversation, everything has to be taken with a grain of salt for that reason, misleading statements in a long debate tend to survive the response.

I have a feeling that the low view count of this debate has more to do with people not being familiar enough with the format to think of adding it to their usual rotation of lon; form audio content- plus maybe the fact that it's usually been described as a single 18 hour debate rather than a series of 1-3 hour debates. If this sort of thing was a lot mor common, I think we'd see some series with far higher view counts.

REPLY (1) S

Radar Mar 28, 2024

You create a compelling image of this, I'm sold.

REPLY S

everythingism Mar 31, 2024

Agree and I think Saar should take credit for coming up with a compelling format for debati complex issues like the origin of COVID. True he lost on this particular issue but the outcor was consistent with the majority view of experts in the field based on recent polling.

If there were a series of debates like this I'd expect Rootclaim to win some and lose others either way it's much more compelling and will reach a wider audience when you get to actu watch the two sides debate.

REPLY S

CB Mar 28, 2024

Fantastic post - I've shifted my opinion quite a bit in response.

One thing that sticks for me is this: it didn't take long to find the civets responsible for the earlie SARS outbreaks. Why has this proved so challenging here?

The troll in me wants the answer to be that it was zoonosis that occurred in a lab, and then it lea out of the lab.

REPLY (2) S

DavidK74 Mar 28, 2024

"The troll in me wants the answer to be that it was zoonosis that occurred in a lab, and ther leaked out of the lab."

I've thought about that, too. If that happened, what would we call it - zoonotic or lab leak?

And what if the virus evolved through inadvertent natural passaging in the lab? Lots of weir animals in cages, lots of humans around with not much in the way of safety precautions, plu thousands virus samples collected from all over the place being handled. Sounds like a rea witches brew.

Thus, a natural emergence, in the sense that it wasn't the result of any deliberate human manipulation. But also something that would not have happened if the lab hadn't been ther doing the types of things they did.

REPLY (1) S

Leo Abstract Mar 28, 2024

I hope it turns out they were studying raccoon-dog behavior in a psych lab and got bit

REPLY S

Kenny Easwaran Mar 28, 2024

The Chinese government prevented all research by outside investigators for a year, which made it very hard either way.

I'm a bit surprised that they didn't consider the zoonosis-in-a-lab hypothesis, because that the one that I had thought was most probable before I went back to the wet market hypoth

REPLY S

DavidK74 Mar 28, 2024

Thank you for this excellent post. It's one of the best things I've read on the internet ever.

I remain in the Lab Leak camp, where I started. I'm of course not 100% certain, and I don't think possible evidence would ever emerge that would make me 100% certain. I really don't like peop who claim Lab Leak is a "conspiracy theory" and am inclined to discount everything they say.

A question:

Should we accept claims by the Chinese Government and technical/academic/policy experts in field at face value? Aren't they all hopelessly conflicted? Even if what they say is logically sound aren't there myriad ways in which they could be fudging the analysis a bit, even unknowingly? W value have researchers doing gain-of-function work on viruses provided to the world? What is tl likelihood someone in the field is going to publish findings saying the risk reward in their field ju isn't worth it and all their skills are worthless because the work should be shut down?

Another question:

What do those maps of the outbreak in the wet market purport to show? The location where someone was exposed to COVID (how can that be known)? Some point that a person who got infected passed through (what about all the other locations they visited)? Whatever they claim t showing, how accurate could their measure be? Based on interviews? How hard would it be to fudge the data a bit - for example, an infected person tells a researcher all the places they've be in the last week, and of the dozens of locations the researcher chooses the wet market to recor

REPLY (1)

S

Radar Mar 28, 2024

I agree with you about this: " I really don't like people who claim Lab Leak is a "conspiracy theory" and am inclined to discount everything they say."

REPLY

S

Charles Pye Mar 28, 2024

First of all- awesome post! I loved all of this. Saar for setting up this challenge, Peter for taking h up on it, the judges for being fair, and you for summarizing the results- all great stuff.

As for the weird coincidence of the Wuhan Institute of Virology being located so close to the epicenter of the breakout- is it? Everyone knew that wet markets were a potential risk center for spreading diseases. See for example: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7141584/ about the 2003 Avian Flue outbreak. That's the same year that China approved construction of biosafety level 4 lab at WIV.

They built the lab there because it was a good place to gather research data, and maybe also because the local government was strongly interested in that sort of research to protect themselves. It's the same reason that the "Center for the Study of Active Volcanoes" is in Hawa the "Earthquake Engineering Research Institute" is in California, and the "Hurricane Resilience Research Institute" is in Houston. Of *course* they are there where the threat is greatest. And g of-function research was just the most effective means of researching that kind of virus, if you to research it before it actually starts spreading.

You might ask, if they knew the threat of wet markets, why didn't they just shut them down? My understanding is that China doesn't have that kind of state power. There are too many poor, rur people who really want to eat meat, and still can't afford ranch-raised animals. If you try and sh down the wet markets they'll just start selling them in secret someplace even worse.

REPLY

S

warty dog Mar 28, 2024

seems like we should put some thought into hypotheses that prevent the coincidence

could Chinese authorities manipulate the evidence to have a wet market excuse?

maybe the raccoon-dogs intentionally sent their trojan dog to Wuhan to frame the virologists

REPLY S

Argentus Mar 28, 2024 *Edited*

Nitpick on that poll asking Americans if they think lab leak is true. (Before this) I'd have had to answer this as "probably true" but my probably was that I was at like 60% lab leak and 40% zoonosis. 40% chance of zoonosis is still pretty dang high and doesn't really seem that indicati acceptance of the bad epistemic edifice that accreted around lab leak. I mean 30%ish saying it' "definitely true" is a problem but "probably true" seems to be doing a lot of work.

Also, good post. Quite markedly shifted my opinion.

Another looooooong series of posts that markedly shifted my opinion on a controversy with a lo noise recently was this series:https://radleybalko.substack.com/p/the-retconning-of-george-flo

Edit Also, both these posts and that parliamentary procedure kid that's been mentioned so mı recently has made me think that the best model for thinking we have isn't heuristics or Bayesiar reasoning or the wisdom of crowds. It's probably just smart guy who does like a ludicrously unreasonable amount of homework.

REPLY (2) S

Radar Mar 28, 2024

This has been my thinking too: " It's probably just smart guy who does like a ludicrously unreasonable amount of homework."

REPLY S

Radar Mar 28, 2024

Thank you for linking to the Radley Balko piece. I just read the whole thing and am so glad t know it's there.

REPLY S

Robert Kennedy Mar 28, 2024

I think this quote stood out to me as off model:

> Zero COVID era Chinese outbreaks were concentrated in wet markets because they received infected animal products. We know why there was an outbreak in the Xinfadi Market in Beijing: i was because the seafood stall got frozen fish from some non-Zero-COVID country, the fish had COVID particles on it, and the vendor got infected and spread it to everyone else. Something lik

this is true for the other Chinese wet market based outbreaks we know about it. So this makes t opposite point you think it does: wet markets start outbreaks because there are infected goods being sold there. Then the virus spreads through the wet market at a completely normal rate.

I definitely know that China ascribed COVID outbreaks to things like "the person in the plane on previous flight left covid residue". China was desperate to explain every outbreak so they could continue attempting to achieve a horrible and impossible goal. I don't actually believe that is ho the virus was spreading? In particular, surface contact spread and the idea that you should wet your boxes quickly fell out of style. I think the ubiquity of spreads getting first observed at wet markets during covid zero is strong evidence that even if covid is zoonotic but first entered the population by like, bats during a vacation, you would still have first observed it in the wet marke

I filled this out with others' hidden, and ended up weighing that even less than Saar did, which probably means I'm overly dismissive.

https://docs.google.com/spreadsheets/d/1IOHp6PINEbqizpbO7liXLEQTsIuNRpqx/edit?usp=sharing&ouid=113816084183231439418&rtpof=true&sd=true

REPLY (1) S

Kenny Easwaran Mar 28, 2024

I think that surface contact spread and the idea that you should wipe your boxes very natur fell out of style, because it's probably several orders of magnitude less likely a mode of transmission than airborne directly from a person. But when you're in covid-zero condition and the rest of the world is in full pandemic, then suddenly "several orders of magnitude le likely" is still the only way for anyone to get infected.

Sort of like how, to a first approximation, you can think of a gas stove, or a candle, as basic having zero respiratory issues compared to a wood burning indoor stove. But once you've eliminated the wood burning indoor stove, those last few indoor flames, even if they're ver clean, may be your main respiratory risk.

REPLY (1) S

Aristophanes Mar 30, 2024

"But when you're in covid-zero conditions... then suddenly "several orders of magnitu less likely" is still the only way for anyone to get infected."

Are there any clearly documented examples by a trustworthy source (e.g. not the CCP in another country) of infection due to residue on frozen package after international transport? Just off the top of my head, Australia, New Zealand and Taiwan all had "zer covid" regimes for an extended period and had multiple outbreaks. None afaik was du frozen packages. They were all "failures in the quarantine process" of one sort or anot

It's pretty suspicious that China would be the main source for "spread driven by impor frozen packages" claims given tha is reasonable to t this is the sort of thing that perfe aligns with their desire to shift blame for Covid onto someone else (even more narrowl ignoring that motive, "our quarantine of incoming travellers failed" is much more egg c face than "we were tricked by those dastardly frozen food packages coming from thos imperialist bastards in XYZ").

And thus, like Robert Kennedy, I think it's reasonable to infer from this that large public markets are probably decent locations for superspreading. (Indoor location with a lot c foot traffic is a prime potential site for a virus outbreak - doesn't seem shocking).

REPLY (1)

Kenny Easwaran Mar 30, 2024

Good point! The Australia and New Zealand comparisons seem important. If their outbreaks were all traceable to quarantine failures, then that's pretty good eviden

REPLY (1)

Aristophanes Mar 30, 2024

Thanks. To be clear / maintain epistemic humility,

(a) I may be misremembering - every outbreak I know of is either believed to due to / was traced to some quarantine infection/escape, but my memory cou be faulty and my recollection of outbreaks is not exhaustive. (e.g. worker in quarantine hotel caught it from infected quarantined traveller, then took it ho with them etc).

(b) logically there should be outbreaks that happened that the authorities ne noticed (e.g. one person is infected by some source, they never infect anyon else - which happens >50% of the time - so the outbreak dies before it is noticed), about which we naturally know little

(c) there was an outbreak in New Zealand that initially had some speculation about imported frozen food transmission (if I understand correctly, justified c "well, the Chinese are claiming this can happen), but this was ruled out withi few days.

REPLY

Jeffrey Soreff Mar 28, 2024

I hope this isn't a derail: In one of the previous ACX posts' comments, someone posted a link to page explaining the case for gain-of-function research. I'm embarrassed that I didn't either follc the link or bookmark it. Does anyone have that link? Many Thanks!

REPLY S

Steve Byrnes Mar 28, 2024

Way back in 2020 there was an article https://www.independentsciencenews.org/commentaries proposed-origin-for-sars-cov-2-and-the-covid-19-pandemic/ , which I read after George Churc tweeted it (!) (without comment or explanation). Their proposal (they call it "Mojiang Miner Passage" theory) in brief was that it WAS a lab leak but NOT gain-of-function. Rather, in April 2(six workers in a "Mojiang mine fell ill from a mystery illness while removing bat faeces. Three of six subsequently died." Their symptoms were a perfect match to COVID, and two were very sick more than four months.

The proposal is that the virus spent those four months adapting to life in human lungs, including (presumably) evolving the furin cleavage site. And then (this is also well-documented) samples these miners were sent to WIV. The proposed theory is that those samples were put in a freezer WIV for a few years while WIV was constructing some new lab facilities, and then in 2019 researchers pulled out those samples for study and infected themselves.

I like that theory! I've like it ever since 2020! It seems to explain many of the contradictions brou up by both sides of this debate—it's compaible with Saar's claim that the furin cleavage site is v different from what's in nature and seems specifically adapted to humans, but it's also compatik with Peter's claim that the furin cleavage site looks weird and evolved. It's compatible with Saar claim that WIV is suspiciously close to the source of the outbreak, but it's also compatible with Peter's claim that WIV might not have been set up to do serious GoF studies. It's compatible wit the data on differences between COVID and other previously-known viruses. Etc. (As I understa it.)

Old as this theory is, the authors are still pushing it and they claim that it's consistent with all th evidence that's come out since then (see https://jonathanlatham.net/). But I'm sure not remotel expert, and would be interested if anyone has opinions about this. I'm still confused why it's nev been much discussed.

REPLY (1) S

hwold Mar 28, 2024 *Edited*

It looks like it's not even a coronavirus ?
https://en.wikipedia.org/wiki/M%C3%B2ji%C4%81ng_virus

REPLY (1) S

Steve Byrnes Mar 28, 2024

Good find! The wiki article is the product of raging 8000-word debate on the talk page
https://en.wikipedia.org/wiki/Talk:M%C3%B2ji%C4%81ng_virus

REPLY (1)

Steve Byrnes Mar 28, 2024

Update: I read the talk page! My understanding is that the miners got sick from Disease Agent A, and then later someone took a sample from the same cave (NO from the miners) and found Disease Agent B. The Wikipedia article you found is a B, not A.

It's POSSIBLE that A=B, but neither the Wikipedia article nor anyone else is claimi that A=B. There's actually some evidence that A≠B, but that evidence didn't make into the main wiki article, partly because the WIV master's thesis discussing the miners was declared not to be a "reliable source" per Wikipedia standards, and pa because the anonymous person who created the article was very aggressively opposed to providing fodder to what they called "baseless sinophobic paranoia" "conspiracy theories" 🙄

REPLY (1)

hwold Mar 28, 2024

I think the pertinent article linked by wikipedia is https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9744119/, which seems to be explicit rebuttal to that theory.

REPLY (1)

Steve Byrnes Mar 28, 2024 *Edited*

Thanks! That link isn't much evidence though, right? Like if we're gonna the Bayes thing, I dunno, likelihood ratio 1.1 maybe? Like, if a biologist is accused of stealing a car, and then the biologist announces that he did h own DNA analysis in his own lab and found results that exonerate himsel well, we haven't learned very much from that. :-P

I'm much more swayed by e.g. the fact that RaTG13 is from the Mojiang mine.

Let me know if I'm misunderstanding or if you find anything else :)

REPLY (1)

hwold Mar 29, 2024 *Edited*

Yes, they could easily lie here. You have to decide for yourself what weight to put to those words. Johnathan Latham explicitly says they lying in https://jonathanlatham.net/faucis-covid-origin-swat-team-versus-the-mojiang-miner-passage-theory/ ("Other criticisms of th MMP theory"). I'm somewhat sympathetic to that : "they died of fungus" is clearly nonsense and can easily interpreted as the researchers saying to us "we have to lie to you but we're not happy about that so we are going to pick the least believable lie".

The big gap in the MMP theory is BANAL-*. I hope you'll forgive me can't find the exact quote, but in the initial MMP theory Johnathan explictly said something in the line of "one thing that will be a big problem for the MMP theory is if the virus hunt yields a genetically closer ancestor to SARS-CoV-2 than RaTG13". And we found one : BANAL-52. Johnathan handwave it by saying "we found it in north L and north Laos is geographically the same thing as Mojiang".

But we don't know where exactly the mine is in Mojiang county (eyeballing google maps, it's 300 km from north Mojiang to south Mojiang). More problematic, the paper that announced the discover BANAL-52 never said it was found in north Laos ; indeed they say th all BANAL clades were found in the same site, "site 1", which was in east Laos, close to (again, eyeballing the map) Pak Lay, which, agai eyeballing Google Maps, is close to 800 km to Mojiang county.

This is a big problem for MMP theory, I haven't seen a good answer that.

But to be fair, it's a big problem for the Wuhan wet market theory to It's mostly a point for the Chinese official theory, "it comes out of China".

REPLY (1) S

Steve Byrnes Mar 29, 2024

Thanks!

I can do this! https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9744119/ says t mine is in Tongguan town, and meanwhile the supplementary information of https://www.nature.com/articles/s41586-022-04532-4 gives longitude & latitude of site 1.

The straight-line distance is... drumroll ... 531 km.

I don’t have a good sense of whether I should think of 531 km a lot” or “a little” ¯_(ツ)_/¯ Like, there exist species of bat that migrate 1000km twice a year, but also species that stay much closer to home. Here’s a paper that tagged 9 bats and found “maximum radial distance” from the tagging site of 17-248km f the different individual bats. https://link.springer.com/article/10.1007/s10393-010-0332-z It probably very different for different species. I don’t know how t think about this, I really know very little about bats or virology :

I agree with your basic point that BANAL-* is more than zero evidence against MMP, so thanks for that. Just not sure how m

REPLY S

Rob K Mar 28, 2024

This was incredibly informative. Thanks to the participants for doing it, and to Scott for summar and publicizing it. I hadn't thought about COVID origins in a while, but I would have told you it w the tossup zone, and digging into this a bit puts me closer to Scott's 90-10.

I wish this was a highly replicable method for illuminating tricky topics, but I kind of suspect that conditions here (two reasonable and sincere people who believed an impartial jury of smart, cur neutral observers would be won over to their side, and who were able to agree on who constitut smart curious neutral observer) are tough to arrive at. I hope I'm wrong!

REPLY S

Joshua Persons Mar 28, 2024

"some rando who nobody had ever heard of"

Not to be too facetious, but perhaps Rootclaim should have calculated the probability of the existence of a debate savant before building an attractor for one.

REPLY S

Samuli Pahalahti Mar 28, 2024

The big problem is that we still have very few details about all this. How is that possible after all these years?

The Chinese authorities haven't been very transparent. Other governments haven't demanded transparency from China. Why not? They just accept that China doesn't want to give any details

The pandemic was a global catastrophe. It killed a ton of people and caused enormous econom losses.

The people deserve to know how it all happened. Secrecy is unacceptable.

The main reason why I'm leaning towards the lab leak is the severe censorship about the topic. came from the individuals who strongly supported the zoonosis hypothesis. Usually people whc desperately want to hide something have something to hide.

REPLY (2)

NA Mar 28, 2024

I think most leaders know it was a lab leak. They just don't want to deal with the political shitshow that will come out of making it official. It has the potential to disrupt the fickle relations with China and as a result, significant amount of global trade

REPLY

Kenny Easwaran Mar 28, 2024

Every government demanded transparency from China. They refused. What more do you w

China was very interested in covering up the wet market aspect of it for a long time, and on allowed investigators to come a year later.

REPLY (1)

Samuli Pahalahti Mar 29, 2024

Nobody really demanded anything. They just asked nicely and China said no. And that it. There was no real pressure against China to provide information.

Everyone else was happy that China didn't share any details. That is a bit suspicious.

REPLY (1)

Kenny Easwaran Mar 29, 2024

No one was happy that China didn't share any details. I'm not sure where you wer getting the idea that they were. The WHO did try to put pressure on China, but Ch has more power over the WHO than vice versa, so they just couldn't put all that m pressure on them.

REPLY (1)

Samuli Pahalahti Mar 29, 2024

Nobody put any REAL pressure towards China or any other suspicious party l EcoHealth Alliance.

WHO put extremely weak pressure on China, it's like they didn't even try.

I don't remeber seeing any politician being genuinely enraged for China hidin much important information. Given that the negative effect of the pandemic \ enormous, I would have expected to see lots of powerful people saying "you must show us everything you have on this topic, or else...". But nobody did th

REPLY (1)

S

Kenny Easwaran Mar 29, 2024

What comes after "or else..."? What pressure could the WHO put on Chir during the pandemic, when they needed Chinese cooperation especially

REPLY (1)

S

Samuli Pahalahti Mar 29, 2024

Of course it's not only WHO but all the governments.

The pandemic had an enormous negative effect on the population c the world. People deserve to know what happened. The governmen who represent the people should find out what happened but for sc reason they don't even try.

If it's not enough to ask nicely, it's not hard to come up with harder ways to create pressure.

For example, setting some sanctions would definitely help. Similar t which are in effect against Russia now. They would be even more effective against China because it's so depending on imports from other countries. A few months of blocked fuel imports would give a incentive to start being more open.

REPLY

S

Redwoodburl Mar 28, 2024

"Hey, this bat we just collected on one of periodic field trips to yunnan had an interesting viral s with some really interesting structure around how it's furin site works and implications for protei folding"

"We should probably research it, is there any grant money we can use?"

"I think the Americans just announced some new initiatives?"

"Great, let's apply. Write a proposal that says we will investigate Furin site placement and it's im on virology."

....

“Did we get the funding?”

“No”

“But the grant reviewers didn’t know about the bat, right? Just that we proposed doing the furin stuff?”

“Yea, we’ve changed the names of the ones we found and generally obfuscated that we have th specific one in house”

“Should we just do the research anyway?”

“… … … yea, probably, could be interesting”

“We haven’t really done this before, and maybe we don’t have the right safety procedures and protocols, specifically to make sure it doesn’t escape?”

“No. But this particular strain can’t even last that long in the lab on its own. So what’s the worst could happen”

REPLY (1)

Écorché Mar 28, 2024

Five out of the six organizations on the DEFUSE proposal are American.

REPLY

Stephen Pimentel Mar 28, 2024

I have no strong opinion about the Lab Leak hypothesis. I have a very strong opinion that Bayesi reasoning does not, in practice, help with deciding such matters.

Early in the post, Scott offers the following. After reading the entire long post, I believe that it is, fact, the correct overall summary:

> But the joke goes that you do Bayesian reasoning by doing normal reasoning while muttering “Bayes, Bayes, Bayes” under your breath. Nobody - not the statisticians, not Nate Silver, certain not me - tries to do full Bayesian reasoning on fuzzy real-world problems. They’d be too hard to model. You’d make some philosophical mistake converting the situation into numbers, then end much worse off than if you’d tried normal human intuition.

REPLY (2)

The Ancient Geek Mar 30, 2024

It's nice to be able to say so. Try saying that on LessWrong.

REPLY

polscistoic Mar 31, 2024

A very accurate observation of Scott there.

REPLY S

Russell Hogg Mar 28, 2024

Well I'm still convinced it was a lab leak but at least now I know I'm probably wrong.

REPLY (1) S

Radar Mar 28, 2024

What a great reply!

REPLY S

Edmund Bannockburn Mar 28, 2024 *Edited*

Disclaimer: I have not watched the full videos. Apologies if this point is addressed at length.

To what extent are both parties relying on CCP-curated information for their data on the early cases? Was there a discussion of how (un) trustworthy that information might be?

REPLY (1) S

Kenny Easwaran Mar 28, 2024

I think at least some of the consideration is that whatever curation was going on in Decemb 2019 is unlikely to have any evidence about the actual origin, since the CCP had no reason suspect it was related to the virology institute.

REPLY S

contrapositive Mar 28, 2024

If you assign 1/10,000 probability of that particular wet market being the location of the first maj outbreak conditional on the lab leak hypothesis, you should also calculate the probability of that particular location under the zoonotic origin hypothesis. The population of Wuhan is <1% of the population of China (maybe the whole world is a better reference class, though rural areas and developed countries are going to have lower risk), for example, and that's before you account fc the fact that zoonotic origin could also cause major outbreaks at other locations within each city in all I'm not sure "covid began to break out at this particular wet market near WIV" is terribly str evidence one way or the other.

Another piece of evidence I don't see discussed here is the lack of openness from China, and relatedly the lack of evidence of a zoonotic source. If China had information against the lab leak hypothesis they would likely release it, so the relative lack of information we have is more likely given a lab leak imo.

REPLY S

Tophattingson Mar 28, 2024

For all the talk of Furin cleavage sites, there's one thing that confuses me. Could we have had a less deadly, probably not even noticed) coronavirus pandemic if there was no Furin cleavage sit My understanding of gain of function here is that there are multiple proteases that can do the cleaving even without Furin, but Furin does the job faster, and therefore sometimes outruns the immune system in a way COVID wouldn't do without that extra cleavage site. But that raises the question of why we got an outbreak of a virus with that Furin cleavage site, instead of a hypothe precursor that lacks it. The existence of a Furin cleavage site is considered unusual, so for Zoon would you expect a precursor virus without it to be far more likely to cross over into humans firs and therefore prevent COVID from being a major issue in the first place?

REPLY (1) S

Grognoscente Mar 28, 2024 *Edited*

Shi's team, in 2018, published a paper reporting serological evidence of bat-human corona spillover around the caves in Yunnan province. Doesn't seem like this strain/these strains would've been infective enough to kick off a pandemic, but if this sort of transmission was happening regularly enough for long enough, then the people around these caves could've definitely served as a natural "gain-of-function" experiment.

Paper here: https://link.springer.com/article/10.1007/s12250-018-0012-7

REPLY S

TTAR Mar 28, 2024

Sometimes you can tell one of Scott's posts is going to be a banger just from the quality of the in the opening section header alone

REPLY S

ChrisBoston Mar 28, 2024

Unfortunately it took me 16 hours to read this article 🥺

REPLY S

Rokko Mar 28, 2024

The notorious fact is that there were only 3 labs in the world at that time doing gain of function technology in the world...2 of those were im US and 3rd one in Wuhan..GOF was banned by Oba

administration so the only destination to outsource technology was wuhan lab...🤗 😚 even fam
defuse grant for the almost same virus creation by GOF was proposed and refused for wuhan la
darpa im 2018...immpossible coincidencenes happen😅 🤣

REPLY

HaraldN Mar 28, 2024

"Still, it's awkward to use "conspiracy theory" as an insult when the conspiracies were real. May better question is whether lab leak is "pseudoscience".

The argument against: lots of smart people and experts believed it was a lab leak. There were al those virologists giving 50-50 odds in their internal conversations. Even Peter says he started o leaning lab leak, back in 2021 when everyone was talking about it.

The argument in favor: since 2021, experts (and Peter) have shifted pretty far in favor of zoonos They've been convinced by new work - the identification of early cases, the wet market surveys genetic analysis. "

I have admittedly not looked into the details, and I cannot claim to truly know how to handle filte evidence, but:

If the experts (mostly) all agreed to have a conspiracy pushing for A, and then all the evidence t turned up after that pointed to A, this really doesn't make me feel like I have to raise my probabi of A all that much. For all *I* know for every published result supporting A there are 2 unpublish results supporting B, or claims that have not been investigated because experts (correctly) beli they would turn out to support B.

REPLY

Stuart Buck Mar 28, 2024

One issue I have with the opening presentation in the first video (I have watched 38 minutes so is that he lists, for example, about a dozen different theories about how the virus could have be made, and his slide says, "it's not possible for all these theories to be correct." So what? If there one true explanation for something, the fact that 11 other people came up with a false explanati for it doesn't diminish the probability of the first explanation being true. Otherwise, I could make probability of any true idea go down just by creating a bunch of false theories.

REPLY (1)

The Ancient Geek Mar 29, 2024 *Edited*

What you should do it reserve some probability mass for the theories you have never heard or that no one has even thought of. The probabilities of the set of theories that happen to e at time T don't have to sum to 1, in general.

REPLY (1)

S

polscistoic Mar 31, 2024

This is a good and important point, thanks.

REPLY

S

Yug Gnirob Mar 28, 2024

Skipped the whole lot and am working purely from Scott's post-transcript summary.

>So the real p(wet market|lab leak) isn’t the 1/10,000 chance a pandemic arising in a random pla hits the wet market, but the (higher?) probability that there’s something wrong with Peter’s argument.

This looks like an excuse to assign whatever numbers you like to any point. If you're readjusting of your stats, surely you should be readjusting all the other parts with the same readjustment metric, and thus reach a conclusion indistinguishable from the original formula.

It does sound like Saar lost mostly because he got caught flat-footed, but considering he set up system it still looks bad for his methodology. Surely he used it to calculate a proper debate setu beforehand, and it underperformed.

REPLY

S

Jay Orne Mar 28, 2024

I learned more about this debate and the arguments in favor of each side in the hour and a half reading this post than I did in any previous discussion of these arguments. Thank you for the ric summary, and while I agree with your arguments about why this kind of thing probably can't hap very often, I really wish we'd see more of these kinds of rich debates

REPLY (1)

S

Radar Mar 28, 2024

Same!

REPLY

S

Some Guy Mar 28, 2024

A systemized version of this plus community notes is my basic model for how all news should w But with reputation scoring in there to smooth out some of these transaction costs for actors wl have proven themselves credible.

REPLY

S

Andy G Mar 28, 2024 *Edited*

“Dumb” question that I did not see answered anywhere, and not mentioned much at all even in these comments to date: does “lab leak” also incorporate the possibility that it was created in a and *deliberately* circulated in the Chinese public?

And if done deliberately is an option within “lab leak”, doesn’t that throw off the odds by several OOMs, given that for someone deliberately wanting the virus out there, but wanting to cover the tracks a bit, “releasing” it in a wet market that had all those animals would be a fairly high proba thing to do (in addition to being a very good way to ensure spread)?

Is deliberately put out into the world its own separate possibility, or is it incorporated into “lab le And if the latter, what is faulty about my reasoning that it throws off the supposed unlikelihood c the outbreaks being centered in the wet market by many OOMs?

REPLY (1) S

Comment removed Mar 28, 2024

Comment removed

REPLY (1) S

Andy G Mar 28, 2024 *Edited*

Really? Seems to me all it would take would be a single Chinese bureaucrat - high, mid or even low-ranking (the last of which seems admittedly less likely) - who decided the benefits of the knowledge gleaned from human trials would outweigh the risks. And m not understanding how high the risks were. The Nazis had “rational justification” for th we people with decent western liberal morals thought were far more ‘insane’.

And if this case is in fact covered within “lab leak” then it changes completely the logi that the wet market would be a 1 in 10,000 chance to something more on the order of 10…

REPLY S

uhoh Mar 28, 2024

Surprised to see no mention of Nick Patterson's latest: https://npatterson.substack.com/p/yet-more-on-covid-origins

REPLY (1) S

Martin Greenwald, M.D. Mar 28, 2024

Likewise

REPLY S

Radar Mar 28, 2024

I loved this whole thing, thank you for writing it and I'm glad you enjoyed it!

I have no dog in this hunt and don't feel like any important policy-making we have to do hinges (this outcome. We know we need to keep labs safe and we know we need to deal with zoonotic r And I still found it riveting to read, as well as all the comments here. I think this lands in the genr competency porn and I'm here for it.

A meta-observation: I started today 50-50 on this question. Right after reading it, I was briefly a 90-10 zoonotic. An hour or two later, I'd say it slipped to 75-25, and now I'm landing maybe at 6 40. That's an interesting experience just to watch over the course of a day. I don't know, but I th if I read it all over again, it might goose me back up to 80-20, temporarily.

When I ask myself why looking at the debate as presented swings me a lot but then that regress over hours, I would summarize that as "mistrust". Mistrust of the institutions that produce information, mistrust of the arguments presented, mistrust of the biasing potential of the debate format. That's my emotional/psychological bias and so the further I get from direct contact with elements of the debate (which I found persuasive), then that default bias starts to take over mor

While my intuition listens along to the content of the debate, it also has some deeply-held opini about the trustworthiness (or lack thereof) of experts. I have my hobbyhorse example of this arc guidance to parents for what position to put their infants down to sleep -- which has changed tl times since my adulthood. It's not that anyone was misleading anyone, but that things that look absolutely true to science in one era turn out to be absolutely wrong in the next era and that's h science proceeds. We don't know what we don't know, and that makes all my reactions to these kinds of competing stories pretty uncertain. And it also makes me marvel at other people's certainty.

I don't know, but I think this might have some relevance for questions people ask about how to persuade people about the importance of large scale risks based on a mish-mash of reasoning a data. Where and how mistrust shows up is a huge issue.

REPLY (1) S

polscistoic Mar 31, 2024

"Where and how mistrust shows up is a huge issue."

This is indeed a core issue. And it can be studied systematically.

REPLY S

ClipMonger Mar 28, 2024

> Fourth, for the first time it made me see the coronavirus as one of God’s biggest and funniest jokes. Think about it. Either a zoonotic virus crossed over to humans fifteen miles from the bigg

coronavirus laboratory in the Eastern Hemisphere. Or a lab leak virus first rose to public attentic right near a raccoon-dog stall in a wet market. Either way is one of the century's biggest coincidences, designed by some cosmic joker who wanted to keep the debate stayed acrimonic for years to come.

Scott messed up. The lab was built near wet markets because wet markets were a threat, and G research probably was conducted there because leaks could be blamed on wet markets.

REPLY (1) S

Scott Alexander Mar 29, 2024
Author

The lab is fifteen miles of dense cityscape away from the wet market. There are thousands buildings between the lab and the wet market. There's no way "sit around, wait for somethi to leak, surely it will surface at the wet market" would be a reasonable strategy.

REPLY (1) S

Simon stats Mar 31, 2024

Also, the lab was not built near wet markets because wet markets were a threat. This i completely confused.

REPLY S

Chris Buck Mar 28, 2024

The new notes USRTK obtained through FOIA in January 2024 aren't trivial. I had been pretty m on the fence about the lab leak hypothesis, but the new evidence is so overwhelmingly damning literally made me nauseous reading it. The notes explicitly propose generating a consensus sequence with an inserted furin site. They propose using six BsmBI sites to assemble the infecti recombinant - confirming a 2023 hypotheses that the spacing of the six BsmBI sites in SARS-C had a suspicious spacing. The notes proposed passaging the recombinant virus in human cell li under BSL2 conditions.

It's a mistake to casually dismiss new smoking guns, all caps or otherwise.

REPLY S

Ben Landau-Taylor Mar 28, 2024

"Why did the older strain start spreading later? Probably the virus crossed from bats into raccoo dogs on some raccoon-dog farm out in the country. It spread in the raccoon-dogs for a while, racking up mutations, including the (less mutated) Lineage A strain and the (slightly more mutat Lineage B strain. Then several raccoon-dogs were taken to Wuhan for sale, including one with

Lineage A and another with Lineage B. The one with Lineage B passed its virus to humans earlie
Then 3-4 days later, the Lineage A one passed its virus to humans."

So apparently the zoonosis story requires *two separate* zoonotic transmissions, from the sam location and within the same week, but from separate viral lineages? Am I understanding this rig And if so, then can someone please explain why both sides are just passing over this claim as a minor aside rather than a devastating hole in the theory?

I am not an expert in the field, at all, but this intuitively strikes me as *really* implausible.

REPLY (2)

Kenny Easwaran Mar 28, 2024

The lab leak hypothesis seems like it either needs two leaks of its own, or a long enough pe of spread to produce this mutation while somehow not leading to the explosion of hospitals until late January.

REPLY (1)

Écorché Mar 28, 2024

B is only two mutations away from A, and one of those mutations is silent. It could have happened in the same person.

Zhang et al 2024 has new early samples from Shanghai with a bunch of intermediates between A and B, consistent with a single spillover. I think their proposed phylogeny ro at an A0 strain which is one mutation closer than A to the bat viruses.

REPLY (2)

Comment removed Mar 28, 2024

Comment removed

REPLY (1)

Écorché Mar 28, 2024

I haven't seen anything about the bug he refers to, I would be interested to se reference supporting his claim. I'm sure he's referring to something real but h interpretation seems overconfident.

REPLY (1)

Écorché Mar 28, 2024

Ok, I'm guessing he's talking about this line in Pekar et al 2022: "We identified numerous instances of C/C and T/T genomes sharing rare mutations with lineage A or lineage B viruses, often sequenced in the sa laboratory, indicating that these intermediate genomes are likely artifact contamination or bioinformatics".

This is somewhat more cautious than Peter's "we can dismiss both" so maybe he was referring to something else?

The new paper (it's Lv et al 2024 contrary to my earlier ref, Zhang is the senior author) mentions this in a couple places but don't seem to think it affects their result.

REPLY

Chris Mar 28, 2024

That seems most plausible to me but it blows a huge hole in everybody's timeline since it means cov2 was spreading around in humans before any of the cases flag by the authorities. For a disease with a 10+ day incubation period and a relatively rate of hospitalization though it could have easily been spreading for a month or s around young healthy people before anyone started looking for it.

REPLY

Scott Alexander Mar 29, 2024

Author

See responses above at https://www.astralcodexten.com/p/practically-a-book-review-rootclaim/comment/52707601

REPLY

Radar Mar 28, 2024

Side note about the use of "conspiracy theory" in this context. It's got these two meanings, one being "nutty, unfounded belief" and another being "a theory about how people coordinated acti secretly behind the scenes." There's a Venn diagram where flat earthers believe an endless stre of government officials are conspiring to keep the truth from us.

It doesn't seem accurate to me to call the lab leak idea conspiracy theorizing in the "nutty, unfounded belief" sense. Smart people in very recent years, including in this debate, disagree. When you accuse the other side in that situation of "conspiracy theorizing", you're just engagin ad hominem attacks which add no useful information.

Some lab leak people do believe that there was behind-the-scenes conspiring that happened, b so do some zoonotic supporters. There are more people on the lab leak side who do seem to be more of the nutty, unfounded type of believers. But it's a distraction to look at them, because th evidence in favor of lab leak is not like the evidence in favor of the earth being flat, etc.

Sometimes too the people we call conspiracy theorists are just people who have been slower to catch up to the preponderance of evidence, given that science is a process that unfolds over ye I don't think it's helpful to call people in this category conspiracy theorists.

Strong arguments were made on both sides of this. Peter's arguments look stronger. But as a person who doesn't have their hands on all the raw data, I wouldn't say that he "debunked" the other side. He just presented some other information that was more persuasive.

Anyway, I guess this is a wish for not calling people "conspiracy theorists" as it relates to things scientists themselves are still debating.

REPLY S

Karl Gallagher Mar 28, 2024

The whole Session 2 issue seems irrelevant to me. The simplest lab leak scenario is a virus bein collected in the wild, brought to the lab, then escaping in guano on a tech's scrubs or somethin similar. No need for genetic engineering of the virus at all.

The argument seems to have been "natural virus in wet market" versus "engineered virus from l while neglecting other scenarios.

REPLY (2) S

Radar Mar 28, 2024

This kind of speaks to what's the relevance of this debate after all, other than all the politic heat that's been generated around it.

Why do we really care? (I mean that as a sincere question rather than dismissively)

REPLY (2) S

Karl Gallagher Mar 28, 2024

Zoonotic origin means we need to fund virologists more to be prepared for the next na virus to come out of the wild.

Lab leak means we need to put a choke chain on virologists to make them practice bet lab hygiene and stop inventing nastier viruses.

So it's an significant impact on policy questions. (Having more virologists practicing tc higher standards, alas, is probably not a viable option.)

REPLY (1) S

Radar Mar 28, 2024

We already know that both of these are big risks worthy of tons of attention. Lab l have already killed lots of people. Pandemics even more.

REPLY (1)

Melvin Mar 28, 2024

This is true.

In a sensible world, the uncertainty about whether the virus came through th door or the window would be seen as a good reason to close both the door a the window; in the real political world we live in it's being used as an excuse t leave both the window and the door open.

In an even more ideal world the rest of the world would demand trillions in reparations from the Chinese government.

REPLY

Moon Moth Mar 28, 2024

> Why do we really care?

Aside from the obvious questions about whether we need to improve lab safety, and whether gain-of-function research caused a problem for us (yet)...

And aside from Saar's underlying motive, which appears to be an attempt to implemer practical Bayesian reasoning...

Mostly because the discussion picked up political heat. Which warped the "official narrative", in ways that are still relevant. It's important to know which of our official scientific sources will follow along with tribal groupthink, and which will base their decisions on data. (Or rather, where on the spectrum each individual source falls.) Anc this leads to a question of what the data actually are: what did we know, and when did know it, and what kind of mistakes were made along the way, and why were those mistakes made, and what if anything can we do about it in the future.

REPLY (1)

Radar Mar 28, 2024

Thank you, nicely said. I find my opinions about all this swinging around. Which I t mostly means I don't have very strong opinions about it.

REPLY (1)

Moon Moth Mar 29, 2024

Yes, the irony on my part is that I actually do want to "trust the science", but also am slightly familiar with how the sausage gets made. And when we have scientists who are also public figures, making claims with far too much certai that makes me skeptical of everything they say: it's being a public figure and engaging in public relations*, not being a scientist and engaging in science. G me something with error bars, and explicit limits on our certainty, and I'm mu more likely to trust it. It's not a perfect heuristic, but I came to it the hard way And it does give me a bit of moral high ground, in that if people would just sto trying to score political points and be honest, I'd be on their side.

* i.e., propaganda

REPLY (1)

polscistoic Mar 31, 2024 *Edited*

"And when we have scientists who are also public figures, making claims with far too much certainty"

...It's a selection effect, at least in my country.

Journalists call around till they find some scientist that says what the journalists want to say themselves but do not have the academic cred to And/or till they find some scientist who is willing to say something drama or controversial about the topic. Those scientists receive new calls. The ones who offer error bars do not.

I do not blame journalists. Most live and die by clicks these days. Those hate the system leave journalism, creating a new selection effect where remaining are those who like to live by clicks.

In the old days, newspapers could print a few long-winded serious articl each issue, since no-one knew for sure how many - or few - that read th (The Germans for some strange cultural reason still have some newspap like that. It also helps to have a potential pool of 82 million readers.) But after online publishing, editors and owners have accurate data on how o such pieces are read. Which has been a death knell to serious science journalism.

...admittedly, I am talking here of the social sciences (broadly defined). T natural sciences is somewhat better placed.

Instead, we have substack and a few other places where the global audi of people who are actually interested in nuanced stuff (with error bars) i large enough that it can fund at least some writers who have nurished a truth, whatever it is, shall set you free" attitude to their subject matter.

.

REPLY

S

Kenny Easwaran Mar 28, 2024

Yeah, it's weird to me that they didn't mention the third hypothesis, which I thought was considered as plausible as either of the other two initially. But I think that Session 2 deals w lot of evidence that had been taken by some to be extremely strong, and it only supported gain-of-function version, not the lab-collection-accident version.

REPLY

S

Andrew Erickson Mar 28, 2024

"As mentioned earlier, the DEFUSE grant was rejected."

I do not know how many people here are familiar with how the federal grant system (whether NI DoD) for biomedicine works, but if one claims that the only way experiments get done is if 1) the are proposed in a grant and 2) that grant is funded, they are either unfamiliar with how the syste works (like Peter) or else engaging in intentional misdirection (like the many virologists who hav made this claim and certainly know better). As a general rule, the proposed experiments in a biomedical grant fall (often roughly equally) into 4 categories: 1) work you have already done bu not published and inot included as preliminary data (you have to include just the right amount o data to justify the proposed experiments/money) ; 2) work you never intend to actually do but w looks good on the application; 3) work you do during the 1-2 years the grant is under review but decision has been rendered; 4) work you actually will end up doing during the funding period.

That the DEFUSE grant itself was not funded is drastically less important (frankly not important all) from the perspective of a Bayesian trying to figure out the origin than the fact they proposec making the precise modification that is so striking about Covid and has not been seen in any clc relatives. David Baltimore gets this, Nick Patterson gets this, and I think anyone with knowledge both how these grants work as well as the molecular biology gets this.

REPLY (3)

S

Radar Mar 28, 2024

I wondered this as well, it seemed like non-evidence to me given how this works.

REPLY

S

DamienLSS Mar 29, 2024

I agree, several of my friends are in grant-seeking areas and universally this has been the reported experience. Most times your current work is being funded by your last grant, your future work is being funded by your current grant, and you current grant was based on you past work.

REPLY (1)

S

polscistoic Mar 31, 2024

Related, a saying by an old colleague: "Collecting the data takes 90 per cent of the available time. Reporting the findings takes the other 90 per cent".

REPLY

S

Jeffrey Soreff Mar 29, 2024

>I do not know how many people here are familiar with how the federal grant system (whetl NIH or DoD) for biomedicine works

I am _not_ familiar with it, and I appreciate the comment!

>1) work you have already done but not published and inot included as preliminary data (yc have to include just the right amount of data to justify the proposed experiments/money) ;

Now that you mention it, that sounds _very_ familiar from other areas of research as well!

REPLY

S

David Bahry Mar 28, 2024

Saar raised the issue of ascertainment bias, but undersold it. There's no question imo: the case search *was* heavily focused on HSM, and the HSM sample search *was* heavily focused on th southwest corner. The Chinese authorities who did the search have been clear on this (Bahry, 2(Table 1 and Fig. 1).

E.g. we now know that the raccoon dog stall was sampled 10 times, with 5/10 positive; while the pool of blood outside a beef stall in the east wing was only sampled once, with 1/1 positive (Liu e al., 2023; cf. Bahry, 2023, Fig. 1). That, not the racoon dog stall, was the highest positivity ratio. Peter's heatmap was based on a paper by some Western scientists who had no idea how many times each stall had been sampled, and had instead speculated that every sampled stall had be sampled equally (Worobey et al., 2022).* After Jesse Bloom also showed there's no positive association between raccoon dog and SARS-CoV-2 genetic material, nothing implicates them, r even in a God's-joke way (Bloom, 2023).

China CDC director George Gao has also emphasized, when asked by the host of a BBC podcas that this overwhelming ascertainment bias means you can't assume genuine case clustering arc HSM (https://www.bbc.co.uk/sounds/play/m001ng7c, 24:00–25:12).

References

Bahry, D. (2023). Rational discourse on virology and pandemics. mBio 14: e00313-23. https://doi.org/10.1128/mbio.00313-23

Bloom, JD. (2023). Association between SARS-CoV-2 and metagenomic content of samples fro the Huanan Seafood Market. Virus Evolution 9: vead050. https://doi.org/10.1093/ve/vead050

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Worobey, M. et al. (2022). The Huanan Seafood Wholesale Market in Wuhan was the early epice of the COVID-19 pandemic. Science 377: 951–959. https://doi.org/10.1126/science.abp8715

*This zoonosis-theory paper is one of many by a prominent coauthor network centered on "Proximal origin" authors Andersen, Holmes, Rambaut, and Garry.

REPLY (2)

David Bahry Mar 28, 2024

*edit: Peter's figure is the version from that co-author network's preprint (Crits-Christoph e al., 2023, https://zenodo.org/doi/10.5281/zenodo.7754298), not the paper (Worobey et al., 2022). (It also includes the partial metagenomic data that they scooped from Liu, Gao et al getting in some trouble with GISAID for.) It's still obfuscatory, acting as though the SC2-negative samples weren't *also* concentrated in the sw corner.

REPLY

Nicholas Halden Mar 28, 2024

Would love to hear Peter's response to this.

REPLY (1)

tgof137 Mar 28, 2024

I do not believe that the pool of cow's blood in the east side of the market is the sourc the covid pandemic based on 100% of its samples (1 out of 1) coming back positive.

Otherwise, I think I adequately answered all the questions about ascertainment bias at debate.

REPLY (2)

David Bahry Mar 29, 2024 *Edited*

Nobody's saying it came from the cow's blood (just like Bloom wasn't saying it ca from fish). The point is that the Worobey et al. / Crits-Christoph et al. heatmap is a obfuscation, based on their making a baseless assumption that was already know be baseless and is now known to be false. At present nothing genuinely points to raccoon dogs.

The only adequate answer to the clear, known, overwhelming ascertainment bias the early case search is to say "Wow, you're right—the ascertainment bias *is* cle known, and overwhelming, so we can't appeal to apparent clustering as evidence.

(For the HSM samples, now that we know how many samples were taken from eac sampled stall, it probably *is* possible to properly redo the Worobey et al. / Crist-Christoph et al. heat map, modifying their code to use the positivity ratio. I haven' done it myself because I'm not good at coding. It could either be done with the measured positivity ratios, e.g. 5/10=50% and 1/1=100%; or using pseudocount smoothing like in my figure, e.g. [5+0.077]/[10+1]≈46% and [1+0.077]/[1+1]≈54%

Note: Worobey et al. / Crits-Christoph et al. haven't properly redone the heat map either; instead, they've since chosen a new way to automatically downplay the les sampled stalls away from the southwest corner, by mapping p-values instead of positivity [https://www.biorxiv.org/content/10.1101/2023.09.13.557637v1#commer 6279754825].)

REPLY (1)

David Bahry Mar 29, 2024

For anyone who might want to try, here's their github for Worobey et al 2022 https://github.com/sars-cov-2-origins/huanan-market

REPLY

Simon stats Mar 31, 2024

Can you just say why George Gao said they focused too much on the market. If w looking for strong evidence - the guy in charge of the case search said there was ascertainment bias both within environmental sampling in the market and in the c of Wuhan. Is he just very confused? What is going on here?

REPLY

Nicholas Halden Mar 28, 2024

Question for Peter: don't you think you're being a little credulous with respect to the early case map, given an obvious and known effort from WHO/China to support anything but LL?

Separately--great job. To go into this with nothing but your own research and bet a good portio your net worth against some rich guy, and win, is something from a movie. You should be proud sure.

REPLY (1)

Kenny Easwaran Mar 28, 2024

China very much opposed the wet market hypothesis early on as well. That's why they didr
allow any investigation of the wet market for a long time.

REPLY (2)

Nicholas Halden Mar 28, 2024

Peter claims the Worobey study's map is from "before ascertainment bias", but I just d understand how. If the first known case was from the market, and with the knowledge SARS zoonotic origin, you don't even need to attribute any bad faith to get ascertainm bias around the market. Once the LL hypothesis existed (and I don't really know when was first proposed, or when the CCP thought it might be a possibility, you also have ba faith to contend with, even if their official story was propaganda about an American lat

REPLY

Simon stats Mar 31, 2024 *Edited*

No they didn't. George Gao said it was from the wet market in Jan 2020

REPLY

Nicholas Halden Mar 28, 2024

What should we make of this study, which found the presence of covid in Brazilian wastewater i late 2019?

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7938741/

REPLY (2)

Scott Alexander Mar 28, 2024 *Edited*

Author

There are lots of studies vaguely like this. I think we should be skeptical, because COVID ha doubled every 3.5 days in every population it's been exposed to.

Suppose there was a person with COVID in Brazil on November 27th. The first confirmed ca in the Americas was January 21st. By January 21st, the one COVID case in Brazil should ha grown into ... 55 days divided by 3.5 = 16 doubling times, 2^16 = 65,000 cases. There's no to miss 65,000 cases of COVID, especially when the next week that's quadrupled to 260,0(

So you would have to argue that COVID fizzled out sometime after November 27 - eg one person had it, and didn't pass it to anyone else. That's possible for one case - maybe that c virus just had bad luck, or the person quarantined. But it quickly because implausible once there's more than 2 or 3 cases.

So the question you have to ask is whether there was some way that there were only three cases of COVID ever in Brazil (until it officially arrived from China), and yet some scientist s got very lucky and found one of those three cases in the wastewater.

I would rather believe they made some kind of error, which also fits with the fact that noboc has any explanation for why a Chinese bat virus should be in Brazil before it's in China.

REPLY (4)

S

Comment deleted Mar 29, 2024 *Edited*

Comment deleted

Scott Alexander Mar 29, 2024

Author

I agree the outbreak probably started late November, which I think does work witł numbers you give.

See my comment at https://www.astralcodexten.com/p/practically-a-book-review rootclaim/comment/52714955 , which said November 27, although I think your numbers suggest something more like November 21. See also the graph Peter toc from Pekar of when he thinks the epidemic began: https://miro.medium.com/v2/resize:fit:720/format:webp/1*duXQpxzXkveoFHWO4 Q.png

REPLY

S

DamienLSS Mar 29, 2024

Maybe this has been answered elsewhere, but isn't it awkward to be relying on Chinese death reports? I had understood that was one of the areas that was most suspect about their records - they consistently downplayed the amount of covid i their population to give the impression that the governmental response was good and/or that there was no problem that even needed a response. Basically deny th there's a problem until it can't be covered up, then cover it up as much as possibl after that. I thought that was common even outside this particular debate, that nobody really believes that the Chinese govt handled covid so well that their population death rates were a tiny fraction of everywhere else.

REPLY (2)

S

Biff Wiss Mar 29, 2024

I think (and may be wrong here) that there would be some lag time before the Chinese authorities noticed there was a problem to cover up. The first week(

increased death counts would look like statistical noise, or just the usual annu
fluctuations between the relatively-less-morbid summer season and relativel
more-morbid winter season. So while the "deaths from Covid" statistics can
(and probably are) fudged downwards, we would still expect to see a "hiccup
the "overall deaths" reports released prior to January that, with hindsight, isr
simply seasonal fluctuations.

REPLY

Thomas Stearns Mar 30, 2024

The idea that all these data involved in this debate are suspect is a fair point.
personal view is that excess deaths is possibly the most robust way of lookin
the early pandemic in Wuhan, but I might be mistaken about that.

REPLY

tgof137 Mar 29, 2024

That's a good guess, and sound reasoning. If covid was in Brazil in November 2019, wl
isn't it also in the wastewater in other cities around the world? Why just there? Where a
all the cases? All the deaths?

I had 20 more slides prepared, in case Saar brought up any of these outside of Wuhan
issues. Since he never contested them, I ended up turning them into a blog post:

https://medium.com/p/e54c358736bb

The specific issue with the Brazilian study is probably lab contamination -- they tried
sequencing some of the covid in the wastewater and it shared mutations with B.1 strai
that weren't otherwise seen until later in 2020. The authors theorized that meant that i
was circulating worldwide before Wuhan (one lab in Italy also found something similar,
probably for the same reason). But the B.1 mutations made the virus better adapted to
humans, so it would need to then travel to Wuhan and revert back to something less w
adapted and then mutate into B.1 again.

A far more likely theory is that the 1 sample from Brazil and 1 sample from Italy were
processed in labs where someone was a little bit sick.

REPLY (2)

Écorché Mar 29, 2024

I think one of the Italy studies tested samples from a range of dates, with positivit
showing up maybe in mid-December? It's harder to explain that as contamination

REPLY (1)

Simon stats Mar 29, 2024

I think they are better explained by false positives, right? It doesn't really add given the lack of an outbreak in italy

REPLY (1) S

Écorché Mar 29, 2024

False positives after some date but not before? I agree that it is hard to reconcile with the lack of an outbreak.

Here's the paper, it's actually harder to believe than I thought: https://pubmed.ncbi.nlm.nih.gov/36029839/

I don't follow this story, but someone said this paper was the strongest c the ones suggesting an early international spread.

REPLY (1) S

Biff Wiss Mar 29, 2024

> False positives after some date but not before?

Not trying to Texas Sharpshoot, but I can think of some plausible explanations:

1. The stored samples are physically sorted chronologically, becaus this is a research lab and research labs like to be organized and alphabetical order wouldn't make sense in this case. Material from a "actual, for-real positive" sample contaminates its neighbors.

2. The work of studying the samples is passed to multiple people. T work is divided into chronological batches. By a stroke of luck, a positive-but-asymptomatic employee happens to handle and contaminate the batch of samples immediately preceding the "actu for-real positive" batch.

3. The samples are stored "bottom to top", chronological order, and processed "top to bottom". Residue from an actual, for-real positive sample is left behind in lab equipment.

4. (comedy option) All the samples are processed by one person in chronological order. None of the samples originally contained covid the one person processing them caught covid at some point and be contaminating the batches as they worked on them. All the positive false.

Basically, jokes aside, it becomes a lot more likely (or at least less-unlikely) when you consider the on-the-ground human aspects of la
work and sample storage.

REPLY (1)

S

Aristophanes Apr 2, 2024

This feels like a good real life example why it is extremely hard the LR from any specific piece of information to ever be extrem large, say 10000. Above some value, "if I am wrong, it is most l this is due not to bad luck, but due to a fundamental error in m model of the world". In this case, for example, rather than testi "did these samples have Covid" what we really are measuring i "did laboratory testing of samples, including various forms of human error, detect Covid". When priors are very small, false positives are relatively likely.

There is an analogy to frequent statistics here. Even with the standard "correct" interpretation of a p-value, p is not "the probability of something this extreme occurring by chance if th is no true effect". It's more like "the probability of something th extreme occurring by chance if there is no true effect and we a modelling the DGP correctly, for example assuming that there a no omitted Zs that are conditionally correlated with the X varial of interest". It doesn't tell you "how likely is this result to go aw I control for additional truly-important variables". So when a p-value is very small, if the result *is* wrong, it is much more likel be wrong not due to sampling luck, but due to misspecification violation of the identifying assumption.

REPLY

S

Nicholas Halden Mar 29, 2024

Informative blog post, thank you.

REPLY

S

Écorché Mar 29, 2024 *Edited*

I'm surprised that you have that kind of trust in a continuous epidemiological model wi only a couple parameters, surely things are noisy at small population sizes? As an exar

from the low end, Pekar et al 2022's modeling suggested 8 spillovers of which 6 died (
There is clearly some point at which COVID does not double every 3.5 days.

There is also documented bias from the case reporting criteria changing over time. Wouldn't cases of a previously unknown disease be undercounted at first?

The experts seem to be using Monte Carlo for this rather than fitting an exponential.

REPLY (2) S

Scott Alexander Mar 29, 2024
Author

I agree at small population sizes things are more iffy, which is why I'm saying the (
don't rule out a very small COVID epidemic, but they do rule out a sizeable one. I guessed "not more than 2 or 3 cases", but you could convince me the real numbe
was 10 cases or something. I wouldn't believe 100.

I would bet those 6/8 failed spillovers didn't reach the 3 case threshold, certainly the 10 or 100.

REPLY (1) S

Écorché Mar 29, 2024 *Edited*

I suspect you are underestimating the threshold.

Pekar's cutoff for a successful outbreak was ~500?

EDIT: Pekar says "failed introductions produced a mean of 2.06 infections an
0.10 hospitalizations", congratulations :-)

REPLY S

Thomas Stearns Mar 29, 2024

Sure. Complex models get you published in a fancy journal. They don't necessaril
give you correct answers. I mean, 8 spillovers? In Wuhan, but no where else? Doe
any one seriously believe that?

REPLY (1) S

Écorché Mar 29, 2024

Yeah, Pekar et al 2022 is likely to be retracted based on the work of twitter ar
@nizzaneela.

But I don't think the basic spread model is controversial and apparently they generated lots of runs with lineages that died out completely.

REPLY (1)

S

Thomas Stearns Mar 29, 2024

All of the many errors uncovered in Pekar et al. were in the direction favc zoonosis, which is noteworthy.

If you don't trust a doubling time model, see if you can plug numbers int the above compartment model predicting a such an enormous excess d figure by 1/23. I couldn't do it in a back half of november time frame with unrealistic assumptions about the CFR or R0.

REPLY (1)

S

Écorché Mar 29, 2024

A priori there is no continuous model that I would trust enough to extrapolate backwards in time, through all the discrete behavior aro the spillover, and come up with a credible start time down to a week

I'm not any kind of expert in the field, but isn't that a very suspect u of modelling?

REPLY (1)

S

Thomas Stearns Mar 30, 2024

You can seed the compartment model with any reasonable nur of “initial” cases you want if you think very early spread violate SIR assumptions. AFAICT, it doesn’t really change the conclusi that the outbreak occurred early in november rather than later.

REPLY (1)

S

Écorché Mar 30, 2024

I thought the idea here was to get some evidence about th
origin by inferring a spillover date, which you can't do by skipping the early dynamics.

If you're just setting a bound on the late side, the continuc
model seems a lot more reasonable. But I wouldn't be sho
if the spillover had happened in August, given how little da
we have. Of course late November is very attractive if you
working from the assumption that the market outbreak wa
also the spillover.

REPLY S

Evesh U. Dumbledork Mar 29, 2024

I don't understand how the doubling time of 3.5 days can be anything other than noise when the number of cases is in the low single digits

REPLY S

tgof137 Mar 29, 2024

Answered below:

https://www.astralcodexten.com/p/practically-a-book-review-rootclaim/comment/5271661

REPLY S

Vitor Mar 28, 2024

This is a highly interesting topic. Unfortunately I don't have the energy tonight to get deep into a
the great points brought up in the post. But I want to share a few scattered thoughts in babble mode:

(1) Taking a 5% chance of dying on a climb is pure insanity. Maaaybe taking a 5% *lifetime* risk
worth it if you really love climbing. Hardly an endorsement for risking $100k at 4%.

(2) It's a terrible idea to offer a standing bet on your true beliefs. You're literally doing adverse selection and end up with a sneaky rules lawyer as your debate "partner".

(3) It seems that Peter is just a good debater, of the "debate club" variety. Laser focused on wir
the debate, rather than finding truth. I was repulsed by the dirty arguments made, for example t
quibbling about whether to call the first outbreak a "superspreader event" or not. Regardless of
completely irrelevant doubling times Peter brings up multiple times, a place like the Wuhan marl
clearly is a mechanism for turning a handful of cases into 80k cases. I've only skimmed the

summary of the debate, so please point me to where Peter tackles the real underlying issue h on.

(4) Re: orders of magnitude. There is no limit to how badly you can fuck up a bayesian estimate. example, Peter's 1 in 10'000 probability of the first infected person working at the wet market. That's just a uniform prior! It's so, so bad. The easiest way to debunk this is to take these odds, start listing conclusions that would need to follow if they were actually correct: "viruses are equ likely to spread to everyone regardless of their profession and their position in the social graph" um no, everyone with an inch of common sense (or kids) knows that viruses spread where peop gather: schools, workplaces, and yes, markets. The vendor at that market didn't need to be the merely among the first *few*, and then keep coming to work while contagious.

(4a) It strikes me this might be a valuable method to flesh out. Let's call it the *reverse inference test". I don't know why, I just feel like it latches onto my intuitions more cleanly than trying to ref a claim in the same direction it's made.

(5) Probability cascades are bullshit. To me, this is the classic example of "science as attire"; yo pretend to be scientific by multiplying numbers together, but what you're really doing is spinning narrative. It's extremely hard to come up with an exhaustive enumeration of posibilities. MECE, a it's sometimes called (mutually exclusive, collectively encompassing). Or for the mathy people, equivalence relation on the possible states of reality. If you doubt this, go take an advanced algorithms class and report back. I'm not joking... repeated encounters with impartial yet implac entities (compilers, test suites, theorem provers, etc) will drill this into you like nothing else. Pet argument style is that of a college freshman, and any competent math mentor would beat it out him before it rots his brain. To the extent Peter is aware of this, it just reinforces (3).

(5a) to clarify the above point, the typical mistakes made is multiplying together probabilities th aren't independent, and to implicitly assume that the only way to reach state X is the specific pa towards X you've laid out. These can make a 50% event seem like a 1% event, or the other way around.

(6) Explicit bayesian calculations in the absence of a *formal model* are a horrible idea. It just ta confusion and multiplies it by orders of magnitude. What I've learned in my time as a scientist is the relationship between the real world and the model you're working on, and how to transport assumptions and conclusions back and forth between them, is one of the hardest things to get right. And we don't really have a systematic way of teaching this skill.

(7) If you wanted to blackpill me on the applicability of bayesian calculations to everyday life, yo succeeded. In particular, I see no reason to update my beliefs regarding the lab leak hypothesis response to this debate. I'm still 50%+ in favor.

Wow, that was a lot more than I thought. Thanks for reading, and I hope it was helpful, even in th low-effort format.

REPLY (2) S

MicaiahC Mar 28, 2024 *Edited*

We disagree in the large, but +1 to this post because every object level point is exactly corr and that's way more important to me than agreement.

May you be blessed with good luck.

REPLY (1) S

Vitor Mar 28, 2024

Thanks! we should really continue our discussion about AI at some point!

REPLY (1) S

MicaiahC Mar 28, 2024

Actually, I have a question, what specifically in advanced algorithms are you thinki about when you say that it'll help with models? I'd like to implement a couple of th things as an exercise you see.

(Personal guess it's some kind of probabilistic algorithm, or just seeing a bunch o greedy algorithms that looks optimal... but actually doesn't even converge. I woul guess that I'm not thinking expansive enough but seeing the gap in my prediction your answer would be illustrative of what I'm missing)

REPLY (1) S

Vitor Mar 29, 2024

For one, combinatorial algorithms. Stuff like sat solving, computational geom and so on. The state space boggles the mind. Graph algorithms, including st like reductions of various problems to max flow.

One of the most interesting classes I took in my master's was models of computation. We had to program algorithms on Turing machines, cellular automata, tile completion systems, and so on. The interesting part about it is we could have our solutions automatically checked. Not sure if there's equiva stuff out there on the internet.

There's so much more. Stuff like computer graphics or physics simulations, where the operations are very elementary, but you can't just debug by lookin a single code path. You kind of have to think on a systematic basis and judge output as a whole.

That's just off the top of my head.

REPLY

S

Aristophanes Mar 30, 2024

Very good points. (4) and (5) are particularly relevant.

REPLY

S

Sandeep Mar 28, 2024

I know I am being ridiculous, but has anyone made a 2-3000 word summary of this post? Would very useful for semi-literates like me.

REPLY (1)

S

Yug Gnirob Mar 29, 2024

Scott did, it's the last part of this post.

REPLY (1)

S

Sandeep Mar 29, 2024

That is the conclusion, I was hoping for a summary of the arguments.

REPLY

S

Neike Taika-Tessaro Mar 28, 2024

My brain skipped a beat when I read the summary that Peter dismissed one of the later market spreads as "that didn't come from a human source, there were COVID particles on the fish". Tha feels eerily lab leak mechanics adjacent and I found myself slightly confused why he thought tha was likely but COVID from the lab coming to the wet market and then spreading from there unlik Probably if I watched the videos, I would be less confused about this, and I'm not trying to make argument, just registering my confusion.

Generally, I have no strong position on this question at all and a low prior on that the things I thir about it are true. I thought this article was a fantastic deep-dive! I realise I could get even more information if I watched the videos, but I'm not really a video person. Thank you so much for thi

REPLY (1)

S

trophy Mar 28, 2024

Something about that whole line of reasoning felt a bit off to me. If the explanation is due tc non-COVID-zero imported goods, shouldn't we expect to see a similar occurrence of outbr with other goods importers. Why did it specifically happen in a wet market?

REPLY (1)

S

Écorché Mar 29, 2024

Frozen fish is the leading "anywhere but here" theory, whose popularity is almost entir
restricted to the government of China. Peter doesn't actually believe in it.

REPLY S

LesHapablap Mar 28, 2024

Regarding the 1 in 10000 chance of being discovered in a wet market if lab leak:

That is flawed. You might as well assign 1 in 365 that it was discovered December 30th. You car
it is flawed because if it had been discovered in a school, Saar would assign 1/10000. Or in a restaurant, 1/10000. Or a bakery, 1/10000. He gets a free 1/10000 wherever because any public place can be differentiated by at least 1/10000. “Only 1/10000 Wuhans work at Wuhan IHOPs!”

REPLY (1) S

tgof137 Mar 29, 2024

That was discussed in the 3rd debate. You can't assign high numbers unless there's some importance. If someone gets murdered in one building in Wuhan, and there are 10 million buildings, that's not a 1 in 10 million coincidence.

On the other hand, there are only 4 markets in Wuhan selling wild animals (out of a giant cit
10 million plus people). Huanan market is the one selling the most animals (about 40% of a
the wild animals coming into Wuhan), but it's otherwise not the biggest market or anywhere
near the most crowded place in Wuhan.

So if you're trying to weigh the odds between "wild animals caused it" and "the lab caused
then it is of significance that it first showed up a the place where the wild animals were sold
opposed to 10,000 other places it could have first showed up in Wuhan.

REPLY (2) S

LesHapablap Mar 29, 2024 *Edited*

You’re right, I was misinterpreting how they were using bayes theorem here

REPLY S

Thegnskald Apr 1, 2024

That's not the complete reference class, however. Had the virus first showed up in a zo
which treated wild animals, or which took wild animals for its exhibits, and the chain of custody involved intersected with the region in question, that would also be treated as equally valid evidence for zoonosis. If it first showed up in a restaurant which used products from the wet market, or somebody who made purchases there, that might als

count as part of the reference class. Anybody involved in the logistics of moving anima and anybody who visited a farm (there's no reason to limit ourselves to raccoon dogs) The workplace or school of anybody who visited a farm. The workplace or school of anybody who traveled to the province in question. The raccoon dog farm already has t degrees of separation, so any plausible path with two degrees of separation is part of reference class - and that's an enormous reference class.

Note that the reference class for zoonosis isn't limited to Wuhan, and, indeed, we wou expect it to first show up there given a zoonotic origin.

Once you start critically examining the actual reference class to which the wet market belongs, without arbitrarily limiting it to particular observed qualities, it is pretty clear i actually enormous.

Whereas the laboratory is almost a reference class in and of itself; there are a few part matches, so it's not entirely its own reference class, but it is something like unique in a that the wet market absolutely is not.

REPLY S

David Karger Mar 28, 2024

You nod at this briefly in the conclusion, but what I find fascinating about this debate is that the what truly happened doesn't seem to matter at all! If an oracle arrived and told us the source, I think it would (or should) change our planning at all. The investigation of COVID has revealed th there are huge risks from *both* lab leaks and zoonosis that we need to invest significant effort reducing, and which one actually happened in this case doesn't change that.

REPLY (1) S

Écorché Mar 29, 2024

It seems to me that it matters. When you are estimating whether the next pandemic will co from nature or a lab, the result here should affect your prior.

What matters more immediately is whether to even continue investigating the origin of CO The most visible advocates for a zoonotic spillover have been hopefully declaring the case closed for some time now. But if COVID really was related to the DEFUSE research, there is huge amount of evidence within the reach of US investigators and it would be a shame not look at it.

REPLY S

trophy Mar 28, 2024

Maybe I'm missing something, but it seems like Peter's argument hinges on two separate cases zoonosis within days of each other in the exact same location and the same mutation? Or mayb misunderstand that point.

REPLY (2)

Scott Alexander Mar 28, 2024
Author

Hopefully Peter will show up and explain, but my impression of his case is something like:

- Raccoon-dog on a farm catches COVID from bats

- Lots of raccoon-dogs on that farm get infected

- Several raccoon-dogs from that farm are brought to Wuhan for sale

- This version of COVID is already good at infecting humans

- The virus spills from two of the raccoon dogs to humans, for the same reason that if a bu of infected Chinese visited America, it might spill from two of the Chinese over to America because the virus spreads quickly in a naive population.

REPLY (2)

Ben Landau-Taylor Mar 29, 2024
So the story is that both strains arose on a farm, and then it crossed to zero humans a farm, but swiftly infected a whole bunch at the wet market? Seems pretty tenuous.

REPLY (2)

everythingism Mar 29, 2024
It's entirely possible there were spillovers elsewhere that died out. Before the pandemic the CCP had encouraged rural people to get involved in farming animal make money. It was part of Xi Jinping's drive to "end absolute poverty" (in reality threshold of what "absolute poverty" meant was a shifting target, but China did g many people a way to make a living this way). So you had people in very remote regions raising animals, in some cases near bat caves, then shipping them into bi cities to be sold.

REPLY

Scott Alexander Mar 29, 2024
Author

Pandemics generally don't start in rural areas, because the average person has m fewer daily contacts, lowering R to the point where they're unsustainable. I think t was a study showing that a bunch of animal handlers had SARS antibodies (suggesting they'd gotten the disease) when all known SARS outbreaks were in ci

REPLY

S

Simon stats Mar 31, 2024

There is a lot wrong with this causal chain.

1. Everyone *on both sides* including the Worobey group and the WIV, agrees that the ancestral virus is in Yunnan or SE Asia. Miller keeps denying this but it is so totally at o with all of the evidence, that you have to ask questions of him here.

2. The claim therefore has to be that the raccoon dog was infected in Yunnan and take a farm in Hubei where it acquired the furin cleavage site in Oct or Nov 2019. There is n direct evidence of this, and raccoon dogs roam wild in Hubei, so it seems weird for the to ship them from Yunnan. There is no evidence that raccoon dogs can sustain such intense transmission to cause acquisition of the furin site as happens with avian influe and lab experiments suggest that they cannot do this. Freuling et al found an R0 of 1 ir their experimental infection of raccoon dogs.

3. Raccoon dogs have never been found infected outside of a lab, unlike eg mink.

4. If lots of raccoon dogs on a farm get infected, why were there no outbreaks anywhe except Wuhan city where the farm was supplying? If you are envisioning thousands of infected raccoon dogs on a farm, this is a big BF in favor of the lab. It only really works you are envisioning about 100-200 raccoon dogs on a farm, but then you are conditior on a very specific farm size, and we have evidence that the typical farm is an order of magnitude larger.

5. The raccoon dogs were wild caught in Hubei. Wang et al 2022 tested 15 wild caugh raccoon dogs at the suppliers of the market in Jan 2020 and they were found negative Around 38 raccoon dogs were sold at four markets per month. So, this is half of the inventory. The raccoon dogs photographed at the market were agreed by the Worobey group to be wild caught in Hubei. Maybe you think half of the raccoon dogs were topp up from farms in Hubei. There is no evidence for this. If they were topped up from fur farms, they would have been in eastern China, even further from the ancestral bat viru

6. So, the claim is now that these ~10 raccoon dogs are responsible for the outbreak. is an extremely small fraction of the total wildlife trade in China, and conditions the zoonosis hypothesis on an extremely specific proposition. To justify this, one would ne extremely strong countervailing evidence of raccoon dogs being the intermediate host But what we do have is extremely small traces of covid in their cages and a negative correlation of raccoon dog genetic material with covid. Whereas for the viruses that th

animals in the market were actually infected by, like bamboo rats covs or canine covs, there is a strong correlation. Miller's claim that 'we have down to the stall' is just wrong

The collective BFs in favor of this causal chain are extremely weak, and in fact rule out raccoon dogs.

REPLY (1)

EMSKE Phytochem Apr 3, 2024

Just noting this comment implicitly references Jesse Bloom's work. (twitter, githu

REPLY

tgof137 Mar 29, 2024

Yeah, it would be:

some infected animals are shipped to the market.

some of those animals get people sick.

2 of those infections go on to infect significant numbers of other people.

The odds of there being 2 spillovers as opposed to 1 are about the same, once you have the right components in place. Eric asked a very similar question in the first week debate, if you want to take some time to watch.

REPLY (1)

Simon stats Mar 29, 2024

Could you lay out from start to finish how the ancestral virus got into the human population. e.g. is it something like:

Ancestral bat virus in Yunnan infects raccoon dog

Some of these taken to farm in Hubei

Intense transmission leads to acquisition of FCS

Within a month of that, 20 raccoon dogs take to Wuhan in November

8 spillovers occur

2 catch on

Is that right?

REPLY

Jeffrey Soreff Mar 28, 2024

Many Thanks!

>The six estimates span twenty-three orders of magnitude. Even if we remove Peter (who's kinc trolling), the remaining estimates span a range of ~7 OOMs. And even if we remove Saar (limitin the analysis to neutral non-participants), we're still left with a factor-of-50 difference.

ouch, ouch, _ouch_, _OUCH_.

And I thought that the single order of magnitude disagreement between the superforcasters an the domain experts on AI risk was bad... (as I'd commented in https://www.astralcodexten.com/p/mantic-monday-31124/comment/51596120)

In a way, the scatter in estimated here is even more disturbing, because AI risk unavoidably invo estimates of future events, many with only weak analogies to known past events. _This_ estima all about events which _do_ have strong analogs to known past events.

I don't know what to suggest. One large class of worries is indeed treating some probabilities as uncorrelated when one could indeed expect correlations. E.g. as you said, researchers modifyin viruses in ways intended to be less researcher-like and more nature-like in order to anticipate natural changes.

One other thing that generally bothers me is the blanks in a lot of the probability calculations wh one side estimated an effect and the other side just didn't discuss it. Maybe it would have been helpful to have a final pass where all sides gave estimates for _all_ of the factors?

REPLY S

Alea Diarrhea Mar 28, 2024

Does peter have a blog for us to follow his future work?

REPLY (1) S

tgof137 Mar 29, 2024

https://medium.com/@tgof137

There's an option to subscribe via e-mail, if you don't like that site.

I could try making a mirror on substack, have just never gotten around to it.

REPLY S

Steve Sailer Mar 28, 2024

So, I should go back to pitching my children's animated TV series "Wuhanimals" about a raccoo dog, a bat, and a pangolin who are friends?

REPLY (1) S

tgof137 Mar 29, 2024

South park had it right, all along:

https://www.youtube.com/watch?v=4-aeJEOLhCA

REPLY S

Petrel Mar 28, 2024 *Edited*

I truly hate to be a meta-commentator refusing to engage with the meat and bones of the debat but I'm still puzzled by why this matters enough for the debate to be worth at least 100000 doll and a few dozen hours of a few intelligent people's time (probably a lot more time and money, actually, given the extent to which all of this was discussed in other sources, of varying levels of sophistication, that I don't follow).

Would it help anyone in particular, even people who lost loved ones or whose businesses did no survive the economic consequences of the pandemic, to know whether the virus was zoonotica originated from a WIV leak?

I doubt there would be meaningful political or economic consequences either way. I'm moderate sure virological research is still being funded about the same way it did before the pandemic, I'n bit more sure wet markets are still as unsanitary as they were before the pandemic, I'm quite su trade/political relations/economic dependence of the world on Chinese economy and industry is about the same as it was before the pandemic, and I'm very sure human beings still die the sam way they did before (and during) the pandemic.

REPLY (2) S

sards3 Mar 29, 2024

How much time time and energy has been spent debating whether the Holocaust happene claimed, or what were the root causes of the American Civil War, or what caused the Great Depression, or how the pyramids at Giza were constructed? People like to discuss and understand important events in world history, even if there are no direct political or econon consequences hinging on the outcomes of these discussions.

REPLY S

John Schilling Mar 29, 2024

Regardless of whether or not this pandemic originated in a lab, it's still the case that GoF research is potentially quite dangerous, its claimed benefits quite dubious, and we certainl should be having a serious debate about whether or not it should be more strictly regulate than it is. If COVID hadn't happened, we could make that a dispassionate theoretical debat But here and now, we're stuck with a debate involving people who remember and care abo

COVID. And the way real people actually work, if one of the things they "remember" about COVID is that a bunch of conspiracy theorists claimed that it came from GoF research, but were later shown to be lying liars who all voted for Donald Trump, then we are not going to a fair discussion about how GoF research ought to be regulated going forward.

The same may be true of wet markets, but the case is I think weaker there because the ani reservoirs will still exist and will still interact with local human populations, and it's not clea how much benefit there would be to shutting down the wet markets if it just delays the rem village -> cosmopolitan city step a bit.

But so long as both hypotheses are credible, we need to keep making it clear that both hypotheses are credible and arguing against anyone who claims to have "proven" that it wa 99.99...% certain to have been one or the other. And if both are credible but one is genuine more probable than the other, then we want to know that so we better know how to allocat political capital in a world where we may not be able to shut down both the wet markets an GoF labs,

REPLY (1)

polscistoic Mar 31, 2024 *Edited*

...since the main applied point of the debate is to shame CCP into making sure neither markets nor laboratory slip-ups cause the next Big Pandemic, a conclusion that the we market and the lab leak hypothesis are equally likely explanations of the Covid pandem would be the ideal (applied) outcome of the debate.

...that is, if your main motivation for caring about these probabilities in the first place, i a call to action to limit major risks of similar pandemics in the future.

REPLY

Anonymous Mar 28, 2024

Great write up, and incredibly intellectually stimulating. One thing that struck me is how much harder it is to understand the past than to predict the future! If Rootclaim is so good, it should p itself on the prediction markets, instead of relying on judges. This also solves the problem of financial imbalance, challenging people to high stakes debates is like bullying someone at the p table when you have a pile of chips.

REPLY (2)

tgof137 Mar 29, 2024

And then when those people win, you tell them it was rigged and they have to bet you agai otherwise it doesn't count.

Super obnoxious.

REPLY S

polscistoic Mar 31, 2024

"One thing that struck me is how much harder it is to understand the past than to predict tl future"

Are you serious?

REPLY S

Biorealism77 Mar 28, 2024

Importantly, Miller incorrectly claimed the N501Y mutation would result from passage in hACE2 mice (mixed them up with BALB/c mice). The major papers Miller relied on have been seriously challenged since the debate. See Stoyan and Chiu (2024), Weissman (2024), Bloom (2023) anc et al (2024). Overall the circumstantial evidence makes lab v plausible:

1. Chinese researchers Botao & Lei Xiao observed lab origin was likely given the nearest known relatives to SARS-CoV-2 were far from Wuhan. Wuhan Institute of Virology (WIV) sampled SARS related bat coronaviruses where the nearest relatives are found in Yunnan, Laos and Vietnam ~1500km away. They refuse to share their records.

2. Patrick Berche, DG at Institut Pasteur in Lille 2014-18, notes you would expect secondary outbreaks if it arose via the live animal trade. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10234839/

3.Molecular data: Only sarbecovirus with a furin cleavage site. Well adapted to human ACE2 cel Low genetic diversity indicating a lack of prior circulation (Berche 2023). The CGG-CGG arginin codon usage is particularly unusual but used in synthetic biology.

Restriction site SARS-CoV-2 BsaI/BsmBI restriction map falls neatly within the ideal range for a reverse genetics system and used previously at WIV and UNC. Ngram analysis of the codon usa per Professor Louis Nemzer https://twitter.com/BiophysicsFL/status/1667232580255490053?t=lJgitS5cw364ioclzVWxaA&s=19

The SARS2 backbone is very low in CG and CpG. While the 12-nt insert that gives it the FCS is extremely high in both. Almost as if it was some kind of chimera of a consensus sequence and a codon-optimized polybasic cleavage site? https://twitter.com/BiophysicsFL/status/1752800486837678377?t=EplRgyybJVaPgeMP5xdstA&s=19

https://www.biorxiv.org/content/10.1101/2022.10.18.512756v1 https://link.springer.com/article/10.1007/s10311-021-01211-0?fbclid=IwAR1HMUMtLIAFOFppVasQDeoIAYrVhP8j4YoPO4wnaTOUiKLsllZl_oKryOw

4. DEFUSE full proposal: virus 20% different from SARS1, consensus seq assembled with 6 segments, without disrupting coding seq, BsmBI order, FCS. SARS2: 20% different than SARS1, evenly spaced fragments w BsmBI and BsaI restriction sites, FCS.

Jesse Bloom, Jack Nunberg, Robert Townley, Alexandre Hassanin have observed this workflow could have lead to SARS-CoV-2. Work often begins before funding sought or goes ahead anywa

5. Market cases were all lineage B. Lv et al (2024) indicates there was a single point of emergen and A came before B. So market cases not the primary cases. See also Bloom (2021), Kumar et (2022). Peter Ben Embarek said there were likely already thousands of cases in Wuhan in Decer 2019.https://t.co/50kFV9zSb6

https://www.ncbi.nlm.nih.gov/pmc/articles/pmid/34398234/

https://academic.oup.com/bioinformatics/article/38/10/2719/6553661

6. Evidence for lineage A in the market is based on a low quality sample according to Liu et. al. (2023).

7. Bloom (2023) shows market samples do not support market origin. There is also no evidence transmission in the claimed susceptible animals elsewhere. https://academic.oup.com/ve/advan article/doi/10.1093/ve/vead089/7504441

8. Lineage A and B only two mutations apart. François Ballox, Bloom and Virginie Courtier-Orgo note this is unlikely to reflect two separate animal spillovers as opposed to incomplete case ascertainment of human to human transmission (Bloom 2021).

9. Sampling bias. George Gao, Chinese CDC head at the time, acknowledged to the BBC stating they may have focused too much on and around the market and missed cases on the other side the city. David Bahry outlines the documented bias. Michael Weissman has shown this mathematically.

https://journals.asm.org/doi/10.1128/mbio.00313-23

https://academic.oup.com/jrsssa/advance-article-abstract/doi/10.1093/jrsssa/qnae021/7632556

10. Spatial statistics experts show the Worobey claim the market was the early epicentre was flawed.

https://academic.oup.com/jrsssa/advance-article-abstract/doi/10.1093/jrsssa/qnad139/7557954

11. Wuhan used as a control for a 2015 serological study on SARS-related bat coronaviruses due its urban location.

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6178078/

12. Superspreader events also seen at wet markets in Beijing and Singapore (Xinfadi and Jurong

13. WIV refuse to share their records with NIH who terminated subaward in 2022. Wider suspen over biosafety concerns. https://www.bloomberg.com/news/articles/2023-07-18/us-suspends-wuhan-institute-funds-over-covid-stonewalling

14. PLA involvement at WIV and MERS research prior to SARS-COV-2. MERS features several similarities with SARS-CoV-2.

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7022351/

15. SARS1 leaked several times and SARS-COV-2 has leaked from a BSL-3 lab in Taiwan.

16. Unpublished infectious clone identified from Wuhan contradicting arguments such reverse genetics systems would be published.

https://www.biorxiv.org/content/10.1101/2023.02.12.528210v1.full

REPLY S

Biorealism77 Mar 28, 2024

Miller incorrectly claimed the N501Y mutation would result from passage in hACE2 mice (he mix them up with BALB/c mice). Since the debate the key papers that Miller relied on have been bac undermined:

1. Worobey et al featured on Retraction Watch after spatial statistics experts eviscerated their c the Huanan Seafood Market was early epicenter. https://academic.oup.com/jrsssa/advance-artic abstract/doi/10.1093/jrsssa/qnad139/7557954

2. Lv et al (2024) found new intermediate genomes so the multiple spillover theory is unlikely (it anyway given lineage A and B are only two mutations apart). Single point of emergence is more likely with lineage A coming first. So market cases are not the primary cases (all lineage B). Thei findings are consistent with Caraballo-Ortiz (2022), Bloom (2021).

t.co/50kFV9zSb6

3. Jesse Bloom showed again the market samples don't support market origin. t.co/rorquFs1wm

4. Michael Weissman shows mathematically ascertainment bias in the early case data the judge relied on (George Gao, Chinese CDC head at the time, acknowledged this to the BBC last year - they focused too much on and around the market and may have missed cases on the other side the city).

https://academic.oup.com/jrsssa/advance-article-abstract/doi/10.1093/jrsssa/qnae021/7632556

5. Account that identified errors in Pekar et. al. leading to an erratum last year has found anothe significant error. Single spillover looks more likely. t.co/GAPihZu51P

REPLY S

ClipMonger Mar 28, 2024

> I have a weird urge to visit Wuhan as a tourist, see the Wuhan Institute of Virology, stroll throu the Huanan Central Seafood Market (unfortunately closed), maybe eat a raccoon-dog.

Scott N Alexander N plans N to N eat N Raphtalia N CONFIRMED

REPLY S

Em Jay Mar 28, 2024

Thank you for an incredible write-up and analysis.

I am actually shocked at how poorly Saar fares here. The vast majority of his evidence is basical things were different in a way we can't know, then our conclusions would also be different, there lab leak." E.g. "if COVID spread rate was lower in the beginning", "if an infected WIV worker spre to the wet market then isolated", and so on. That's it, really? I was expecting so much more actu evidence. And then what data points they do provide (e.g. "Mr Chen", 90 cases, intermediates, have apparently valid reasons why they're not evidence. I started out 60/40 zoonotic, but basec this I feel I have no choice but to update to 99/1.

I feel the real problem with this setup/framework is that a specific position is taken, with overly-precise numbers, and the goal of the debate is to simply justify that final percentage. You see tl with Saar's "but the judges just did the math wrong" sour grapes complaint.

In the IT and infosec world, we have the FAIR method--a similar method for building quantified models of risks. The real value of these isn't that you convinced yourself you have only a 1-in-1C chance of being compromised by a state actor in the next 10 years... it's the shared understand of the risks your business thinks are important, how you think those risks are correlated, how yc think they're evolving over time, etc. That enables your organization to act from shared goals, a not get bogged down in interminable debates day-to-day. The final numbers are fine, especially you use confidence intervals, but you know it's only a snapshot in time using your best data anc estimates as of the last time you underwent the exercise; by the time you update in a year/quart you will have more data and different estimates. Sometimes, risks that weren't even on your mo last cycle, are now the top priority because of what's happening in the world (or because you gained new information that you missed previously). It's a fact of life.

With that in mind, I think Saar fares better if his arguments are framed as reducing absolute confidence in zootonic origin, and not the reason why LL is the only possible explanation (which obviously isn't, even from Saar's own arguments). So yeah, I was only joking about updating to ! in favor of Z--I think there's enough doubt thrown in from Saar's evidence alone that it could no possibly be more than 80%. But Peter/Z still wins this version of the debate, and Saar's over-confidence in "no way it's anything other than LL, here's the math to prove it, and if you disagre then you're doing the math wrong" does border on pseudoscience here.

To that end, I think Scott's final takeaways are the most important. If we have any shared goals anymore as a society, then we should devote resources to better preparing for pandemics from origins, and in also improving government/organizational transparency so it's easier next time tc understand what went wrong. As it is, either a natural disaster can be ignored because "teh government conspiracy," or else somebody got away with murder; either way, the debates that focus solely on convincing everybody that one's favored position is the only explanation means we're not more prepared for future disasters than we were with this one, we're just more dividec and easier to manipulate.

REPLY (1) S

polscistoic Mar 31, 2024

"The final numbers are fine, especially if you use confidence intervals, but you know it's on snapshot in time using your best data and estimates as of the last time you underwent the exercise; by the time you update in a year/quarter, you will have more data and different estimates. Sometimes, risks that weren't even on your model last cycle, are now the top pri because of what's happening in the world (or because you gained new information that you missed previously). It's a fact of life."

Thanks for this.

REPLY

Sudip Paul Mar 29, 2024

I found this debate to be very valuable. I started with 70:30 odds on lab leak's side (I didn't do a extensive research on COVID origins) but after this debate that shifted to 5:95 (i.e., favoring zoonosis). I wish there are more such debates on contentious topics.

REPLY

Matt Reardon Mar 29, 2024

The Rootclaim model seems pretty sensitive to modest differences in probabilities; I'm guessing factor of having too many variables so small differences across all of them add up to a big effec tried to average the six guesses giving Saar the least weight and add some of my own intuition a got (I think) a modest pro-lab leak outcome.
https://docs.google.com/spreadsheets/d/14OkrvJ21qCqQcXINQnMO4-a1Xi-npidF/edit#gid=2057332411

REPLY

Michael Watts Mar 29, 2024

> About half of cancer patients lose their hair, and Putin hasn't, so we'll divide by two.

This leapt out as methodologically unsound. How do you know whether Putin's lost his hair? He are some possible scenarios:

1. Putin has never had cancer or anything cancer-like. His hair is fine.

2. Putin has had chemotherapy. He wears, or has worn, prosthetic hair to hide this fact.

Note that these two scenarios are perceptually identical. You have no way to tell one from the o

If the question you're evaluating is "Is Putin covering up a case of cancer?", his visible hair provi no information at all, because all scenarios predict visible hair. In the "no" case, the hair is irrele and in the "yes" case, the hair is deceptive.

It's kind of like making an adjustment to your evaluation of "Is this a coral snake, or is it a king snake?" for the observation that the snake in question has red and yellow stripes. That observat was already part of the premise of the question; you're double counting (among other weird mistakes) if you adjust for it again.

REPLY (2)

S

Comment deleted Mar 29, 2024

Comment deleted

Michael Watts Mar 30, 2024

?

REPLY (1)

S

Comment deleted Mar 31, 2024

Comment deleted

Michael Watts Apr 1, 2024

Not your finest argument.

Haven't you ever watched a movie? You know how the same actor will show u different movies with different hairstyles? (Or... the same movie, with differer hairstyles?)

Generally those are wigs. It isn't possible to tell the difference between a wig real hair; often they are real hair, just not real hair that the person to whom th are attached grew.

If you had the opportunity to physically manipulate Putin's hair, you might be to demonstrate that something was off. But you don't.

REPLY

S

polscistoic Mar 31, 2024 *Edited*

Good illustration.

It reminds me of a Taleb quip directed against only-thinks-within-the-box-rationalists, in or his books. Someone rolls a dice nine times. It comes up with sice (six) each time. He asks a statistics professor how likely it will come up with six also in the tenth roll. The professor answers "1/6 chance". He then asks Fat Tony, a street-smart mafioso. Fat Tony answers: 10 percent chance. Why? "Obviously, the dice is loaded."

REPLY

S

Michael Watts Mar 29, 2024

> Another of Saar's concerns with the verdict was that Peter was an extraordinary debater, to th point where it could have overwhelmed the signal from the evidence.

I've mentioned this before in the ACX comments, but this is such a great lead-in that I'll say aga that this is why I didn't make a habit of reading Overcoming Bias. Eliezer Yudkowsky used that b to make a lot of excellent points in an entertaining and well-written fashion.

And I got the sense, reading his work, that sounding persuasive was a bigger part of his strateg than being right, meaning that reading him was dangerous. So I enjoyed a number of pieces, bu didn't follow him or seek out his work.

REPLY

LesHapablap Mar 29, 2024

My suspicion here is that using estimated probabilities instead of probability distributions leads the same issue as for the Fermi paradox, in that your result is extremely sensitive to your assumptions and you end up with extreme outcomes on either side like we have here: 10^-25 ar 534 are both clearly wrong.

REPLY (2)

Bruce Cleaver Mar 29, 2024

This is a very cogent observation. Like Eric Drexler et al. in 'Dissolving the Fermi Paradox' tl realized using point estimates instead of entire distributions lead to weird & paradoxical res Good job!

REPLY

polscistoic Mar 31, 2024

If you have expanded on this somewhere, I would be interested in the reference.

REPLY

ARD62 Mar 29, 2024

My overwhelming feeling while reading this was a sense that a lot of the important evidence is it subject to long debates and explanations that are being glossed over. Is a certain length of prot sequence likely or unlikely to be done by a mutation in the wild? What is the right way to constru priors about wet market and virus lab locations? Everyone was just stating certain things confidently, while providing detailed documentation for other things.

In the end, "evidence for my claims" seems to have a fractal quality sometimes. The closer you l the more evidence you need to back up your use of evidence. It's turtles all the way down.

REPLY

Jacob Steel Mar 29, 2024

While I'm terribly sympathetic to most of Wilf's points, one thing I absolutely agree with him on i the superiority of written to real-time/spoken debate, because the things it rewards correlate mu more strongly with being right.

REPLY

S

Anonymous Mar 29, 2024 *Edited*

It seems to me that Saar's evidence-weighing analysis is basically "outside view" with extra step

As I see it, Saar's analysis hinges on the point that the zoonosis hypothesis contains a coincider that is very unlikely to be explained by some out-of-model error (which I agree with), whereas th lab leak hypothesis only contains a coincidence that could very plausibly be an out-of-model er agree with this as well). This sets up a barrier that is almost impossible for any amount of mere digging into the details to overcome, as that details-digging is all itself quite vulnerable to out-o model errors. And so the end result is an outside-view judgement that no amount of inside-view analysis can outweigh.

This surely seems like an error to me, though those with different positions on the outside-versu inside-view debate might disagree.

REPLY (1)

S

polscistoic Mar 31, 2024

Could you specify/elaborate on what you mean by "inside view" versus "outside view"? (or a reference?)

REPLY

S

Jacob Mar 29, 2024

I genuinely do not understand why there is not more talk about the third option: that one of the many bat cave spelunkers from Wuhan caught COVID-19 and brought it back to Wuhan.

The WIV and the market were fairly far apart from each other, as noted in the debate. The Wuha CDC is *right next to the market*. And while it doesn't seem that the Wuhan CDC was conductin gain of function research, their employees were fanning out all over China going into bat caves.

From this Washington Post article - https://www.washingtonpost.com/world/asia_pacific/coronavirus-bats-china-wuhan/2021/06/02/772ef984-beb2-11eb-922a-c40c9774bc48_story.html

"In the video, the researchers scale the cavern wall, their headlamps ghostly blue.

“If our skin is exposed, it can easily come in contact with bat excrement and contaminated matt which means this is quite risky,” says Tian Junhua ["associate chief technician in the Wuhan CD

pest-control department, but he has a reputation as a swaggering adventurer in his work with b
and insects"], one of the bat hunters.

...

The video was released by national science authorities and Chinese state broadcaster CCTV on Dec. 10, 2019, and circulated on social media. It's a high-quality production, designed to promo China's world-leading viral research.

...

Tian and his team from the Wuhan Center for Disease Control and Prevention are filmed catchin horseshoe and pipistrelle bats and collecting samples of guano, in search of new bat-borne diseases and the basis of new vaccines. Tian talks about the need for caution. "It is while discovering new viruses that we are most at risk of infection," he says, though he is shown hand sample vials without wearing full protective gear.

...

In 2017, Tian told the state-run Wuhan Evening News he once forgot personal protective equipm and was splattered with bat urine, leading him to quarantine at home for two weeks. On multiple occasions, bat blood squirted onto his skin while he was trying to grasp the animals with a clam he told the paper."

So the questions would be - exactly when did the Wuhan CDC move to be close by the market? move finished on December 2, 2019 - how long was the move and how was it conducted? Does science require an intermediate animal or could it have jumped directly from bats to humans wit the cave exploration?

REPLY (1) S

EMSKE Phytochem Apr 3, 2024

USAID seems to have been sufficiently aligned with your concerns by late 2023 to end the DEEP VZN program:

https://www.science.org/content/article/u-s-cancels-program-aimed-identifying-potential-pandemic-viruses

REPLY (1) S

Jacob Apr 3, 2024

Thanks, very interesting! I'd like your comment if I could.

REPLY S

Ruben C. Arslan Mar 29, 2024

I am now scared that the next pandemic will be started by debaters on either side doing gain of function research to shore up the evidence for auxiliary points in their debate.

REPLY S

Jonathan Ray Mar 29, 2024

There's so much stuff on this page that Chrome is having difficulty rendering it even on a high-e desktop. Consider closing comments and creating a separate page for that.

REPLY (1) S

Isaac King Mar 29, 2024

This is an issue on most of Scott's posts; Substack just isn't designed to handle lots of comments. Given that this has been an issue for years, it seems unlikely they care to fix it. S was much better in this respect.

REPLY (1) S

Biff Wiss Mar 29, 2024

Substack just isn't designed to handle, period. It's unbelievable that a text-based site could possibly run so poorly, and yet here we are, on computers so powerful they can emulate computers *emulating* computers from twenty five years ago, chugging alon worse than Slashdot twenty five years ago, Livejournal twenty years ago, Blogspot fifte years ago, or Reddit ten years ago.

Sites with lots of text and comments are a *solved issue*. I shouldn't have to literally download Firefox just to browse the comments for more than ten minutes without cras (on Firefox it "only" crashes every twenty minutes or so).

REPLY (1) S

EMSKE Phytochem Apr 3, 2024

(I don't remember slashdot ever struggling as much as substack is here...)

REPLY (1) S

Biff Wiss Apr 5, 2024

I remember when "nested" comments first came out (different from "threade view, which was more like old-school UBB) it could get a little bit laggy... on a Pentium II. A computer far less powerful than the cheapest burner phone out there today. And it still wouldn't crash, per se.

Substack is just embarrassing.

REPLY S

John Schilling Mar 29, 2024

At the outset of the pandemic, I was at about 15% lab leak, 85% natural origin, and thought it unimportant because I didn't think we'd ever have enough evidence to do better than guess. By start of this year, after way more than 15 hours of studying way more evidence than I imagined v would ever have, I was at 80% lab leak, 20% natural origin, After this, maybe I should update to lab leak, 30% natural origin. But I'm going to indulge in a bit of emotion and make that 25% lab l 100% annoyed.

Annoyed in that it seems like the rationalist and rationalist-adjacent community, parts of which a least I value and respect, has decided that on this one issue we're going to flip a $100K coin to decide what to believe, and I suspect anyone who dissents will at best be quietly dismissed as a ignoramus by much of the community. I will grant that by paying $100K, we got a coin that is *slightly* biased towards the truth,

Seriously, one-on-one personal realtime debate is a *lousy* way to find the truth. There's a reas we don't have debates as a central feature of e.g. scientific conferences. Throwing in high-value bets doesn't improve things. The process selects for the most skilled and charismatic debater, a in some cases for the favor of biased judges, with the truth mostly just serving as a tiebreaker.

In this case, there was no tie to break. Miller, by all accounts including this one, is an exceptiona capable debater. Wilf, while clearly skilled as an entrepreneur and possibly as an analyst, does n seem to be very good at in-person debate. From this very thorough account, Wilf completely fai to raise some of the very strong points in favor of the lab leak hypothesis, failed to call out Mille some of his indefensibly weak points, and agreed to a structure where he'd open by spending th first third of the debate on the very weak assertion, "maybe it didn't come from the Wet Market' even though that is in no way necessary to the lab leak case. Or maybe he did raise the strong points and call out Miller's weak points, but in a way that neither Scott nor the judges noticed.

As for Scott shifting is odds from 50-50 to 90-10 on the basis of this debate, I think that require p<0.20 for "Wilf isn't very good at this and left the winning arguments at home", which I don't th is defensible. You'd need a statistically significant sample of debates to be confident you have s both sides well-argued.

I will continue to value and respect the rationalist and rationalist-adjacent community, particular Scott's corner. But it has a clear blind spot in its enthusiasm for clever and novel truth-seeking mechanisms (bets, prediction markets, debates, etc) on the basis of their cleverness and novelt without adequate regard for their in many cases obvious flaws.

REPLY (1)

Scott Alexander Mar 30, 2024

Author

I'm not taking the results of this debate too seriously. I'm taking the fact that the debate exposed me to the arguments on both sides, that I spent a long time grilling the debaters o their arguments, that I read their followups and blog posts, and then that I read many other blog posts and a couple of papers on this topic seriously. Plus the surveys of experts and superforecasters.

REPLY (2)

Écorché Mar 30, 2024

The post has long cut and pasted sections of unchallenged borderline dishonest arguments from Peter. The quotes are so long that sometimes I had to double check whether it was you or Peter talking. Many of the arguments come from a paper which i flunking a careful replication attempt, not mentioned. Weissman's probabilities are not included because the items didn't match up with the rows of a spreadsheet built from Peter's analysis or something. There is not much evidence of research beyond the det here.

REPLY

John Schilling Mar 30, 2024

I'm glad to hear that and, knowing you, not terribly surprised. But this,

"For what it’s worth, I was close to 50-50 before the debate, and now I’m 90-10 in fav zoonosis"

does kind of read like it was just the debate. And, as I said, I don't think it would be reasonable to make that kind of shift on the basis of just a debate, because it requires unwarranted confidence in the process and participants. You say that the debate expo you to the arguments on both sides, but in reality it only exposed you to a subset of th arguments. Ideally, that would be a comprehensive subset including all of the strong a relevant arguments on both sides, but I am skeptical that is the case here. And I'm cer that you have at least some knowledge of missing arguments from your other readings but most people (including most rationalists) don't.

Unfortunately, as I see this debate spread out into the broader internet, it is largely bei stripped of nuance and turned into "Two really smart people held a very thorough deb on COVID origins, and now all the smart people agree that it's almost certainly natural zoonotic origin, nothing more to see here, move along".

REPLY S

Isaac King Mar 29, 2024

> The paper that claimed that defined how well COVID was adapted

Typo

REPLY (1) S

Isaac King Mar 29, 2024

> Something like this is true for the other Chinese wet market based outbreaks we know ab it.

Extraneous "it" at the end.

REPLY (1) S

Isaac King Mar 29, 2024

> the furin cleavage site really does stand on a genetic map

I think this is supposed to be "stand out"?

REPLY (1) S

Isaac King Mar 29, 2024

> and which which Viewers Like You did not appreciate

Duplicate "which".

REPLY (1) S

Isaac King Mar 29, 2024

> something that everyone already everyone else had already considered

Repeated phrase

REPLY S

Tristan Mar 29, 2024

If it turns out it was natural origin — as I am more likely to now believe — it's sad it will be that m harder to end gain of function research. Even if the lab did not leak the pandemic, the risk that t

lab would leak a pandemic eventually was still too damn high. Hopefully this event will nonethel
lead people to take the risks more seriously.

I've spent some time in virologist subreddits since covid, and I find their attitude towards these
to be terrifying. They remain very blasé about it. It appears to be deep in their culture, similar to
civil engineers are culturally committed to designing for cars, and architects are culturally again
ornamentation.

REPLY

Isaac King Mar 29, 2024

> So for example, suppose there are fifty things about a virus. You should expect at least one of those to have a one-in-fifty coincidence by pure chance.

Minor math error here I think: You should expect *exactly* one, not at least one. It's equally likel be more than one or less than one.

REPLY

kyb Mar 29, 2024 *Edited*

Thank you to Saar for running this in good faith, and thank you to Scott for a great write up. One the key things that stood out to me while reading the analysis was that there are a few relatively simple factual items that were disagreed on that might be relatively easy to chase down if we weren't worried about governments trying to obfuscate things. There's also a few simple pieces information that could shift my thinking a bit too, like where do the lab workers live? I wouldn't expect a lab leak outbreak to be centered around the lab, but around one of its employees home

I noticed in myself a high risk of falling victim to a classic 'david and goliath' story too, which is potentially a very bad bias. Almost everyone reading that first section is going to be rooting for Peter, and enjoy it everytime he seems to win a point.

REPLY (1)

Blackjack Mar 29, 2024

I also found myself with the David vs Goliath bias, despite having no opinions on COVID ori and not particularly caring about it.

Despite this, Saar's arguments (as described here) often ended up fairly weak due to him leaning on what seems like a lot of "bad claims" about things like the pre market cases that Peter easily dismissed just by following the logical consequences that should have occurec they were true. I think this ends up with him showing weaker than his argument really was.

Also agree, huge props to Saar for doing the thing in good faith, and hopefully this taught h lesson that when someone bets a large % of their NW versus your proportionally small bet, they will prepare 10-100x as hard as you will (because they have so much more to lose), an the fact that they took the bet in the first place implies they are likely already exceptionally good at whatever you are betting on.

REPLY S

Warmek Mar 29, 2024

I must admit that I don't care nearly as much about whether COVID was zoonotic or not, as I do about the government paying social media sites to enforce a particular orthodoxy about the question.

REPLY S

Juanita del Valle Mar 29, 2024

Some of this sounds like an “Encyclopedia Brown” level of argument. E.g. the idea that because University of North Carolina said they’d do furin site insertion in a grant application from years before the pandemic, therefore the WIV cannot have done any insertions of their own.

Research institution capabilities, and researcher ambitions, and the real world generally, are a lo more complicated than that reasoning implies (I say this despite putting little weight personally COVID being engineered).

Likewise with the idea that the WIV would have fully released the details of all viruses they have sequenced.

I haven’t watched the debate though, so maybe it was better than Scott makes it sound.

REPLY S

JohanL Mar 29, 2024

It strikes me that whether you should debate theories you think are crazy depends on how popu they are. Consider:

1/10000 believe in theory A. You debate someone arguing A, and you win 99% of the audience (considering what humans are like, this is super impressive). But now, 1/100 of the audience believes in the crazy. And *no-one* can convince 99.99%.

66% believe in theory B. Now when you debate someone arguing theory B, even if you only convince *half* the audience, things have improved!

This, of course, assumes that the people believing what you think is crazy can be convinced in t first place. This might well be true about lab leaks or *possibly* even UFOs; it will *not* be true about creationism (evidence isn't even a factor in why someone is a creationist).

REPLY S

JohanL Mar 29, 2024

The JFK assassination is another of those cosmic jokes. Sensible people should not believe in conspiracy theories about it. But dear Lord, was Lee Harvey Oswald a weirdo, the rare person w had been involved with *all* the kinds of people who _might_ be suspected of conspiracies; anc murdered before he had a chance to testify; and with an investigation that couldn't have seeme more suspicious and cover-up:y if it tried.

REPLY S

Big Worker Mar 29, 2024

Really fantastic read. I already leaned towards zoonosis but this removed most of the remaining doubt. It so valuable to see one side put up their strongest arguments and then see if the other can knock them down, which in this case they did pretty dramatically.

REPLY S

Alaina Drake Mar 29, 2024

Absolutely fantastic.

REPLY S

polscistoic Mar 29, 2024

"… And so on and so forth, until we end up with the final calculation: 86% chance Putin doesn't cancer, too bad."

What Rootclaim is doing is interesting. But should not such hard-core Bayesianism also include sensitivity analyses? That is, investigate how robust the conclusion is (here: "86 percent chance at least for minor-to-moderate changes in priors, age-related cancer statistics, and the like?

Using sensitivity tests in cost-benefit analyses is usually recommended. It appears that Bayesia could benefit from such tests, for the same reason.

It would also quell the skepticism people like me automatically feel when encountering predictio spelled out in fine-grained percentages. (Plus, why stop at percentages? Why not add a few decimal places as well: 86,xy percent chance Putin doesn't have cancer– that would make the prediction even more "accurate"….)

Do not get me wrong – I respect what Rootclaim is doing. But I would be (genuinely) interested i why Rootclaim does not add/construct some kind of confidence interval around their percentag estimates. (And why Bayesians in general do not do this.)

REPLY S

Jim Haslam Mar 29, 2024

SARS2 leaked from a Wuhan lab but it's not Chinese junk

https://jimhaslam.substack.com/p/lets-be-blunt-sars2-leaked-from-a

REPLY

sclmlw Mar 29, 2024

This post caused me to strongly update against Bayesian methods. Maybe it's the impossibility seeing accurate priors, or the tendency to engage in cherry-picked absurd probability estimates maybe it's the qualitative difference between statistical probabilities prospectively estimated vs retrospectively measured. I no longer have confidence this method produces more signal than noise.

For example, what prior do you set for an outbreak in WSM? You might think that's low, but give multiplicity you're not really looking at a prior probability anymore, are you? Both a spillover eve and a lab leak might have come from any number of places. We'd tell some plausible-sounding story about why that place was unique, then set absurd priors as a fence around that story. But heart it's still a post-hoc story, and all the prior-setting in the world will only serve to reinforce y story with more plausible-sounding stories. This seems to be what Saar is fighting against in ma adjustments for out of model variance, but that seems like the wrong approach.

The better approach is probably abandoning Bayes - at least explicitly/strictly. I'm not yet convi about the whole Bayesian enterprise, but it feels like it's on much more shaky ground for me toc than it ever has since I first learned about it.

REPLY (1)

polscistoic Mar 31, 2024 *Edited*

"This post caused me to strongly update against Bayesian methods."

This is also my major takeaway from the post & from the comment section.

I sometimes teach students who enter occupations where they have real power over other humans. Such as child protection officers, who in my country have a lot of influence when deciding in a child should be taken from its parents and moved to foster care.

I have toyed with the idea to include some "real" Bayesianism in their curriculum, with rega calculate the risk of various forms of child abuse in various types of families. (As of now, I c make references to what I half-jokingly label "everyday bayesianism", which is basically wh Scott talks about early in the post - also captured by the expression "inference to the best explanation".)

This post, and this extensive comments section, has made me update (as they say here) strongly against this idea. Few things are more dangerous than professionals with real pow believing that "science" can give them analytical tools that makes if quite certain when sufficiently serious child abuse goes on, that drastic measures are called for (and can be defended in the courts).

Putting numbers on inherently uncertain things creates professional peace of mind. It is a temptation in all professions given power from the state to dramatically interfere in the lives and health of other human beings. So no wonder that "Bayesianism" is an increasingly pop catch-word in the helping & caring professions, as elsewhere. But judging from this blog pc comments, "real" bayesianism is a danger, not a help, in reducing type I errors (not interfer when one should) and type II errors (interfering when one should not).

REPLY (1)

S

sclmlw Mar 31, 2024

Well said.

After thinking it over more thoroughly, I'm not sure the problem is that Bayesian thinkir wrong, per se. The problem is that some of us have been convinced to apply it more broadly than it should be. (Here I'm thinking of Scott's "P(A|B) = [P(A)*P(B|A)]/P(B), al rest is commentary.")

Simple heuristic: before you try to apply Bayesian reasoning, ask yourself, can you accurately fill out the equation? "Not quite, but I think I can estimate-" No. The answer no. Move along, or you'll be getting yourself into trouble, trying to apply (even loosely) principles that will only lead you to error.

REPLY (1)

S

polscistoic Mar 31, 2024 *Edited*

That's a useful rule of thumb, I'll remember it.

REPLY

S

David Joshua Sartor Mar 29, 2024

"maybe you should lower your disbelief in each hypothesis to something more reasonable, like ! But now the chance that the sun rises tomorrow is 0.99^100, aka 36%. Seems bad."

All these hypotheses are very correlated; extreme probabilities of base events are not required, extreme probabilities of conditionals.

'If all 99 other hypotheses are false, and I'm not making some silly maths mistake or whatever, t hypothesis 100 is 99.999999% false' is not necessarily overconfident even for a human.

Right?

REPLY

S

bart Mar 30, 2024

I have done my own extensive research and believe the evidence is compelling that there was a leak. IMO, Peter has a real axe to grind and his strong personal conviction against the lab leak hypothesis has made him blind to common sense.

The odds that the virus evolved free of human-engineering are probably much less than in 1/10,(Peter has no idea how to figure those odds and frankly no one else does either. The actions of t Chinese government in stonewalling are far more convincing to me that there was a lab leak. If t Chinese government didn't have something to hide why be so opaque???

REPLY S

Aftabeh Mar 30, 2024

A few thoughts:

For an adversarial setup (such as debates or US court cases) to converge on something close t truth, the case for each side needs to be represented at a much more similar level than was don here. Even if an innocent man is accused of a crime (so he *knows* that the truth will set him fre it is still generally recommended that they hire a competent lawyer to represent their side, as opposed to defending themselves. This is probably not as important in countries with a less adversarial court system.

Calling Peter a better debater doesn't quite capture what happened here. I don't think there was great difference in rhetorical skill, but there was an extreme difference in both the level of preparedness and the commitment to the cause. This case involves 50+ pieces of circumstantial evidence of various degrees of strength, some favoring lab leak and some favoring zoonosis. All the ones favoring lab leak were heavily contested, but at least to me it did seem like the same was true for many of the ones favoring zoonosis. This was probably a mixtur a few things. Saar didn't seem to not know enough of the precise details to effectively push bac against some of the bolder zoonosis arguments, such as how confident we should be that we actually have a good picture of the first cluster. In other words, he conceded far too much, allow the zoonosis side to look much stronger than I suspect many people still believe it is. Secondly, don't think Saar fully understood what his "role" in this debate was. Instead of being a soldier fc side, he mostly stuck with his 2-3 year old argument for what he thought the "truth" was. Peter, the other hand, seemed eager to actually win the debate, so he made the maximally strong argument for his cause, not granting anything, as opposed to trying to converge on the "truth". for every piece of evidence, he was able to put on the "maximally pro zoonosis" spin, perhaps resulting in his absurdly low final lab leak probability.

It is perhaps interesting to think about how this compares to adversarial collaborations, which ar popular in our community. Using the language of Julia Galef, one can perhaps think of an advers collaboration as a "dance" between two scouts. Similarly, a debate can perhaps be thought of a battle between two soldiers. The incentive is to win, not to find the truth. The rootclaim debate looked like a battle between a soldier and a scout, with the predictable result.

REPLY S

EMSKE Phytochem Mar 30, 2024

(Disclosure that I support in BiosafetyNow, but these are my views and don't necesarily reflect those of BN)

The Twitter origins debate, which Scott understandably shies away from (ref. his most recent tweets, and ofc I certainly don't blame him) represented the richest, most boisterous most tumultuous, most rancorous, most acrimonious form of the origins debate I've ever encounterec online. The current ACX substack post & its comments don't reflect the Twitter debate's divers of opinions, evidence, and individual credibilities IMO.

The key items I feel are insufficiently highlighted (even if they are mentoned at least once) from scan of the comments section:

• the value of 'adverse inference' facing:

  ° PRC's gov

  ° the most outspoken virologists (+ 1 evolutionary biologist, + 1 disease ecologist, and yes not virologists) especially when relevant materials turned up from their side only following subpoen

• No progenitor or 'sister' virus with an MRCA dated to shortly before outbreak has been found i animal populations despite extensive sampling.

• I don't feel any post-outbreak (and some near-pre-outbreak) data out of the PRC should be accepted on faith; rather it & all conclusions drawn therefrom should be placed in its own little suspect epistemic box. By contrast, any data in non-PRC hands pre-outbreak is 'gold'. The remaining quadrant, data & analysis produced in the ordinary course ex-PRC but post-outbreak tainted by the need to avoid 'rocking the geopolitical boat'. Often we can't reference such data for example underlies post-outbreak intel findings, like that of the ODNI (atop DoE, FBI, CIA DI so I can only give it the 'silver' label treatment.

And transcending all these quadrants, verified data that turned up ex-PRC from leaks, FOIAs, sequences reconstructed from SDAs, etc. is 'platinum'.

REPLY (1) S

EMSKE Phytochem Apr 2, 2024

* ~~SDA~~s SRA's - Sequence Read Archive

REPLY S

Biorealism77 Mar 30, 2024

1. Peer reviewed papers since the debate that show ascertainment bias in early case data and intermediate genomes that undermine the multiple spillover theory indicating lineage A arose fir so the market cases are not the primary cases (Weissman 2024, Lv et al (2024), Stoyan and Chi (2024).

2. Miller's claims about the furin cleavage site are dubious:

Miller incorrectly claimed the N501Y mutation would result from passage in hACE2 mice (he mix them up with BALB/c mice). WIV was performing in vivo experiments in transgenic (human ACE: expressing) mice and civets in 2018 and 2019 in SARS-like CoVs.

I think both sides may have also overlooked the MERS furin cleavage site shares several structu and functional similarities. The sequence looks quite similar. In 2019 WIV researchers were invol in MERS research. Yusen Zhou who died in mysterious circumstances in May 2020 and was invo in an early Covid-19 vaccine was also involved in this research that involved manipulating the FC t.co/7zcSUPR60T

Broad Institute biologist Alina Chan also observes the S1/S2 FCS PRRA insertion in SARS-CoV-2 generates a Class IIS restriction enzyme site (BsaXI). This was used by WIV and Ralph Baric at U previously.

Dr Andreas Martin Lisewski discusses similarities with a MERS infectious clone described in 201 here. The argument no one would engineer an FCS like this overlooks the MERS example.

t.co/fAVUlJu0TK

REPLY (1) S

Scott Alexander Mar 31, 2024

Author

You've posted approximately this same comment three times on here now, plus five or six t on Twitter. If you post it further, I will ban you.

REPLY S

Mike Rhodes Mar 30, 2024

I think it's wrong to characterize this as a tragedy for Saar and Rootclaim. Sure the contest didn validate their methods -- but I think their purpose was bigger than just winning internet argumen Their goal is basically furthering scientific knowledge. Every model and method is flawed, but so are still useful. I see this as a heroic story, and Saar's stubbornness doesn't negate his accomplishment. By gracefully hosting and funding this exercise, he's made us all better off and advanced our knowledge of important issues. And that is the most important thing.

REPLY S

Jon Mar 30, 2024

So the Chinese want us to think that they caused a global pandemic?

Although not infallible, humans are actually pretty good at ascertaining the intentions of enemie and potential enemies, and that is why it is a universal human inclination. Being oblivious to the likely motivations and intentions of other people is a big evolutionary disadvantage.

REPLY S

Jesse Mar 30, 2024

> Newtonian mechanics wasn't pseudoscience when Newton discovered it, but if someone argu for it today (against relativity), that would be pseudoscientific.

I disagree strongly with the ontological assumptions here. Newtonian mechanics and relativity a both great theories. Newtonian mechanics is superior for terrestrial, human-scale phenomena because it provides a simpler representation of the physical laws. Relativity is superior at speed approaching c because the complexity is justified by the accuracy gained. To claim that Einsteir Newton for modeling the trajectory of a baseball would be the same error as saying that Newtor Einstein for modeling a black hole: both arguments would be domain-of-applicability errors.

I self-identify as a flat earther, because for 99% of daily life, the flat earth theory (viz., conceptualizing the Earth as a flat plane) is simpler and more useful than conceptualizing it as a globe. The error made by "actual" flat Earthers is not that their theory is bad - they have an excellent and very useful theory - it's that they misunderstand their theory's domain of applicab Flat Earth theory gets you to the grocery store; global earth theory gets you to the moon.

REPLY (1) S

Jesse Mar 30, 2024

To state my core point a bit more explicitly: the "best" theory (for any particular purpose/problem) is that which most simply encodes all information that's relevant (to the particular thing being studied). In some cases this is Newtonian mechanics; in others, it's relativity.

REPLY

S

David Manheim Mar 31, 2024

Quoting a comment I made on a lesswrong:

I think most points here are good points to make, but I also think it's useful as a general caution against this type of exercise being used as an argument at all! So I'd obviously caution against anyone taking your response itself as a reasonable attempt at an estimate of the "correct" Baye factors, because this is all very bad epistemic practice! Public explanations and arguments are social claims, and usually contain heavily filtered evidence (even if unconsciously). Don't do this public.

That is, this type of informal Bayesian estimate is useful as part of a ritual for changing your owr mind, when done carefully. That requires a significant degree of self-composure, a willingness t change one's mind, and a high degree of justified confidence n your own mastery of unbiased reasoning.

Here, though, it is presented as an argument, which is not how any of this should work. And in tl case, it was written by someone who already had a strong view of what the outcome should be, repeated publicly frequently, which makes it doubly hard to accept the implicit necessary claim it was performed starting from an unbiased point at face value! At the very least, we need stron evidence that it was not an exercise in motivated reasoning, that the bottom line wasn't written before the evaluation started - which statement is completely missing, though to be fair, it woul unbelievable if it had been stated.

https://www.lesswrong.com/posts/NrKQGyggC7jcersuJ/on-coincidences-and-bayesian-reasoni as-applied-to-the?commentId=HnxnDuqwucFAsG6bP

REPLY

S

Shane Apr 1, 2024

Counterposition here- given the verified attempts of multiple authorities to shape the "narrative around covid, plus the powerful incentives for them to do so, does this mean that the sum of evidence available to assess the situation is probably unreliable and unfit for the purpose of this debate? The added layer of information control typical of the Chinese government makes the situation even more opaque.

Maybe this isn't a debate which can be settled based on the available information. Maybe that is design.

The real question for me is this- why is any of this important in the long run? The obvious answe that it has some bearing on how the world might respond to future pandemics. If it is yet anothe zoonoses then all the efforts to date to prevent them didn't prevent this one and probably wont prevent the next one, so maybe we should attempt to apply democratic pressure to make governments do more. I'm not sure they can do much more either way.

If laboratory research is behind covid, then this is also not completely unprecedented (though tl level of genetic engineering might be somewhat new). Again, the technology isnt something tha can be easily regulated by governments. Nucleotide sequence synthesis is relatively centrally controlled by a few large companies with the technical capacity, but that is only one of many wa to generate genetically modified pathogens. Again, we could attempt to apply democratic press to democratic governments, but again that would likely have limited impact.

The only thing that really shines through with the whole covid affair was the government and he authority overreach, panic, incompetence and opportunism. Once again none of this is new. The corporate capture of the medical system has been progressing for decades. At best the whole c affair is just another reminder that authorities do not often have the interests of the individual as their main priority, and even when they do their capacity to deliver functional outcomes is usuall highly limited. All trends point to this situation worsening in the future.

Novel pathogens will likely always remain potential black swans, which means that most of the t authorities will over react to their emergence relative to the actual risk posed. Over time this will probably build up broad social resistance to government responses until a genuinely threatening pandemic emerges. The relative difference in perceived threat between pathogens that are nove and fast moving versus those that are familiar and slow (such as tuberculosis) is also worth contemplating. Our information ecosystem thrives on novelty. Nobody is going to buy a newspa that reports the same steady rate of slow deaths from tuberculosis or malaria. No government minister is going to win the fanatical devotion of the masses by heroically fighting such unexcitir killers.

REPLY S

David Bahry Apr 1, 2024 *Edited*

>"Peter seems to have a photographic memory for every detail of every study he’s ever read. H has some kind of 3D model in his brain of Wuhan, the wet market, and how all of its ventilation ducts and drains interacted with each other."

Mostly I think it's just keeping up with the Standard Arguments™ , from papers and preprints by crew (Holmes, Andersen, Garry, Rambaut, Worobey, Pekar, Rasmussen, Débarre, Crits-Christop Neil etc.—the "Proximal origin" authors and their collaborators). They're the group of Western virologists who ~yearly come out with a new paper finally proving market zoonosis for real this t (Andersen et al. 2020, Holmes et al. 2021, Worobey et al. 2022, Pekar et al. 2022, Crits-Christo al. 2023a, Crits-Christoph et al. 2023b, etc.) and get a ~yearly media circus about it. Maybe he read some pop-sci books like Chan & Ridley's "Viral" for pro-lab (2021) and Quammen's

"Breathless" for pro-zoo (2023). Probably also the Chinese HSM-sampling paper (Liu et al., 202 and WHO-China joint study (2021), following zoo crew in how to interpret them.

E.g. Peter's heat map slide is just Fig. 1 of (Crits-Christoph et al., 2023a), which is an elaboratio Fig. 4 of (Worobey et al., 2022); and his "drains" argument is from Fig. 2C of (Crits-Christoph et 2023b).*

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*The heat map is misleading: it was based on implausibly assuming all sampled stalls were sam equally; in fact the wildlife-corner stalls were sampled far more heavily (Liu et al., 2023). For a critique of dismissing ascertainment bias, see my (Bahry, 2023). For a critique of Crits-Christop

al. 2023a, see Bloom (2023). For a critique of their later, also-misleading heatmap (Crits-Christ
et al. 2023b Fig. 2a), see (https://www.biorxiv.org/content/10.1101/2023.09.13.557637v1#comm
6279754825).

I find the "drains" argument mildly interesting (Crits-Christoph et al., 2023b Fig. 2c): nonzero evidence, but just one real datapoint. (The two downstream of the racoon dog stall are *in gene downstream of a big chunk of the western wing and the entire western wing respectively, so thc mean little.])

REPLY (1)

deusexmachina Apr 3, 2024

I guess I don't quite understand your point. Scott points out that Peter has an eidetic memc for the arguments in this debate, and you respond: "Well, he just memorized the studies an arguments, especially those that support his view."

Aren't you saying the same thing as Scott? Surely, we wouldn't expect Peter to actually aut the papers himself. So what else is there to do than defending his position using other peo work?

And if Peter's understanding of the research is actually much shallower than it looks to a layperson (which is I think what you're saying), that seems fairly embarassing for Rootclair no?

REPLY (1)

David Bahry Apr 3, 2024 *Edited*

My point is that Peter was well-prepared—but it was the high end of the normal range preparedness for someone who'd been following the topic, not savant-level (compare Scott's "I can't imagine how many mutations it would take to make me even a fraction competent as Peter was").

E.g. to make the drains argument, he didn't need a working 3D model of every brick ar rivet in the market; he just needed to follow zoo crew on Twitter, see one of them share Crits-Christoph et al. 2023b, read it make the drains argument, and see its Fig. 2C.

Note that this comment isn't intended to disparage Peter, just to bring things down to Earth—and to give some pointers on where the arguments came from so they can be s in their original publications, so readers can evaluate them further, etc.

(It is meant to disparage zoo crew though lol)

REPLY (1)

David Bahry Apr 3, 2024

I agree that Rootclaim should have been more prepared

REPLY

S

Michael Weissman Apr 2, 2024

Scott- I can roughly translate my factors into the grouping you use in your table.

Net priors using your grouping)

me: 12 you: 16

combo of all HSM/lineage factors

me: 1 you 0.002

FCS-ish

me 25 you 25

reasons WIV wouldn't do it

me: in priors you: 0.17

cover up success

you 0.5 me: did it?

other factor restriction enzyme pattern

me 70 you NA

The RE pattern is new, and I think you'll find my argument at least reasonable.

The big disagreement is about Worobey/Pekar. If I were to argue for ZW, I'd stay the hell away fr Pekar, whose errors are now verging on something worse than "wrong".

Worobey does correctly point out that there was a spreading event at HSM, but so many specifi (lineage, internal DNA-RNA correlations, Wuhan share of relevant species (if any) don't fit that Z has to rely on the non-market versions, which actually were generally considered before hand t have more of the priors than a market version- including by DEFUSERs.

REPLY (1)

S

Michael Weissman Apr 2, 2024

Why don't I then get stronger LL odds? There's a lot of uncertainty in the priors. I do "robus Bayes" to take that into account, integrating over a broad fat-tailed distribution of priors. O tail keeps ZW in the game. Changing likelihood factors makes much less difference in the fi odds than you might expect, because the enormous likelihood ratio has already been dilute down by the uncertain priors.

REPLY S

Catmint Apr 3, 2024

Lots of people in the comments seem to have additional evidence one way or another that they want to add in. Maybe we need a coronavirus version of TalkOrigins.

REPLY S

Cjw Apr 3, 2024

I just cannot get myself to the point where I take anything being stated about this as reliable. Fo 3 years, everything that was being said about COVID by "experts" was either on what Zvi calls Simulacra level 2 (the public health establishment; saying X, despite ~X, to make you believe X a act in a way they want you to) or on what he calls Simulacra level 3 (Twitter users saying someth not because THEY believe it's true, or even that they expect YOU to believe it, but just to signal they're good non-racist Science-lovers). So even while I find Peter's points as you've summariz them to be pretty reasonable, he hurts himself IMHO by making the characterization of lab leak "conspiracy theory", because accusing an opponent of that also happens to be great tribal signaling. And the data on which all of this is premised comes from sources like the WHO which clearly willing to lie for a variety of reasons.

So despite finding Peter's arguments here pretty reasonable, I can't get over the hump. I also, w as dishonest as the PH establishment, would be inclined to just say it was a lab leak to induce p fear of gain-of-function research to help eliminate it. I can live with that having been wrong for t one virus, but still wanting to ban such research because next time it may NOT be wrong, but si most people don't reason that way then zoonosis being true is quite a blow to efforts to end this dangerous stuff. So let's give public health a dose of its own medicine and keep a nice healthy percentage on lab leak regardless.

REPLY (1) S

Nyctereutes Dec 1, 2025 *Edited*

Note that your characterisation of the data having been sourced from the WHO is incorrect they were actually sourced from the Chinese government, not the WHO. The WHO experts were not allowed to do any independent research during their short trip to Wuhan, with mo: their time being consumed with irrelevant activities.

REPLY

Quantum Quokka Apr 4, 2024 *Edited*

Even after this excellent post, I find Michael Weissman's v5.7 analysis from March (https://michaelweissman.substack.com/p/an-inconvenient-probability-v57) to be the most compelling information available online to date, and it contains significant evidence updates fro his v5.6 post that you linked in the write-up from January. I've kept an open mind through the ye about Covid origins and found the Rootclaim debate to be especially interesting and cause for shifting my probabilities, but I have not encountered anything that has so significantly updated ı probabilities than Michael's latest v5.7 analysis does. I'm curious if people have read that analys full and what everyone's thoughts are.

You seem to initially be interested specifically in Michael's Bayesian probabilities to compare his results to your table (which I think he posted recently in the comments above). I understand tha you likely have dedicated dozens of hours to this post already and reading through a massive updated analysis is likely not especially appealing, but just looking at the probabilities alone doe disservice to the immense amount of research, evidence, and analysis that is included in Micha write-up. For what it's worth, I've read an incredible amount of high-effort research and debate this topic throughout years of following this, and Michael's research/analysis is the first one tha find convincing. In addition to more convincing Bayesian logic, his research includes crucial compelling evidence/information that came out after the debate (or was available but dismissed potentially misrepresented at the time of the debate). I am trying to be as fair to both sides as possible and I honestly can't see how someone can read Michael's v5.7 post in full and come av with a 90% probability of Zoonosis. In my mental model, the two most likely reasons are that eit people haven't read Michael's 5.7 analysis due to low exposure of a small blog, or I am fundamentally misunderstanding something.

Either way, I want to know what people think. If Michael's post isn't as convincing as it seems to I'd like to know why so I can update. If it is convincing, I'd like others to know so they can updat @Scott, I historically have placed a lot of value on your opinions and I'm genuinely interested in what your thoughts are here.

REPLY (1)

Nyctereutes Dec 1, 2025

I agree about Weissman's blog, but have you seen this from January 2025? It focusses on j 4 factors deemed to be least subjective by the author:

https://www.nber.org/system/files/working_papers/w33428/w33428.pdf

REPLY S

JimB2 Apr 4, 2024

Thankyou for writing this up.

Personally, I don't have a lot of emotional/ideological investment in the issue so wouldn't sit thrc the debate but it great to get a detailed enough write up of the main arguments and their proble I guess I should be over it, but I remain amazed by the level and vehemence of the responses he and elsewhere. Lab leaks remain possible whether or not it happened here so there's an ongoin risk anyway. There are plenty of risks. In the long term, I'm actually more worried about the intentional release of an engineered virus than an accidental leak.

REPLY S

Biorealism77 Apr 8, 2024

The Rootclaim response in case it hasn't already been shared. Although the main issue to me is several papers have superseded the key papers relied on for the debate anyway. https://blog.rootclaim.com/covid-origins-debate-response-to-scott-alexander/

REPLY S

Easton Smith Apr 15, 2024 *Edited*

Peter had more at stake than Saar.

100k is about 0.05% of Saar's net worth. I assume it is a much much larger share of Peter's net worth. More to lose. More to gain.

The incentive for debate performance is massively skewed toward Peter in this case. Worth noti when analyzing the truth-seeking utility of this format.

Sorry if this is noted elsewhere, it is a big comment section.

REPLY (2) S

David Walker May 9, 2024

Big comment section? Nah, it's barely 100,000 words. You should be able to knock it off in day if you don't break for lunch.

REPLY S

Nyctereutes Dec 1, 2025

The debate for $100k format is also clearly flawed, in that it presupposes that only people \
a spare $100k can validly debate a historical epidemiological issue.

REPLY S

Groke Toffle Apr 17, 2024

This was such an interesting read. There are a lot of smart people here in the comments continu the discussion, but I just wanted to chime in and say THANK YOU for writing something so compelling to read for somebody (me) who has no prior knowledge or real opinion on the subject(s). Legend ⭐

REPLY S

Bob Jones Jun 4, 2024

I believe it is absolutely impossible that an animal or animal specimen transported to Wuhan lab study caused the pandemic.

I only believe that an animal or animal specimen transported to the Wuhan wet market to be EAT could have possibly caused the pandemic. Obviously.

I believe the Chinese govt is LYING when they say no pangolins, bats or racoon dogs were sold a the market at that time. I believe the Chinese govt is LYING that the only COVID found in the ma were on fish, who can't get COVID .

I absolutely believe the Chinese government when they say with exact certitude when and wher the earliest cases were, so I can see my dots on the map. The cases neatly surround the fish on map where the COVID was found.

I believe the Chinese govt when they say they took down their virus database in Sept because o fear of hackers. Cause you know, hacking had just been invented. I also believe the Chinese gov when they say they increased safety protocols at Wuhan just because. I understand the Chinese govt blocking attempts to get to the bottom of this and silencing doctors and scientist who tired warn the world.

I believe the former head of the CDC, the FBI and the Department of Energy are all wrong on the beliefs of the Covid orgin.

REPLY (1) S

Nyctereutes Dec 1, 2025

See my latest comment, the whole debate did not take into account the credibility of the Chinese authorities as a witness, which is ridiculous, given how unreliable Chinese governn data about even the most innocent things are.

The only logical approaches are either to trust everything China said, in which case it came from frozen food, or only trust verifiable data, i.e., none of the case data upon which the ma origin relies.

REPLY S

Grant Apr 15, 2025

Like other commenters here, I'm perplexed as to why serialized passage wasn't discussed? Someone said it wouldn't have worked in humanized mice. Alright, how about another animal?

If a new virus appears right next to New Virus Research lab, but we don't see how it could have been created with known techniques, that doesn't mean zoonosis. It might just mean New Virus Research Lab used something new or unusual in its creation.

REPLY S

Jonathan Ray Jul 27, 2025

so even peter's priors estimate an almost 1% chance per year that WIV starts a pandemic with a leak? That's pretty atrocious. If they had to buy insurance against that do do gain of function research, there'd be no gain of function research.

REPLY S

Evan Totten Oct 28, 2025

I'm not sure how much I trust rootclaim solely given their prediction of Usain Bolt not using PED Everyone dopes. Everyone. The sub-pro runners that are getting to the trials but not making the olympics are doping. Even if only one person was doping, we would know the winner was dopin BECAUSE THE PERSON WHO DOPES WINS. Idk about the rest of rootclaim, but this is a braind take for them to take.

REPLY (1) S

Evan Totten Oct 28, 2025

This was a very unprofessional comment, I try to only stick to logic. However, as a runner, t claim was so ridiculous that it got me a little bit annoyed.

REPLY S

Nyctereutes Nov 30, 2025

Witness Credibility Analysis: PRC Government

The analysis in the debate, though valuable, does not take account the veracity of the data prov by government authorities in the China/WHO report and other sources. Certain datasets, such a the zero infections amongst 600+ WIV staff, a statistical impossibility, and no animals traded at market during late 2019 are universally acknowledged to be fabrications. If not, both the zoonos and lab leak theories would have been conclusively dismissed, and everyone would believe the frozen food origin promoted by the Chinese government. Clearly, Chinese authorities had no problem with lying to the WHO about critical data that they considered inconvenient, so a long-standing legal principal applies, “falsus in uno, falsus in omnibus”.

https://www.oxfordreference.com/display/10.1093/acref/9780195369380.001.0001/acref-9780195369380-e-764

Knowing that two important datasets in the China/WHO report were entirely false means we shc not place great faith on any other dataset in the same report, absent independent corroboratior Any arguments based on data provided by a witness who had already lied about two important matters would be torn to shreds in a court case, and the origins investigation should be no diffe The market origin theory greatly depends on the accuracy of the case data, for example the spa clustering around the market and the Wuhan CDC in the next block, and many of the earliest ca: being at the market, but unfortunately, the purpose of the report they were sourced from was tc make the Chinese government look good, not to pursue scientific truth, and these unverified da were not reliable, as demonstrated below. If the virus was zoonotic, it did not originate at the market.

Case Study: Case Data

A case study of the accuracy of the earliest cases enables us to estimate how accurate the whc case dataset was. For the 8th December non-market-linked index case in the China/WHO repor Worobey concluded the date was incorrect, based on a local TV station report with an anonymiz interview with someone who claimed to be the index patient, but had originally visited the hospi on 8th December for a dental problem. During their visit to China, the WHO team interviewed someone that China said was the index patient and found nothing untoward, so apparently, a loc TV station was better at epidemiology than China's top scientists and the WHO experts who interviewed the patient. As this seems unlikely, the conclusion is that authorities lied to the WH(about the case data, and even got an actor to pretend to be the index patient for the WHO interview.

The first version of the case data on 30th January 2020 had a December 1st non-market-linked index case:

https://www.thelancet.com/article/S0140-6736(20)30183-5/fulltext#:~:text=discontinuation%20of%20isolation.-,Data%20collection,-

We%20reviewed%20clinical

The large team of highly-qualified authors took comprehensive measures to ensure data accura as listed in the paper, but despite this, the December 1st case was dropped from subsequent datasets. This paper included a further 2 non-market linked cases on 10th December, which we also apparently mistakes, so, including the 8th December case, 4/5 of the first reported cases turned out to be incorrect. In the absence of independent corroborating data, there is no reasor suspect that the other 170 or so of the early cases were any more correct, and it would be wron draw any conclusions from such data.

Coda: Including Timing in Case Map as well as Location Indicates Market Origin Unlikely

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Notwithstanding the above witness credibility assessment, it is interesting to analyse the publisl data, to draw appropriately caveated, cautious conclusions. A major problem with the Worobey (2022) spatial analysis claiming to have proven the origin based on clustering around the Wuhar CDC/Market was that it only included locations, not timings. The classical Snow (1866) study of cholera cases in clustered around a water pump was valid as cholera is caught from contaminat water, a static source, whereas after a handful of zoonotic jumps, all COVID-19 cases were cauç from humans circulating around the city, not from the handful of racoon dogs at the market that were the alleged source. Noting that case transmission was in almost all cases from infected humans, and taking into account case timing, Levin (2025)'s spaciotemporal analysis suggestec market origin was unlikely:

https://www.nber.org/system/files/working_papers/w33428/w33428.pdf

REPLY S

Nyctereutes Feb 17

"Either way, you're accepting that a 1-in-10,000 freak coincidence happened. Isn't it more likely you've bungled your analysis?" (FCS versus first cases at market coincidences).

To resolve this apparent contradiction, observe that the source of the 1/10000 freak coincidenc favour of the natural origin is case data published by Chinese politicians who were trying to mak themselves look good, so it has, at best, the same chance of being correct as other COVID data published by Chinese politicians, such as the death count, i.e. approximately 0% chance of actu being correct.

On the other hand, the source for the 1/10000 freak coincidence in favour of lab leak is the genc itself, so is 100% likely to be correct.

So, the bungled step of the analysis is assuming that the case data published by Chinese politic is accurate enough to draw any conclusions from. Since that is the only factor in favour of natur origin, it's obvious what the source was, once you take into account the reliability of the data.

REPLY S